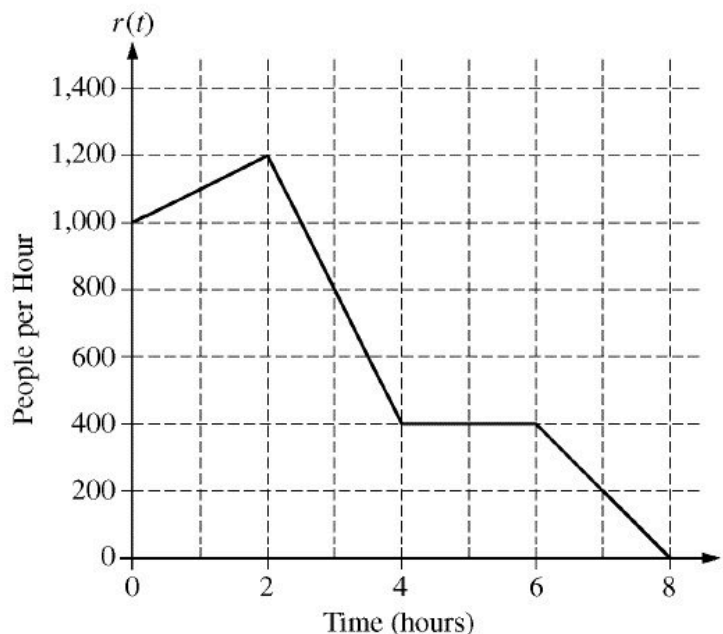


4-1-1

1.



There are 700 people in line for a popular amusement-park ride when the ride begins operation in the morning. Once it begins operation, the ride accepts passengers until the park closes 8 hours later. While there is a line, people move onto the ride at a rate of 800 people per hour. The graph above shows the rate, $r(t)$, at which people arrive at the ride throughout the day. Time t is measured in hours from the time the ride begins operation.

- How many people arrive at the ride between $t = 0$ and $t = 3$? Show the computations that lead to your answer.
- Is the number of people waiting in line to get on the ride increasing or decreasing between $t = 2$ and $t = 3$? Justify your answer.
- At what time t is the line for the ride the longest? How many people are in line at that time? Justify your answers.
- Write, but do not solve, an equation involving an integral expression of r whose solution gives the earliest time t at which there is no longer a line for the ride.

Part A

1 point is earned for integral

1 point is earned for the answer

$$\int_0^3 r(t) dt = 2 \cdot \frac{1000 + 1200}{2} + \frac{1200 + 800}{2} = 3200 \text{ people}$$



0	1	2
---	---	---

4-1-1

The student response earns two of the following points:

1 point is earned for integral

1 point is earned for the answer

$$\int_0^3 r(t) dt = 2 \cdot \frac{1000 + 1200}{2} + \frac{1200 + 800}{2} = 3200 \text{ people}$$

Part B

1 point is earned for answer with reason

The number of people waiting in line is increasing because people move onto the ride at a rate of 800 people per hour and for $2 < t < 3$, $r(t) > 800$.



0	1
---	---

The student response earns all of the following points:

1 point is earned for answer with reason

The number of people waiting in line is increasing because people move onto the ride at a rate of 800 people per hour and for $2 < t < 3$, $r(t) > 800$.

Part C

1 point is earned for identifying $t = 3$

1 point is earned for the number of people in line

1 point is earned for justification

$$r(t) = 800 \text{ only at } t = 3$$

For $0 \leq t < 3$, $r(t) > 800$. For $3 < t \leq 8$, $r(t) < 800$.

Therefore, the line is longest at time $t = 3$.

There are $700 + 3200 - 800 \cdot 3 = 1500$ people waiting in line at time $t = 3$.

4-1-1



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for identifying $t = 3$

1 point is earned for the number of people in line

1 point is earned for justification

$r(t) = 800$ only at $t = 3$

For $0 \leq t < 3$, $r(t) > 800$. For $3 < t \leq 8$, $r(t) < 800$.

Therefore, the line is longest at time $t = 3$.

There are $700 + 3200 - 800 \cdot 3 = 1500$ people waiting in line at time $t = 3$.

Part D

1 point is earned for $800t$

1 point is earned for integral

1 point is earned for the answer

$$0 = 700 + \int_0^t r(s)ds - 800t$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for $800t$

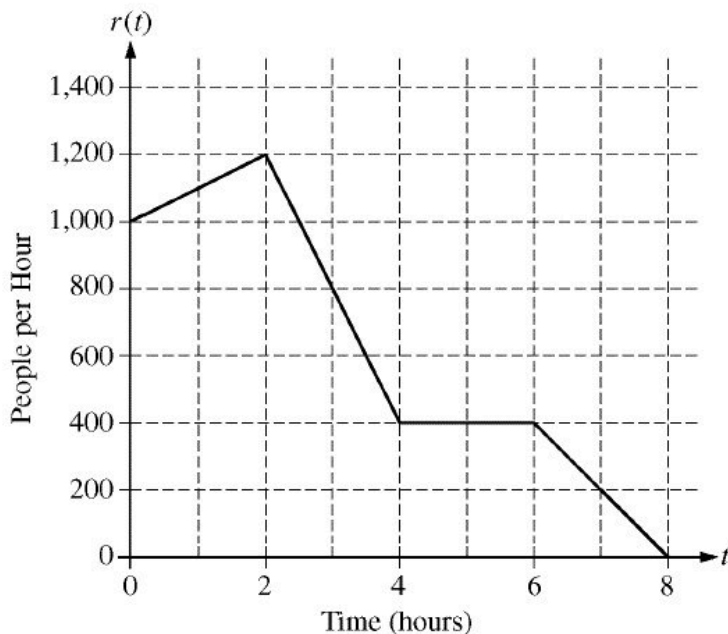
1 point is earned for integral

1 point is earned for the answer

4-1-1

$$0 = 700 + \int_0^t r(s)ds - 800t$$

2.



There are 700 people in line for a popular amusement-park ride when the ride begins operation in the morning. Once it begins operation, the ride accepts passengers until the park closes 8 hours later. While there is a line, people move onto the ride at a rate of 800 people per hour. The graph above shows the rate, $r(t)$, at which people arrive at the ride throughout the day. Time t is measured in hours from the time the ride begins operation.

Is the number of people waiting in line to get on the ride increasing or decreasing between $t=2$ and $t=3$? Justify your answer.

Part B

1 point is earned for answer with reason

The number of people waiting in line is increasing because people move onto the ride at a rate of 800 people per hour and for $2 < t < 3$, $r(t) > 800$.



0

1

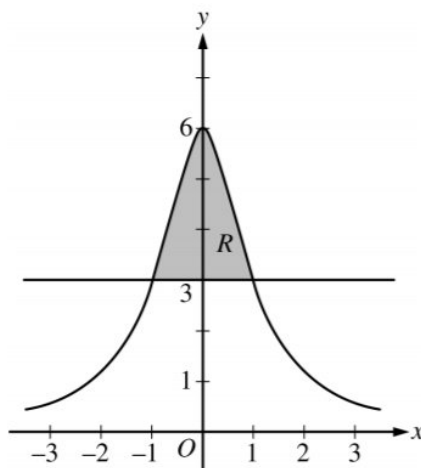
The student response earns all of the following points:

1 point is earned for answer with reason

The number of people waiting in line is increasing because people move onto the ride at a rate of 800 people per hour and for $2 < t < 3$, $r(t) > 800$.

4-1-1

3. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.



Let f be the function defined by $f(x) = \frac{6}{1+x^2}$. Let R be the shaded region bounded by the graph of f and the horizontal line $y = 3$, as shown in the figure above.

- Find the area of R .
- Write, but do not evaluate, an integral expression for the volume of the solid generated when R is rotated about the horizontal line $y = 7$.
- Let $h(x)$ be the vertical distance between the point $(x, f(x))$ and the horizontal line $y = 3$. Find the rate of change of $h(x)$ with respect to x at $x = 2$.

Part A**1 point is earned for:** integral**1 point is earned for:** antiderivative**1 point is earned for:** answer

$$f(x) = 3 \Rightarrow x = -1 \text{ and } x = 1$$

$$\begin{aligned} \int_{-1}^1 (f(x) - 3) \, dx &= \int_{-1}^1 \left(\frac{6}{1+x^2} - 3 \right) \, dx \\ &= (6 \tan^{-1} x - 3x) \Big|_{-1}^1 \\ &= (6 \tan^{-1} 1 - 3) - (6 \tan^{-1}(-1) + 3) \\ &= \left(6 \cdot \frac{\pi}{4} - 3 \right) - \left(6 \cdot \left(-\frac{\pi}{4} \right) + 3 \right) \\ &= 3\pi - 6 \end{aligned}$$

The area of R is $3\pi - 6$.

4-1-1



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for: integral

1 point is earned for: antiderivative

1 point is earned for: answer

$$f(x) = 3 \Rightarrow x = -1 \text{ and } x = 1$$

$$\begin{aligned} \int_{-1}^1 (f(x) - 3) \, dx &= \int_{-1}^1 \left(\frac{6}{1+x^2} - 3 \right) \, dx \\ &= (6 \tan^{-1} x - 3x) \Big|_{-1}^1 \\ &= (6 \tan^{-1} 1 - 3) - (6 \tan^{-1}(-1) + 3) \\ &= \left(6 \cdot \frac{\pi}{4} - 3 \right) - \left(6 \cdot \left(-\frac{\pi}{4} \right) + 3 \right) \\ &= 3\pi - 6 \end{aligned}$$

The area of R is $3\pi - 6$.

Part B

2 points are earned for: integrand

1 point is earned for: limits and constant

$$\text{Volume} = \pi \int_{-1}^1 ((7-3)^2 - (7-f(x))^2) \, dx$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for: integrand

1 point is earned for: limits and constant

$$\text{Volume} = \pi \int_{-1}^1 ((7-3)^2 - (7-f(x))^2) \, dx$$

4-1-1

Part C

1 point is earned for: $h(x)$

1 point is earned for: $h'(x)$

1 point is earned for: $h'(2)$

$$h(x) = 3 - f(x) = 3 - \frac{6}{1+x^2} \text{ for } x > 1$$

$$h'(x) = \frac{12x}{(1+x^2)^2} \text{ for } x > 1$$

$$h'(2) = \frac{12 \cdot 2}{5^2} = \frac{24}{25}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for: $h(x)$

1 point is earned for: $h'(x)$

1 point is earned for: $h'(2)$

$$h(x) = 3 - f(x) = 3 - \frac{6}{1+x^2} \text{ for } x > 1$$

$$h'(x) = \frac{12x}{(1+x^2)^2} \text{ for } x > 1$$

$$h'(2) = \frac{12 \cdot 2}{5^2} = \frac{24}{25}$$

4-1-1

4.  When a certain grocery store opens, it has 50 pounds of bananas on a display table. Customers remove bananas from the display table at a rate modeled by

$$f(t) = 10 + (0.8t) \sin\left(\frac{t^3}{100}\right) \text{ for } 0 < t \leq 12,$$

where $f(t)$ is measured in pounds per hour and t is the number of hours after the store opened. After the store has been open for three hours, store employees add bananas to the display table at a rate modeled by

$$g(t) = 3 + 2.4 \ln(t^2 + 2t) \text{ for } 3 < t \leq 12,$$

where $g(t)$ is measured in pounds per hour and t is the number of hours after the store opened.

- (a) How many pounds of bananas are removed from the display table during the first 2 hours the store is open?
- (b) Find $f'(7)$. Using correct units, explain the meaning of $f'(7)$ in the context of the problem.
- (c) Is the number of pounds of bananas on the display table increasing or decreasing at time $t = 5$? Give a reason for your answer.
- (d) How many pounds of bananas are on the display table at time $t = 8$?

Part A

1 point is earned for integral

1 point is earned for answer

$$\int_0^2 f(t) dt = 20.051175$$

20.051 pounds of bananas are removed from the display table during the first 2 hours the store is open.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for integral

1 point is earned for answer

$$\int_0^2 f(t) dt = 20.051175$$

20.051 pounds of bananas are removed from the display table during the first 2 hours the store is open.

Part B

4-1-1

1 point is earned for value

1 point is earned for meaning

$$f'(7) = -8.120 \text{ (or } -8.119)$$

After the store has been open 7 hours, the rate at which bananas are being removed from the display table is decreasing by 8.120 (or 8.119) pounds per hour per hour.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for value

1 point is earned for meaning

$$f'(7) = -8.120 \text{ (or } -8.119)$$

After the store has been open 7 hours, the rate at which bananas are being removed from the display table is decreasing by 8.120 (or 8.119) pounds per hour per hour.

Part C

1 point is earned for considers $f(5)$ and $g(5)$

1 point is earned for answer with reason

$$g(5) - f(5) = -2.263103 < 0$$

Because $g(5) - f(5) < 0$, the number of pounds of bananas on the display table is decreasing at time $t = 5$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for considers $f(5)$ and $g(5)$

1 point is earned for answer with reason

$$g(5) - f(5) = -2.263103 < 0$$

Because $g(5) - f(5) < 0$, the number of pounds of bananas on the display table is decreasing at time $t = 5$.

4-1-1

Part D

2 points are earned for integrals

1 point is earned for answer

$$50 + \int_3^8 g(t)dt - \int_0^8 f(t)dt = 23.347396$$

23.347 pounds of bananas are on the display table at time $t = 8$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for integrals

1 point is earned for answer

$$50 + \int_3^8 g(t)dt - \int_0^8 f(t)dt = 23.347396$$

23.347 pounds of bananas are on the display table at time $t = 8$.

4-1-1

5. When a certain grocery store opens, it has 50 pounds of bananas on a display table. Customers remove bananas from the display table at a rate modeled by

$$f(t) = 10 + (0.8t) \sin\left(\frac{t^3}{100}\right) \text{ for } 0 < t \leq 12,$$

where $f(t)$ is measured in pounds per hour and t is the number of hours after the store opened. After the store has been open for three hours, store employees add bananas to the display table at a rate modeled by

$$g(t) = 3 + 2.4 \ln(t^2 + 2t) \text{ for } 3 < t \leq 12,$$

where $g(t)$ is measured in pounds per hour and t is the number of hours after the store opened.



Find $f'(7)$. Using correct units, explain the meaning of $f'(7)$ in the context of the problem.

General

1 point is earned for: value

1 point is earned for: meaning

$$f'(7) = -8.120 \text{ (or } -8.119)$$

After the store has been open 7 hours, the rate at which bananas are being removed from the display table is decreasing by 8.120 (or 8.119) pounds per hour per hour.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for: value

1 point is earned for: meaning

$$f'(7) = -8.120 \text{ (or } -8.119)$$

After the store has been open 7 hours, the rate at which bananas are being removed from the display table is decreasing by 8.120 (or 8.119) pounds per hour per hour.

4-1-1

6. On a certain workday, the rate, in tons per hour, at which unprocessed gravel arrives at a gravel processing plant is modeled by $G(t) = 90 + 45\cos(t^2/18)$, where t is measured in hours and $0 \leq t \leq 8$. At the beginning of the workday ($t = 0$), the plant has 500 tons of unprocessed gravel. During the hours of operation, $0 \leq t \leq 8$, the plant processes gravel at a constant rate of 100 tons per hour.



(a) Find $G'(5)$. Using correct units, interpret your answer in the context of the problem.

Part A

1 point is earned for $G'(5)$

1 point is earned for the interpretation with units

$G'(5) = -24.588$ (or -24.587) The rate at which gravel is arriving is decreasing by 24.588 (or 24.587) tons per hour per hour at time $t = 5$ hours.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $G'(5)$

1 point is earned for the interpretation with units

$G'(5) = -24.588$ (or -24.587) The rate at which gravel is arriving is decreasing by 24.588 (or 24.587) tons per hour per hour at time $t = 5$ hours.

4-1-1

7.

t (minutes)	0	4	9	15	20
$W(t)$ (degrees Fahrenheit)	55.0	57.1	61.8	67.9	71.0

The temperature of water in a tub at time t is modeled by a strictly increasing, twice-differentiable function W , where $W(t)$ is measured in degrees Fahrenheit and t is measured in minutes. At time $t=0$, the temperature of the water is 55°F . The water is heated for 30 minutes, beginning at time $t=0$. Values of $W(t)$ at selected times t for the first 20 minutes are given in the table above.

Use the data in the table to estimate $W'(12)$. Show the computations that lead to your answer. Using correct units, interpret the meaning of your answer in the context of this problem.

Part A

1 point is earned for the estimate

1 point is earned for the interpretation with units

$$W'(12) \approx \frac{W(15) - W(9)}{15 - 9} = \frac{67.9 - 61.8}{6}$$

$$= 1.017 \text{ (or } 1.016)$$

The water temperature is increasing at a rate of approximately 1.0170 F per minute at time

$t=12$ minutes.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the estimate

1 point is earned for the interpretation with units

$$W'(12) \approx \frac{W(15) - W(9)}{15 - 9} = \frac{67.9 - 61.8}{6}$$

$$= 1.017 \text{ (or } 1.016)$$

The water temperature is increasing at a rate of approximately 1.017 $^\circ\text{F}$ per minute at time $t=12$ minutes.

4-1-1

t (days)	$W(t)$ (°C)
0	20
3	31
6	28
9	24
12	22
15	21

The temperature, in degrees Celsius (°C), of the water in a pond is a differentiable function W of time t . The table above shows the water temperature as recorded every 3 days over a 15-day period.

8. Use the data from the table to find an approximation for $W'(12)$. Show the computations that lead to your answer. Indicate units of measure.

Part A

1 point is earned for the difference quotient

1 point is earned for the answer (with units)

Difference quotient; e.g.

$$W'(12) \approx \frac{W(15) - W(12)}{15 - 12} = -\frac{1}{3}^{\circ}\text{C/day, or}$$

$$W'(12) \approx \frac{W(12) - W(9)}{12 - 9} = -\frac{2}{3}^{\circ}\text{C/day, or}$$

$$W'(12) \approx \frac{W(15) - W(9)}{15 - 9} = -\frac{1}{2}^{\circ}\text{C/day}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the difference quotient

1 point is earned for the answer (with units)

Difference quotient; e.g.

$$W'(12) \approx \frac{W(15) - W(12)}{15 - 12} = -\frac{1}{3}^{\circ}\text{C/day, or}$$

$$W'(12) \approx \frac{W(12) - W(9)}{12 - 9} = -\frac{2}{3}^{\circ}\text{C/day, or}$$

$$W'(12) \approx \frac{W(15) - W(9)}{15 - 9} = -\frac{1}{2}^{\circ}\text{C/day}$$

4-1-1

9. A student proposes the function P , given by $P(t) = 20 + 10te^{(-t/3)}$, as a model for the temperature of the water in the pond at time t , where t is measured in days and $P(t)$ is measured in degrees Celsius. Find $P'(12)$. Using appropriate units, explain the meaning of your answer in terms of water temperature.

Part C

1 point is earned for $P'(12)$ (with or without units)

1 point is earned for the interpretation

$$P'(12) = 10e^{-t/3} - \frac{10}{3}te^{-t/3} \Big|_{t=12}$$

$$= -30e^{-4} = -0.549^\circ\text{C/day}$$

This means the the temperature is decreasing at the rate of 0.549 C/day when $t = 12$ days.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $P'(12)$ (with or without units)

1 point is earned for the interpretation

$$P'(12) = 10e^{-t/3} - \frac{10}{3}te^{-t/3} \Big|_{t=12}$$

$$= -30e^{-4} = -0.549^\circ\text{C/day}$$

This means the the temperature is decreasing at the rate of 0.549 C/day when $t = 12$ days.

4-1-1

10. The rate at which people enter an amusement park on a given day is modeled by the function E defined by

$$E(t) = \frac{15600}{(t^2 - 24t + 160)}$$

The rate at which people leave the same amusement park on the same day is modeled by the function L defined by

$$L(t) = \frac{9890}{(t^2 - 38t + 370)}$$

Both $E(t)$ and $L(t)$ are measured in people per hour and time t is measured in hours after midnight. These functions are valid for $9 \leq t \leq 23$, the hours during which the park is open. At time $t = 9$, there are no people in the park.

 Let $H(t) = \int_9^t (E(x) - L(x))dx$ for $9 \leq t \leq 23$. The value of $H(17)$ to the nearest whole number is 3725. Find the value of $H'(17)$, and explain the meaning of $H(17)$ and $H'(17)$ in the context of the amusement park.

Part C

1 point is earned for value $H'(17)$

2 points are earned for the meanings

1: meaning of $H(17)$

1: meaning of $H'(17)$

<-1> if no reference to $t = 17$

$$H'(17) = E(17) - L(17) = -380.281$$

There were 3725 people in the park at $t = 17$. The number of people in the park was decreasing at the rate of approximately 380 people/hr at time $t = 17$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for value $H'(17)$

2 points are earned for the meanings

1: meaning of $H(17)$

1: meaning of $H'(17)$

4-1-1

<-1> if no reference to $t = 17$

$$H'(17) = E(17) - L(17) = -380.281$$

There were 3725 people in the park at $t = 17$. The number of people in the park was decreasing at the rate of approximately 380 people/hr at time $t = 17$.

11. The number of insects in a certain population at time t days is modeled by the function P with first derivative $P'(t) = 0.3t^2 + 12t + 210$. At time $t = 0$, the number of insects in the population is 40.
- I. At time $t = 10$, the number of insects in the population is 2840.
 - II. At time $t = 10$, the number of insects in the population is increasing at a rate of 360 insects per day.
 - III. At time $t = 10$, the rate of change of the number of insects in the population is increasing at a rate of 18 insects per day per day.

Which of the following statements are true?

- (A) I only
- (B) II only
- (C) III only
- (D) I, II, and III



12. There is no snow on Janet's driveway when snow begins to fall at midnight. From midnight to 9 A.M., snow accumulates on the driveway at a rate modeled by $f(t) = 7te^{\cos t}$ cubic feet per hour, where t is measured in hours since midnight. Janet starts removing snow at 6 A.M. ($t = 6$). The rate $g(t)$, in cubic feet per hour, at which Janet removes snow from the driveway at time t hours after midnight is modeled by

$$g(t) = \begin{cases} 0 & \text{for } 0 \leq t < 6 \\ 125 & \text{for } 6 \leq t < 7 \\ 108 & \text{for } 7 \leq t \leq 9. \end{cases}$$



Find the rate of change of the volume of snow on the driveway at 8 A.M

Part B

1 point is earned for rate of change or answer

Rate of change is $f(8) - g(8) = -59.582$ or -59.583 cubic feet per hour



0	1
---	---

The student response earns all of the following points:

1 point is earned for rate of change or answer

Rate of change is $f(8) - g(8) = -59.582$ or -59.583 cubic feet per hour

4-1-1

13. A 12,000-liter tank of water is filled to capacity. At time $t = 0$, water begins to drain out of the tank at a rate modeled by $r(t)$, measured in liters per hour, where r is given by the piecewise-defined function

$$r(t) = \begin{cases} \frac{600t}{t+3} & \text{for } 0 \leq t \leq 5 \\ 1000e^{-0.2t} & \text{for } t > 5 \end{cases}$$



Find $r'(3)$. Using correct units, explain the meaning of that value in the context of this problem.

Part C

1 point is earned for $r'(3)$

1 point is earned for meaning of $r'(3)$

$r'(3)=50$ The rate at which water is draining out of the tank at time $t=3$ hours is increasing at 50 liters/ hour².



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $r'(3)$

1 point is earned for meaning of $r'(3)$

$r'(3)=50$

The rate at which water is draining out of the tank at time $t=3$ hours is increasing at 50 liters/ hour².

4-1-1

14.

t (minutes)	0	1	2	3	4	5	6
$C(t)$ (ounces)	0	5.3	8.8	11.2	12.8	13.8	14.5

Hot water is dripping through a coffeemaker, filling a large cup with coffee. The amount of coffee in the cup at time t , $0 \leq t \leq 6$, is given by a differentiable function C , where t is measured in minutes. Selected values of $C(t)$, measured in ounces, are given in the table above.

The amount of coffee in the cup, in ounces, is modeled by $B(t) = 16 - 16e^{-0.4t}$. Using this model, find the rate at which the amount of coffee in the cup is changing when $t = 5$.

Part D

1 point is earned for $B'(t)$

1 point is earned for $B'(5)$

$$B'(t) = -16(-0.4)e^{-0.4t} = 6.4e^{-0.4t}$$

$$B'(5) = 6.4e^{-0.4(5)} = \frac{6.4}{e^2} \text{ ounces/min}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $B'(t)$

1 point is earned for $B'(5)$

$$B'(t) = -16(-0.4)e^{-0.4t} = 6.4e^{-0.4t}$$

$$B'(5) = 6.4e^{-0.4(5)} = \frac{6.4}{e^2} \text{ ounces/min}$$

4-1-1

15. A storm washed away sand from a beach, causing the edge of the water to get closer to a nearby road. The rate at which the distance between the road and the edge of the water was changing during the storm is modeled by $f(t) = \sqrt{t} + \cos t - 3$ meters per hour, t hours after the storm began. The edge of the water was 35 meters from the road when the storm began, and the storm lasted 5 hours. The derivative of $f(t)$ is $f'(t) = \frac{1}{2\sqrt{t}} - \sin t$.



Using correct units, interpret the value $f'(4) = 1.007$ in terms of the distance between the road and the edge of the water.

Part B

1 point is earned for interpretation of $f'(4)$

1 point is earned for units

Four hours after the storm began, the rate of change of the distance between the road and the edge of the water is increasing at a rate of 1.007 meters/ hours².



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for interpretation of $f'(4)$

1 point is earned for units

Four hours after the storm began, the rate of change of the distance between the road and the edge of the water is increasing at a rate of 1.007 meters/ hours².

4-1-1

The wind chill is the temperature, in degrees Fahrenheit ($^{\circ}\text{F}$), a human feels based on the air temperature, in degrees Fahrenheit, and the wind velocity v , in miles per hour (mph). If the air temperature is 32°F , then the wind chill is given by $W(v)=55.6-22.1v^{0.16}$ and is valid for $5 \leq v \leq 60$.

16.  Find $W'(20)$. Using correct units, explain the meaning of $W'(20)$ in terms of the wind chill.

Part A

1 point is earned for the value

1 point is earned for the explanation

$W'(20)=-22.1*0.16*20^{-0.84}=-0.285$ or -0.286 When $v=20$ mph, the wind chill is decreasing at $0.286^{\circ}\text{F}/\text{mph}$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the value

1 point is earned for the explanation

$W'(20)=-22.1*0.16*20^{-0.84}=-0.285$ or -0.286

When $v=20$ mph, the wind chill is decreasing at $0.286^{\circ}\text{F}/\text{mph}$.

4-1-1

17.  Over the time interval $0 \leq t \leq 4$ hours, the air temperature is a constant 32°F . At time $t=0$, the wind velocity is $v=20$ mph. If the wind velocity increases at a constant rate of 5 mph per hour, what is the rate of change of the wind chill with respect to time at $t=3$ hours? Indicate units of measure.

Part C

1 point is earned for $\frac{dv}{dt} = 5$

1 point is earned if uses $v(3)=35$ **or** uses $v(5)=20+5t$

1 point is earned for answer

1 point is earned if uses units in (a) and (c)

$$\left. \frac{dW}{dt} \right|_{t=3} = \left(\frac{dW}{dv} \cdot \frac{dv}{dt} \right) \Big|_{t=3}$$

$$=W'(35) \cdot 5 = -0.892 \text{ } ^{\circ}\text{F/hr}$$

OR

$$W = 55.6 - 22.1(20 + 5t)0.16$$

$$\left. \frac{dW}{dt} \right|_{t=3} = -0.892 \text{ } ^{\circ}\text{F/hr}$$

Units of $^{\circ}\text{F}/\text{mph}$ in (a) and $^{\circ}\text{F}/\text{hr}$ in (c)



0	1	2	3	4
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The student response earns four of the following points:

1 point is earned for the integrand

1 point is earned if uses $v(3)=35$ **or** uses $v(5)=20+5t$

1 point is earned for answer

1 point is earned if uses units in (a) and (c)

$$\left. \frac{dW}{dt} \right|_{t=3} = \left(\frac{dW}{dv} \cdot \frac{dv}{dt} \right) \Big|_{t=3}$$

$$=W'(35) \cdot 5 = -0.892 \text{ } ^{\circ}\text{F/hr}$$

OR

4-1-1

$$W = 55.6 - 22.1(20 + 5t)0.16$$

$$\left. \frac{dW}{dt} \right|_{t=3} = -0.892 \text{ oF/hr}$$

Units of oF/mph in (a) and oF/hr in (c)

18. The velocity v , in meters per second, of a certain type of wave is given by $v(h) = 3\sqrt{h}$, where h is the depth, in meters, of the water through which the wave moves. What is the rate of change, in meters per second per meter, of the velocity of the wave with respect to the depth of the water, when the depth is 2 meters?

(A) $-\frac{3}{4\sqrt{2}}$

(B) $-\frac{3}{8\sqrt{2}}$

(C) $\frac{3}{2\sqrt{2}}$



(D) $\frac{3}{\sqrt{2}}$

(E) $4\sqrt{2}$

19. The number of gallons of water in a storage tank at time t , in minutes, is modeled by $w(t) = 25 - t^2$ for $0 \leq t \leq 5$. At what rate, in gallons per minute, is the amount of water in the tank changing at time $t = 3$ minutes?

(A) 66

(B) 16

(C) -3

(D) -6



4-1-1

20. The rate of consumption of cola in the United States is given by $S(t) = Ce^{kt}$, where S is measured in billions of gallons per year and t is measured in years from the beginning of 1980.

The consumption rate doubles every 5 years and the consumption rate at the beginning of 1980 was 6 billion gallons per year. Find C and k .

Part A

1 point is earned for the correct answer $C = 6$

$$S(t) = Ce^{kt}$$

$$S(0) = 6 \Rightarrow C = 6$$

1 point is earned for the correct answer $12 = 6e^{5k}$

$$S(5) = 12 \Rightarrow 12 = 6e^{5k}$$

1 point is earned for the correct answer $k = \frac{\ln 2}{5}$

$$2 = e^{5k}$$

$$k = \frac{\ln 2}{5}$$

(.138 or .139)



0	1	2	3
---	---	---	---

The student response earns three of the following points:

1 point is earned for the correct answer $C = 6$

$$S(t) = Ce^{kt}$$

$$S(0) = 6 \Rightarrow C = 6$$

1 point is earned for the correct answer $12 = 6e^{5k}$

$$S(5) = 12 \Rightarrow 12 = 6e^{5k}$$

1 point is earned for the correct answer $k = \frac{\ln 2}{5}$

$$2 = e^{5k}$$

$$k = \frac{\ln 2}{5}$$

(.138 or .139)

4-1-1

21. For $t \geq 0$ hours, H is a differentiable function of t that gives the temperature, in degrees Celsius, at an Arctic weather station. Which of the following is the best interpretation of $H'(24)$?
- (A) The change in temperature during the first day
 - (B) The change in temperature during the 24th hour
 - (C) The average rate at which the temperature changed during the 24th hour
 - (D) The rate at which the temperature is changing during the first day
 - (E) The rate at which the temperature is changing at the end of the 24th hour ✓
22. A file is downloaded to a computer at a rate modeled by the differentiable function $f(t)$, where t is the time in seconds since the start of the download and $f(t)$ is measured in megabits per second. Which of the following is the best interpretation of $f'(5) = 2.8$?
- (A) At time $t = 5$ seconds, the rate at which the file is downloaded to the computer is 2.8 megabits per second. (B) (C)(D)
 - (B) At time $t = 5$ seconds, the rate at which the file is downloaded to the computer is increasing at a rate of 2.8 megabits per second per second. ✓
 - (C) Over the time interval $0 \leq t \leq 5$ seconds, 2.8 megabits of the file are downloaded to the computer.
 - (D) Over the time interval $0 \leq t \leq 5$ seconds, the average rate at which the file is downloaded to the computer is 2.8 megabits per second.
23. The height above the ground of a passenger on a Ferris wheel t minutes after the ride begins is modeled by the differentiable function H , where $H(t)$ is measured in meters. Which of the following is an interpretation of the statement $H'(7.5) = 15.708$?
- (A) The Ferris wheel is turning at a rate of 15.708 meters per minute when the passenger is 7.5 meters above the ground.
 - (B) The Ferris wheel is turning at a rate of 15.708 meters per minute 7.5 minutes after the ride begins.
 - (C) The passenger's height above the ground is increasing by 15.708 meters per minute when the passenger is 7.5 meters above the ground.
 - (D) The passenger's height above the ground is increasing by 15.708 meters per minute 7.5 minutes after the ride begins. ✓
 - (E) The passenger is 15.708 meters above the ground 7.5 minutes after the ride begins.

4-1-1

24. The rate at which water leaks from a tank, in gallons per hour, is modeled by R , a differentiable function of the number of hours after the leak is discovered. Which of the following is the best interpretation of $R'(3)$?
- (A) The amount of water, in gallons, that has leaked out of the tank during the first three hours after the leak is discovered
 - (B) The amount of change, in gallons per hour, in the rate at which water is leaking during the three hours after the leak is discovered
 - (C) The rate at which water leaks from the tank, in gallons per hour, three hours after the leak is discovered
 - (D) The rate of change of the rate at which water leaks from the tank, in gallons per hour per hour, three hours after the leak is discovered ✓
25. A tire that is leaking air has an initial air pressure of 30 pounds per square inch (psi). The function $t = f(p)$ models the amount of time t , in hours, it takes for the air pressure of the tire to reach p psi. What are the units for $f'(p)$?
- (A) hours
 - (B) psi
 - (C) psi per hour
 - (D) hours per psi ✓

Answer D

This option is correct. The derivative of the function $t = f(p)$ is the limit of the difference. The units on the dependent variable, t , in the numerator is hours and the units on the independent variable, p , in the denominator is psi. The rate of change, therefore, has units in hours per psi.

26. The function $P(t)$ models the population of the world, in billions of people, where t is the number of years since January 1, 2010. Which of the following is the best interpretation of the statement $P'(1) = 0.076$?
- (A) On February 1, 2010, the population of the world was increasing at a rate of 0.076 billion people per year.
 - (B) On January 1, 2011, the population of the world was increasing at a rate of 0.076 billion people per year. ✓
 - (C) On January 1, 2011, the population of the world was 0.076 billion people.
 - (D) From January 1, 2010 to January 1, 2011, the population of the world was increasing at an average rate of 0.076 billion people per year.
 - (E) When the population of the world was 1 billion people, the population of the world was increasing at a rate of 0.076 billion people per year.

4-1-1

27. On a certain day, the total number of pieces of candy produced by a factory since it opened is modeled by C , a differentiable function of the number of hours since the factory opened. Which of the following is the best interpretation of $C'(3) = 500$?
- (A) The factory produces 500 pieces of candy during its 3rd hour of operation.
 - (B) The factory produces 500 pieces of candy in the first 3 hours after it opens.
 - (C) The factory is producing candy at a rate of 500 pieces per hour, 3 hours after it opens. ✓
 - (D) The rate at which the factory is producing candy is increasing at a rate of 500 pieces per hour per hour, 3 hours after it opens.

Answer C

This option is correct. In the expression $C'(3)$, the 3 represents the value of the independent variable and is therefore the number of hours since the factory opened. $C'(3)$, being the value of a derivative, is the rate of change of C — i.e. the rate at which candy is being produced, in this case three hours after the factory opens. The units for the derivative would be the units of C per unit of time, hence pieces per hour.

28. The rate at which water leaks from a tank, in gallons per hour, is modeled by R , a differentiable function of the number of hours after the leak is discovered. Which of the following is the best interpretation of $R'(3)$?
- (A) The amount of water, in gallons, that has leaked out of the tank during the first three hours after the leak is discovered
 - (B) The amount of change, in gallons per hour, in the rate at which water is leaking during the three hours after the leak is discovered
 - (C) The rate at which water leaks from the tank, in gallons per hour, three hours after the leak is discovered
 - (D) The rate of change of the rate at which water leaks from the tank, in gallons per hour per hour, three hours after the leak is discovered ✓

4-1-1

29. Particle X moves along the positive x -axis so that its position at time $t \geq 0$ is given by $x(t) = 5t^3 - 9t^2 + 7$. Is particle X moving toward the left or toward the right at time $t = 1$? Give a reason for your answer.

Part A

The response can earn up to 2 points:

1 point: For consideration of $x'(1)$

1 point: Correct answer with reason.

$$x'(t) = 15t^2 - 18t$$

$$x'(1) = 15 - 18 = -3$$

Since $x'(1) < 0$, the particle is moving to the left at time $t = 1$



0	1	2
---	---	---

The response earns both of the following points:

1 point: For consideration of $x'(1)$

1 point: Correct answer with reason.

$$x'(t) = 15t^2 - 18t$$

$$x'(1) = 15 - 18 = -3$$

Since $x'(1) < 0$, the particle is moving to the left at time $t = 1$.

4-1-1

30. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

For $0 \leq t \leq 6$ seconds, a screen saver on a computer screen shows two circles that start as dots and expand outward.

(a) At the instant that the first circle has a radius of 9 centimeters, the radius is increasing at a rate of $\frac{3}{2}$ centimeters per second. Find the rate at which the area of the circle is changing at that instant. Indicate units of measure.

(b) The radius of the first circle is modeled by $w(t) = 12 - 12e^{-0.5t}$ for $0 \leq t \leq 6$, where $w(t)$ is measured in centimeters and t is measured in seconds. At what time t is the radius of the circle increasing at a rate of 3 centimeters per second?

(c) A model for the radius of the second circle is given by the function f for $0 \leq t \leq 6$, where $f(t)$ is measured in centimeters and t is measured in seconds. The rate of change of the radius of the second circle is given by $f'(t) = t^2 - 4t + 4$. Based on this model, by how many centimeters does the radius of the second circle increase from time $t = 0$ to $t = 3$?

Part A

1 point is earned for: $\frac{dA}{dt}$

1 point is earned for: answer

1 point is earned for: units

$$A = \pi r^2$$

$$\frac{dA}{dt} = 2\pi r \frac{dr}{dt}$$

$$\left. \frac{dA}{dt} \right|_{r=9} = 2\pi \cdot 9 \cdot \frac{3}{2} = 27\pi$$

When the radius is 9 centimeters, the area is changing at a rate of $27\pi \text{ cm}^2/\text{sec}$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for: $\frac{dA}{dt}$

1 point is earned for: answer

1 point is earned for: units

4-1-1

$$A = \pi r^2$$

$$\frac{dA}{dt} = 2\pi r \frac{dr}{dt}$$

$$\left. \frac{dA}{dt} \right|_{r=9} = 2\pi \cdot 9 \cdot \frac{3}{2} = 27\pi$$

When the radius is 9 centimeters, the area is changing at a rate of $27\pi \text{ cm}^2/\text{sec}$.

Part B

1 point is earned for: $w'(t)$

1 point is earned for: sets $w'(t) = 3$

1 point is earned for: answer

$$w'(t) = (-12)(-0.5)e^{-0.5t} = 6e^{-0.5t}$$

$$6e^{-0.5t} = 3 \Rightarrow e^{-0.5t} = \frac{1}{2} \Rightarrow -0.5t = \ln\left(\frac{1}{2}\right) \Rightarrow t = 2 \ln 2$$

The radius is increasing at a rate of 3 centimeters per second at time $t = 2 \ln 2$ seconds.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for: $w'(t)$

1 point is earned for: sets $w'(t) = 3$

1 point is earned for: answer

$$w'(t) = (-12)(-0.5)e^{-0.5t} = 6e^{-0.5t}$$

$$6e^{-0.5t} = 3 \Rightarrow e^{-0.5t} = \frac{1}{2} \Rightarrow -0.5t = \ln\left(\frac{1}{2}\right) \Rightarrow t = 2 \ln 2$$

The radius is increasing at a rate of 3 centimeters per second at time $t = 2 \ln 2$ seconds.

Part C

1 point is earned for: integral

1 point is earned for: antiderivative

1 point is earned for: answer

4-1-1

$$\begin{aligned}
 \int_0^3 (t^2 - 4t + 4) \, dt &= \left[\frac{1}{3}t^3 - 2t^2 + 4t \right]_{t=0}^{t=3} \\
 &= \left(\frac{1}{3} \cdot 3^3 - 2 \cdot 3^2 + 4 \cdot 3 \right) - \left(\frac{1}{3} \cdot 0^3 - 2 \cdot 0^2 + 4 \cdot 0 \right) \\
 &= 9 - 18 + 12 = 3
 \end{aligned}$$

The radius increases by 3 centimeters from time $t = 0$ to time $t = 3$ seconds.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for: integral

1 point is earned for: antiderivative

1 point is earned for: answer

$$\begin{aligned}
 \int_0^3 (t^2 - 4t + 4) \, dt &= \left[\frac{1}{3}t^3 - 2t^2 + 4t \right]_{t=0}^{t=3} \\
 &= \left(\frac{1}{3} \cdot 3^3 - 2 \cdot 3^2 + 4 \cdot 3 \right) - \left(\frac{1}{3} \cdot 0^3 - 2 \cdot 0^2 + 4 \cdot 0 \right) \\
 &= 9 - 18 + 12 = 3
 \end{aligned}$$

The radius increases by 3 centimeters from time $t = 0$ to time $t = 3$ seconds.

4-1-1

31.

t (minutes)	0	1	2	3	4	5	6
$C(t)$ (ounces)	0	5.3	8.8	11.2	12.8	13.8	14.5

Hot water is dripping through a coffeemaker, filling a large cup with coffee. The amount of coffee in the cup at time t , $0 \leq t \leq 6$, is given by a differentiable function C , where t is measured in minutes. Selected values of $C(t)$, measured in ounces, are given in the table above.

(a) Use the data in the table to approximate $C'(3.5)$. Show the computations that lead to your answer, and indicate units of measure.

(b) Is there a time t , $2 \leq t \leq 4$, at which $C'(t) = 2$? Justify your answer.

(c) Use a midpoint sum with three subintervals of equal length indicated by the data in the table to approximate the value of $\frac{1}{6} \int_0^6 C(t) dt$. Using correct units, explain the meaning of $\frac{1}{6} \int_0^6 C(t) dt$ in the context of the problem.

(d) The amount of coffee in the cup, in ounces, is modeled by $B(t) = 16 - 16e^{-0.4t}$. Using this model, find the rate at which the amount of coffee in the cup is changing when $t = 5$.

Part A

1 point is earned for the approximation

1 point is earned for units

$$C'(3.5) \approx \frac{C(4) - C(3)}{4 - 3} = \frac{12.8 - 11.2}{1} = 1.6 \text{ ounces/min}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the approximation

1 point is earned for units

$$C'(3.5) \approx \frac{C(4) - C(3)}{4 - 3} = \frac{12.8 - 11.2}{1} = 1.6 \text{ ounces/min}$$

Part B

4-1-1

1 point is earned for $\frac{C(4) - C(2)}{4 - 2}$

1 point is earned for the conclusion, using MVT

C is differentiable $\Rightarrow C$ is continuous (on the closed interval)

$$\frac{C(4) - C(2)}{4 - 2} = \frac{12.8 - 8.8}{2} = 2$$

Therefore, by the Mean Value Theorem, there is at least one time t , $2 < t < 4$, for which $C'(t) = 2$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $\frac{C(4) - C(2)}{4 - 2}$

1 point is earned for the conclusion, using MVT

C is differentiable $\Rightarrow C$ is continuous (on the closed interval)

$$\frac{C(4) - C(2)}{4 - 2} = \frac{12.8 - 8.8}{2} = 2$$

Therefore, by the Mean Value Theorem, there is at least one time t , $2 < t < 4$, for which $C'(t) = 2$.

Part C

1 point is earned for the midpoint sum

1 point is earned for the approximation

1 point is earned for the interpretation

4-1-1

$$\begin{aligned}\frac{1}{6} \int_0^6 C(t) dt &\approx \frac{1}{6} [2 \cdot C(1) + 2 \cdot C(3) + 2 \cdot C(5)] \\ &= \frac{1}{6} (2 \cdot 5.3 + 2 \cdot 11.2 + 2 \cdot 13.8) \\ &= \frac{1}{6} (60.6) = 10.1 \text{ ounces}\end{aligned}$$

$\frac{1}{6} \int_0^6 C(t) dt$ is the average amount of coffee in the cup, in ounces, over the time interval $0 \leq t \leq 6$ minutes.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the midpoint sum

1 point is earned for the approximation

1 point is earned for the interpretation

$$\begin{aligned}\frac{1}{6} \int_0^6 C(t) dt &\approx \frac{1}{6} [2 \cdot C(1) + 2 \cdot C(3) + 2 \cdot C(5)] \\ &= \frac{1}{6} (2 \cdot 5.3 + 2 \cdot 11.2 + 2 \cdot 13.8) \\ &= \frac{1}{6} (60.6) = 10.1 \text{ ounces}\end{aligned}$$

$\frac{1}{6} \int_0^6 C(t) dt$ is the average amount of coffee in the cup, in ounces, over the time interval $0 \leq t \leq 6$ minutes.

Part D

1 point is earned for $B'(t)$

1 point is earned for $B'(5)$

$$\begin{aligned}B'(t) &= -16(-0.4)e^{-0.4t} = 6.4e^{-0.4t} \\ B'(5) &= 6.4e^{-0.4(5)} = \frac{6.4}{e^2} \text{ ounces/min}\end{aligned}$$

4-1-1



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $B'(t)$

1 point is earned for $B'(5)$

$$B'(t) = -16(-0.4)e^{-0.4t} = 6.4e^{-0.4t}$$

$$B'(5) = 6.4e^{-0.4(5)} = \frac{6.4}{e^2} \text{ ounces/min}$$

4-1-1

32.  The rate of consumption of cola in the United States is given by $S(t) = Ce^{kt}$, where S is measured in billions of gallons per year and t is measured in years from the beginning of 1980.
- (a) The consumption rate doubles every 5 years and the consumption rate at the beginning of 1980 was 6 billion gallons per year. Find C and k .
- (b) Find the average rate of consumption of cola over the 10-year time period beginning January 1, 1983. Indicate units of measure.
- (c) Use the trapezoidal rule with four equal subdivisions to estimate $\int_5^7 S(t)dt$.
- (d) Using correct units, explain the meaning of $\int_5^7 S(t)dt$ in terms of cola consumption.

Part A

1 point is earned for the correct answer $C = 6$

$$S(t) = Ce^{kt}$$

$$S(0) = 6 \Rightarrow C = 6$$

1 point is earned for the correct answer $12 = 6e^{5k}$

$$S(5) = 12 \Rightarrow 12 = 6e^{5k}$$

1 point is earned for the correct answer $k = \frac{\ln 2}{5}$

$$2 = e^{5k}$$

$$k = \frac{\ln 2}{5}$$

(.138 or .139)



0	1	2	3
---	---	---	---

4-1-1

The student response earns all of the following points:

1 point is earned for the correct answer $C = 6$

$$S(t) = Ce^{kt}$$

$$S(0) = 6 \Rightarrow C = 6$$

1 point is earned for the correct answer $12 = 6e^{5k}$

$$S(5) = 12 \Rightarrow 12 = 6e^{5k}$$

1 point is earned for the correct answer $k = \frac{\ln 2}{5}$

$$2 = e^{5k}$$

$$k = \frac{\ln 2}{5}$$

(.138 or .139)

Part B

1 point is earned for correctly using $[3,13]$ and dividing by $13 - 3$

1 point is earned for the correct integrand

$$\text{Average rate} = \frac{1}{13 - 3} \int_3^{13} 6e^{\frac{\ln 2}{5}t} dt$$

1 point is earned for the correct answer with units

$$0/1 \text{ if not } \frac{1}{b - a} \int_a^b S(t) dt$$

4-1-1

$$= \frac{3}{\ln 2} [e^{2.6 \ln 2} - e^{0.6 \ln 2}] \text{ billion gal/yr}$$

(19.680 billion gal/yr)

Student solves $\frac{S(13) - S(3)}{10} = S'(t)$ for t_0 .

$S(t_0) = 19.680$ billions gal/yr

3/3 if correct.

Otherwise, read by standard where max. 1-0-0.



0	1	2	3
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The student response earns all of the following points:

1 point is earned for correctly using $[3, 13]$ and dividing by $13 - 3$

1 point is earned for the correct integrand

$$\text{Average rate} = \frac{1}{13 - 3} \int_3^{13} 6e^{\frac{\ln 2}{5}t} dt$$

1 point is earned for the correct answer with units

$$0/1 \text{ if not } \frac{1}{b - a} \int_a^b S(t) dt$$

$$= \frac{3}{\ln 2} [e^{2.6 \ln 2} - e^{0.6 \ln 2}] \text{ billion gal/yr}$$

(19.680 billion gal/yr)

Student solves $\frac{S(13) - S(3)}{10} = S'(t)$ for t_0 .

4-1-1

$$S(t_0) = 19.680 \text{ billions gal/yr}$$

3/3 if correct.

Otherwise, read by standard where max. 1-0-0.

Part C

1 point is earned for the correct trapezoidal rule with S , $n = 4$, interval $[5, 7]$

$$\int_5^7 S(t) dt \doteq \frac{1}{4}[S(5) + 2S(5.5) + 2S(6) + 2S(6.5) + S(7)]$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for the correct trapezoidal rule with S , $n = 4$, interval $[5, 7]$

$$\int_5^7 S(t) dt \doteq \frac{1}{4}[S(5) + 2S(5.5) + 2S(6) + 2S(6.5) + S(7)]$$

Part D

1 point is earned for the correct total consumption in a time period

1 point is earned for the correct time period and liquid measure

This gives the total consumption, in billions of gallons, during the years 1985 and 1986.

0/2: "rate of consumption"



0	1	2
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The student response earns all of the following points:

1 point is earned for the correct total consumption in a time period

4-1-1

1 point is earned for the correct time period and liquid measure

This gives the total consumption, in billions of gallons, during the years 1985 and 1986.

0/2: "rate of consumption"

4-1-1

33.  The rate at which people enter an amusement park on a given day is modeled by the function E defined by

$$E(t) = \frac{15600}{(t^2 - 24t + 160)}.$$

The rate at which people leave the same amusement park on the same day is modeled by the function L defined by

$$L(t) = \frac{9890}{(t^2 - 38t + 370)}.$$

Both $E(t)$ and $L(t)$ are measured in people per hour and time t is measured in hours after midnight. These functions are valid for $9 \leq t \leq 23$, the hours during which the park is open. At time $t = 9$, there are no people in the park.

(a) How many people have entered the park by 5:00 P.M. ($t = 17$)? Round your answer to the nearest whole number.

(b) The price of admission to the park is \$15 until 5:00 P.M. ($t = 17$). After 5:00 P.M., the price of admission to the park is \$11. How many dollars are collected from admissions to the park on the given day? Round your answer to the nearest whole number.

(c) Let $H(t) = \int_9^t (E(x) - L(x))dx$ for $9 \leq t \leq 23$. The value of $H(17)$ to the nearest whole number is 3725. Find the value of $H'(17)$, and explain the meaning of $H(17)$ and $H'(17)$ in the context of the amusement park.

(d) At what time t , for $9 \leq t \leq 23$, does the model predict that the number of people in the park is a maximum?

Part A

1 point is earned for limits

1 point is earned for integrand

1 point is earned for the answer

$$\int_9^{17} E(t)dt = 6004.270$$

6004 people entered the park by 5 pm.



0	1	2	3
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The student response earns all of the following points:

1 point is earned for limits

4-1-1

1 point is earned for integrand

1 point is earned for the answer

$$\int_9^{17} E(t) dt = 6004.270$$

6004 people entered the park by 5 pm.

Part B

1 point is earned for the setup

$$15 \int_9^{17} E(t) dt + 11 \int_{17}^{23} E(t) dt = 104048.165$$

The amount collected was \$104,048.

or

$$\int_{17}^{23} E(t) dt = 1271.283$$

1271 people entered the park between 5 pm and 11 pm, so the amount collected was

$$\$15 \cdot (6004) + \$11 \cdot (1271) = \$104,041.$$



0

1

The student response earns all of the following points:

1 point is earned for the setup

$$15 \int_9^{17} E(t) dt + 11 \int_{17}^{23} E(t) dt = 104048.165$$

The amount collected was \$104,048.

4-1-1

or

$$\int_{17}^{23} E(t) dt = 1271.283$$

1271 people entered the park between 5 pm and 11 pm, so the amount collected was

$$\$15 \cdot (6004) + \$11 \cdot (1271) = \$104,041.$$

Part C

1 point is earned for value $H'(17)$

2 points are earned for meanings

1: meaning of $H(17)$

1: meaning of $H'(17)$

<-1> if no reference to $t = 17$

$$H'(17) = E(17) - L(17) = -380.281$$

There were 3725 people in the park at $t = 17$. The number of people in the park was decreasing at the rate of approximately 380 people/hr at time $t = 17$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for value $H'(17)$

2 points are earned for meanings

1: meaning of $H(17)$

1: meaning of $H'(17)$

<-1> if no reference to $t = 17$

$$H'(17) = E(17) - L(17) = -380.281$$

4-1-1

There were 3725 people in the park at $t = 17$. The number of people in the park was decreasing at the rate of approximately 380 people/hr at time $t = 17$.

Part D

1 point is earned for $E(t) - L(t) = 0$

1 point is earned for the answer

$$H'(t) = E(t) - L(t) = 0$$

$$t = 15.794 \text{ or } 15.795$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $E(t) - L(t) = 0$

1 point is earned for the answer

$$H'(t) = E(t) - L(t) = 0$$

$$t = 15.794 \text{ or } 15.795$$

4-1-1

34. A tank contains 30 liters of water at time $t = 0$. For time $0 \leq t \leq 20$ minutes, water flows into the tank at a rate modeled by the function f defined by $f(t) = 2 \ln(t + 3) - 1.6$. For time $20 < t \leq 45$ minutes, water flows into the tank at a rate modeled by the function g defined by $g(t) = \frac{7.5}{\ln t} + 2.2$. Both $f(t)$ and $g(t)$ are measured in liters per minute.

**Part A**

Find the rate of change of g at time $t = 23$ minutes. Show the setup for your calculations, and indicate units of measure.

Part B

How many liters of water are in the tank at time $t = 20$ minutes? Show the setup for your calculations.

Part C

Let w be the function defined by $w(t) = \begin{cases} f(t) & \text{for } 0 \leq t \leq 20 \\ g(t) & \text{for } 20 < t \leq 45 \end{cases}$. Find $\lim_{t \rightarrow 20^+} w(t)$. Is w continuous at $t = 20$? Use the definition of continuity to justify your answer.

Part D

Consider the function w defined in part C. Find the average value of w from time $t = 15$ minutes to time $t = 30$ minutes. Show the setup for your calculations.

Part A Point 1

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- **P1** is earned for a correct value either rounded or truncated to three places after the decimal and for labeling the value as $g'(23)$. An inappropriately rounded answer does not earn the point.
- **P1** can also be earned for the exact value $\frac{-7.5}{23(\ln 23)^2}$.

Model Solution:

$$g'(23) = -0.033$$



0	1
---	---

The response accurately includes the criteria below.

- $g'(23)$

4-1-1

Part A Point 2

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- **P2** is earned for correct units, whether or not they are attached to a numerical value for the rate of change.
- **P2** is also earned for the units “liters/minute².”

Model Solution:

The rate of change in the rate at time $t = 23$ is -0.033 liters per minute per minute.



0	1
---	---

The response accurately includes the criteria below.

- Units

Part B Point 3

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- **P3** is earned with a definite integral containing the correct integrand and correct lower and upper limits, with or without the differential.
- If the differential dt is missing:
 - Writing $\int_0^{20} f(t)$ or $30 + \int_0^{20} f(t)$ earns **P3**.
 - Because $\int_0^{20} f(t) + 30$ may mean $\int_0^{20} f(t)dt + 30$ or $\int_0^{20} (f(t) + 30)dt$:
 - Writing $\int_0^{20} f(t) + 30$ in the presence of a correct answer of 95.641 earns **P3**.
 - Writing $\int_0^{20} f(t) + 30$ in the presence of an answer of 95, 96, 95.6, or 95.64 earns **P3**.
 - Writing $\int_0^{20} f(t) + 30$ in the presence of no answer or an incorrect answer does not earn **P3**.
- Writing $\int f(t)dt$, $30 + \int f(t)dt$, $30 + \int f(t)$, or $\int f(t) + 30$ and never providing the definite integral, does not earn **P3**.

Model Solution:

4-1-1

$$30 + \int_0^{20} f(t)dt = 30 + \int_0^{20} (2 \ln(t+3) - 1.6)dt$$



0	1
---	---

The response accurately includes the criteria below.

- Definite integral

Part B Point 4

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- **P4** is earned for the correct answer, with or without supporting work. A reported answer should be accurate to three places after the decimal point, rounded or truncated. An inappropriately rounded answer does not earn the point, unless an earlier point was not earned due to inappropriate rounding.
- Incorrect or unclear communication between the correct integral and the correct answer is treated as scratch work and is not considered in scoring. The following examples earn **P4** for the correct answer:
 - $\int_0^{20} f(t)dt = 65.641060$ so the number of liters of water is 95.641.
 - $\int_0^{20} f(t)dt = 65.641060 + 30 = 95.641060$ (Incorrect linkage is not considered in scoring.)
 - $\int_0^{20} f(t)dt = 95.641060$ (Incorrect linkage is not considered in scoring.)

Model Solution:

There are 95.641 liters of water in the tank at time $t = 20$ minutes.



0	1
---	---

The response accurately includes the criteria below.

- Answer

Part C Point 5

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

4-1-1

- To earn **P5**, a response must indicate $\lim_{t \rightarrow 20^+} w(t) = 4.704$ (or 4.703), either symbolically or verbally. A reported answer should be accurate to three places after the decimal point, rounded or truncated. An inappropriately rounded answer does not earn **P5**, unless an earlier point was not earned due to inappropriate rounding.
 - A response of “the right-hand limit is 4.704” is sufficient to earn **P5**.

Model Solution:

$$\lim_{t \rightarrow 20^+} w(t) = \lim_{t \rightarrow 20^+} g(t) = 4.703562$$

The limit as t approaches 20 from the right is 4.704 (or 4.703).



0	1
---	---

The response accurately includes the criteria below.

- Right-hand limit

Part C Point 6

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- To earn **P6**, a response must include $w(20) = 4.671$ (or 4.67 or 4.6), $\lim_{t \rightarrow 20^+} w(t) \neq w(20)$, and a conclusion of not continuous (with or without indicating $t = 20$).
- Alternate solution:

$$\lim_{t \rightarrow 20^-} w(t) = \lim_{t \rightarrow 20^-} f(t) = 4.670988$$

Because $\lim_{t \rightarrow 20^-} w(t) \neq \lim_{t \rightarrow 20^+} w(t)$, w is not continuous at $t = 20$.

Such a response is scored analogously to the model solution and earns **P6**.

- Special case: A response of “ $\lim_{t \rightarrow 20^+} w(t) = 4.7$ and $\lim_{t \rightarrow 20^-} w(t) = w(20) = 4.7$, therefore continuous” earns **P6**.

Model Solution:

$$w(20) = f(20) = 4.670988$$

Because $\lim_{t \rightarrow 20^+} w(t) \neq w(20)$, w is not continuous at $t = 20$.

4-1-1



0	1
---	---

The response accurately includes the criteria below.

- Answer with justification

Part D Point 7

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- To earn **P7**, a response must present one or more definite integrals with correct integrands, with or without differentials. Limits of integration are not considered in scoring for **P7**. That is, any of the following earns **P7**: $\int_a^c w(t)dt$, $\int_a^b w(t)dt + \int_b^c w(t)dt$, or $\int_a^b f(t)dt + \int_b^c g(t)dt$, where a , b , and c are real numbers between 0 and 45, inclusive.

Model Solution:

$$\frac{1}{30-15} \int_{15}^{30} w(t)dt$$



0	1
---	---

The response accurately includes the criteria below.

- Definite integral

Part D Point 8

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- To earn **P8**, a response must present a correct definite integral and evidence of division by 15.
 - A response that earned **P7** earns **P8** for $\frac{1}{15} \int_{15}^{30} w(t)dt$, $\frac{1}{15} \left(\int_{15}^{20} w(t)dt + \int_{20}^{30} w(t)dt \right)$, $\frac{1}{15} \left(\int_{15}^{20} f(t)dt + \int_{20}^{30} g(t)dt \right)$, or equivalent.

4-1-1

- Presenting the correct integral and the correct answer will suffice as evidence of division by 15 and will earn **P8**. Work presented between earning **P7** and the correct answer that does not reveal incorrect work done in the calculator will be treated as scratch work and will not be considered in scoring.
 - “ $\int_{15}^{30} w(t)dt = 67.559$, so 4.504” earns **P8** for correct limits of integration and an implied division by 15 made evident by the correct answer.
 - “ $\int_{15}^{30} w(t)dt = \frac{67.559}{15} = 4.504$ ” earns **P8** for correct limits of integration and an implied division by 15 made evident by the correct answer. The incorrect communication between the integral and the answer does not reveal incorrect work done in the calculator. This work is treated as scratch work and is not considered in scoring of **P8**.
- A response that did not earn **P7** is eligible to earn **P8** for $\frac{1}{15} \int_{15}^{30} f(t)dt$ or $\frac{1}{15} \int_{15}^{30} g(t)dt$.

Model Solution:

$$\begin{aligned} & \frac{1}{15} \left(\int_{15}^{20} w(t)dt + \int_{20}^{30} w(t)dt \right) \\ &= \frac{1}{15} \left(\int_{15}^{20} f(t)dt + \int_{20}^{30} g(t)dt \right) \end{aligned}$$



0	1
---	---

The response accurately includes the criteria below.

- Average value formula

Part D Point 9

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- P9** is earned for the correct answer, with or without supporting work. A reported answer should be accurate to three places after the decimal point, rounded or truncated. An inappropriately rounded answer does not earn the point, unless an earlier point was not earned due to inappropriate rounding.

Model Solution:

The average value of w from time $t = 15$ and time $t = 30$ minutes is 4.504 (or 4.503) liters per minute.

4-1-1



0	1
---	---

The response accurately includes the criteria below.

- Answer

35. The cost, in dollars, to paint x square feet of a house is modeled by the differentiable function $C(x)$. Which of the following is the best interpretation of the statement $C'(300) = 6$?

- (A) The cost of painting 6 square feet is \$300.
- (B) The cost of painting 300 square feet is \$6.
- (C) The cost of painting 300 square feet is \$6 per square foot.
- (D) When 300 square feet have been painted, the cost to paint the next square foot is approximately \$6. ✓

Answer D

Correct. In the expression $C'(300)$, the 300 represents the value of the independent variable and is therefore the number of square feet of the house to be painted. $C'(300)$, being the value of a derivative, is the rate of change of the cost C . The units for the derivative would be the units of C per unit of x , hence dollars per square foot. For small values of Δx , $\Delta C \approx C'(300)\Delta x$. So when 300 square feet have already been painted ($x = 300$), painting an additional square foot ($\Delta x = 1$) will cause an additional cost (ΔC) of approximately

$$C'(300)\Delta x = \left(6 \frac{\$}{\text{ft}^2}\right) \cdot (1\text{ft}^2) = \$6.$$

4-1-1

36. The function $g(t)$ models the number of books sold in a store on day t , where t is the number of days after January 1, 2020. Which of the following is the best interpretation of the statement $g'(7) = -11$?
- (A) On January 8, 2020, approximately 11 books were sold.
 - (B) On January 8, 2020, the number of books sold was decreasing at a rate of 11 books per day. ✓
 - (C) On January 8, 2020, the rate at which books were sold was decreasing at a rate of 11 books per day per day.
 - (D) From January 1, 2020, to January 8, 2020, the number of books sold was decreasing at an average rate of 11 books per day.

Answer B

Correct. In the expression $g'(7)$, the 7 represents the value of the independent variable and is therefore the number of days after January 1, 2020, that is January 8, 2020. $g'(7)$, being the value of a derivative, is the rate of change of the number of books sold. The units for the derivative would be the units of g per unit of t , hence books per day. Because $g'(7)$ is negative, on January 8, 2020, the number of books sold is decreasing at a rate of 11 books per day.

37. For a car driven 100 kilometers at a constant speed, the amount of fuel used as a function of the speed is modeled by a differentiable function F . Fuel is measured in liters, and speed is measured in kilometers per hour (km/hr). In this context, which of the following is a correct interpretation of the statement $F'(50) > F'(80)$?
- (A) Driving at 50 km/hr uses more fuel than driving at 80 km/hr.
 - (B) The rate at which 50 liters of fuel is used is greater than the rate at which 80 liters of fuel is used.
 - (C) The rate of change of the speed is greater when driving at 50 km/hr than when driving at 80 km/hr.
 - (D) The rate of change of liters of fuel used with respect to the speed is greater when driving at 50 km/hr than when driving at 80 km/hr. ✓

Answer D

Correct. In the expression $F'(50)$, the 50 represents the value of the independent variable and is therefore the speed of the car in km/hr. $F'(50)$, being the value of a derivative, is the rate of change of F —that is, the rate of change of liters of fuel used with respect to the speed, in this case when the speed is 50 km/hr. Similarly, $F'(80)$ is the rate of change of liters of fuel used with respect to the speed when the speed is 80 km/hr. The statement says that the value of the derivative when the speed is 50 km/hr is greater than when the speed is 80 km/hr.

4-1-1

38. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

The number of mosquitoes in a field after a major rainfall is modeled by the function M defined by $M(t) = -t^3 + 12t^2 + 144t$, where t is the number of days after the rainfall ended and $0 \leq t \leq 18$.

- Using correct units, interpret the meaning of $M'(15) = -171$ in the context of the problem.
- Based on the model, what is the absolute maximum number of mosquitoes in the field over the time interval $0 \leq t \leq 18$? Justify your answer.
- For what values of t is the rate of change of the number of mosquitoes in the field increasing?
- For $0 \leq t \leq 18$, the number of bats in the field is modeled by the differentiable function B , where B is a function of the number of mosquitoes in the field. Based on the models, write an expression for the rate of change of the number of bats in the field at time $t = a$.

Part A

The point requires a time reference, rate, and units.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The student response accurately includes a correct interpretation of $M'(15) = -171$

Solution:

At time $t = 15$ days, the number of mosquitoes in the field is decreasing at a rate of 171 mosquitoes per day.

Part B

Note: Sign charts are a useful tool to investigate and summarize the behavior of a function. By itself a sign chart for $f'(x)$ or $f''(x)$ is not a sufficient response for a justification.

4-1-1

The response is eligible for additional points based on consistent answers using an incorrect $M'(t)$ with a maximum of one computational error.

The fourth point does not require a simplified answer.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3	4
---	---	---	---	---

The student response accurately includes all four of the criteria below.

- ☐ $M'(t)$
- ☐ sets $M'(t) = 0$
- ☐ identifies $t = 12$ as a candidate
- ☐ answer with justification

Solution:

$$M'(t) = -3t^2 + 24t + 144 = -3(t^2 - 8t - 48) = -3(t - 12)(t + 4)$$

$$M'(t) = 0 \Rightarrow t = 12$$

Because $M'(t)$ changes from positive to negative at $t = 12$, M has a relative maximum at $t = 12$.

Because $t = 12$ is the only critical point of M on the interval $0 \leq t \leq 18$, that is the location of a relative maximum. It is also the location of the absolute maximum of M on the interval $0 \leq t \leq 18$.

$$M(12) = -12^3 + 12(12^2) + 144(12) = 144(12) = 1728$$

The absolute maximum number of mosquitoes in the field over the time interval $0 \leq t \leq 18$ is $M(12) = 1728$.

Part C

The third point may be earned based on a consistent answer using an incorrect $M''(t)$ with a maximum of one computational error.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

4-1-1

- ☐ considers $M''(t)$
- ☐ expression for $M''(t)$
- ☐ answer

Solution:

$$M''(t) = -6t + 24 = -6(t - 4)$$

$$M''(t) = -6(t - 4) > 0 \Rightarrow t < 4$$

The rate of change of the number of mosquitoes in the field is increasing for $0 \leq t \leq 4$.

Part D

The point is earned with $B'(M(a)) \cdot M'(a)$

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The student response accurately includes a correct expression.

Solution:

The number of bats in the field at time $t = a$ is given by $B(M(a))$.

Therefore, the rate of change of the number of bats in the field at time $t = a$ is given by

$$B'(M(a)) \cdot M'(a) = B'(-a^3 + 12a^2 + 144a) \cdot (-3a^2 + 24a + 144).$$

4-1-1

39. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

The number of fish in a small bay is modeled by the function F defined by $F(t) = 10(t^3 - 12t^2 + 45t + 100)$, where t is measured in days and $0 \leq t \leq 8$.

- Using correct units, interpret the meaning of $F'(4) = -30$ in the context of the problem.
- Based on the model, what is the absolute minimum number of fish in the bay over the time interval $0 \leq t \leq 8$? Justify your answer.
- For what values of t , $0 \leq t \leq 8$, is the rate of change of the number of fish in the bay decreasing?
- For $0 \leq t \leq 8$, the number of pelicans flying near the bay is modeled by the differentiable function P , where P is a function of the number of fish in the bay. Based on the models, write an expression for the rate of change of the number of pelicans flying near the bay at time $t = c$.

Part A

The point requires a time reference, rate, and units.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The student response accurately includes a correct interpretation of $F'(4) = -30$.

Solution:

At time $t = 4$ days, the number of fish in the bay is decreasing at a rate of 30 fish per day.

Part B

Note: Sign charts are a useful tool to investigate and summarize the behavior of a function. By itself a sign chart for $f'(x)$ or $f''(x)$ is not a sufficient response for a justification.

4-1-1

The response is eligible for additional points based on consistent answers using an incorrect $F'(t)$ with a maximum of one computational error.

The fourth point does not require a simplified answer.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3	4
---	---	---	---	---

The student response accurately includes all four of the criteria below.

- ☐ $F'(t)$
- ☐ sets $F'(t) = 0$
- ☐ identifies $t = 5$ as a candidate
- ☐ answer with justification

Solution:

$$F'(t) = 10(3t^2 - 24t + 45) = 10 \cdot 3(t^2 - 8t + 15) = 30(t - 3)(t - 5)$$

$$F'(t) = 0 \Rightarrow t = 3 \text{ and } t = 5$$

Because $F'(t)$ changes from positive to negative at $t = 3$ and from negative to positive at $t = 5$, consider $t = 0$ and $t = 5$ for the location of the absolute minimum. F has a relative minimum at $t = 5$.

$$F(0) = 1000$$

$$F(5) = 10(5^3 - 12(5)^2 + 45(5) + 100) = 1500$$

The absolute minimum number of fish in the bay over the time interval

$$0 \leq t \leq 8 \text{ is } F(0) = 1000.$$

Part C

The third point may be earned based on a consistent answer using an incorrect $F''(t)$ with a maximum of one computational error.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3
---	---	---	---

4-1-1

The student response accurately includes all three of the criteria below.

- ☐ considers $F''(t)$
- ☐ expression for $F''(t)$
- ☐ answer

Solution:

$$F''(t) = 10(6t - 24) = 60(t - 4)$$

$$F''(t) = 60(t - 4) < 0 \Rightarrow t < 4$$

The rate of change of the number of fish in the bay is decreasing for $0 \leq t \leq 4$.

Part D

The point is earned with $P'(F(c)) \cdot F'(c)$

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The student response accurately includes a correct expression.

Solution:

The number of pelicans flying near the bay at time $t = c$ is given by $P(F(c))$.

Therefore, the rate of change of the number of pelicans flying near the bay at time $t = c$ is given by

$$P'(F(c)) \cdot F'(c) = P'(10(c^3 - 12c^2 + 45c + 100)) \cdot (10(3c^2 - 24c + 45)).$$

4-1-1

40. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

$$W(t) = \begin{cases} \frac{17}{2} - \frac{3}{2} \cos\left(\frac{\pi t}{6}\right) & \text{for } 0 \leq t \leq 6 \\ 10 - \frac{1}{5}(t-6)^2 & \text{for } 6 < t \leq 10 \end{cases}$$

The depth of a river at a certain point is modeled by the function W defined above, where $W(t)$ is measured in feet and time t is measured in hours.

- (a) Find $W'(8)$. Using correct units, explain the meaning of $W'(8)$ in the context of the problem.
- (b) The graph of W is concave down for $3 \leq t \leq 3.5$. Use the line tangent to the graph of W at $t = 3$ to show that $W(3.5) \leq 9$.
- (c) Find $\lim_{t \rightarrow 2} \frac{W(t) - t^3 + \frac{1}{4}}{t-2}$.

Part A

The first 2 points may be earned with the presence of a correct numerical answer.

The second point may be earned with a maximum of one sign error in $W'(t)$.

The third point requires a time reference, rate, and units. The response is eligible for the third point if the second point is earned AND provided the behavior described is consistent with the sign of the approximation, the numerical value, and the time reference.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ $W'(t)$
- ☐ $W'(8)$

4-1-1

☐ interpretation with units

Solution:

For $6 < t < 10$, $W'(t) = -\frac{2}{5}(t - 6)$.

$$W'(8) = -\frac{2}{5}(8 - 6) = -\frac{4}{5}$$

At time $t = 8$ hours, the depth of the river is changing at a rate of $-\frac{4}{5}$ foot per hour.

Part B

The second point may be earned with a maximum of one error in $W'(t)$.

For the second and third points, trigonometric function values do not need to be evaluated.

The third point may be earned if the response is consistent with previous results.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3	4
---	---	---	---	---

The student response accurately includes all four of the criteria below.

- ☐ $W'(t)$
- ☐ $W'(3)$
- ☐ approximation
- ☐ explanation

Solution:

$$W(3) = \frac{17}{2} - \frac{3}{2}\cos\left(\frac{3\pi}{6}\right) = \frac{17}{2}$$

For $0 < t < 6$,

$$W'(t) = \frac{3}{2}\sin\left(\frac{\pi t}{6}\right) \cdot \frac{\pi}{6} = \frac{\pi}{4}\sin\left(\frac{\pi t}{6}\right).$$

$$W'(3) = \frac{\pi}{4}\sin\left(\frac{3\pi}{6}\right) = \frac{\pi}{4}$$

$$W(3.5) \approx \frac{17}{2} + \frac{\pi}{4}(3.5 - 3) = \frac{17}{2} + \frac{\pi}{8}$$

4-1-1

Because the graph of W is concave down for $3 \leq t \leq 3.5$, approximation of $W(3.5)$ is an overestimate of the actual value of $W(3.5)$.

Therefore, $W(3.5) < \frac{17}{2} + \frac{\pi}{8} < \frac{17}{2} + \frac{1}{2} = 9$.

Part C

The first point requires showing that an indeterminate form $\frac{0}{0}$ is present in order to apply L'Hospital's Rule. Any error in mathematical communication (e.g., writing " $= \frac{0}{0}$ " or not using limit notation) impacts the first point.

The second point requires substitution of function values. A simplified answer is not required; trigonometric function values do not need to be evaluated. Any differentiation or computation error impacts the second point.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ applies L'Hospital's Rule
- ☐ answer

Solution:

$$\lim_{t \rightarrow 2} \left(W(t) - t^3 + \frac{1}{4} \right) = W(2) - 2^3 + \frac{1}{4}$$

$$= \frac{17}{2} - \frac{3}{2} \cos\left(\frac{2\pi}{6}\right) - 8 + \frac{1}{4} = 0$$

$$\lim_{t \rightarrow 2} (t - 2) = 0$$

By L'Hospital's Rule,

$$\lim_{t \rightarrow 2} \frac{W(t) - t^3 + \frac{1}{4}}{t - 2} = \lim_{t \rightarrow 2} \frac{W'(t) - 3t^2}{1} = W'(2) - 3 \cdot 2^2$$

$$= \frac{\pi}{4} \sin\left(\frac{2\pi}{6}\right) - 12 = \frac{\pi\sqrt{3}}{8} - 12$$

4-1-1

41. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

$$W(t) = \begin{cases} \frac{32}{5} + \frac{2}{5}\cos\left(\frac{\pi t}{4}\right) & \text{for } 0 \leq t \leq 4 \\ 6 + \frac{1}{8}(t-4)^2 & \text{for } 4 < t \leq 9 \end{cases}$$

The depth of water in a zoo aquarium at a certain point is modeled by the function W defined above, where $W(t)$ is measured in feet and time t is measured in hours.

- (a) Find $W'(7)$. Using correct units, explain the meaning of $W'(7)$ in the context of the problem.
- (b) The graph of W is concave up for $2 \leq t \leq 2.5$. Use the line tangent to the graph of W at $t = 2$ to show that $W(2.5) \geq 6$.
- (c) Find $\lim_{t \rightarrow 2} \frac{W(t) - t^2 - \frac{12}{5}}{t-2}$.

Part A

The first 2 points may be earned with the presence of a correct numerical answer.

The second point may be earned with a maximum of one sign error in $W'(t)$.

The third point requires a time reference, rate, and units. The response is eligible for the third point if the second point is earned AND provided the behavior described is consistent with the sign of the approximation, the numerical value, and the time reference.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ $W'(t)$
- ☐ $W'(7)$

4-1-1

☐ interpretation with unit

Solution:

For $4 < t < 9$, $W'(t) = \frac{1}{4}(t - 4)$.

$$W'(7) = \frac{1}{4}(7 - 4) = \frac{3}{4}$$

At time $t = 7$ hours, the depth of the water in the zoo aquarium is changing at a rate of $\frac{3}{4}$ foot per hour.

Part B

The second point may be earned with a maximum of one error in $W'(t)$.

For the second and third points, trigonometric function values do not need to be evaluated.

The third point may be earned if the response is consistent with previous results. Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3	4
---	---	---	---	---

The student response accurately includes all four of the criteria below.

- ☐ $W'(t)$
- ☐ $W'(2)$
- ☐ approximation
- ☐ explanation

Solution:

$$W(2) = \frac{32}{5} + \frac{2}{5}\cos\left(\frac{2\pi}{4}\right) = \frac{32}{5}$$

For $0 < t < 4$,

$$W'(t) = -\frac{2}{5}\sin\left(\frac{\pi t}{4}\right) \cdot \frac{\pi}{4} = -\frac{\pi}{10}\sin\left(\frac{\pi t}{4}\right).$$

$$W'(2) = -\frac{\pi}{10}\sin\left(\frac{2\pi}{4}\right) = -\frac{\pi}{10}$$

$$W(2.5) \approx \frac{32}{5} - \frac{\pi}{10}(2.5 - 2) = \frac{32}{5} - \frac{\pi}{20}$$

4-1-1

Because the graph of W is concave up for $2 \leq t \leq 2.5$, the tangent line approximation of $W(2.5)$ is an underestimate of the actual value of $W(2.5)$. Therefore, $W(2.5) > \frac{32}{5} - \frac{\pi}{20} > 6$.

Part C

The first point requires showing that an indeterminate form $\frac{0}{0}$ is present in order to apply L'Hospital's Rule. Any error in mathematical communication (e.g., writing " $\frac{0}{0}$ " or not using limit notation) impacts the first point.

The second point requires substitution of function values. A simplified answer is not required; trigonometric function values do not need to be evaluated. Any differentiation or computation error impacts the second point.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ applies L'Hospital's Rule
- ☐ answer

Solution:

$$\begin{aligned}\lim_{t \rightarrow 2} \left(W(t) - t^2 - \frac{12}{5} \right) &= W(2) - 2^2 - \frac{12}{5} \\ &= \frac{32}{5} + \frac{2}{5} \cos\left(\frac{2\pi}{4}\right) - 4 - \frac{12}{5} = 0\end{aligned}$$

$$\lim_{t \rightarrow 2} (t - 2) = 0$$

By L'Hospital's Rule,

$$\begin{aligned}\lim_{t \rightarrow 2} \frac{W(t) - t^2 - \frac{12}{5}}{t - 2} &= \lim_{t \rightarrow 2} \frac{W'(t) - 2t}{1} = W'(2) - 2 \cdot 2 \\ &= -\frac{\pi}{10} \sin\left(\frac{2\pi}{4}\right) - 4 = -\frac{\pi}{10} - 4\end{aligned}$$

4-1-1

42. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (hours)	0	3	8	12	14
$S'(t)$ (centimeters per hour)	2.3	2.1	1.9	1.7	1.6

The depth of snow in a field is given by the twice-differentiable function S for $0 \leq t \leq 14$, where $S(t)$ is measured in centimeters and time t is measured in hours. Values of $S'(t)$, the derivative of S , at selected values of time t are shown in the table above. It is known that the graph of S is concave down for $0 \leq t \leq 14$.

- Use the data in the table to approximate $S''(10)$. Show the computations that lead to your answer. Using correct units, explain the meaning of $S''(10)$ in the context of the problem.
- Is there a time t , for $0 \leq t \leq 14$, at which the depth of snow is changing at a rate of 2 centimeters per hour? Justify your answer.
- At time $t = 8$, the depth of snow is 45 centimeters. Use the line tangent to the graph of S at $t = 8$ to approximate the depth of snow at time $t = 10$. Is the approximation an underestimate or an overestimate of the actual depth of snow at time $t = 10$? Justify your answer.
- The depth of snow, in centimeters, is also modeled by the twice-differentiable function D for $0 \leq t \leq 14$, where $D(t) = 120 - 92e^{-\frac{t}{40}}$ and time t is measured in hours. Use the model to find $D'(10)$. Using correct units, explain the meaning of $D'(10)$ in the context of the problem.

Part A

The first point requires a correct substitution of values. A simplified answer is not required.

The second point requires a time reference, rate, and units. The point may be earned with an incorrect approximation based on one computational error and provided the behavior described is consistent with the sign of the approximation, the numerical value, and the time reference.

4-1-1

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ approximation
- ☐ interpretation with units

Solution:

$$S''(10) \approx \frac{S'(12) - S'(8)}{12 - 8} = \frac{1.7 - 1.9}{4} = -0.05$$

At time $t = 10$ hours, the rate of change of the depth of snow is changing at a rate of -0.05 centimeter per hour per hour.

Part B

The second point may be earned if response contains conditions of IVT without naming the theorem.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ S' is continuous
- ☐ justification using Intermediate Value Theorem

Solution:

S is twice differentiable. $\Rightarrow S'$ is differentiable and thus continuous.

$$S'(8) = 1.9 < 2 < 2.1 = S'(3)$$

By the Intermediate Value Theorem, there is a value of t , for $0 \leq t \leq 14$, such that $S'(t) = 2$.

Part C

The second point may be earned if using an incorrect value for slope, provided the value is labeled as $S'(8)$.

4-1-1

The second point does not require a simplified answer. Substitution of function values is required.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ equation for tangent line
- ☐ approximation
- ☐ overestimate with justification using concavity

Solution:

$$S'(8) = 1.9 \text{ from table}$$

An equation for the tangent line is $y = 45 + 1.9(t - 8)$.

$$S(10) \approx 45 + S'(8)(10 - 8) = 48.8 \text{ centimeters}$$

This approximation is an overestimate because the graph of S is concave down for $8 \leq t \leq 10$.

Part D

The second point requires a time reference, rate, and units. The point may be earned with an incorrect approximation based on one computational error and provided the behavior described is consistent with the sign of the approximation, the numerical value, and the time reference.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ $D'(10)$
- ☐ interpretation with units

Solution:

$$D'(10) = 1.791$$

4-1-1

At time $t = 10$ hours, the depth of snow is changing by 1.791 centimeters per hour.

4-1-1

43. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

On a certain day at a diner, the total number of food orders taken by time t is modeled by the function F that satisfies the differential equation $\frac{dF}{dt} = \frac{48}{\sqrt{3t+1}}$ with $F(1) = 32$, where t is the number of hours since the diner opened and $0 \leq t \leq 8$.

- Write an equation for the line tangent to the graph of F at $t = 1$. Use this equation to approximate the total number of food orders taken by time $t = 2$.
- Find $F''(5)$. Using correct units, interpret the meaning of $F''(5)$ in the context of the problem.
- Find an expression for $F'(t)$. Based on this model, what is the total number of food orders taken by time $t = 8$?

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ tangent line equation
- ☐ approximation

Solution:

$$F(1) = 32$$

$$F'(1) = \frac{48}{\sqrt{3 \cdot 1 + 1}} = 24$$

An equation for the line tangent to the graph of F at $t = 1$ is $y = 32 + 24(t - 1)$.

By time $t = 2$, the total number of food orders taken is approximately $32 + 24(2 - 1) = 56$.

4-1-1

Part B

The third point requires (1) rate of $-\frac{9}{8}$, (2) units of orders per hour per hour, and (3) reference to $t = 5$.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ $F''(t)$
- ☐ $F''(5)$
- ☐ interpretation

Solution:

$$F''(t) = 48\left(-\frac{1}{2}\right)(3t+1)^{-\frac{3}{2}} \cdot 3 = -72(3t+1)^{-\frac{3}{2}}$$

$$F''(5) = -72(3 \cdot 5 + 1)^{-\frac{3}{2}} = -72(16)^{-\frac{3}{2}} = -\frac{72}{64} = -\frac{9}{8}$$

At time $t = 5$, the rate at which food orders are taken is changing at a rate of $-\frac{9}{8}$ orders per hour per hour.

Part C

At most 1 out of 4 points earned [1-0-0-0] if no constant of integration.

The fourth point requires both an expression for $F(t)$ and value for $F(8)$.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3	4
---	---	---	---	---

The student response accurately includes all four of the criteria below.

- ☐ antiderivative
- ☐ constant of integration
- ☐ uses initial conditions
- $F(t)$ and $F(8)$

4-1-1

**Solution:**

$$\frac{dF}{dt} = \frac{48}{\sqrt{3t+1}}$$

$$F(t) = 48 \cdot 2\sqrt{3t+1} \cdot \frac{1}{3} + C = 32\sqrt{3t+1} + C$$

$$F(1) = 32\sqrt{3 \cdot 1 + 1} + C = 64 + C = 32 \Rightarrow C = -32$$

$$F(t) = 32\sqrt{3t+1} - 32$$

The number of food orders taken by time $t = 8$ is $32\sqrt{3 \cdot 8 + 1} - 32 = 128$.

4-1-1

44. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

A book in a popular book series goes on sale at time $t = 0$. The total number of books sold by time t hours is modeled by the function B that satisfies the differential equation $\frac{dB}{dt} = \frac{5}{\sqrt{t+4}} + \frac{1}{9}(t-6)^2$ with $B(12) = 36$, where B is measured in hundreds of books and $0 \leq t \leq 24$.

- Write an equation for the line tangent to the graph of B at $t = 12$. Use this equation to approximate the total number of books sold by time $t = 10$.
- Find $B''(5)$. Using correct units, interpret the meaning of $B''(5)$ in the context of the problem.
- Find an expression for $B(t)$. Based on this model, what is the total number of books sold by time $t = 21$?

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ tangent line equation
- ☐ approximation

Solution:

$$B(12) = 36$$

$$B'(12) = \frac{5}{\sqrt{12+4}} + \frac{1}{9}(12-6)^2 = \frac{5}{4} + 4 = \frac{21}{4}$$

An equation for the line tangent to the graph of B at $t = 12$ is $y = 36 + \frac{21}{4}(t - 12)$.

4-1-1

$$B(10) \approx 36 + \frac{21}{4}(10 - 12) = 36 - \frac{21}{2} = \frac{51}{2} = 25.5$$

The number of books sold by time $t = 10$ is approximately 2550.

Part B

The third point requires (1) rate of $-\frac{17}{54}$, (2) units of hundred books per hour per hour, and (3) reference to $t = 5$.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ $B''(t)$
- ☐ $B''(5)$
- ☐ interpretation

Solution:

$$B''(t) = 5\left(-\frac{1}{2}\right)(t+4)^{-\frac{3}{2}} + \frac{1}{9} \cdot 2(t-6) = -\frac{5}{2(t+4)^{\frac{3}{2}}} + \frac{2}{9}(t-6)$$

$$B''(5) = -\frac{5}{2(5+4)^{\frac{3}{2}}} + \frac{2}{9}(5-6) = -\frac{5}{54} - \frac{2}{9} = -\frac{17}{54}$$

At time $t = 5$, the rate at which books are selling is changing at a rate of $-\frac{17}{54}$ hundred books per hour per hour.

Part C

At most 1 out of 4 points earned [1-0-0-0] if no constant of integration.

The fourth point requires both an expression for $B(t)$ and value for $B(21)$.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3	4
---	---	---	---	---

The student response accurately includes all four of the criteria below.

4-1-1

- ☐ antiderivative
- ☐ constant of integration
- ☐ uses initial conditions
- ☐ $B(t)$ and $B(21)$

Solution:

$$\frac{dB}{dt} = \frac{5}{\sqrt{t+4}} + \frac{1}{9}(t-6)^2$$

$$B(t) = 5 \cdot 2\sqrt{t+4} + \frac{1}{9} \cdot \frac{1}{3}(t-6)^3 + C = 10\sqrt{t+4} + \frac{1}{27}(t-6)^3 + C$$

$$B(12) = 10\sqrt{12+4} + \frac{1}{27}(12-6)^3 + C = 48 + C = 36 \Rightarrow C = -12$$

$$B(t) = 10\sqrt{t+4} + \frac{1}{27}(t-6)^3 - 12$$

$$B(21) = 10\sqrt{21+4} + \frac{1}{27}(21-6)^3 - 12 = 50 + 125 - 12 = 163$$

The number of books sold by time $t = 21$ is 16,300.

4-1-1

45. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (seconds)	0	60	90	120	135	150
$f(t)$ (gallons per second)	0	0.1	0.15	0.1	0.05	0

A customer at a gas station is pumping gasoline into a gas tank. The rate of flow of gasoline is modeled by a differentiable function f , where $f(t)$ is measured in gallons per second and t is measured in seconds since pumping began. Selected values of $f(t)$ are given in the table.

(a) Using correct units, interpret the meaning of $\int_{60}^{135} f(t) \, dt$ in the context of the problem. Use a right Riemann sum with the three subintervals $[60, 90]$, $[90, 120]$, and $[120, 135]$ to approximate the value of $\int_{60}^{135} f(t) \, dt$.

(b) Must there exist a value of c , for $60 < c < 120$, such that $f'(c) = 0$? Justify your answer.

(c) The rate of flow of gasoline, in gallons per second, can also be modeled by

$g(t) = \left(\frac{t}{500}\right) \cos\left(\left(\frac{t}{120}\right)^2\right)$ for $0 \leq t \leq 150$. Using this model, find the average rate of flow of gasoline over the time interval $0 \leq t \leq 150$. Show the setup for your calculations.

(d) Using the model g defined in part (c), find the value of $g'(140)$. Interpret the meaning of your answer in the context of the problem.

Part A

4-1-1

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point the response must reference gallons of gasoline added/pumped and the time interval $t = 60$ to $t = 135$.

To earn the second point at least five of the six factors in the Riemann sum must be correct.

If there is any error in the Riemann sum, the response does not earn the third point.

A response of $(0.15)(30) + (0.1)(30) + (0.05)(15)$ earns both the second and third points, unless there is a subsequent error in simplification, in which case the response would earn only the second point.

A response that presents a correct value with accompanying work that shows the three products in the Riemann sum but does not show all six of the factors and/or the sum process, does not earn the second point but does earn the third point. For example, responses of either $4.5 + 3.0 + 0.75$ or $(0.15)(30), 0.1(30), 0.05(15) \rightarrow 8.25$ earn the third point but not the second.

A response of $f(90)(90 - 60) + f(120)(120 - 90) + f(135)(135 - 120) = 8.25$ earns both the second and the third points.

A response that presents an answer of only 8.25 does not earn either the second or third point.

A response that provides a completely correct left Riemann sum with accompanying work, $f(60)(30) + f(90)(30) + f(120)(15) = 9$, or $(0.1)(30) + (0.15)(30) + (0.1)(15)$ earns 1 of the last 2 points. A response with any errors or missing factors in a left Riemann sum earns neither of the last 2 points.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Interpretation with units
- ☐ Form of Riemann sum
- ☐ Answer

Solution:

$\int_{60}^{135} f(t) dt$ represents the total number of gallons of gasoline pumped into the gas tank from time $t = 60$ seconds to time $t = 135$ seconds.

4-1-1

$$\int_{60}^{135} f(t) dt$$

$$\approx f(90)(90 - 60) + f(120)(120 - 90) + f(135)(135 - 120)$$

$$= (0.15)(30) + (0.1)(30) + (0.05)(15) = 8.25$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point a response must present either $f(120) - f(60) = 0$, $0.1 - 0.1 = 0$ (perhaps as the numerator of a quotient), or $f(60) = f(120)$.

To earn the second point a response must:

- have earned the first point,
- state that f is continuous because f is differentiable (or equivalent), and
- answer “yes” in some way.

A response may reference either the Mean Value Theorem or Rolle’s Theorem.

A response that references the Intermediate Value Theorem cannot earn the second point.



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ $f(120) - f(60) = 0$
- ☐ Answer with justification

Solution:

f is differentiable. $\Rightarrow f$ is continuous on $[60, 120]$.

$$\frac{f(120) - f(60)}{120 - 60} = \frac{0.1 - 0.1}{60} = 0$$

By the Mean Value Theorem, there must exist a c , for $60 < c < 120$, such that $f'(c) = 0$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

4-1-1

The exact value of $\frac{1}{150} \int_0^{150} g(t) dt$ is $\frac{12}{125} \sin(\frac{25}{16})$.

A response may present the average value formula in single or multiple steps. For example, the following response earns both points: $\int_0^{150} g(t) dt = 14.399504$ so the average rate is 0.0959967.

A response that presents the average value formula in multiple steps but provides incorrect or incomplete communication (e.g., $\int_0^{150} g(t) dt = \frac{14.399504}{150} = 0.0959967$) earns 1 out of 2 points.

A response of $\int_0^{150} g(t) dt = 0.0959967$ does not earn either point.

Degree mode: A response that presents answers obtained by using a calculator in degree mode does not earn the first point it would have otherwise earned. The response is generally eligible for all subsequent points (unless no answer is possible in degree mode or the question is made simpler by using degree mode). In degree mode, $\frac{1}{150} \int_0^{150} g(t) dt = 0.149981$ or 0.002618.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Average value formula
- ☐ Answer

Solution:

$$\frac{1}{150-0} \int_0^{150} g(t) dt = 0.0959967$$

The average rate of flow of gasoline, in gallons per second, is 0.096 (or 0.095).

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The exact value of $g'(140)$ is $\frac{1}{500} \cos(\frac{49}{36}) - \frac{49}{9000} \sin(\frac{49}{36})$.

The value of $g'(140)$ may appear only in the interpretation.

To be eligible for the second point a response must present some numerical value for $g'(140)$.

4-1-1

To earn the second point the interpretation must include “the rate of flow of gasoline is changing at a rate of [the declared value of $g'(140)]$ ” and “at $t = 140$ ” (or equivalent).

An interpretation of “decreasing at a rate of -0.005 ” or “increasing at a rate of ” does not earn the second point.

Degree mode: In degree mode, $g'(140) = 0.001997$ or 0.00187 .



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ $g'(140)$
- ☐ Interpretation

Solution:

$$g'(140) \approx -0.004908$$

$$g'(140) = -0.005 \text{ (or } -0.004)$$

The rate at which gasoline is flowing into the tank is decreasing at a rate of 0.005 (or 0.004) gallon per second per second at time $t = 140$ seconds.

4-1-1

46. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

For time $0 \leq t \leq 10$, water is flowing into a small tub at a rate given by the function F defined by $F(t) = \arctan\left(\frac{\pi}{2} - \frac{t}{10}\right)$. For time $5 \leq t \leq 10$, water is leaking from the tub at a rate given by the function L defined by $L(t) = 0.03(20t - t^2 - 75)$. Both $F(t)$ and $L(t)$ are measured in cubic feet per minute, and t is measured in minutes. The volume of water in the tub, in cubic feet, at time t minutes is given by $W(t)$.

- (a) At time $t = 3$, there are 2.5 cubic feet of water in the tub. Write an equation for the locally linear approximation of W at $t = 3$, and use it to approximate the volume of water in the tub at time $t = 3.5$.
- (b) Find $W''(8)$. Using correct units, interpret the meaning of $W''(8)$ in the context of the problem.
- (c) Is there a time t , for $5 < t < 10$, at which the rate of change of the volume of water in the tub changes from positive to negative? Give a reason for your answer.
- (d) The tub is in the shape of a rectangular box that is 0.5 foot wide, 4 feet long, and 3 feet deep. What is the rate of change of the depth of the water in the tub at time $t = 6$?

Part A

The second point may be earned if response contains an incorrect value for slope based on one computational error.

The third point does not require a simplified answer. Substitution of function values is required. Units are not required.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

4-1-1



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ slope
- ☐ equation for tangent line
- ☐ approximation

Solution:

$$W(3) = 2.5$$

$$W'(3) = F(3) = 0.904089$$

$$y = W(3) + W'(3)(t - 3) = 2.5 + 0.904(t - 3)$$

$$W(3.5) \approx W(3) + W'(3)(3.5 - 3) = 2.952 \text{ cubic feet}$$

Part B

The second point requires a time reference, rate, and units. The point may be earned with an incorrect approximation based on one computational error and provided the behavior described is consistent with the sign of the approximation, the numerical value, and the time reference.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ $W''(8)$
- ☐ interpretation with units

Solution:

$$W''(8) = F'(8) - L'(8) = -0.183 \text{ (or } -0.182)$$

At time $t = 8$ minutes, the rate at which the rate of change of the volume of water is changing by -0.183 (or -0.182) cubic feet per minute per minute.

4-1-1

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The student response accurately includes a correct answer with reason.

Solution:

$$W'(t) = F(t) - L(t) = 0 \text{ for } 5 < t < 10.$$

$$W'(t) = 0 \Rightarrow t = 8.149287$$

Yes, there is such a time t , because $W'(t) > 0$ for $5 < t < 8.149$ and $W'(t) < 0$ for $8.149 < t < 10$.

Part D

The first point is earned with $\frac{dV}{dt} = 2\frac{dh}{dt}$.

The third point may be earned with a maximum of one error in the evaluation of $\frac{dV}{dt}$.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ relationship between $\frac{dV}{dt}$ and $\frac{dh}{dt}$
- ☐ $\frac{dV}{dt} \big|_{t=6}$
- ☐ answer

The volume of water in the tub is $V = (0.5)(4)h = 2h$, where h is the depth of water, in feet.

$$\frac{dV}{dt} = 2\frac{dh}{dt}$$

$$\text{At time } t = 6, \frac{dV}{dt} \big|_{t=6} = W'(t) = F(6) - L(6) = 0.500581.$$

4-1-1

Therefore, $\left. \frac{dh}{dt} \right|_{t=6} = \frac{1}{2} \frac{dV}{dt} = 0.250$

4-1-1

47. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

For time $0 \leq t \leq 8$, people arrive at a venue for an outdoor concert at a rate modeled by the function A defined by $A(t) = 0.3 \sin(1.9t) + 0.3 \cos(0.6t) + 1.3$. For time $0 \leq t \leq 1$, no one leaves the venue, and for time $1 \leq t \leq 8$, people leave the venue at a rate modeled by the function L defined by $L(t) = 0.2 \cos(1.9t) + 0.2 \sin t + 0.8$. Both $A(t)$ and $L(t)$ are measured in hundreds of people per hour, and t is measured in hours. The number of people at the venue, in hundreds, at time t hours is given by $P(t)$.

- (a) At time $t = 2$, there are 6 hundred people at the venue. Write an equation for the locally linear approximation of P at $t = 2$, and use it to approximate the number of people at the venue at time $t = 2.5$.
- (b) Find $P''(5)$. Using correct units, interpret the meaning of $P''(5)$ in the context of the problem.
- (c) Is there a time t , for $1 < t < 8$, at which the rate of change of the number of people at the venue changes from negative to positive? Give a reason for your answer.
- (d) A rectangular “standing-only” section at the venue changes size as t increases in order to manage the flow of people. Let x represent the length, in feet, of the section, and let y represent the width, in feet, of the section. The length of the section is increasing at a rate of 6 feet per hour, and the width of the section is decreasing at a rate of 3 feet per hour. What is the rate of change of the area of the section with respect to time when $x = 16$ and $y = 10$? Indicate units of measure.

Part A

The second point may be earned if response contains an incorrect value for slope based on one computational error.

The third point **does not** require a simplified answer. Substitution of function values is required. An answer of 6.201 (or 6.200) alone does not earn the point.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

4-1-1



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ slope
- ☐ equation for tangent line
- ☐ approximation

Solution:

$$P(2) = 6$$

$$P'(2) = A(2) - L(2) = 0.401484$$

$$y = P(2) + P'(2)(t - 2) = 6 + 0.401484(t - 2)$$

$$P(2.5) \approx P(2) + P'(2)(2.5 - 2) = 6.201 \text{ (or 6.200) hundred people (} \approx 620 \text{ OR } \approx 621 \text{ people)}$$

Part B

The second point requires a time reference, rate, and units. The point may be earned with an incorrect approximation based on one computational error and provided the behavior described is consistent with the sign of the approximation, the numerical value, and the time reference.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ $P''(5)$
- ☐ interpretation with units

Solution:

$$P''(5) = A'(5) - L'(5) = -0.679$$

At time $t = 5$ hours, the rate of change of the number of people at the stage is changing at a rate of -0.679 hundred people (≈ -68 people) per hour per hour.

4-1-1

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The student response accurately includes a correct answer with reason.

Solution:

$$P'(t) = A(t) - L(t) = 0 \text{ for } 1 < t < 8.$$

$$P'(t) = 0 \Rightarrow t = 5.811 \text{ (or 5.810), } t = 6.548 \text{ (or 6.547)}$$

Yes, there is such a time t , because $P'(t) < 0$ for $5.811 < t < 6.548$ and $P'(t) > 0$ for $6.548 < t < 8$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ $\frac{dA}{dt} = x \frac{dy}{dt} + y \frac{dx}{dt}$
- ☐ $\frac{dx}{dt} = 6$ and $\frac{dy}{dt} = -3$
- ☐ answer with units

Solution:

Let x represent the length of the section, in feet, and let y represent the width of the section, in feet.

The area of the rectangular section is $A = xy$.

$$\frac{dA}{dt} = x \frac{dy}{dt} + y \frac{dx}{dt}$$

$$\text{When } x = 16 \text{ and } y = 10, \frac{dA}{dt} = x \frac{dy}{dt} + y \frac{dx}{dt} = (16)(-3) + (10)(6) = 12.$$

Therefore, the rate of change of the area of the section with respect to time is 12 square feet per hour.

4-1-1

48. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (days)	0	6	13	18
$M(t)$ (coins per day)	42	38	35	30

A class collected 1-cent coins to raise money for classroom supplies. Students in the class deposited the coins into a container where all the coins were kept until the end of the collection period. For time t days, $0 \leq t \leq 18$, the rate at which coins are being added to the container is modeled by a decreasing, twice-differentiable function M . Values of $M(t)$ at various times t are shown in the table above.

(a) Use a right Riemann sum with the three subintervals indicated by the table to approximate

$\int_0^{18} M(t) \, dt$. Show the computations that lead to your answer.

(b) Is the approximation from part (a) an overestimate or an underestimate for $\int_0^{18} M(t) \, dt$? Give a reason for your answer.

(c) Using correct units, approximate $M'(13)$ using the average rate of change of M over the interval $6 \leq t \leq 18$.

(d) Explain the meaning of $M'(13)$ in the context of the problem.

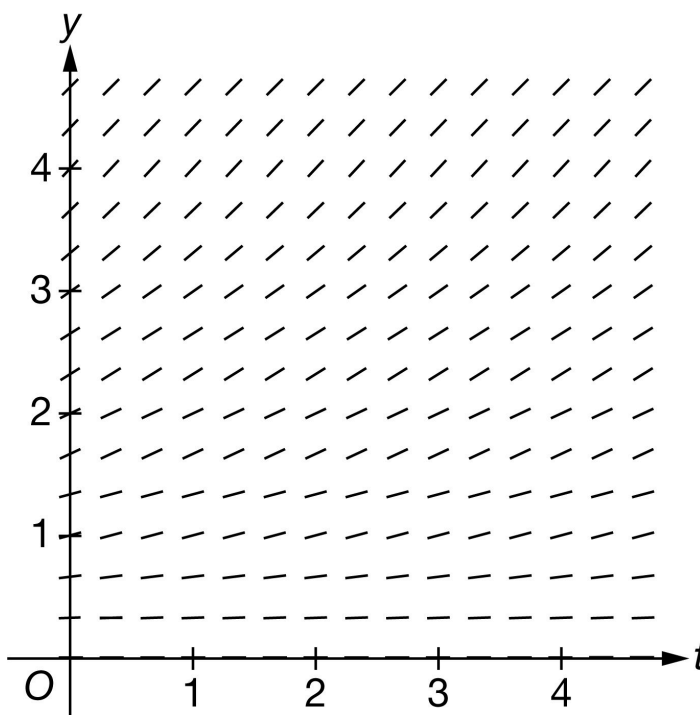
(e) It is known that $M'(6) = -0.5$. Write an equation for the line tangent to the graph of $y = M(t)$ at $t = 6$.

(f) Find the value of $\int_0^{18} M'\left(\frac{t}{3}\right) \, dt$. Show the computations that lead to your answer.

(g) A second class is also collecting 1-cent coins. For $0 \leq t \leq 18$, the total number of coins collected by the second class can be modeled by the decreasing and differentiable function G , where G is a function of t . Assume the weight of each coin is 3 grams. Write an expression for the rate of change, in grams per day, of the total weight of the coins for the second class at $t = 8$ days.

4-1-1

(h) Explain why the following could not be a slope field for the differential equation $\frac{dy}{dt} = G'(t)$ for the function G defined in part (g).

**Part A**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for setting up a right Riemann sum. The setup must be a sum of three products.

A response may still earn this point if there is only one numerical error in the expression for the Riemann sum with values pulled from the table. (See example below).

Responses with the numerical error described above are not eligible for the second point.

Note: either the multiplication or the addition may be indicated implicitly. For example, the response may show appropriate values put into rows and columns, like so.

38	6	228
35	7	245
30	5	+150
		623

or

4-1-1

$(6 - 0)(38)$	$= 228$
$(13 - 6)(35)$	$= 245$
$(18 - 13)(30)$	$= 150$
	623

Implicit responses such as those above are eligible for both points.

A response presenting a left Riemann sum (instead of right), will not earn any points for part (a).

The second point is for the correct numerical result.

In order to earn the second point, there must be supporting calculations.

The numerical values for $M(6)$, $M(13)$, and $M(18)$ must be pulled from the table

Any simplification shown must be correct.

Use the following as an illustration how to score both points. (In most cases, you see scores for which the expression represents the entirety of relevant work shown for the response.)

- A response of $38 \cdot (6 - 0) + 35 \cdot (13 - 6) + 30 \cdot (18 - 13)$ earns both points.
- A response of $6 \cdot M(6) + 7 \cdot M(13) + 5 \cdot M(18)$ earns the first point. Eligible for second point if more work is on the page.
- A response of $6 \cdot M(6) + 7 \cdot M(13) + 5 \cdot M(18) = 623$ earns both points.
- A response of $6 \cdot 38 + 7 \cdot 35 + 5 \cdot 30$ earns both points.
- A response of $6 \cdot 38 + 7 \cdot 35 + 6 \cdot 30$ contains one numerical error. Earns the first point. Not eligible for the second point.
- A response of $6 \cdot 38 + 7 \cdot 35 + 5 \cdot 30 = 641$ earns first point for left side of equation. Does not earn second point due to incorrect simplification.
- A response of $228 + 245 + 150$ or $228 + 245 + 150 = 623$ does not earn the first point because we cannot see the sum of three products. However, this is the correct sum, so we are not seeing an unsupported answer. The second point will be earned.

The response does not need to include “approximately” in any way. We will read $=$ and $\approx$ as the same thing.

Minimalist response: The following expression represents the minimum work that must be seen in a response in order to earn both points. $6 \cdot 38 + 7 \cdot 35 + 5 \cdot 30$



0	1	2
---	---	---

4-1-1

The student response accurately includes **both** of the criteria below.

- ☐ Right Riemann sum
- ☐ Answer

Solution:

$$\int_0^{18} M(t) dt \approx M(6)(6-0) + M(13)(13-6) + M(18)(18-13)$$

$$= 38 \cdot 6 + 35 \cdot 7 + 30 \cdot 5 = 623$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned when the response states that the approximation is an underestimate because M is decreasing.

The response does not need an explicit reference to part (a) or to a right Riemann sum.

A response that appeals to concavity as a reason will NOT earn this point.

The following examples earn the point:

- Underestimate because M is decreasing
- Underestimate because $M'(t) < 0$
- Underestimate because the graph of M is decreasing (we know they are talking about the given function M)
- Underestimate because the function is decreasing. (There is only one function in the problem at this point.)

The following examples do not earn the point:

- Underestimate because M is concave down/up
- Underestimate because the graph/curve is decreasing. Since no graph/curve is provided, we do not know what graph/curve they are talking about.
- Underestimate because the table [or table values] is/are decreasing. The behavior of the values shown in the table is not the same as the behavior of the function M . The table does not tell us what is happening to M between the given points.



0	1
---	---

4-1-1

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

Because M is a decreasing function, and this is a right Riemann sum, this is an underestimate.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for the correct approximation shown with supporting work.

We need to see both a difference and a quotient. They don't have to be together.

The expression does not need to be simplified, but values for $M(18)$ and $M(6)$ must be pulled from the table.

Decimal approximation is not expected, but if presented, then the decimal value must be correct to three decimal places, either rounded or truncated. We do not read anything past the third decimal place.

- If the first three places match -0.666 or -0.667 then the point is earned.
- It does not matter whether any subsequent decimal places match the correct value or not. (For example, the value -0.6672 will earn the point because the first three decimal places match the correct rounded value.)
- Responses with fewer than three decimal places (-0.66 , -0.67 , -0.7 , etc.) will not earn the point

The response does not need to include “approximately” in any way. We will read $=$ and $\approx$ as the same thing.

The second point is for correct units.

Units must be attached to a numerical value (either correct or incorrect).

Can be written using words or a combination of words and symbols. For example:

- Coins per day per day or Coins/day/day
- Coins per day² or Coins/day²
- Coins per day squared

We do not read work shown in part (d) for this point. The units must appear in part (c).

The letter “C” is not a standard abbreviation for “coins.” It will not be interpreted as such.

The response will not be penalized for any attempt to verbally describe their answer after a correct answer has been presented.

4-1-1

Minimalist response: The following expression and units represent the minimum work that must be seen in a response in order to earn both points. $\frac{30-38}{12}$ coins/day²



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$M'(13) \approx \frac{M(18) - M(6)}{18 - 6} = \frac{30 - 38}{12} = -\frac{2}{3} \text{ coin per day per day}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Three components of a correct answer:

- Indicating a rate of a rate
- Including the context of coin collection
- Explicitly referring to the specified point in time ($t = 13$)

Incorrect or missing units will not be penalized as long as there is a clear connection to the context of the problem.

The words “increasing” and “decreasing” are interpreted as modifiers describing one of the specified rates.

If the numerical value from part (c) is used in this interpretation, then we expect that the choice of “increasing” or “decreasing” is consistent with that value.

The following examples use $-\frac{2}{3}$ as the imported value from part (c). If a different [incorrect] value is imported from part (c), then these statements should be consistent with that value. (Remember that we are reading $=$ and $\approx$ as the same thing.)

- “... increasing at a rate of $-\frac{2}{3}$ ” is OK [Units not required in part (d)]. (the point is not yet earned with this phrase alone)
- “... decreasing at a rate of $\frac{2}{3}$ ” is OK [Units not required in part (d)]. (the point is not yet earned with this phrase alone)

4-1-1

- “... decreasing at a rate of $-\frac{2}{3}$ ” is not OK and the point is not earned.

Rate of a rate

To earn this point, the response must explicitly discuss the rate of change of a rate of change. For example:

- The rate of change in the rate at which coins are changing
- The rate of change in coins is changing (or decreasing) at a rate of . . .
- The rate at which the rate of change of coins is changing

The words “increasing,” “decreasing,” or “changing” are not interpreted as rates.

Units are not interpreted as rates. (The phrase “coins per day” is not enough to indicate a rate.)

Slope of tangent line, velocity, and acceleration are not appropriate interpretations for this context.

“Rate of change of M ” is not enough without describing what M represents in this context.

Context

A key component of the context for this explanation is “coins.”

Explicit reference to $t = 13$

Acceptable reference to specific point in time include (but are not limited to):

- On day 13 (where day could be replaced with any other unit of time)
- At $t = 13$ (with or without units of time)
- After 13 days (where days could be replaced with any other unit of time).

Note: we are choosing to assume that “after” is being used in lieu of “at” unless we have additional evidence to the contrary.

If the response is explicitly indicating an interval of time (e.g., “from day 13 on” or “during the first 13 days”), this point will not be earned.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

4-1-1

$M'(13)$ is the rate at which the rate of change of the number of 1-cent coins in the container is changing, in coins per day per day, on day 13.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned with a correct tangent line equation in any form. Any of these equations can be written as is without any explanation or additional supporting work. If rewriting of equations is done (transforming to slope-intercept form, perhaps), it must be done correctly.

- $y = -0.5(t - 6) + 38$

- $y - 38 = -0.5(t - 6)$

- $y = -0.5t + 41$

The key calculus concept being assessed in this part of the problem is that the value of $M'(6)$ is the slope of the tangent line. In that spirit:

- The response will not be penalized for using x instead of t .
- We are accepting $M(t)$ in lieu of y in all forms of the tangent line equation.



0	1
---	---

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$$y = M'(6)(t - 6) + M(6)$$

$$y = -0.5(t - 6) + 38$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with a correct substitution. There are three components of substitution that we need to see:

- Uses $u = \frac{t}{3}$, which may be done implicitly
- Uses the correct constant of 3

4-1-1

· Uses the correct limits for the substitution (0 and 6). These limits may only be seen when evaluating the antiderivative. We don't need to see them as limits on a definite integral.

Either of the expressions $3M\left(\frac{t}{3}\right)\big|_0^{18}$ or $3M(u)\big|_0^6$ will earn the substitution point without any additional supporting work.

Special Handling: Notation Errors

Errors in notation, such as the examples listed here, will not prevent the response from earning the first or second points. These errors will cause the response to be ineligible for the third point.

- Limits for t attached to an integral with u as the variable of integration (or vice versa)
- Improper equals signs, such as an indefinite integral equal to a definite integral
- Reversed limits of integration, for example: $\int_6^0 M'(u) du$

The second point is for the Fundamental Theorem of Calculus.

This point can be earned with any of the following, where k is a nonzero constant and $b = 6$ or $b = 18$. No explanation or additional supporting work is necessary.

- o $kM\left(\frac{t}{3}\right)\big|_0^b$
- o $k(M(b) - M(0))$

This point can be earned without earning the substitution point.

The third point is for the correct numerical value, which is -12 .

Values from the table must be used. $3(M(6) - M(0))$ is not enough to earn the answer point.

The expression $3(38 - 42)$ is sufficient for this point. No simplification is required. Any simplification that is shown must be correct.

Units are not required, and if shown, they need not be correct in order to earn this point.

The response cannot earn this point without also earning first two points.

Minimalist response: The expression $3(38 - 42)$ represents the minimum work that must be seen in a response in order to earn all three points.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

4-1-1

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = \frac{t}{3}$. Then $du = \frac{dt}{3}$.

When $t = 0$, $u = 0$.

When $t = 18$, $u = 6$.

$$\begin{aligned}\int_0^{18} M' \left(\frac{t}{3} \right) dt &= 3 \int_0^6 M' (u) du \\ &= 3 (M (6) - M (0)) \\ &= 3 (38 - 42) = -12\end{aligned}$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct expression at $t = 8$ earns the point. It may be written as $3 \cdot G'(8)$ or $3 \cdot \frac{dG}{dt} \Big|_{t=8}$.

The expression $3 \cdot G'(t)$ is not sufficient because it does not use $t = 8$.

The response must use G (not M or any other function name) in order to earn the point.



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Expression

Solution:

An expression for the rate of change, in grams per day, of the total weight of the coins for the second class at $t = 8$ days is $3G'(8)$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

4-1-1

This point is earned for a correct statement identifying the distinction between the differential equation

$$\frac{dy}{dt} = G'(t) \text{ and the given slope field.}$$

In order to earn the point, the response must state that the values of the differential equation are negative while the slopes in the given slope field are positive or 0 (nonnegative).

Some ways to say that the values of differential equation are negative:

- G is decreasing
- $G'(t) < 0$
- G has negative slopes

When discussing the slopes in the slope field, the response must explicitly refer to the signs of the slopes.

The following examples could earn the point:

- The slope field has nonnegative slopes.
- The slopes shown are positive or 0.
- The slope field contains positive slopes.
- “The graph” or “The field” could be used in lieu of “The slope field.”

The following examples will not earn the point:

- The slope field is increasing/positive.
- The slope field looks like it represents an increasing function.
- The slope field represents a quadratic [or exponential] function.
- “It” (rather than “the slope field”)

Statements like “ G is a decreasing function, so the signs of the slopes in the slope field should be negative” can earn this point because “should be” is indicating contrast with what is known.

The response will not be penalized for saying that the slopes in the slope field are “always” positive or “all” positive.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

4-1-1**Solution:**

G is a decreasing function, so $\frac{dy}{dt} = G'(t) \leq 0$. However, in the given slope field, the slopes of the line segments for $y > 0$ are positive.

4-1-1

49. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (days)	0	4	10	18
$M(t)$ (number of 1-cent coins)	0	200	540	900

A class collected 1-cent coins to raise money for classroom supplies. Students in the class deposited the coins into a container where all the coins were kept until the end of the collection period. For time t days, $0 \leq t \leq 18$, the total number of coins in the container is modeled by an increasing, twice-differentiable function M . Values of $M(t)$ at various times t are shown in the table above.

(a) Using correct units, approximate $M'(10)$ using the average rate of change of M over the interval $4 \leq t \leq 18$.

(b) Explain the meaning of $M'(10)$ in the context of the problem.

(c) It is known that $M'(4) = 48$. Write an equation for the line tangent to the graph of $y = M(t)$ at $t = 4$.

(d) Use a right Riemann sum with the three subintervals indicated by the table to approximate

$\int_0^{18} M(t) \, dt$. Show the computations that lead to your answer.

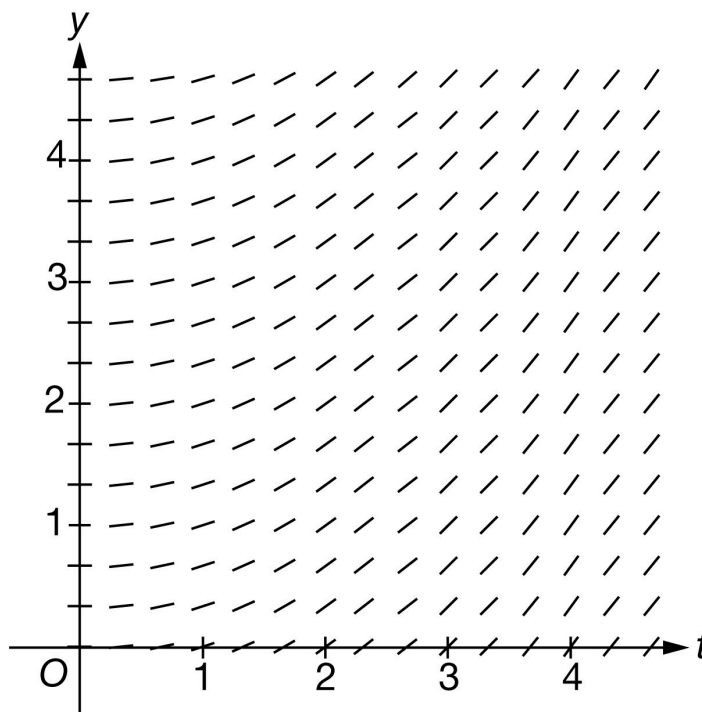
(e) Is the approximation from part (d) an overestimate or an underestimate for $\int_0^{18} M(t) \, dt$? Give a reason for your answer.

(f) Find the value of $\int_0^6 M'(3t) \, dt$. Show the computations that lead to your answer.

(g) For $0 \leq t \leq 18$, the total number of coins collected can also be modeled by the differentiable function N , where N is a function of t . Assume the weight of each coin is 2.5 grams. Write an expression for the rate of change, in grams per day, of the total weight of the coins at $t = 9$ days.

4-1-1

(h) Explain why the following slope field cannot represent the differential equation $\frac{dy}{dt} = 0.4y$.

**Part A**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for the correct approximation shown with supporting work. The supporting work must be at least a difference and a quotient. Correct values must be taken from the table to earn the point. The difference quotient does not need to be simplified.

Examples that earn the first point include:

$$\cdot \frac{900 - 200}{18 - 4}$$

$$\cdot \frac{M(18) - M(4)}{14} = \frac{700}{14}$$

$$\cdot \frac{M(18) - M(4)}{14} = 50$$

Examples that do not earn the first point include:

$$\cdot \frac{M(18) - M(4)}{14}$$

$$\cdot \frac{M(18) - M(4)}{18 - 4}$$

Minimal responses:

4-1-1

· $\frac{900-200}{14}$ or $\frac{700}{18-4}$ earns first point

· Neither $\frac{700}{14}$ nor 50, with no supporting work, will earn this point

The second point is for correct units

The correct units must be attached to a numerical value: either correct or incorrect.

The units may be written using words or a combination of words and symbols. For example:

- Coins per day
- Pennies per day
- Cents per day
- Coins/day

The response will not be penalized for any verbal attempt to discuss the answer after the correct answer has been presented.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$M'(10) \approx \frac{M(18)-M(4)}{18-4} = \frac{900-200}{14} = 50 \text{ coins per day}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Three components of a correct answer:

- Indicating a rate of change
- Including the context of coin collection
- Referring to the specific point in time, $t = 10$

4-1-1

Students are not penalized for incorrect units in this part. (Whether correct or incorrect, units have no effect on the score in this part.)

Responses may use “changing” or “increasing” as a modifier as long as the value found in part (a) is not brought into part (b) in an incorrect manner. Example: “The number of coins is increasing at a rate of 50 coins per day...” is okay.

Acceptable references to a specific point in time include (but are not limited to):

- On day 10 and after 10 days
- At $t = 10$
- At $t = 10$ weeks or any incorrect unit of time

If the response clearly indicates an interval of time (e.g., “from day 10 on,” “during the first 10 days,” or “for $t = 0$ to $t = 10$ ”), the point is not earned.

Examples that earn the point include:

- $M'(10)$ is the rate of change of the number of coins per day at time $t = 10$
- The rate of change of number of coins is 50 coins per day on day 10
- $M'(10)$ is the rate of change in the number of coins at time $t = 10$ in coins per *month*
- 50 is the rate of increase in the number of coins per day on day 10

Examples that do not earn the point include:

- $M'(10)$ is the average rate of change of the total number of coins per day on day 10
- $M'(10)$ is the total number of coins per day per day on day 10
- The rate of change is 50 coins per day (Missing “at $t = 10$ ”)
- $M'(10)$ is the slope of the tangent line at $t = 10$
- Rate of change of M at time $t = 10$
- Coins per day increases on day 10



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

4-1-1

Solution:

$M'(10)$ is the rate at which the number of 1-cent coins in the container is changing, in coins per day, on day 10.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned with a correct tangent line equation in any form. The use of x in place of t is acceptable. Some examples include

· $y - 200 = 48(t - 4), y - 200 = 48(x - 4)$

· $y = 48(t - 4) + 200, y = 48(x - 4) + 200$

· $y = 48t + 8, y = 48x + 8$

The key calculus concept being assessed in this part of the problem is that the value of $M'(4)$ is the slope of the tangent line. In that spirit, we are accepting both $y = \dots$ and $M(t) = \dots$, but not $M(4) = \dots$

· $M(t) = 48(t - 4) + 200$ earns the point

· However, $M(4) = 48(t - 4) + 200$ does not earn the point



0	1
---	---

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$$y = M'(4)(t - 4) + M(4)$$

$$y = 48(t - 4) + 200$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for setting up a right Riemann sum. The setup must be a sum of three products.

· A response may still earn this point if there is only one numerical error in the expression for the Riemann sum with values pulled from the table. (See example below).

4-1-1

- Responses with the numerical error described above are not eligible for the second point.

Note: either the multiplication or the addition may be indicated implicitly. For example, the response may show appropriate values put into rows and columns. Implicit responses are eligible for both points.

A response presenting a left Riemann sum (instead of right), will not earn any points for part (d).

The second point is for the correct numerical result.

- In order to earn the second point, there must be supporting calculations.
- The numerical values for $M(4)$, $M(10)$, and $M(18)$ must be pulled from the table.
- Any simplification shown must be correct.

Use the following as an illustration to score both points. (In most cases, the expression on the left side represents the entirety of relevant work shown for the response.)

- A response of $200 \cdot (4 - 0) + 540 \cdot (10 - 4) + 900 \cdot (18 - 10)$ earns both points.
- A response of $4M(4) + 6M(10) + 8M(18)$ earns the first point. Eligible for second point if more work is on the page.
- A response of $4M(4) + 6M(10) + 8M(18) = 11,240$ earns both points.
- A response of $4 \cdot 200 + 6 \cdot 540 + 8 \cdot 900$ earns both points.
- A response of $4 \cdot 200 + 10 \cdot 540 + 8 \cdot 900$ contains one numerical error and earns the first point, but is not eligible for the second point.
- A response of $4 \cdot 200 + 6 \cdot 540 + 8 \cdot 900 = 10,240$ earns the first point for left side of equation but does not earn second point due to incorrect simplification.
- A response of $800 + 3240 + 7200$ or $800 + 3240 + 7200 = 11,240$ does not earn the first point because we cannot see the sum of 3 products. However, this is the correct sum, so we see an answer with supporting work. The second point will be earned.

The response does not need to include “approximately” in any way. We will read $=$ and $\approx$ as the same thing.

Minimalist response: The following expression represents the minimum work that must be seen in a response in order to earn both points.

$$4 \cdot 200 + 6 \cdot 540 + 8 \cdot 900$$



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

4-1-1

- ☐ Right Riemann sum
- ☐ Answer

Solution:

$$\int_0^{18} M(t) dt \approx M(4)(4-0) + M(10)(10-4) + M(18)(18-10)$$
$$= 200 \cdot 4 + 540 \cdot 6 + 900 \cdot 8 = 11,240$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is earned when the response states that the approximation is an overestimate because M is increasing.

The response does not need an explicit reference to the approximation in part (d), nor does there need to be an answer given in part (d).

The following statements are sufficient for earning this point:

- Overestimate because M is increasing
- Overestimate because the function is increasing

Note: At this point, there is only one function being discussed in this problem so we will assume “the function” is M .

When a response includes additional information about concavity, interpret that as a lack of understanding about the question, so this will not earn the point. When the response references a graph or the given table rather than the function M , the point is not earned.

Examples that earn the point:

- The right Riemann sum overestimates because the function is increasing.
- The sum is an overestimate because $M(t)$ is increasing.
- The sum is an overestimate because $M'(t) > 0$.

Examples that do not earn the point:

- The sum is an overestimate because $M'(t)$ is increasing. ($M'(t)$ increasing is not known.)
- The right Riemann sum overestimates because the graph is increasing.
- The right Riemann sum overestimates because the function is increasing and concave up.

4-1-1



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

Because M is an increasing function, and this is a right Riemann sum, the answer from part (d) is an overestimate.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with a correct substitution. There are three components of substitution that need to be seen:

- Uses $u = 3t$, which may be done implicitly
- Uses the correct constant of $\frac{1}{3}$
- Uses the correct limits for the substitution (0 and 18). These limits may only be seen when evaluating the antiderivative. These don't need to be seen as limits on a definite integral.
- Either of the expressions below will earn the substitution point without any additional supporting work:

$$\frac{1}{3} M(3t) \Big|_0^6 \text{ or } \frac{1}{3} M(u) \Big|_0^{18}$$

Special Handling: Notation Errors

Errors in notation, such as the examples listed here, will not prevent the response from earning the first or second points. These errors will cause the response to be ineligible for the third point.

- Limits for t attached to an integral with u as the variable of integration (or vice versa),
- Improper equals signs, such as an indefinite integral equal to a definite integral
- Reversed limits of integration, for example: $\int_{18}^0 M'(u) du$

The second point is for the Fundamental Theorem of Calculus. This point can be earned with any of the following, where k is a nonzero constant and $b = 6$ or $b = 18$. No explanation or additional supporting work is necessary.

$$\cdot kM(3t) \Big|_0^b$$

4-1-1

$$\cdot k(M(b) - M(0))$$

This point can be earned without earning the substitution point.

The third point is for the correct numerical value, which is 300.

Values from the table must be used. $\frac{1}{3}[M(18) - M(0)]$ is not enough to earn the answer point.

The expression $\frac{1}{3}(900 - 0)$ is sufficient for this point. No simplification is required. Any simplification that is shown must be correct.

Units are not required, and if shown, they need not be correct in order to earn this point.

The response cannot earn this point without also earning the FTC point.

Minimalist response: The expression $\frac{1}{3}(900)$ represents the minimum work that must be seen in a response in order to earn all three points.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = 3t$. Then $du = 3 dt$.

When $t = 0$, $u = 0$.

When $t = 6$, $u = 18$.

$$\begin{aligned} \int_0^6 M'(3t) dt &= \frac{1}{3} \int_0^{18} M'(u) du \\ &= \frac{1}{3}(M(18) - M(0)) \\ &= \frac{1}{3}(900 - 0) = 300 \end{aligned}$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

4-1-1

This point is earned for a correct expression at $t = 9$. No supporting work or justification is required.

The correct expression of $2.5M'(9)$ or $2.5N'(9)$ earns this point.

No justifications or explanations are required.

Any incorrect statement such as, $M'(9) = 2.5N'(9)$, does not earn this point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Expression

Solution:

An expression for the rate of change, in grams per day, of the total weight of the coins at $t = 9$ days is $2.5N'(9)$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned for a correct statement identifying the distinction between two things: the differential equation, $\frac{dy}{dt} = 0.4y$, and the given slope field.

The response does not need to explicitly reference both the differential equation and the given slope field., but it must explicitly reference one and implicitly reference the other.

Acceptable responses include:

- Identifying an inconsistency between the differential equation and the slope field at one point, e.g., at the point $(4, 0)$, the slope of the segment should be 0, but it is positive.
- Referencing a pattern of slopes, and explaining why that pattern does not fit the given differential equation.

A response that lacks even an implicit reference to the one of two things being compared will not earn the point.

For example, merely stating either $\frac{dy}{dt}$ is positive, or that the slope field is positive discretely does not earn the point.

- “The values for y are greater than or equal to zero so $\frac{dy}{dt} \geq 0$ ” lacks any reference to the slope field and will not earn the point.
- “All the slopes in the slope field are positive” lacks any reference to the differential equation and will not earn the point.

4-1-1

Vague answers which have the potential to be incorrect will not earn the point. Some examples are:

- “The field is increasing”
- Responses that state a specific slope value on the slope field for a general value of x or y that is not correct for all points with that x or y value do not earn the point. (e.g. “the slope field shows a slope of 0 when $y = 4$ ” is an incorrect statement)

A response may correctly solve or attempt to solve the differential equation and then provide a correct argument without reference to their solution. Such a response would earn the point.

A response might actually solve the differential equation and provide a correct argument on the resulting function.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

When $y = 0$, $\frac{dy}{dt} = 0$. However, in the given slope field, for $t > 0$, the slopes of the line segments for $y = 0$ are nonzero.

4-1-1

50. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (days)	0	5	10	15	20
$M(t)$ (number of 1-cent coins)	0	300	500	870	1100

A class collected 1-cent coins to raise money for classroom supplies. Students in the class deposited the coins into a container where all the coins were kept until the end of the collection period. For time t days, $0 \leq t \leq 20$, the total number of coins in the container is modeled by a twice-differentiable function M . Values of $M(t)$ at various times t are shown in the table above.

(a) Use a midpoint Riemann sum with the two subintervals indicated by the table to approximate

$$\int_0^{20} M(t) \, dt.$$

Show the computations that lead to your answer.

(b) Find the value of $\int_0^5 M'(3t) \, dt$. Show the computations that lead to your answer.

(c) Using correct units, approximate $M'(15)$ using the average rate of change of M over the interval $10 \leq t \leq 20$.

(d) Explain the meaning of $M'(15)$ in the context of the problem.

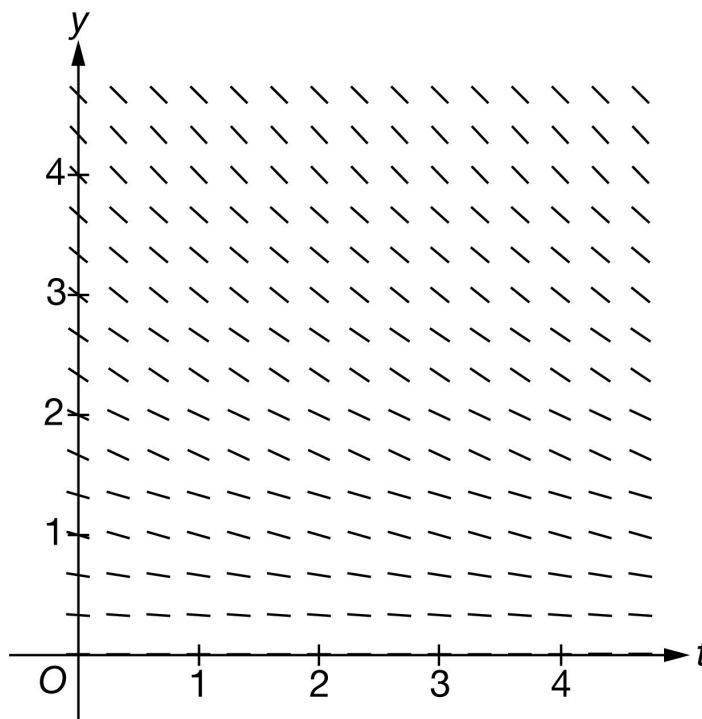
(e) It is known that $M'(10) = 55$. Write an equation for the line tangent to the graph of $y = M(t)$ at $t = 10$.

(f) It is known that $M''(t) > 0$ for $10 \leq t \leq 15$. If $M(14)$ is approximated using the equation from part (e), would the approximation be an overestimate or underestimate for the actual value of $M(14)$? Give a reason for your answer.

(g) For $0 \leq t \leq 20$, the total number of coins collected can also be modeled by the differentiable function R , where R is a function of t . Assume the weight of each coin is 3.5 grams. Write an expression for the rate of change, in grams per day, of the total weight of the coins at $t = 11$ days.

4-1-1

(h) Explain why the following slope field cannot represent the differential equation $\frac{dy}{dt} = 0.3y$.

**Part A**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point (Form of midpoint sum): Philosophy: The response should provide clear evidence adding the area of two rectangles whose heights are found using the midpoint of the interval.

Acceptable evidence:

$$\cdot 10 \cdot (M(5) + M(15))$$

$$\cdot 10 \cdot M(5) + 10 \cdot M(15)$$

$$\cdot (10 - 0) \cdot 300 + (20 - 10) \cdot 870$$

$$\cdot 10 \cdot (300 + 870) \text{ or } 10 \cdot 300 + 10 \cdot 870$$

To earn this point the response can have at most one mistake in the expression or equation. So, expressions of the form $\frac{10}{b} \cdot (M(5) + M(15))$, for any nonzero value of b would earn the point. Note: The value $\frac{10}{b}$ might be distributed across the sum.

Expressions of the form $\frac{a}{b} \cdot (M(5) + M(15))$ for $a \neq 10$ will not earn this point.

Once the first point is earned, it cannot be lost.

Second Point (Answer)

4-1-1

To earn the point, the answer must be correct. It does not need to be simplified, but if it is, the simplification must be correct.

The answer point cannot be earned unless the first (Form of midpoint sum) point was earned.

Minimal complete answers that earn both points:

$$\cdot 10 \cdot (300 + 870)$$

$$\cdot 10 \cdot 300 + 10 \cdot 870$$



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Form of midpoint sum

☐ Answer

Solution:

$$\begin{aligned} \int_0^{20} M(t) dt &\approx M(5)(10 - 0) + M(15)(20 - 10) \\ &= 300 \cdot 10 + 870 \cdot 10 = 11,700 \end{aligned}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point (Substitution) Philosophy: This point is earned when the response has all three components of the substitution done correctly.

Uses $u = 3t$, perhaps implicitly (any variable could be used for u)

Uses the correct constant of $\frac{1}{3}$

Uses the correct limits for their substitution.

· The limits might not be seen until the final evaluation, and at that point, response may have returned to the variable t .

· Limits for t attached to an integral involving u or vice versa will not earn this substitution point but could still earn the answer point, if the numerical answer is correct.

4-1-1

Second point: (Fundamental Theorem of Calculus) Philosophy: This point is earned when the FTC has been applied correctly to either $\int M'(3t) dt$ or $\int M'(u) du$.

The point is earned for any of the following:

- The equation $\int M'(3t) dt = M(3t)$ or $\int M'(u) du = M(u)$
- $\frac{1}{3}M(3t)|_0^5$ or $\frac{1}{3}M(u)|_0^{15}$
- $M(15) - M(0)$, $M(15)$, or $M(0)$

This point can be earned if the substitution point is not.

Note: $\int_a^b M'(u) du = M'(b) - M'(a)$ does not earn the point. The FTC point is for the response communicating that the antiderivative of M' is M .

Third Point (Answer)

Must be the correct numerical answer.

Must have earned the second point for the FTC.

Might have not earned the first point for substitution because of an incorrect use of limits, but will still earn the third point for the correct answer.

$(M(15) \text{ and } M(0) \text{ must be evaluated; the expression } \frac{1}{3}(M(15) - M(0)) \text{ by itself does not earn the third point.}$

The answer may be unsimplified, but if simplified it must be done correctly.

Scoring common responses:

The minimal work required to earn all three points is $\frac{1}{3}(870 - 0)$ or $\frac{1}{3} \cdot 870$

The following responses would earn the first two points and be eligible for the answer point if the response continues:

- $\frac{1}{3}M(3t)|_0^5$
- $\frac{1}{3}(M(15) - M(0))$

Responses of only the following would earn only the second point:

- $M(15) - M(0)$
- $M(5) - M(0)$

4-1-1

$$\cdot M(15)$$

$$\cdot \int_0^5 M'(u) \, du = M(5)$$

$$\cdot M(3t)|_0^5 \text{ or } M(u)|_0^{15}$$

The following responses would earn no points:

· 290 in the absence of one of the first two points.

$$\cdot \int_0^5 M'(3t) \, dt = M(5)$$



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = 3t$. Then $du = 3 \, dt$.

When $t = 0$, $u = 0$.

When $t = 5$, $u = 15$.

$$\int_0^5 M'(3t) \, dt = \frac{1}{3} \int_0^{15} M'(u) \, du$$

$$= \frac{1}{3}(M(15) - M(0))$$

$$= \frac{1}{3}(870 - 0) = 290$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point (Approximation)

The response must be a difference and a quotient.

4-1-1

The answer does not need to be simplified, but if it is simplified it must be done correctly.

The minimal work that must be shown to earn this point is either $\frac{1100-500}{10}$ or $\frac{600}{20-10}$.

Second point (Units)

The units must be attached to a correct or incorrect value.

The response will not be penalized for any verbal attempt to discuss the answer after the correct approximation with units is presented.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$M'(15) \approx \frac{M(20) - M(10)}{20 - 10} = \frac{1100 - 500}{10} = 60 \text{ coins per day}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The interpretation must include three components:

- indicate the rate of change
- the specific context of the number of coins in the container
- the specific point in time, $t = 15$

Responses do not need to include correct units or any units.

The word “increasing” may be used in place of the word “changing”.

The response is not earned for “the rate of coins deposited on day 15 “ as this does not indicate a change in the rate of deposit.

“the rate of change of M “ without specifying that M is the number of coins deposited, does not earn the point.

Acceptable ways to provide the specific point in time:

4-1-1

- On day 15 or after 15 days.
- At $t = 15$ (missing the unit of time)
- On week 15 (incorrect unit of time)

A response that clearly indicates an interval of time (e.g. “during the first 15 days” or “from day 15 on”) does not earn the point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

$M'(15)$ is the rate at which the number of 1-cent coins in the container is changing, in coins per day, on day 15.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The use of $M(t)$ in place of y earns the point.

The use of x in place of t earns the point.

Tangent lines do not need to be simplified, but any simplification must be done correctly.

Any correct form of the tangent line earns the point. For example:

- $y = 500 + 55(t - 10)$
- $y - 500 = 55(t - 10)$
- $y = 55t - 550 + 500$
- $y = 55t - 50$



0	1
---	---

The student response accurately includes the criteria below.

4-1-1

☐ Tangent line equation

Solution:

$$y = M'(10)(t - 10) + M(10)$$

$$y = 55(t - 10) + 500$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the point the response must include $M''(t) > 0$ and underestimate.

“ $M(t)$ is concave up” can be used instead of $M''(t) > 0$.

Information about $M'(t)$ by itself without reference to $M''(t) > 0$ is not sufficient and does not earn this point.

“Underestimate because it is concave up” is not sufficient by itself because of the use of “it”. The response must clearly, directly or indirectly, indicate to what “it” is referring.

If a response makes the correct conclusion and states, $M''(t) > 0$, and then goes on and makes an incorrect statement or a statement that cannot be verified will not earn this point.

For example: “ $M'(t)$ is concave up, underestimate” does not earn this point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

Because $M''(t) > 0$, the tangent line approximation would produce an underestimate.

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct expression of only $3.5R'(11)$ earns this point.

Only expression $3.5M'(11)$ also earns this point.

No justifications or explanations are required.

4-1-1

Any incorrect statement, for example $R'(11) = 3.5M'(11)$, does not earn this point.



0

1

The student response accurately includes the criteria below.

☐ Expression

Solution:

The rate of change, in grams per day, of the total weight of the coins at time $t = 11$ days is $3.5R'(11)$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Philosophy: Response clearly references what part of the differential equation is different from the slope field and the response needs only to clearly and explicitly reference one of them.

The response does not need to explicitly reference both the differential equation and the given slope field.

The distinction can be made with explicit reference to one of the two things and an implicit reference to the other

For example:

· “ $\frac{dy}{dt}$ is positive, not negative”. We will infer that negative is referring to the slopes shown in the slope field.

· “This would be correct for a negative derivative, but $\frac{dy}{dt}$ is positive.” We will read “This” as “the slope field” in this sentence.

· “All of the slopes in the slope field are negative, not positive.” We will infer that the values of $\frac{dy}{dt}$ at points in the first quadrant are positive.

A response that lacks even an implicit reference to either the differential equation or the slope field will not earn the point. For example:

· “ $\frac{dy}{dt}$ is positive” lacks any reference to the slope field and so will not earn the point.

· “All the slopes in the graph are negative” lacks any reference to the differential equation, and so will not earn the point.

· Vague answers such as “The slope field has negative points” or “The field is decreasing” will not earn the point.

4-1-1



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

When $y > 0$, $\frac{dy}{dt} > 0$. However, in the given slope field, the slopes of the line segments for $y > 0$ are negative.

4-1-1

51. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

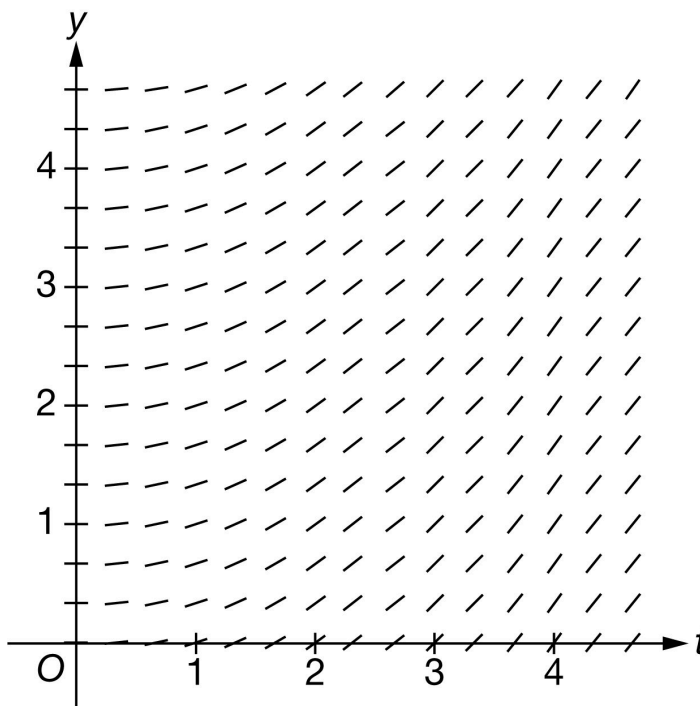
t (days)	0	6	18
$M(t)$ (coins per day)	40	54	46

A class collected 1-cent coins to raise money for classroom supplies. Students in the class deposited the coins into a container where all the coins were kept until the end of the collection period. For time t days, $0 \leq t \leq 18$, the rate at which coins are being added to the container, in coins per day, is modeled by a twice-differentiable function M . Values of $M(t)$ at various times t are shown in the table above.

- (a) Using correct units, approximate $M'(6)$ using the average rate of change of M over the interval $0 \leq t \leq 18$.
- (b) Explain the meaning of $M'(6)$ in the context of the problem.
- (c) The function $C(t)$ models the number of coins in the container at time t days, where $C'(t) = M(t)$. It is known that $C(6) = 300$. Write an equation for the line tangent to the graph of $y = C(t)$ at $t = 6$.
- (d) Find the value of $\int_0^6 M'(3t) \, dt$. Show the calculations that lead to your answer.
- (e) Use a trapezoidal sum with the two subintervals indicated by the table to approximate $\int_0^{18} M(t) \, dt$. Show the calculations that lead to your answer.
- (f) A second class is also collecting 1-cent coins. For $0 \leq t \leq 18$, the rate at which coins are being collected by the second class can be modeled by the differentiable function B , where B is a function of t . Assume the weight of each coin is 3 grams. Write an expression in terms of M and B for the rate of change, in grams per day, of the total weight of the coins collected by both classes at $t = 7$ days.

4-1-1

(g) Explain why the following could not be a slope field for the differential equation $\frac{dy}{dt} = 0.3y$.

**Part A**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for the correct approximation shown with supporting work.

Answers without supporting work do not earn this point. Supporting work must be at least a difference and a quotient.

Response does not need to be simplified, but any simplification must be done correctly.

If a decimal is presented, it must be correct to three decimal places. We do not read anything past the third decimal place.

The second point is earned for the correct units.

Units must be attached to a numerical value (either correct or incorrect).

Can be written using words or a combination of words and symbols.

The response will not be penalized for any verbal attempt to discuss the answer after the correct presentation of the approximation.

The minimum response needed to earn both points is: $\frac{46-40}{18}$ coins per day per day.

4-1-1



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$M'(6) \approx \frac{M(18) - M(0)}{18 - 0} = \frac{46 - 40}{18} = \frac{1}{3} \text{ coin per day per day}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This part is intended to be independent from part (a), so we do not want or need to see any reference to the value obtained in (a). If that value is referenced, conclusion must be consistent.

Three components of a correct answer:

- Indicating a rate of a rate (look for the word rate to be used correctly twice)
- Including the context of coin collection
- Referring to the specific point in time

Units will not be considered for this point. They do not need to be present, and if they are, they do not need to be correct.

Any incorrect statements make the response ineligible for the point. For example:

- $M'(6)$ is the average (or an approximation of, or an estimate of) rate of change of
- $M'(6)$ is the acceleration (or rate of change of velocity) of

“Increasing” or “decreasing” may be used in place of “changing” as long as the value found in part (a) is not brought into part (b) in an incorrect manner. Beware of phrases such as “decreasing at a negative rate.”

- Example: $M'(6)$ is the rate of increase of the rate of change of the number of coins at 6 days. (earns the point)
- Example: $M'(6)$ means the number of coins per day is increasing by 6 coins/day/day. (does not earn the point)

Acceptable references to time include (but are not limited to):

- On day 6

4-1-1

- At $t = 6$ (with correct units of time, without correct units of time, or even with incorrect units of time)
- After $t = 6$ (with correct units of time, without correct units of time, or even with incorrect units of time)

If the response clearly indicates an interval of time (e.g. “from day 0 to day 6”) the response will not earn the point.

Examples that earn the point include:

- $M'(6)$ represents the rate at which the rate of change of the number of coins is changing on day 6.
- $M'(6)$ is the rate of change in the rate of change of the number of coins at 6 days.
- $M'(6)$ represents the rate at which the rate of change of the number of coins is increasing at $t = 6$.
- $M'(6)$ equals the answer from part (a) means the rate of change in the number of coins per day is increasing at a rate of (answer from part (a)) at $t = 6$.
- $M'(6)$ represents the rate at which the rate of change of the number of coins is decreasing on the sixth day.

Examples that do not earn the point include:

- $M'(6)$ represents the rate at which the number of coins is changing on day 6.
- $M'(6)$ is the rate of change in the number of coins per day at $t = 6$ days
- The rate of change in the number of coins per day is increasing at $t = 6$.
- The rate of change in the rate of change of the number of coins per day is increasing at $t = 6$.
- $M'(6) = -\frac{1}{3}$ means that the rate of change in the number of coins is decreasing at a rate of $-\frac{1}{3}$ at $t = 6$. (double negative).



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

$M'(6)$ is the rate at which the rate of change of the number of 1-cent coins in the container is changing, in coins per day per day, on day 6.

Part C

4-1-1

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is earned with a correct tangent line equation in any form. Equivalent forms include (but are not limited to):

- $y = 54(t - 6) + 300$

- $y - 300 = 54(t - 6)$

- $y = 54t - 24$

The student may write x in place of t and still earn the point.

No work is required to earn the point.

We will accept $C(t)$ in lieu of y in the equation of the tangent line.



0	1
---	---

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$$y = C'(6)(t - 6) + C(6)$$

$$y = 54(t - 6) + 300$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with a correct substitution.

Two things are required to award this point:

- Coefficient of $\frac{1}{3}$ correctly applied to an integral or an antiderivative
- Evidence that the correct limits of 0 to 18 will be or have been applied

Examples that will earn the first point (these are just a few):

- $\frac{1}{3} \int_0^{18} M'(u) \, du$

4-1-1

$$\cdot \frac{1}{3} M(u) \Big|_0^{18} \text{ or } \frac{1}{3} M(3t) \Big|_0^6$$

Responses that include misstatements such as using t -values for limits with u as the variable of integration in an intermediate step which is later corrected may still earn this point. Intermediate errors in bounds that are subsequently corrected will disqualify the student from earning the third point. If the bounds are never appropriately handled, the response cannot earn the first point. Scratch work that is not connected to the solution will not be read.

Examples:

$$\cdot \int_0^6 M'(3t) dt = \frac{1}{3} \int_0^6 M'(u) du = \frac{1}{3} M(3t) \Big|_0^6 \text{ earns the first point, but will not earn the third point.}$$

$$\cdot \int_0^6 M'(3t) dt = \frac{1}{3} \int_0^6 M'(u) du = \frac{1}{3} [M(6) - M(0)] \text{ does not earn the first point.}$$

The second point is for the Fundamental Theorem of Calculus, demonstrating the knowledge that

$$\int_a^b M'(u) du = M(u) \Big|_a^b = M(b) - M(a)$$

This point can be earned with either $kM(3t) \Big|_a^b$, $kM(t) \Big|_a^b$, $k(M(b) - M(a))$ where k is a nonzero constant [not necessarily correct], $a = 0$, and $b = 6$ or $b = 18$.

A response can earn this point without earning the first point.

$$\frac{1}{3} [M(18) - M(0)] \text{ earns the first two points.}$$

The third point is for the correct numerical value. Our answer only.

Values from the table must be used (i.e., $\frac{1}{3} [M(18) - M(0)]$ is not sufficient to earn the third point).

If the response includes any linkage error, (i.e., the use of equal signs to equate two expressions that are not equal), the third point will not be earned. If an indefinite integral is linked with a definite integral the response will not be penalized.

$$\cdot \int_0^6 M'(3t) dt = \frac{1}{3} \int M'(u) du = \frac{1}{3} M(3t) \Big|_0^6 = \dots \text{ is eligible for the third point.}$$

Eligibility: Must have earned the first two points.

The minimum response needed to earn all three points is: $\frac{1}{3}(46 - 40)$.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

4-1-1

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = 3t$. Then $du = 3 dt$.

When $t = 0$, $u = 0$.

When $t = 6$, $u = 18$.

$$\begin{aligned}\int_0^6 M'(3t) dt &= \frac{1}{3} \int_0^{18} M'(u) du \\ &= \frac{1}{3}(M(18) - M(0)) \\ &= \frac{1}{3}(46 - 40) = 2\end{aligned}$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the form of a trapezoidal sum.

Point is earned for the setup of the trapezoidal sum. Numerical values do not need to be pulled from the table for this point, but Δt values must be presented. Responses that contain one incorrect numerical value in the trapezoidal sum setup are still eligible to earn the first point, but will not earn the second.

Response may present a left Riemann sum and a right Riemann sum, then average the two. Again, one incorrect numerical value will not disqualify the response from earning this first point.

Response may compute trapezoids as areas of rectangles and triangles.

Trapezoidal areas may be given in a graph. In order to earn this point, the response must clearly show the computations leading to the values presented in the graph, and must show that these areas are added to obtain the final approximation value.

A response using subtraction instead of addition will not earn the point (e.g.,

$$\frac{1}{2}[6(M(0) - M(6)) + 12(M(6) - M(18))]$$

earns no points).

Response utilizing “Trapezoidal Rule” will not earn this point, unless this represents only one incorrect numerical values.

$\cdot \frac{1}{2}(6)[M(0) + 2M(6) + M(18)]$ is equivalent to $\frac{1}{2}[6(M(0) + M(6)) + 6(M(6) + M(18))]$, thus has only one incorrect numerical value (the second interval width is 6 instead of 12), so can earn the point.

4-1-1

$\cdot \frac{1}{2}(9) [M(0) + 2M(6) + M(18)]$ has two incorrect numerical values (both interval widths are incorrect) so will not earn the point.

The second point is for the correct numerical result.

Our answer only. Response does not need to be simplified, but if it is, it must be simplified correctly.

The answer must be accompanied with supporting work. Supporting work may appear in a graph, with clearly labeled areas. It is possible to earn the second point without the first if the supporting work is not sufficient to earn the first point.

Pictures without clear area computations and a clear summation of quantities can earn at most the second point.

Special case: A positive constant multiple of a complete and correctly expressed trapezoidal sum (with the correct values pulled from the table) will earn only the first point:

$\cdot k \left[\left(\frac{40+54}{2} \right) \cdot 6 + \left(\frac{54+46}{2} \cdot 12 \right) \right]$ for $k > 0$ earns the first but not the second point.

The minimum response needed to earn both points: $\left(\frac{94}{2} \right) \cdot 6 + \left(\frac{100}{2} \right) \cdot 12$



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Form of trapezoidal sum
- ☐ Answer

Solution:

$$\begin{aligned} \int_0^{18} M(t) dt &\approx \left(\frac{M(0)+M(6)}{2} \right) (6-0) + \left(\frac{M(6)+M(18)}{2} \right) (18-6) \\ &= \left(\frac{40+54}{2} \right) \cdot 6 + \left(\frac{54+46}{2} \right) \cdot 12 = 882 \end{aligned}$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for a correct constant multiple or correct sum.

Response must contain the constant 3 as a factor of the expression, and/or the correct sum $B(7) + M(7)$.

If the sum is incorrect, then the constant factor of 3 must be applied either to a sum that involves only $B(t)$ and $M(t)$, $B'(t)$ and $M'(t)$, or integrals of $B(t)$ and $M(t)$, or to one of the correct terms of the sum $B(7)$ and

4-1-1

$M(7)$, with no additional incorrect terms.

Examples that earn the point include:

- $3[M(t) + B(t)]$
- $3 \int [M(t) + B(t)] dt$
- $3[M' + B']$
- $3B(7)$

Examples that do not earn the point include:

- $3[M(t) + B'(t)]$
- $3 \int M(t) dt$
- $3M' + B'$
- $3B(7) + t$

Special case: $3M(7) + B(7)$ is an error in parentheses, so will earn the first point but is not eligible to earn the second point.

If the constant multiple is missing or incorrect, the sum must be correct. $k[B(7) + M(7)]$ for any nonzero constant k will earn the first point.

The second point is for the correct expression.

$3B(7) + 3M(7)$ (or an equivalent expression) earns both points.

Inclusion of units (correct or incorrect) will not affect the score.

A completely correct expression with one additional constant term added/subtracted would earn just the first point. For example, $3B(7) + 3M(7) + 3(2)$ earns only the first point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Constant multiple or sum
- ☐ Expression

Solution:

4-1-1

An expression for the rate of change, in grams per day, of the total weight of the coins collected by both classes at time $t = 7$ days is $3B(7) + 3M(7)$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is earned for a correct statement identifying specific inconsistencies between the given differential equation and the given slope field. There are many such correct statements, such as:

- The response references a specific point on the y -axis (such as $(0, 4)$), and indicates that the given slope field shows a slope of zero, yet the differential equation would yield a positive slope.
- The response references a pattern of slopes, and explains why that pattern does not fit our differential equation (e.g. “the slope field shows slopes that change as t changes, but our differential equation depends only on y , so slopes should remain constant for a given value of y as t changes.”)

Identifying a mismatch between the differential equation and the slope field at one point is sufficient to earn the point.

Response is not penalized for referencing “ x ” instead of “ t ”.

If the response contains any incorrect statement, the response will not earn the point.

Vague answers which have the potential to be incorrect will not earn the point, nor will responses that present information beyond what we are given. For example:

- Responses that specify that the slope field shows a particular nonzero slope at a particular point will not earn the point, as this cannot actually be determined visually (i.e., the response should not indicate that at $(1, 2)$ the slope shown on the slope field is 1)
- Responses that state a specific slope value on the slope field for a general value of x or y that are not correct for all points with that x or y value do not earn the point. (e.g. “the slope field shows a slope of 0 when $y = 4$ ” is an incorrect statement)

A response may involve a solution to the differential equation, with an argument based on the resulting function. If done correctly, and presented with a correct argument, this will earn the point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

4-1-1

When $y = 0$, $\frac{dy}{dt} = 0$. However, in the given slope field, for $t > 0$, the slopes of the line segments for $y = 0$ are nonzero.

4-1-1

52. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (minutes)	0	10	20	30	40
$N(t)$ (number of 5-cent coins remaining)	1200	1000	750	400	0

A child has collected 5-cent coins and is preparing them to deposit in the bank by counting them into bags. The child begins with 1200 coins. For time t minutes, $0 \leq t \leq 40$, the total number of coins remaining to prepare is modeled by a decreasing, twice-differentiable function N . Values of $N(t)$ at various times t are shown in the table above.

(a) Use a midpoint Riemann sum with the two subintervals indicated by the table to approximate

$\int_0^{40} N(t) \, dt$. Show the computations that lead to your answer.

(b) Find the value of $\int_0^{10} N'(2t) \, dt$. Show the computations that lead to your answer.

(c) Using correct units, approximate $N'(20)$ using the average rate of change of N over the interval $10 \leq t \leq 30$.

(d) Explain the meaning of $N'(20)$ in the context of the problem.

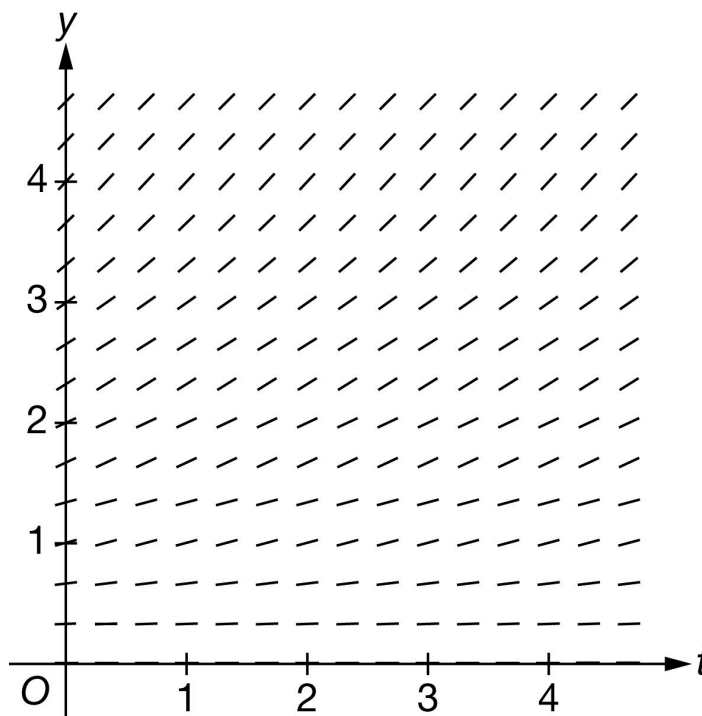
(e) It is known that $N'(10) = -20$. Write an equation for the line tangent to the graph of $y = N(t)$ at $t = 10$.

(f) It is known that $N''(t) < 0$ for $10 \leq t \leq 20$. If $N(17)$ is approximated using the equation from part (e), would the approximation be an overestimate or underestimate for the actual value of $N(17)$? Give a reason for your answer.

(g) For $0 \leq t \leq 40$, the total number of 5-cent coins remaining can also be modeled by the differentiable function G , where G is a function of t . Assume the weight of each coin is 6 grams. Write an expression for the rate of change, in grams per minute, of the total weight of the 5-cent coins remaining at $t = 37$ minutes.

4-1-1

(h) Explain why the following slope field cannot represent the differential equation $\frac{dy}{dt} = -0.4y$.

**Part A**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point (Form of midpoint sum): Philosophy: The response should provide clear evidence adding the area of two rectangles whose heights are found using the midpoint of the interval.

Acceptable evidence:

$$\cdot 20 \cdot (N(10) + N(30)) \text{ or } 20 \cdot N(10) + 20 \cdot N(30)$$

$$\cdot (20 - 0) \cdot 1000 + (40 - 20) \cdot 400$$

$$\cdot 20 \cdot (1000 + 400) \text{ or } 20 \cdot 1000 + 20 \cdot 400$$

To earn this point the response can have at most one mistake in the expression or equation. So, expressions of the form $\frac{20}{b} \cdot (N(10) + N(30))$, for any nonzero value of b would earn the point. Note: The value $\frac{20}{b}$ might be distributed across the sum.

Expressions of the form $\frac{a}{b} \cdot (N(10) + N(30))$ for $a \neq 20$ will not earn this point.

Once the first point is earned, it cannot be lost.

Second Point (Answer)

4-1-1

To earn the point, the answer must be correct. It does not need to be simplified, but if it is, the simplification must be correct.

The answer point cannot be earned unless the first (midpoint sum) point was earned.

Minimal complete answers that earn both points:

$$\cdot 20 \cdot (1000 + 400)$$

$$\cdot 20 \cdot 1000 + 20 \cdot 400$$



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Form of midpoint sum
- ☐ Answer

Solution:

$$\begin{aligned} \int_0^{40} N(t) dt &\approx N(10)(20 - 0) + N(30)(40 - 20), \\ &= 1000 \cdot 20 + 400 \cdot 20 = 28,000 \end{aligned}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point (Substitution) Philosophy: This point is earned when the response has all three components of the substitution done correctly.

Uses $u = 2t$, perhaps implicitly (any variable could be used for u)

Uses the correct constant of $\frac{1}{2}$

Uses the correct limits for their substitution.

· The limits might not be seen until the final evaluation, and at that point, response may have returned to the variable t .

· Limits for t attached to an integral involving u or vice versa will not earn this substitution point but could still earn the answer point, if the numerical answer is correct.

4-1-1

Second point: (Fundamental Theorem of Calculus) Philosophy: This point is earned when the FTC has been applied correctly to either $\int N'(2t) dt$ or $\int N'(u) du$.

The point is earned for any of the following:

- The equation $\int N'(2t) dt = N(2t)$ or $\int N'(u) du = N(u)$
- $\frac{1}{2}N(2t)|_0^{10}$ or $\frac{1}{2}N(u)|_0^{20}$
- $N(20) - N(0)$, $N(20)$, or $N(0)$

This point can be earned if the substitution point is not.

Note: $\int_a^b N'(u) du = N(b) - N(a)$ does not earn the point. The FTC point is for the response communicating that the antiderivative of N' is N .

Third Point (Answer)

Must be the correct numerical answer.

Must have earned the second point for the FTC.

Might have not earned the first point for substitution because of an incorrect use of limits, but will still earn the third point for the correct answer.

$N(20)$ and $N(0)$ must be evaluated; the expression $\frac{1}{2}(N(20) - N(0))$ by itself does not earn the third point.

The answer may be unsimplified, but if simplified it must be done correctly.

Scoring common responses:

The minimal work required to earn all three points is $\frac{1}{2}(750 - 1200)$ or $\frac{1}{2}(750) - \frac{1}{2}(1200)$.

The following responses would earn the first two points and be eligible for the answer point if the response continues:

- $\frac{1}{2}N(2t)|_0^{10}$
- $\frac{1}{2}(N(20) - N(0))$

Responses of only the following would earn only the Second point:

- $N(20) - N(0)$
- $N(10) - N(0)$

4-1-1

$$\cdot \int_0^{10} N'(u) \, du = N(10) - N(0)$$

$$\cdot N(2t)|_0^{10} \text{ or } N(u)|_0^{20}$$

The following responses would earn no points:

$$\cdot -225 \text{ in the absence of one of the first two points.}$$

$$\cdot \int_0^{10} N'(2t) \, dt = N(10) - N(0)$$



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = 2t$. Then $du = 2 \, dt$.

When $t = 0$, $u = 0$.

When $t = 10$, $u = 20$.

$$\int_0^{10} N'(2t) \, dt = \frac{1}{2} \int_0^{20} N'(u) \, du$$

$$= \frac{1}{2}(N(20) - N(0))$$

$$= \frac{1}{2}(750 - 1200) = -225$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point (Approximation)

The response must be a difference and a quotient.

The answer does not need to be simplified, but if it is simplified it must be done correctly.

4-1-1

The minimal work that must be shown to earn this point is either $\frac{400-1000}{20}$ or $\frac{-600}{30-10}$.

Second point (Units)

The units must be attached to a value (either correct or incorrect).

The response will not be penalized for any verbal attempt to discuss the answer after the correct approximation with units is presented.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$N'(20) \approx \frac{N(30)-N(10)}{30-10} = \frac{400-1000}{20} = -30 \text{ coins per minute}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The interpretation must include three components:

- indicate the rate of change
- the specific context of the number of coins remaining
- the specific point in time, $t = 20$.

Responses do not need to include correct units or any units.

The word “decreasing” may be used in place of the word “changing”.

The response is not earned for:

- “the rate of coins remaining at minute 20 ” as this does not indicate a change in the rate of deposit.
- “decreasing at a rate of -30 coins per minute”

“the rate of change of N ” without specifying that N is the number of coins remaining, does not earn the point.

Acceptable ways to provide the specific point in time:

4-1-1

- At minute 20 or after 20 minutes.
- At $t = 20$ (missing the unit of time)
- At 20 seconds (incorrect unit of time)

A response that clearly indicates an interval of time (e.g. “during the first 20 minutes” or “from minute 20 on”) does not earn the point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

$N'(20)$ is the rate at which the number of 5-cent coins remaining is changing, in coins per minute, at time $t = 20$ minutes.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The use of $N(t)$ in place of y earns the point.

The use of x in place of t earns the point.

Tangent lines do not need to be simplified, but any simplification must be done correctly.

Any correct form of the tangent line earns the point. For example:

- $y = 1000 - 20(t - 10)$
- $y - 1000 = 20(t - 10)$
- $y = -20t + 200 + 1000$
- $y = -20t + 1200$



0	1
---	---

The student response accurately includes the criteria below.

4-1-1

☐ Tangent line equation
Solution:

$$y = N'(10)(t - 10) + N(10)$$

$$y = -20(t - 10) + 1000$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point the response must include $N''(t) < 0$ and overestimate.

“ $N(t)$ is concave down” can be used instead of $N''(t) < 0$.

$N'(t)$ by itself is not sufficient and does not earn this point

“Overestimate because it is concave down” is not sufficient by itself because of the use of "it". The response must clearly, directly or indirectly, indicate to what "it" is referring.

If a response makes the correct conclusion and states $N''(t) < 0$ and then goes on and makes an incorrect statement or a statement that cannot be verified will not earn this point.

For example: “ $N'(t)$ is concave down, overestimate” does not earn this point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason
Solution:

Because $N''(t) < 0$, the tangent line approximation would produce an overestimate.

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct expression of only $6 \cdot G'(37)$ earns this point.

Only expression $6 \cdot N'(37)$ also earns this point.

No justifications or explanations are required.

4-1-1

Any incorrect statement such as, $G'(37) = 6 \cdot N'(37)$, does not earn this point.



0

1

The student response accurately includes the criteria below.

☐ Expression

Solution:

The rate of change, in grams per minute, of the total weight of the 5-cent coins remaining at $t = 37$ minutes is $6G'(37)$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Philosophy: Response clearly references what part of the differential equation is different from the slope field and the response needs only to clearly and explicitly reference one of them.

The response does not need to explicitly reference both the differential equation and the given slope field.

The distinction can be made with explicit reference to one of the two things and an implicit reference to the other

For example:

· “ $\frac{dy}{dt}$ is negative, not positive”. We will infer that positive is referring to the slopes shown in the slope field.

· “This would be correct for a positive derivative, but $\frac{dy}{dt}$ is negative.” We will read “This” as “the slope field” in this sentence.

· “All of the slopes in the slope field are positive, not negative.” We will infer that the values of $\frac{dy}{dt}$ at points in the first quadrant are negative.

A response that lacks even an implicit reference to either the differential equation or the slope field will not earn the point. For example:

· “ $\frac{dy}{dt}$ is negative” lacks any reference to the slope field and so will not earn the point.

· “All the slopes in the graph are positive” lacks any reference to the differential equation, and so will not earn the point.

· Vague answers such as “The slope field has positive points” or “The field is increasing” will not earn the point.

4-1-1



0

1

The student response accurately includes the criteria below.

☐

Explanation

Solution:

When $y > 0$, $\frac{dy}{dt} < 0$. However, in the given slope field, the slopes of the line segments for $y > 0$ are positive.

4-1-1

53. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

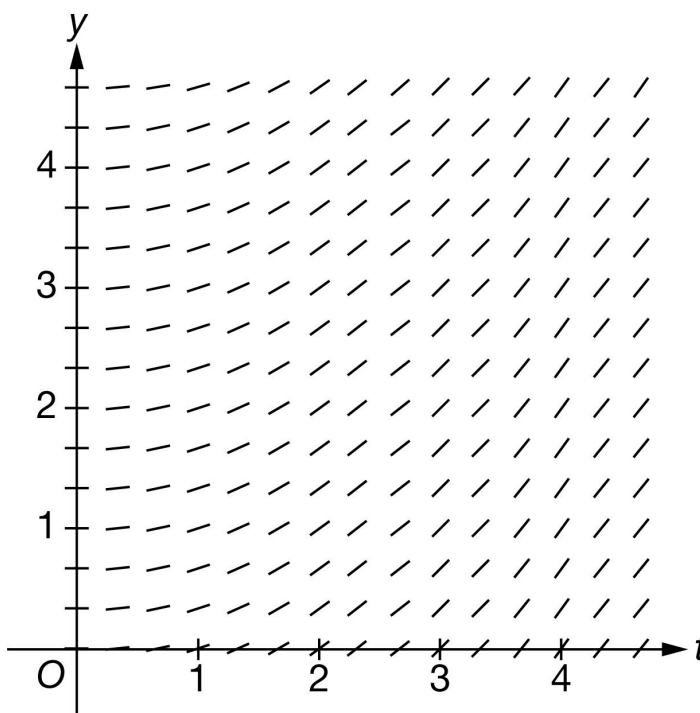
t (minutes)	0	10	40
$N(t)$ (coins per minute)	-10	-16	-8

A child has collected 5-cent coins and is preparing them to deposit in the bank by putting them into bags. For time t minutes, $0 \leq t \leq 40$, the rate of change in the number of coins remaining to prepare, in coins per minute, is modeled by a twice-differentiable function N . Values of $N(t)$ at various times t are shown in the table above.

- (a) Using correct units, approximate $N'(10)$ using the average rate of change of N over the interval $0 \leq t \leq 40$.
- (b) Explain the meaning of $N'(10)$ in the context of the problem.
- (c) The function $C(t)$ models the number of coins remaining to be put into bags at time t minutes, where $C'(t) = N(t)$. It is known that $C(10) = 400$. Write an equation for the line tangent to the graph of $y = C(t)$ at $t = 10$.
- (d) Find the value of $\int_0^{10} N'(4t) \, dt$. Show the calculations that lead to your answer.
- (e) Use a trapezoidal sum with the two subintervals indicated by the table to approximate $\int_0^{40} N(t) \, dt$. Show the calculations that lead to your answer.
- (f) A second child is preparing another deposit of 5-cent coins. For $0 \leq t \leq 40$, the rate of change in the number of coins remaining for the second child to prepare, in coins per minute, can be modeled by the differentiable function F , where F is a function of t . Assume the weight of each coin is 5 grams. Write an expression in terms of N and F for the rate of change, in grams per minute, in the combined weight of the 5-cent coins remaining for the two children to prepare at $t = 11$ minutes.

4-1-1

(g) Explain why the following slope field cannot represent the differential equation $\frac{dy}{dt} = 0.4y$.

**Part A**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for the correct approximation shown with supporting work.

Answers without supporting work do not earn this point. Supporting work must be at least a difference and a quotient.

Response does not need to be simplified, but any simplification must be done correctly.

If a decimal is presented, it must be correct to 3 decimal places. We do not read anything past the third decimal place.

The second point is earned for the correct units.

Units must be attached to a numerical value (either correct or incorrect).

Can be written using words or a combination of words and symbols.

The response will not be penalized for any verbal attempt to discuss the answer after the correct presentation of the approximation.

The minimum response needed to earn both points is: $\frac{-8+10}{40}$ coins per minute per minute

4-1-1



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$N'(10) \approx \frac{N(40) - N(0)}{40 - 0} = \frac{-8 - (-10)}{40} = \frac{1}{20} \text{ coin per minute per minute}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This part is intended to be independent from part (a), so we do not want or need to see any reference to the value obtained in (a). If that value is referenced, conclusion must be consistent.

Three components of a correct answer:

- Indicating a rate of a rate (look for the word rate to be used correctly twice)
- Including the context of coin collection
- Referring to the specific point in time

Units will not be considered for this point. They do not need to be present, and if they are, they do not need to be correct.

Any incorrect statements make the response ineligible for the point. For example:

- $N'(10)$ is the average (or an approximation of, or an estimate of) rate of change of
- $N'(10)$ is the acceleration (or rate of change of velocity) of

“Increasing” or “decreasing” may be used in place of “changing” as long as the value found in part (a) is not brought into part (b) in an incorrect manner. Beware of phrases such as “decreasing at a negative rate.”

· Example: $N'(10)$ is the rate of increase of the rate of change of the number of coins at 10 minutes. (earns the point)

· Example: $N'(10)$ means the number of coins per minute is increasing by $\frac{1}{20}$ coins/min/min (does not earn the point)

4-1-1

Acceptable references to time include (but are not limited to):

- At the tenth minute
- At $t = 10$ (with correct units of time, without correct units of time, or even with incorrect units of time)
- After $t = 10$ (with correct units of time, without correct units of time, or even with incorrect units of time)

If the response clearly indicates an interval of time (e.g. “over the first ten minutes”) the response will not earn the point.

Examples that earn the point include:

- $N'(10)$ represents the rate at which the rate of change of the number of coins is changing at minute 10.
- $N'(10)$ is the rate of change in the rate of change of the number of coins at 10 minutes.
- $N'(10)$ represents the rate at which the rate of change of the number of coins is increasing at $t = 10$.
- $N'(10)$ equals (answer from part (a)) means the rate of change in the number of coins per minute is increasing at a rate of (answer from part (a)) at $t = 10$.
- $N'(10)$ represents the rate at which the rate of change of the number of coins is decreasing after 10 10 minutes.

Examples that do not earn the point include:

- $N'(10)$ represents the rate at which the number of coins is changing at minute 10.
- $N'(10)$ is the rate of change in the number of coins per minute at 10 minutes.
- The rate of change in the number of coins per minute is increasing at $t = 10$.
- The rate of change in the rate of change of the number of coins per minute is increasing at $t = 10$.
- $N'(10) = -\frac{1}{20}$ means that the rate of change in the number of coins is decreasing at a rate of $-\frac{1}{20}$ after 10 minutes. (double negative)



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

$N'(10)$ is the rate at which the rate of change of the number of 5-cent coins remaining is changing, in coins per minute per minute, at time $t = 10$ minutes.

4-1-1

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is earned with a correct tangent line equation in any form. Equivalent forms include (but are not limited to):

$$\cdot y = -16(t - 10) + 400$$

$$\cdot y - 400 = -16(t - 10)$$

$$\cdot y = -16t + 560$$

The student may write x in place of t and still earn the point.

No work is required to earn the point.

We will accept $C(t)$ in lieu of y in the equation of the tangent line.



0	1
---	---

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$$y = C'(10)(t - 10) + C(10)$$

$$y = -16(t - 10) + 400$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with a correct substitution.

Two things are required to award this point:

- Coefficient of $\frac{1}{4}$ correctly applied to an integral or an antiderivative
- Evidence that the correct limits of 0 to 40 will be or have been applied

Examples that will earn the first point (these are just a few):

4-1-1

$$\cdot \frac{1}{4} \int_0^{40} N'(u) \, du$$

$$\cdot \frac{1}{4} N(u) \Big|_0^{40} \text{ or } \frac{1}{4} N(4t) \Big|_0^{10}$$

Responses that include misstatements such as using t -values for limits with u as the variable of integration in an intermediate step which is later corrected may still earn this point. Intermediate errors in bounds that are subsequently corrected will disqualify the student from earning the third point. If the bounds are never appropriately handled, the response cannot earn the first point. Scratch work that is not connected to the solution will not be read.

Examples:

$$\cdot \int_0^{10} N'(4t) \, dt = \frac{1}{4} \int_0^{10} N'(u) \, du = \frac{1}{4} N(4t) \Big|_0^{10} \text{ earns the first point, but will not earn the third point.}$$

$$\cdot \int_0^{10} N'(4t) \, dt = \frac{1}{4} \int_0^{10} N'(u) \, du = \frac{1}{4} [N(10) - N(0)] \text{ does not earn the first point.}$$

The second point is for the Fundamental Theorem of Calculus, demonstrating the knowledge that

$$\int_a^b N'(u) \, du = N(u) \Big|_a^b = N(b) - N(a)$$

This point can be earned with either $kN(4t) \Big|_a^b$, $kN(t) \Big|_a^b$, or $k(N(b) - N(a))$ where k is a nonzero constant [not necessarily correct], $a = 0$, and $b = 10$ or $b = 40$.

A response can earn this point without earning the first point.

$$\frac{1}{4} [N(40) - N(0)] \text{ earns the first two points.}$$

The third point is for the correct numerical value. Our answer only.

Values from the table must be used (i.e., $\frac{1}{4} [N(40) - N(0)]$ is not sufficient to earn the third point).

If the response includes any linkage error, (i.e., the use of equal signs to equate two expressions that are not equal), the third point will not be earned. If an indefinite integral is linked with a definite integral the response will not be penalized.

$$\cdot \int_0^{10} N'(4t) \, dt = \frac{1}{4} \int N'(u) \, du = \frac{1}{4} N(4t) \Big|_0^{10} = \dots \text{ is eligible for the third point.}$$

Eligibility: Must have earned the first two points.

The minimum response needed to earn all three points is: $\frac{1}{4}(-8 + 10)$.

4-1-1



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = 4t$. Then $du = 4 dt$.

When $t = 0$, $u = 0$.

When $t = 10$, $u = 40$.

$$\begin{aligned}\int_0^{10} N'(4t) dt &= \frac{1}{4} \int_0^{40} N'(u) du \\ &= \frac{1}{4}(N(40) - N(0)) \\ &= \frac{1}{4}(-8 - (-10)) = \frac{1}{2}\end{aligned}$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for setting up a trapezoidal sum

Point is earned for the setup of the trapezoidal sum. Numerical values do not need to be pulled from the table for this point, but Δt values must be presented. Responses that contain one incorrect numerical value in the trapezoidal sum setup are still eligible to earn the first point, but will not earn the second.

Response may present a left Riemann sum and a right Riemann sum, then average the two. Again, one incorrect numerical value will not disqualify the response from earning this first point.

Response may compute trapezoids as areas of rectangles and triangles.

Trapezoidal areas may be given in a graph. In order to earn this point, the response must clearly show the computations leading to the values presented in the graph, and must show that these areas are added to obtain the final approximation value.

A response using subtraction instead of addition will not earn the point

4-1-1

(e.g. $\frac{1}{2}[10(N(0) - N(10)) + 30(N(10) - N(40))]$ earns no points.)

Response utilizing “Trapezoidal Rule” will not earn this point, unless this represents only one incorrect numerical value.

$\cdot \frac{1}{2}(10)[N(0) + 2N(10) + N(40)]$ is equivalent to $\frac{1}{2}[10(N(0) + N(10)) + 10(N(10) + N(40))]$
 $\frac{1}{2}[10(N(0) + N(10)) + 10(N(10) + N(40))]$, thus has only one incorrect numerical value (the second interval width is 10 instead of 30), so can earn the point.

$\cdot \frac{1}{2}(20)[N(0) + 2N(10) + N(40)]$ has two incorrect numerical values (both interval widths are incorrect) so will not earn the point.

The second point is for the correct numerical result.

Our answer only. Response does not need to be simplified, but if it is, it must be simplified correctly.

The answer must be accompanied with supporting work. Supporting work may appear in a graph, with clearly labeled areas. It is possible to earn the second point without earning the first point, if the supporting work is not sufficient to earn the first point.

Pictures without clear area computations and a clear summation of quantities can earn at most the second point.

Special case: A positive constant multiple of a complete and correctly expressed trapezoidal sum (with the correct values pulled from the table) will earn the first, but not the second point:

$\cdot k \left[\left(\frac{-10-16}{2} \right) \cdot 10 + \left(\frac{-16-8}{2} \cdot 30 \right) \right]$ for $k > 0$ earns only the first point.

The minimum response needed to earn both points: $\left(\frac{-26}{2} \right) \cdot 10 + \left(\frac{-24}{2} \right) \cdot 30$



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Form of trapezoidal sum
- ☐ Answer

Solution:

$$\int_0^{40} N(t) dt \approx \left(\frac{N(0)+N(10)}{2} \right) (10-0) + \left(\frac{N(10)+N(40)}{2} \right) (40-10)$$

$$= \left(\frac{-10-16}{2} \right) \cdot 10 + \left(\frac{-16-8}{2} \right) \cdot 30 = -490$$

4-1-1

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for a correct constant multiple or correct sum.

Response must contain the constant 5 as a factor of the expression, and/or the correct sum $N(11) + F(11)$.

If the sum is incorrect, then the constant factor of 5 must be applied either to a sum that involves only $N(t)$ and $F(t)$, $N'(t)$ and $F'(t)$, or integrals of $N(t)$ and $F(t)$ or to one of the correct terms of the sum $N(11)$ or $F(11)$, with no additional incorrect terms.

Examples that earn the point include:

$$\cdot 5 [N(t) + F(t)]$$

$$\cdot 5 \int [N(t) + F(t)] dt$$

$$\cdot 5 [N' + F']$$

$$\cdot 5N(11)$$

Examples that do not earn the point include:

$$\cdot 5 [N(t) + F'(t)]$$

$$\cdot 5 \int N(t) dt$$

$$\cdot 5N' + F'$$

$$\cdot 5N(11) + t$$

Special case: $5N(11) + F(11)$ is an error in parentheses, so will earn the first point but is not eligible to earn the second point.

If the constant multiple is missing or incorrect, the sum must be correct. $k[N(11) + F(11)]$, for any nonzero constant k will earn the first point.

The second point is for the correct expression.

$5[N(11) + F(11)]$ (or an equivalent expression) earns both points.

Inclusion of units (correct or incorrect) will not affect the score.

A completely correct expression with one additional constant term added/subtracted would earn the first point only. For example: $5[N(11) + F(11)] + 5(2)$ earns only the first point.

4-1-1



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Constant multiple or sum
- ☐ Expression

Solution:

The rate of change, in grams per minute, in the combined weight of the 5-cent coins remaining for the two children to prepare at $t = 11$ minutes is $5N(11) + 5F(11)$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is earned for a correct statement identifying specific inconsistencies between the given differential equation and the given slope field. There are many such correct statements, such as:

- The response references a specific point on the y -axis (such as $(0, 4)$), and indicates that the given slope field shows a slope of zero, yet the differential equation would yield a positive slope.
- The response references a pattern of slopes, and explains why that pattern does not fit our differential equation (e.g. “the slope field shows slopes that change as t changes, but our differential equation depends only on y , so slopes should remain constant for a given value of y , as t changes.”)

Identifying a mismatch between the differential equation and the slope field at one point is sufficient to earn the point.

Response is not penalized for referencing “ x ” instead of “ t ”.

If the response contains any incorrect statement, the response will not earn the point.

Vague answers which have the potential to be incorrect will not earn the point, nor will responses that present information beyond what we are given. For example:

- Responses that specify that the slope field shows a particular nonzero slope at a particular point will not earn the point, as this cannot actually be determined visually (i.e., the response should not indicate that at $(1, 2)$ the slope shown on the slope field is 1)
- Responses that state a specific slope value on the slope field for a general value of x or y that are not correct for all points with that x or y value do not earn the point. (e.g. “the slope field shows a slope of 0 when $y = 4$ ” is an incorrect statement)

4-1-1

A response may involve a solution to the differential equation, with an argument based on the resulting function. If done correctly, and presented with a correct argument, this will earn the point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

When $y = 0$, $\frac{dy}{dt} = 0$. However, in the given slope field, for $t > 0$, the slopes of the line segments for $y = 0$ are nonzero.

4-1-1

54. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (minutes)	0	10	35	40
$N(t)$ (coins per minute)	-8	-11	-13	-15

A child has collected 5-cent coins and is preparing them to deposit in the bank by counting them into bags. For time t minutes, $0 \leq t \leq 40$, the rate of the total number of coins remaining to prepare, in 5-cent coins per minute, is modeled by a decreasing, twice-differentiable function N . Values of $N(t)$ at various times t are shown in the table above.

(a) Use a left Riemann sum with the three subintervals indicated by the table to approximate

$$\int_0^{40} N(t) \, dt. \text{ Show the calculations that lead to your answer.}$$

(b) Is your answer from part (a) an overestimate or an underestimate for $\int_0^{40} N(t) \, dt$? Give a reason for your answer.

(c) Using correct units, approximate $N'(10)$ using the average rate of change of N over the interval $0 \leq t \leq 35$.

(d) Explain the meaning of $N'(10)$ in the context of the problem.

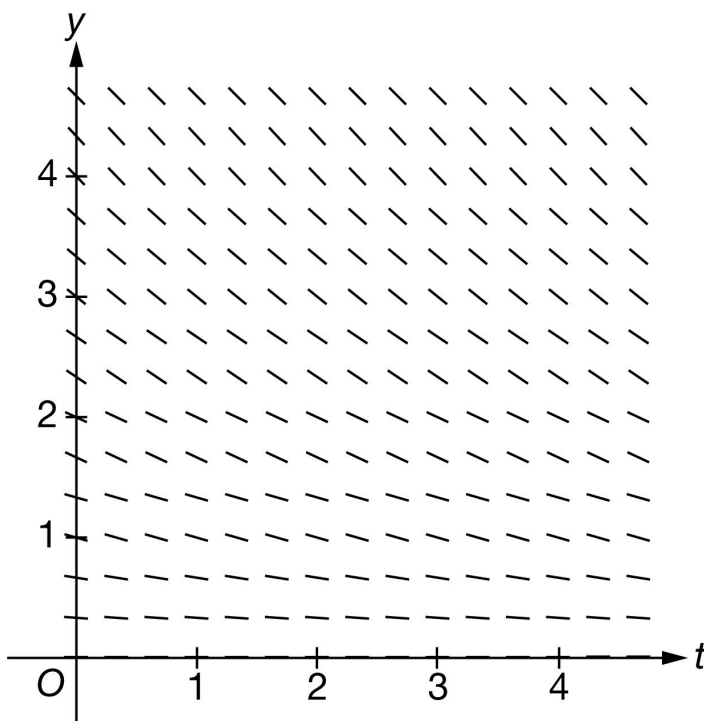
(e) It is known that $N'(35) = -0.2$. Write an equation for the line tangent to the graph of $y = N(t)$ at $t = 35$.

(f) Find the value of $\int_0^{40} N'\left(\frac{t}{4}\right) \, dt$. Show the calculations that lead to your answer.

(g) A second child is also preparing 5-cent coins for deposit. For $0 \leq t \leq 40$, the total number of 5-cent coins prepared by the second child for deposit can be modeled by the increasing, differentiable function F , where F is a function of t . Assume the weight of each coin is 5 grams. Write an expression for the rate of change, in grams per minute, of the total weight of the 5-cent coins remaining for the second child at $t = 23$ minutes.

4-1-1

(h) Explain why the following could not be a slope field for the differential equation $\frac{dy}{dt} = F'(t)$ for the function F defined in part (g).

**Part A**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for setting up a left Riemann sum. The setup must be a sum of three products.

A response may still earn this point if there is only one numerical error in the expression for the Riemann sum with values pulled from the table. (See example below).

Responses with the numerical error described above are not eligible for the second point.

Note: either the multiplication or the addition may be indicated implicitly. For example, the response may show appropriate values put into rows and columns, like so.

10	(-8)	-80
25	(-11)	-275
5	(-13)	+(-65)
		-420

or

4-1-1

$(10 - 0)(-8)$	$= -80$
$(35 - 10)(-11)$	$= -275$
$(40 - 35)(-13)$	$= -65$
	-420

Implicit responses such as those above are eligible for both points.

A response presenting a right Riemann sum (instead of left), will not earn any points for part (a).

The second point is for the correct numerical result.

In order to earn the second point, there must be supporting calculations.

The numerical values for $N(0)$, $N(10)$, and $N(35)$ must be pulled from the table.

Any simplification shown must be correct.

Use the following as an illustration of how to score both points. (In most cases, you see scores for which the expression on the left side represents the entirety of relevant work shown for the response.)

- A response of $(-8) \cdot (10 - 0) + (-11) \cdot (35 - 10) + (-13) \cdot (40 - 35)$ earns both points.
- A response of $10 \cdot N(0) + 25 \cdot N(10) + 5 \cdot N(35)$ earns the first point. Eligible for second point if more work is on the page.
- A response of $10 \cdot N(0) + 25 \cdot N(10) + 5 \cdot N(35) = -420$ earns both points.
- A response of $10 \cdot (-8) + 25 \cdot (-11) + 5 \cdot (-13)$ earns both points.
- A response of $10 \cdot (-8) + 35 \cdot (-11) + 5 \cdot (-13)$ contains one numerical error. Earns the first point. Not eligible for the second point.
- A response of $10 \cdot (-8) + 25 \cdot (-11) + 5 \cdot (-13) = -400$ earns first point for left side of equation. Does not earn second point due to incorrect simplification.
- A response of $(-80) + (-275) + (-65)$ or $(-80) + (-275) + (-65) = -420$ does not earn the first point because we cannot see the sum of three products. However, this is the correct sum, so we are not seeing an unsupported answer. The second point will be earned.

The response does not need to include “approximately” in any way. We will read $=$ and $\approx$ as the same thing.

Minimalist response: The following expression represents the minimum work that must be seen in a response in order to earn both points.

$$10 \cdot (-8) + 25 \cdot (-11) + 5 \cdot (-13)$$

4-1-1



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Left Riemann sum
- ☐ Answer

Solution:

$$\int_0^{40} N(t) dt \approx N(0)(10 - 0) + N(10)(35 - 10) + N(35)(40 - 35)$$

$$= (-8) \cdot 10 + (-11) \cdot 25 + (-13) \cdot 5 = -420$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned when the response states that the approximation is an overestimate because N is decreasing.

The response does not need an explicit reference to part (a) or to a left Riemann sum.

A response that appeals to concavity as a reason will NOT earn this point.

The following table contains some examples of what WILL or will NOT earn the point.

The following examples earn the point:

- Underestimate because N is decreasing
- Underestimate because $N'(t) < 0$
- Underestimate because the graph of N is decreasing (we know they are talking about the given function N)
- Underestimate because the function is decreasing. (There is only one function in the problem at this point.)

The following examples do not earn the point:

- Underestimate because N is concave down/up
- Underestimate because the graph/curve is decreasing. Since no graph/curve is provided, we do not know what graph/curve they are talking about.
- Underestimate because the table [or table values] is/are decreasing. The behavior of the values shown in the table is not the same as the behavior of the function N . The table does not tell us what is happening to N between the

4-1-1

given points.



0	1
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The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

Because N is a decreasing function, and this is a left Riemann sum, this is an overestimate.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for the correct approximation shown with supporting work.

We need to see both a difference and a quotient. They don't have to be together.

The expression does not need to be simplified, but values for $N(35)$ and $N(0)$ must be pulled from the table.

Decimal approximation is not expected, but if presented, then the decimal value must be correct to three decimal places, either rounded or truncated. We do not read anything past the third decimal place.

- If the first three places match -0.142 or -0.143 then the point is earned.
- It does not matter whether any subsequent decimal places match the correct value or not. (For example, the value -0.1424 will earn the point because the first three decimal places match the correct value.)
- Responses with fewer than 3 decimal places (-0.14 , -0.1 , etc.) will not earn the point

The response does not need to include “approximately” in any way. We will read $=$ and $\approx$ as the same thing.

The second point is for correct units.

Units must be attached to a numerical value (either correct or incorrect).

Can be written using words or a combination of words and symbols. For example:

- Coins per minute per minute or Coins/min/min
- Coins per min^2 or Coins/ min^2
- Coins per minute squared

We do not read work shown in part (d) for this point. The units must appear in part (c).

4-1-1

The letter “ C ” is not a standard abbreviation for “Coins.” It will not be interpreted as such.

The response will not be penalized for any attempt to verbally describe their answer after a correct answer has been presented.

Minimalist response: The following expression and units represent the minimum work that must be seen in a response in order to earn both points. $\frac{-13+8}{35}$ coins/min² or $\frac{-13-(-8)}{35}$ coins/min²



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$N'(10) \approx \frac{N(35)-N(0)}{35-0} = \frac{-13-(-8)}{35} = -\frac{1}{7} \text{ coin per minute per minute}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Three components of a correct answer:

- Indicating a rate of a rate
- Including the context of coin preparation (coins remaining)
- Explicitly referring to the specified point in time ($t = 10$)

Incorrect or missing units will not be penalized as long as there is a clear connection to the context of the problem.

The words “increasing” and “decreasing” are interpreted as modifiers describing one of the specified rates.

If the numerical value from part (c) is used in this interpretation, then we expect that the choice of “increasing” or “decreasing” is consistent with that value.

The following examples use $\frac{-1}{7}$ as the imported value from part (c). If a different [incorrect] value is imported from part (c), then these statements should be consistent with that value. (Remember that we are reading $=$ and $\approx$ as the same thing.)

4-1-1

- “... increasing at a rate of $\frac{-1}{7}$ ” is OK [Units not required in part (d)] (the point is not earned with this phrase alone)
- “... decreasing at a rate of $\frac{-1}{7}$ ” is OK [Units not required in part (d)] (the point is not earned with this phrase alone)
- “... decreasing at a rate of $\frac{-1}{7}$ ” is not OK and the point is not earned.

Note: the context is complicated. $N(t)$ is describing the rate of change of the remaining unprepared coins. The values in the table indicate that the number of unprepared coins is decreasing more rapidly as time passes. Common sense might tell us that this equates to the idea that the rate at which coins are prepared is increasing as time passes. Explaining the meaning using “coins prepared” rather than “coins remaining” will not prevent the response from earning this point.

Rate of a rate

To earn this point, the response must explicitly discuss the rate of change of a rate of change. For example:

- The rate of change in the rate at which coins are changing
- The rate of change in coins is changing/increasing/decreasing at a rate of . . .
- The rate at which the rate of change of coins is changing

The words “increasing,” “decreasing,” or “changing” are not interpreted as rates.

Units are not interpreted as rates. (The phrase “coins per minute” is not enough to indicate a rate.)

Slope of tangent line, velocity, and acceleration are not appropriate interpretations for this context.

“Rate of change of N ” is not enough without describing what N represents in this context.

Context

A key component of the context for this explanation is “coins.”

Any misstatement about coins prepared vs. coins remaining will not prevent the response from earning this point.

Explicit reference to $t = 10$

Acceptable references to a specific point in time include (but are not limited to):

- At minute 10 (where minute could be replaced with any other unit of time)
- At $t = 10$ (with or without units of time)
- After 10 minutes (where minutes could be replaced with any other unit of time).

4-1-1

Note: we are choosing to assume that “after” is being used in lieu of “at” unless we have additional evidence to the contrary.

If the response is explicitly indicating an interval of time (e.g., “from minute 10 on” or “during the first 10 minutes”), this point will not be earned.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

$N'(10)$ is the rate at which the rate of change in the number of 5-cent coins remaining is changing, in coins per minute per minute, at minute 10.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned with a correct tangent line equation in any form. Any of these equations can be written as is without any explanation or additional supporting work. If rewriting of equations is done (transforming to slope-intercept form, perhaps), it must be done correctly.

· $y = -0.2(t - 35) - 13$

· $y + 13 = -0.2(t - 35)$

· $y = -0.2t - 6$

The key calculus concept being assessed in this part of the problem is that the value of $N'(35)$ is the slope of the tangent line. In that spirit:

- The response will not be penalized for using x instead of t .
- We are accepting $N(t)$ in lieu of y in all forms of the tangent line equation.



0	1
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The student response accurately includes the criteria below.

☐ Tangent line equation

4-1-1

Solution:

$$y = N'(35)(t - 35) + N(35)$$

$$y = -0.2(t - 35) - 13$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with a correct substitution. There are 3 components of substitution that we need to see:

- Uses $u = \frac{t}{4}$, which may be done implicitly
- Uses the correct constant of 4
- Uses the correct limits for the substitution (0 and 10). These limits may only be seen when evaluating the antiderivative. We don't need to see them as limits on a definite integral.

Either of the expressions $4N\left(\frac{t}{4}\right)\bigg|_0^{40}$ or $4N(u)\bigg|_0^{10}$ will earn the substitution point without any additional supporting work.

Special Handling: Notation Errors

Errors in notation, such as the examples listed here, will not prevent the response from earning the first or second points. These errors will cause the response to be ineligible for the third point.

- Limits for t attached to an integral with u as the variable of integration (or vice versa)
- Improper equals signs, such as an indefinite integral equal to a definite integral
- Reversed limits of integration, for example: $\int_{10}^0 N'(u) du$

The second point is for the Fundamental Theorem of Calculus.

This point can be earned with any of the following, where k is a nonzero constant and $b = 10$ or $b = 40$. No explanation or additional supporting work is necessary.

- $kN\left(\frac{t}{4}\right)\bigg|_0^b$
- $k(N(b) - N(0))$

This point can be earned without the substitution point.

The third point is for the correct numerical value, which is -12 .

Values from the table must be used. $4(N(10) - N(0))$ is not enough to earn the answer point.

4-1-1

The expression $4(-11 - (-8))$ is sufficient for this point. No simplification is required. Any simplification that is shown must be correct.

Units are not required, and if shown, they do not need to be correct to earn this point.

The response cannot earn this point without also earning first two points.

Minimalist response: The expression $4(-11 - (-8))$ represents the minimum work that must be seen in a response in order to earn all three points.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = \frac{t}{4}$. Then $du = \frac{dt}{4}$.

When $t = 0$, $u = 0$.

When $t = 40$, $u = 10$.

$$\int_0^{40} N' \left(\frac{t}{4} \right) dt = 4 \int_0^{10} N' (u) du$$

$$= 4(N(10) - N(0))$$

$$= 4(-11 - (-8)) = -12$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Philosophy for this part: This question was difficult due to the fact that the new function introduced here represents the total number of coins prepared by the second child, but the question asks about the total weight of the coins remaining.

The key calculus concept being assessed in this part of the problem is to understand that the rate of change in the total weight of the coins (either prepared or remaining) at time $t = 23$ could be written as an expression involving

4-1-1

a constant multiple of $F'(23)$. In that spirit, either of the following expressions will earn the point: $5F'(23)$ or $-5F'(23)$.

Notations such as $5 \cdot \frac{dF}{dt} \Big|_{t=23}$ may also be used and earn the point.

The expression must use the time $t = 23$, so $5F'(t)$ is not sufficient to earn the point.



0	1
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The student response accurately includes the criteria below.

☐ Expression

Solution:

Let $C(0)$ be the total number of coins available to be prepared at time $t = 0$. Let $R(t)$ be a model for the number of coins remaining at time t with $R(t) = C(0) - F(t)$. The rate of change of the number of coins remaining at time t , in grams per minute, is $R'(t) = -F'(t)$. Therefore, the rate of change of the total weight of the 5-cent coins remaining for the second child at $t = 23$ minutes, in grams per minute, is $-5F'(23)$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned for a correct statement identifying the distinction between the differential equation

$\frac{dy}{dt} = F'(t)$ and the given slope field.

In order to earn the point, the response must:

State that the values of the differential equation are positive while the slopes in the given slope field are negative or 0 (non-positive).

Some ways to say that the values of differential equation are positive:

- F is increasing
- $F'(t) > 0$
- F has positive slopes

When discussing the slopes in the slope field, the response must explicitly refer to the signs of the slopes.

The following examples could earn the point:

- The slope field has nonpositive slopes.

4-1-1

- The slopes shown are negative or 0.
- The slope field contains negative slopes.
- “The graph” or “The field” could be used in lieu of “The slope field.”

The following examples will not earn the point:

- The slope field is decreasing/negative.
- The slope field looks like it represents a decreasing function.
- The slope field represents a quadratic [or exponential] function.
- “It” (rather than “the slope field”)

Statements like “ F is an increasing function, so the signs of the slopes in the slope field should be positive” can earn this point because “should be” is indicating contrast with what is known.

The response will not be penalized for saying that the slopes in the slope field are “always” negative or “all” negative.



0	1
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The student response accurately includes the criteria below.

☐ Explanation

Solution:

F is an increasing function, so $\frac{dy}{dt} = F'(t) \geq 0$. However, in the given slope field, the slopes of the line segments for $y > 0$ are negative.

4-1-1

55. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

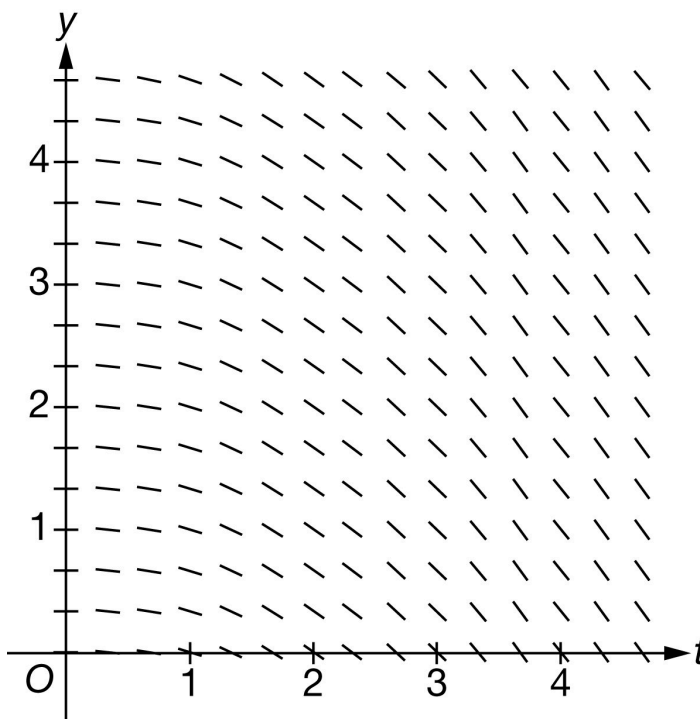
t (minutes)	0	10	35	40
$N(t)$ (number of 5-cent coins)	1000	810	200	0

A child has collected 5-cent coins and is preparing them to deposit in the bank by counting them into bags. The child begins with 1000 coins. For time t minutes, $0 \leq t \leq 40$, the total number of coins remaining to prepare is modeled by a decreasing, twice-differentiable function N . Values of $N(t)$ at various times t are shown in the table above.

- (a) Using correct units, approximate $N'(35)$ using the average rate of change of N over the interval $10 \leq t \leq 40$.
- (b) Explain the meaning of $N'(35)$ in the context of the problem.
- (c) It is known that $N'(10) = -20$. Write an equation for the line tangent to the graph of $y = N(t)$ at $t = 10$.
- (d) Use a right Riemann sum with the three subintervals indicated by the table to approximate $\int_0^{40} N(t) \, dt$. Show the computations that lead to your answer.
- (e) Is the approximation from part (d) an overestimate or an underestimate for $\int_0^{40} N(t) \, dt$? Give a reason for your answer.
- (f) Find the value of $\int_0^{10} N'(4t) \, dt$. Show the computations that lead to your answer.
- (g) For $0 \leq t \leq 40$, the total number of 5-cent coins remaining can also be modeled by the differentiable function $\bar{F}$, where F is a function of t . Assume the weight of each coin is 5 grams. Write an expression for the rate of change, in grams per minute, of the total weight of the 5-cent coins remaining at $t = 11$ minutes.

4-1-1

(h) Explain why the following slope field cannot represent the differential equation $\frac{dy}{dt} = -0.3y$.

**Part A**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for the correct approximation shown with supporting work. The supporting work must be at least a difference and a quotient. Correct values must be taken from the table to earn the point. The difference quotient does not need to be simplified.

Examples that earn the first point include:

$$\cdot \frac{0-810}{40-10}$$

$$\cdot \frac{N(40)-N(10)}{30} = \frac{-810}{30}$$

$$\cdot \frac{N(40)-N(10)}{30} = -27$$

An example that does not earn the first point is $\frac{N(40)-N(10)}{30}$.

Minimal response that earns the first point: $\frac{0-810}{40-10}$ or $\frac{0-810}{30}$

· Neither $\frac{-810}{30}$ nor -27 , with no supporting work, will earn the point.

The second point is for correct units

4-1-1

The correct units must be attached to a numerical value: either correct or incorrect.

The units may be written using words or a combination of words and symbols. For example:

- Coins per minute
- Nickels per minute,
- 5-cents per minute
- Coins/minute

The response will not be penalized for any verbal attempt to discuss the answer after the correct answer has been presented.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$N'(35) \approx \frac{N(40) - N(10)}{40 - 10} = \frac{0 - 810}{30} = -27 \text{ coins per minute}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Three components of a correct answer:

- Indicating a *rate of change*
- Including the context of coin collection
- Referring to the specific point in time, $t = 35$

The response is not penalized for incorrect units in this part. (Whether correct or incorrect, units have no effect on the score in this part.)

Responses may use “changing” or “decreasing” as a modifier as long as the value found in part (a) is not brought into part (b) in an incorrect manner. Example: “the number of coins is decreasing at a rate of 27 coins per day...” is okay, but “the number of coins is decreasing at a rate of -27 coins per day...” would not earn the point.

4-1-1

Acceptable references to a specific point in time include (but are not limited to):

- On minute 35 and after 35 minutes
- At $t = 35$
- At $t = 35$ hours or any incorrect unit of time

If the response clearly indicates an interval of time (e.g., “from minute 35 on,” “during the first 35 minutes,” or “for $t = 0$ to $t = 35$ ”), the point is not earned.

Examples that earn the point include:

- $N'(35)$ is the rate of change of the number of coins per minute at time $t = 35$
- The rate of change of number of coins is -27 coins per minute on minute 35
- $N'(35)$ is the rate of change in the number of coins at time $t = 35$ in coins per minute
- 27 is the rate of decrease in the number of coins per minute on minute 35

Examples that do not earn the point include:

- $N'(35)$ is the average rate of change of the total number of coins per minute on minute 35
- $N'(35)$ is the total number of coins per minute on minute 35
- The rate of change is -27 coins per minute (Missing “at $t = 35$ ”)
- $N'(35)$ is the slope of the tangent line at $t = 35$
- Rate of change of N at time $t = 35$
- Coins per minute decreases at minute 35



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

$N'(35)$ is the rate at which the number of 5-cent coins remaining is changing, in coins per minute, at time $t = 35$ minutes.

Part C

4-1-1

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned with a correct tangent line equation in any form. The use of x in place of t is acceptable.

· $y = -20(t - 10) + 810, y = -20(x - 10) + 810$

· $y - 810 = -20(t - 10), y - 810 = -20(x - 10)$

· $y = -20t + 1010, y = -20x + 1010$

The key calculus concept being assessed in this part of the problem is that the value of $N'(10)$ is the slope of the tangent line. In that spirit, we are accepting $N'(t) = \dots$ in lieu of $y = \dots$, but we are not accepting $N'(10) = \dots$

Note that $N(t) = -20(t - 10) + 810$ earns the point, but $N(10) = -20(t - 10) + 810$ does not earn the point.



0	1
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The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$$y = N'(10)(t - 10) + N(10)$$

$$y = -20(t - 10) + 810$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for setting up a right Riemann sum. The setup must be a sum of three products.

· A response may still earn this point if there is only one numerical error in the expression for the Riemann sum with values pulled from the table. (See example in the table below).

· Responses with the numerical error described above are not eligible for the second point.

Note: either the multiplication or the addition may be indicated implicitly. For example, the response may show appropriate values put into rows and columns. Implicit responses are eligible for both points.

A response presenting a left Riemann sum (instead of right), will not earn any points for part (d).

The second point is for the correct numerical result.

4-1-1

- In order to earn the second point, there must be supporting calculations.
- The numerical values for $N(10)$, $N(35)$, and $N(40)$ must be pulled from the table
- Any simplification shown must be correct.

Use the following as an illustration how to score both points. (In most cases, you see scores for which the expression on the left side represents the entirety of relevant work shown for the response.)

- A response of $810 \cdot (10 - 0) + 200 \cdot (35 - 10) + 0 \cdot (40 - 35)$ earns both points.
- A response of $10 \cdot N(10) + 25 \cdot N(35) + 5 \cdot N(40)$ earns the first point. Eligible for second point if more work is on the page.
- A response of $10 \cdot N(10) + 25 \cdot N(35) + 5 \cdot N(40) = 13,100$ earns both points.
- A response of $10 \cdot 810 + 25 \cdot 200 + 5 \cdot 0$ earns both points.
- A response of $10 \cdot 810 + 15 \cdot 200 + 5 \cdot 0$ contains one numerical error. Earns the first point. Not eligible for the second point.
- A response of $10 \cdot 810 + 25 \cdot 200 + 5 \cdot 0 = 12,100$ earns first point for left side of equation. Does not earn second point due to incorrect simplification.
- A response of $8100 + 5000 + 0$ or $8100 + 5000 + 0 = 13,100$ does not earn the first point because we cannot see the sum of three products. However, this is the correct sum, so we see an answer with supporting work. The second point will be earned.

The response does not need to include “approximately” in any way. We will read $=$ and $\approx$ as the same thing.

Minimalist response: The following expression represents the minimum work that must be seen in a response in order to earn both points.

$$10 \cdot 810 + 25 \cdot 200 + 5 \cdot 0$$



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Right Riemann sum
- ☐ Answer

Solution:

$$\int_0^{40} N(t) dt \approx N(10)(10 - 0) + N(35)(35 - 10) + N(40)(40 - 35)$$

4-1-1

$$= 810 \cdot 10 + 200 \cdot 25 + 0 \cdot 5 = 13,100$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned when the response states that the approximation is an underestimate because N is decreasing.

The response does not need an explicit reference in part (d). The following statements are sufficient for earning this point:

- Underestimate because N is decreasing.
- Underestimate because the function is decreasing

Note: At this point, there is only one function being discussed in this problem, so we will assume “the function” is N .

When a response includes additional information about concavity, we interpret that as a lack of understanding about the question, so this will not earn the point. When the response references a graph or the given table rather than the function N , the point is not earned.

Examples that earn the point:

- The right Riemann sum underestimates because the function is decreasing.
- The sum is an underestimate because $N(t)$ is decreasing.
- The sum is an underestimate because $N'(t) < 0$.

Examples that do not earn the point:

- The sum is an underestimate because $N'(t)$ is decreasing. ($N'(t)$ decreasing is not known)
- The right Riemann sum underestimates because the graph is decreasing.
- The right Riemann sum underestimates because the function is decreasing and concave up.



0	1
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The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

Because N is a decreasing function, and this is a right Riemann sum, the answer from part (d) is an underestimate.

4-1-1

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with a correct substitution. There are 3 components of substitution that we need to see:

- Uses $u = 4t$, which may be done implicitly
- Uses the correct constant of $\frac{1}{4}$
- Uses the correct limits for the substitution (0 and 40). These limits may only be seen when evaluating the antiderivative. We don't need to see them as limits on a definite integral.
- Either of the expressions below will earn the substitution point without any additional supporting work:

$$\frac{1}{4} N(4t) \Big|_0^{10} \text{ or } \frac{1}{4} N(u) \Big|_0^{40}$$

Special Handling: Notation Errors

Errors in notation, such as the examples listed here, will not prevent the response from earning the first or second points. These errors will cause the response to be ineligible for the third point

- Limits for t attached to an integral with u as the variable of integration (or vice versa),
- Improper equals signs, such as an indefinite integral equal to a definite integral
- Reversed limits of integration, for example: $\int_{40}^0 N'(u) du$

The second point is for the Fundamental Theorem of Calculus. This point can be earned with any of the following, where k is a nonzero constant and $b = 10$ or $b = 40$. No explanation or additional supporting work is necessary.

$$\circ kN(4t) \Big|_0^b$$

$$\circ k(N(b) - N(0))$$

This point can be earned without earning the substitution point.

The third point is for the correct numerical value, which is -250 .

Values from the table must be used. $\frac{1}{4}[N(40) - N(0)]$ is not enough to earn the answer point.

The expression $\frac{1}{4}(0 - 1000)$ is sufficient for this point. No simplification is required. Any simplification that is shown must be correct.

Units are not required, and if shown, they need not be correct in order to earn this point.

4-1-1

The response cannot earn this point without also earning the FTC point.

Minimalist response: The expression $\frac{1}{4}(-1000)$ represents the minimum work that must be seen in a response in order to earn all three points.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = 4t$. Then $du = 4 dt$.

When $t = 0$, $u = 0$.

When $t = 10$, $u = 40$.

$$\begin{aligned}\int_0^{10} N'(4t) dt &= \frac{1}{4} \int_0^{40} N'(u) du \\ &= \frac{1}{4}(N(40) - N(0)) \\ &= \frac{1}{4}(0 - 1000) = -250\end{aligned}$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned for a correct expression at $t = 11$. No supporting work or justification is required.

The correct expression of just $5F'(11)$ or $5N'(11)$ earns this point.

No justifications or explanations are required.

Any incorrect statement, such as $N'(11) = 5F'(11)$, does not earn this point.



0	1
---	---

4-1-1

The student response accurately includes the criteria below.

☐ Expression

Solution:

The rate of change, in grams per minute, of the total weight of the 5 -cent coins remaining at $t = 11$ minutes is $5F'(11)$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned for a correct statement identifying the distinction between two things: the differential equation, $\frac{dy}{dt} = -0.3y$, and the given slope field.

The response does not need to explicitly reference both the differential equation and the given slope field., but it must explicitly reference one and implicitly reference the other.

Acceptable responses include

- Identifying an inconsistency between the differential equation and the slope field at one point , e.g., at the point $(4, 0)$, the slope of the segment should be 0, but it is negative.
- Referencing a pattern of slopes, and explaining why that pattern does not fit the given differential equation.

A response that lacks even an implicit reference to one of the two things being compared will not earn the point.

For example, merely stating either $\frac{dy}{dt}$ is negative, or that the slope field is negative does not earn the point.

- “The values for y are less than or equal to zero so $\frac{dy}{dt} \leq 0$ ” lacks any reference to the slope field and will not earn the point.
- “All the slopes in the slope field are negative” lacks any reference to the differential equation and will not earn the point.

Vague answers which have the potential to be incorrect will not earn the point. Some examples are:

- “The field is decreasing”
- Responses that state a specific slope value on the slope field for a general value of x or y that is not correct for all points with that x or y value do not earn the point. (e.g. “the slope field shows a slope of 0 when $y = 4$ ” is an incorrect statement)

A response may correctly solve or attempt to solve the differential equation and then provide a correct argument without reference to their solution. Such a response would earn the point.

4-1-1

A response might actually solve the differential equation and provide a correct argument on the resulting function.



0

1

The student response accurately includes the criteria below.

☐

Explanation

Solution:

When $y = 0$, $\frac{dy}{dt} = 0$. However, in the given slope field, for $t > 0$, the slopes of the line segments for $y = 0$ are nonzero.

4-1-1

56. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (days)	0	5	9
$S(t)$ (boxes per day)	-70	-30	-16

A student organization is selling boxes of snacks to raise money for a school trip. For time t days, $0 \leq t \leq 9$, the rate of change in the number of boxes of snacks remaining to sell is modeled by an increasing, twice-differentiable function S . Values of $S(t)$ at various times t are shown in the table above.

- (a) Using correct units, approximate $S'(5)$ using the average rate of change of S over the interval $0 \leq t \leq 9$.
- (b) Explain the meaning of $S'(5)$ in the context of the problem.
- (c) The function $B(t)$ models the number of boxes remaining to sell at time t days, where $B'(t) = S(t)$. It is known that $B(5) = 240$. Write an equation for the line tangent to the graph of $y = B(t)$ at $t = 5$.

- (d) Find the value of $\int_0^{4.5} S'(2t) \, dt$. Show the calculations that lead to your answer.

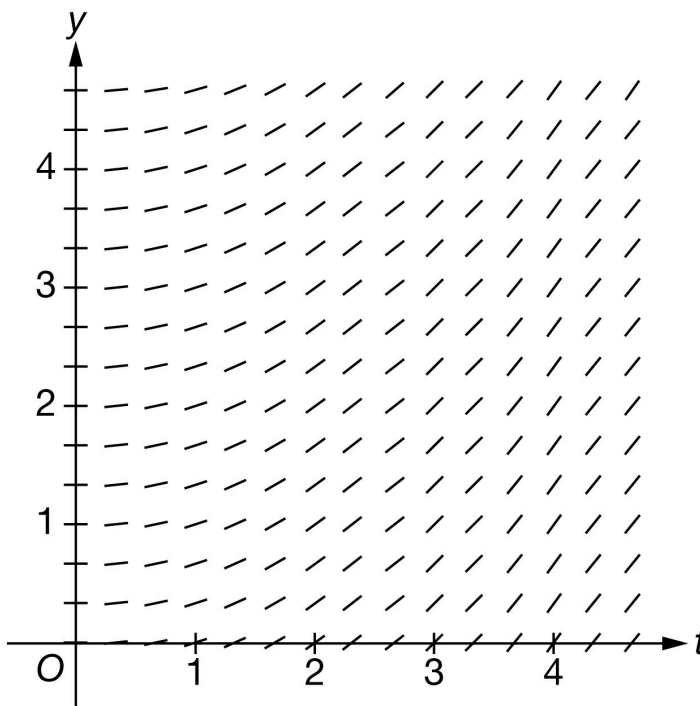
- (e) Use a trapezoidal sum with the two subintervals indicated by the table to approximate $\int_0^9 S(t) \, dt$.

Show the calculations that lead to your answer.

- (f) Another student organization is also selling boxes of snacks. For $0 \leq t \leq 9$, the rate of change in the number of boxes of snacks remaining for the second student organization to sell is modeled by a decreasing, differentiable function M , where M is a function of t . Assume the weight of each box of snacks is 7 ounces. Write an expression in terms of M and S for the rate of change, in ounces per day, of the combined weight of the boxes of snacks remaining for the two organizations to sell at $t = 8$ days.

4-1-1

(g) Explain why the following could not be a slope field for the differential equation $\frac{dy}{dt} = 0.25y$.

**Part A**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for the correct approximation shown with supporting work.

Answers without supporting work do not earn this point. We need to see both a difference and a quotient for the supporting work.

Response does not need to be simplified, but any simplification must be done correctly.

If a decimal is presented, it must be correct to three decimal places. We do not read anything past the third decimal place.

The second point is earned for the correct units.

Units must be attached to a numerical value (either correct or incorrect).

Can be written using words or a combination of words and symbols.

The response will not be penalized for any verbal attempt to discuss the answer after the correct presentation of the approximation.

The minimum response needed to earn both points is $\frac{-16+70}{9}$ boxes per day per day

4-1-1



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$S'(5) \approx \frac{S(9) - S(0)}{9 - 0} = \frac{-16 - (-70)}{9} = 6 \text{ boxes per day per day}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This part is intended to be independent from part (a), so we do not want or need to see any reference to the value obtained in (a). If that value is referenced, conclusion must be consistent.

Three components of a correct answer:

- Indicating a rate of a rate (look for the word rate to be used correctly twice)
- Including the context of boxes
- Referring to the specific point in time

Units will not be considered for this point. They do not need to be present, and if they are, they do not need to be correct.

Any incorrect statements make the response ineligible for the point. For example:

- $S'(5)$ is the average (or an approximation of, or an estimate of) rate of change of . . .
- $S'(5)$ is the acceleration (or rate of change of velocity) of . . .

“Increasing” or “decreasing” may be used in place of “changing” as long as the value found in part (a) is not brought into part (b) in an incorrect manner. Beware of phrases such as “decreasing at a negative rate.”

- Example: $S'(5)$ is the rate of increase of the rate of change of the number of boxes at 5 days. (earns the point)
- Example: $S'(5)$ means the number of boxes per day is increasing by 5 boxes/day/day. (does not earn the point)

Acceptable references to time include (but are not limited to):

- On day 5

4-1-1

- At $t = 5$ (with correct units of time, without correct units of time, or even with incorrect units of time)
- After $t = 5$ (with correct units of time, without correct units of time, or even with incorrect units of time)

If the response clearly indicates an interval of time (e.g. “from day 0 to day 5”) the response will not earn the point.

Examples that earn the point include:

- $S'(5)$ represents the rate at which the rate of change of the number of boxes is changing on day 5.
- $S'(5)$ is the rate of change in the rate of change of the number of boxes at 5 days.
- $S'(5)$ represents the rate at which the rate of change of the number of boxes is increasing at $t = 5$.
- “ $S'(5) = (\text{answer from part (a)})$ ” means the rate of change in the number of boxes per day is increasing at a rate of (answer from part (a)) at $t = 5$.
- $S'(5)$ represents the rate at which the rate of change of the number of boxes is decreasing on the fifth day.

Examples that do not earn the point include:

- $S'(5)$ represents the rate at which the number of boxes is changing on day 5.
- $S'(5)$ is the rate of change in the number of boxes per day at $t = 5$ days
- The rate of change in the number of boxes per day is increasing at $t = 5$.
- The rate of change in the rate of change of the number of boxes per day is increasing at $t = 5$.
- $S'(5) = -6$ means that the rate of change in the number of boxes is decreasing at a rate of -6 at $t = 5$ (double negative).



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

$S'(5)$ is the rate at which the rate of change in the number of boxes per day is changing, in boxes per day per day, on day 5.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

4-1-1

The point is earned with a correct tangent line equation in any form. For example:

$$y = -30(t - 5) + 240$$

$$y - 240 = -30(t - 5)$$

$$y = -30t + 390$$

The student may write x in place of t and still earn the point.

No work is required to earn the point.

We will accept $B(t)$ in lieu of y in the equation of the tangent line.



0	1
---	---

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$$y = B'(5)(t - 5) + B(5)$$

$$y = -30(t - 5) + 240$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with a correct substitution.

Two things are required as evidence of correct substitution:

- Coefficient of $\frac{1}{2}$ correctly applied to an integral or an antiderivative
- Evidence that the correct limits of 0 to 9 will be applied

Examples that will earn the first point (these are just a few):

$$\cdot \frac{1}{2} \int_0^9 S'(u) \, du$$

$$\cdot \frac{1}{2} S(u) \Big|_0^9 \text{ or } \frac{1}{2} S(2t) \Big|_0^{4.5}$$

4-1-1

Responses that include misstatements such as using t -values for limits with u as the variable of integration in an intermediate step which is later corrected may still earn this point. Intermediate errors in bounds that are subsequently corrected will disqualify the student from earning the third point. If the bounds are never appropriately handled, the response cannot earn the first point. Scratch work that is not connected to the solution will not be read.

Examples:

$$\int_0^{4.5} S'(2t) dt = \frac{1}{2} \int_0^{4.5} S'(u) du = \frac{1}{2} S(2t) \Big|_0^{4.5} \text{ earns the first point, but will not earn the third point.}$$

$$\int_0^{4.5} S'(2t) dt = \frac{1}{2} \int_0^{4.5} S'(u) du = \frac{1}{2} [S(4.5) - S(0)] \text{ does not earn the first point.}$$

The second point is for the Fundamental Theorem of Calculus, demonstrating the knowledge that

$$\int_a^b S'(u) du = S(u) \Big|_a^b = S(b) - S(a)$$

This point can be earned with either $kS(2t) \Big|_a^b$, $kS(t) \Big|_a^b$, or $k(S(b) - S(a))$ where k is a nonzero constant [not necessarily correct], $a = 0$, and $b = 4.5$ or $b = 9$.

The response can earn this point without earning the first point.

$$\frac{1}{2} [S(9) - S(0)] \text{ earns the first two points.}$$

The third point is for the correct numerical value. Our answer only.

Values from the table must be used (i.e., $\frac{1}{2} [S(9) - S(0)]$ is not sufficient to earn the third point).

If the response includes any linkage error, (i.e., the use of equal signs to equate two expressions that are not equal), the third point will not be earned. If an indefinite integral is linked with a definite integral the response will not be penalized.

$$\int_0^{4.5} S'(2t) dt = \frac{1}{2} \int S'(u) du = \frac{1}{2} S(2t) \Big|_0^{4.5} = \dots \text{ is eligible for the third point.}$$

Eligibility for the third point: Must have earned the first two points.

The minimum response needed to earn all three points is: $\frac{1}{2}(-16 + 70)$.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

☐ Substitution

4-1-1

- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = 2t$. Then $du = 2 dt$.

When $t = 0$, $u = 0$.

When $t = 4.5$, $u = 9$.

$$\begin{aligned}\int_0^{4.5} S'(2t) dt &= \frac{1}{2} \int_0^9 S'(u) du \\ &= \frac{1}{2}(S(9) - S(0)) \\ &= \frac{1}{2}(-16 - (-70)) = \frac{1}{2}(54) = 27\end{aligned}$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the form of a trapezoidal sum

Point is earned for the setup of the trapezoidal sum, with one numerical error allowed. Numerical values do not need to be pulled from the table for this point, but Δt values must be presented. Responses that contain one incorrect numerical value in the trapezoidal sum setup are still eligible to earn the first point, but will not earn the second.

Response may present a left Riemann sum, a right Riemann sum, then average the two. Again, one incorrect numerical value will not disqualify the response from earning the first point.

Response may compute trapezoids as areas of rectangles and triangles.

Trapezoidal areas may be given in a graph. In order to earn this point, the response must clearly show the computations leading to the values presented in the graph, and must show that these areas are added to obtain the final approximation value.

A response using subtraction instead of addition will not earn the point (e.g.

$$\frac{1}{2}[5(S(0) - S(5)) + 4(S(5) - S(9))]$$

earns no points).

Response utilizing “Trapezoidal Rule” will not earn this point, unless this represents only one error in the numerical values.

$\cdot \frac{1}{2}(5)[S(0) + 2S(5) + S(9)]$ is equivalent to $\frac{1}{2}[5(S(0) + S(5)) + 5(S(5) + S(9))]$, has only one incorrect numerical value (the second interval width is 5 instead of 4), so can earn the point.

4-1-1

$\cdot \frac{1}{2}(4.5) [S(0) + 2S(5) + S(9)]$ has two incorrect numerical values (both interval widths are incorrect) so will not earn the point.

The second point is for the correct numerical result.

Our answer only. Response does not need to be simplified, but if it is, it must be simplified correctly.

The answer must be accompanied with supporting work. Supporting work may appear in a graph, with clearly labeled areas. It is possible to earn the second point without earning the first if the supporting work is not sufficient to earn the first point.

Pictures without clear area computations and a clear summation of quantities can earn at most the second point.

Special case: A positive constant multiple of a complete and correctly expressed trapezoidal sum (with the correct values pulled from the table) will earn the first point but not the second:

$\cdot k \left[\left(\frac{-70-30}{2} \right) \cdot 5 + \left(\frac{-30-16}{2} \right) \cdot 4 \right]$ for $k > 0$ earns just the first point.

The minimum response needed to earn both points is: $\left(\frac{-100}{2} \right) \cdot 5 + \left(\frac{-46}{2} \right) \cdot 4$



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Form of trapezoidal sum
- ☐ Answer

Solution:

$$\begin{aligned} \int_0^9 S(t) dt &\approx \left(\frac{S(0)+S(5)}{2} \right) (5-0) + \left(\frac{S(5)+S(9)}{2} \right) (9-5) \\ &= \left(\frac{-70-30}{2} \right) \cdot 5 + \left(\frac{-30-16}{2} \right) \cdot 4 = -342 \end{aligned}$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for a correct constant multiple or correct sum.

Response must contain the constant 7 as a factor of the expression, and/or the correct sum $M(8) + S(8)$.

If the sum is incorrect, then the constant factor of 7 must be applied either to a sum that involves only $M(t)$ and $S(t)$, $M'(t)$ and $S'(t)$, or integrals of $M(t)$ and $S(t)$, or to one of the correct terms of the sum $M(8)$ or

4-1-1

$S(8)$, with no additional incorrect terms.

Examples that earn the point include:

· $7[M(t) + S(t)]$

· $7 \int [M(t) + S(t)] dt$

· $7[M' + S']$

· $7S(8)$

Examples that do not earn the point include:

· $7[M(t) + S'(t)]$

· $7 \int M(t) dt$

· $7M' + S'$

· $7S(8) + t$

Special case: $7M(8) + S(8)$ is an error in parentheses, so will earn the first point but is not eligible to earn the second point.

If the constant multiple is missing or incorrect, the sum must be correct. $k[M(8) + S(8)]$ for any nonzero constant k will earn the first point.

The second point is for the correct expression.

$7M(8) + 7S(8)$ (or an equivalent expression) earns both points.

Inclusion of units (correct or incorrect) will not affect the score.

A completely correct expression with one additional constant term added/subtracted would earn a just the first point. For example, $7M(8) + 7S(8) + 7(2)$ earns only the first point.



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Constant multiple or sum
- ☐ Expression

Solution:

4-1-1

The rate of change, in ounces per day, of the combined weight of the boxes of snacks remaining for the two organizations to sell at $t = 8$ days is $7M(8) + 7S(8)$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is earned for a correct statement identifying specific inconsistencies between the given differential equation and the given slope field. There are many such correct statements, such as:

- The response references a specific point on the y -axis (such as $(0, 4)$), and indicates that the given slope field shows a slope of zero, yet the differential equation would yield a positive slope.
- The response references a pattern of slopes, and explains why that pattern does not fit our differential equation (e.g. “the slope field shows slopes that change as t changes, but our differential equation depends only on y , so slopes should remain constant for a given value of y as t changes.”)

Identifying a mismatch between the differential equation and the slope field at one point is sufficient to earn the point.

Response is not penalized for referencing “ x ” instead of “ t ”.

If the response contains any incorrect statement, the response will not earn the point.

Vague answers which have the potential to be incorrect will not earn the point, nor will responses that present information beyond what we are given. For example:

- Responses that specify that the slope field shows a particular nonzero slope at a particular point will not earn the point, as this cannot actually be determined visually (i.e., the response should not indicate that at $(2, 2)$ the slope shown on the slope field is 1)
- Responses that state a specific slope value on the slope field for a general value of x or y that are not correct for all points with that x or y value do not earn the point. (e.g. “the slope field shows a slope of 0 when $y = 4$ ” is an incorrect statement)

A response may involve a solution to the differential equation, with an argument based on the resulting function. If done correctly, and presented with a correct argument, this will earn the point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

4-1-1

When $y = 0$, $\frac{dy}{dt} = 0$. However, in the given slope field, for $t > 0$, the slopes of the line segments for $y = 0$ are nonzero.

4-1-1

57. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (days)	0	3	6	9	12
$S(t)$ (number of boxes of snacks remaining)	600	500	440	240	20

A student organization is selling boxes of snacks to raise money for a school trip. The student organization begins with 600 boxes of snacks and sells them over a twelve-day period. For time t days, $0 \leq t \leq 12$, the total number of boxes of snacks remaining to sell is modeled by a twice-differentiable function S . Values of $S(t)$ at various times t are shown in the table above.

(a) Use a midpoint Riemann sum with the two subintervals indicated by the table to approximate

$$\int_0^{12} S(t) \, dt.$$

Show the computations that lead to your answer.

(b) Find the value of $\int_0^3 S'(3t) \, dt$. Show the computations that lead to your answer.

(c) Using correct units, approximate $S'(9)$ using the average rate of change of S over the interval $6 \leq t \leq 12$.

(d) Explain the meaning of $S'(9)$ in the context of the problem.

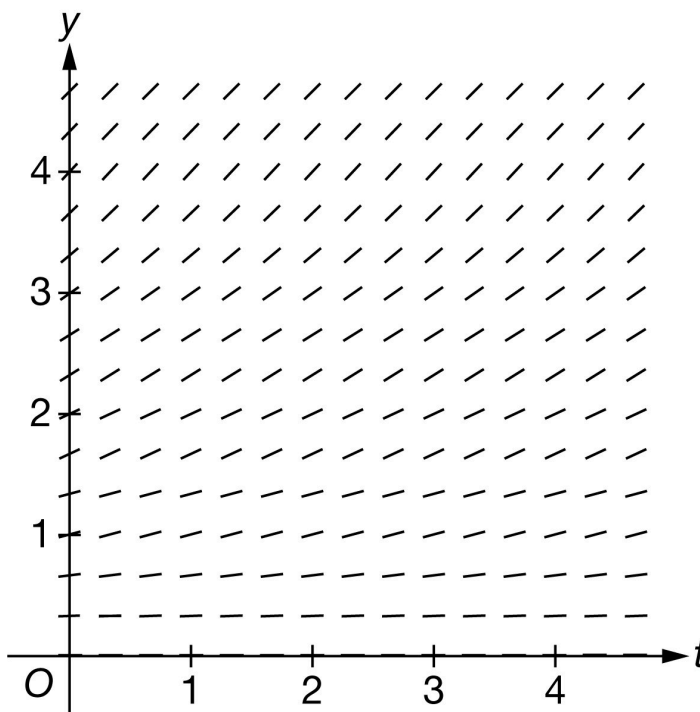
(e) It is known that $S'(6) = -40$. Write an equation for the line tangent to the graph of $y = S(t)$ at $t = 6$.

(f) It is known that $S''(t) < 0$ for $6 \leq t \leq 9$. If $S(8)$ is approximated using the equation from part (e), would the approximation be an overestimate or underestimate for the actual value of $S(8)$? Give a reason for your answer.

(g) For $0 \leq t \leq 12$, the total number of boxes of snacks remaining can be modeled by the differentiable function Q , where Q is a function of t . Assume the weight of each box is 8 ounces. Write an expression for the rate of change, in ounces per day, of the total weight of the boxes of snacks remaining at $t = 5$ days.

4-1-1

(h) Explain why the following slope field cannot represent the differential equation $\frac{dy}{dt} = -0.25y$.

**Part A**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point (Form of midpoint sum): Philosophy: The response should provide clear evidence adding the area of two rectangles whose heights are found using the midpoint of the interval.

Acceptable evidence:

$$\cdot 6 \cdot (S(3) + S(9)) \text{ or } 6 \cdot S(3) + 6 \cdot S(9)$$

$$\cdot (6 - 0) \cdot 500 + (12 - 6) \cdot 240$$

$$\cdot 6 \cdot (500 + 240) \text{ or } 6 \cdot 500 + 6 \cdot 240$$

To earn this point the response can have at most one mistake in the expression or equation. So, expressions of the form $\frac{6}{b} \cdot (S(3) + S(9))$, for any nonzero value of b would earn the point. Note: The value $\frac{6}{b}$ might be distributed across the sum.

Expressions of the form $\frac{a}{b} \cdot (S(3) + S(9))$, for $a \neq 6$ will not earn this point.

Once the first point is earned, it cannot be lost.

Second Point (Answer)

To earn the point, the answer must be correct. It does not need to be simplified, but if it is, the simplification must be correct.

4-1-1

The answer point cannot be earned unless the first (midpoint sum) point was earned.

Minimal complete answers that earn both points:

$$\cdot 6 \cdot (500 + 240)$$

$$\cdot 6 \cdot 500 + 6 \cdot 240$$



0	1	2
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The student response accurately includes **both** of the criteria below.

☐ Form of midpoint sum

☐ Answer

Solution:

$$\int_0^{12} S(t) dt \approx S(3)(6 - 0) + S(9)(12 - 6)$$

$$= 500 \cdot 6 + 240 \cdot 6 = 4440$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point (Substitution) Philosophy: This point is earned when the response has all three components of the substitution done correctly.

Uses $u = 3t$, perhaps implicitly (any variable could be used for u)

Uses the correct constant of $\frac{1}{3}$

Uses the correct limits for their substitution.

· The limits might not be seen until the final evaluation, and at that point, response may have returned to the variable t .

· Limits for t attached to an integral involving u or vice versa will not earn this substitution point but could still earn the answer point, if the numerical answer is correct.

Second point: (Fundamental Theorem of Calculus) Philosophy: This point is earned when the FTC has been applied correctly to either $\int S'(3t) dt$ or $\int S'(u) du$.

The point is earned for any of the following:

4-1-1

· The equation $\int S'(3t) dt = S(3t)$ or $\int S'(u) du = S(u)$

· $\frac{1}{3}S(3t)\big|_0^3$ or $\frac{1}{3}S(u)\big|_0^9$

· $S(9) - S(0)$, $S(9)$, or $S(0)$

This point can be earned if the substitution point is not.

Note: $\int_a^b S'(u) du = S(b) - S(a)$ does not earn the point. The FTC point is for the response communicating that the antiderivative of S' is S .

Third Point (Answer)

Must be the correct numerical answer.

Must have earned the second point for the FTC.

Might have not earned the first point for substitution because of an incorrect use of limits but will still earn the third point for the correct answer.

$S(9)$ and $S(0)$ must be evaluated; the expression $\frac{1}{3}(S(9) - S(0))$ by itself does not earn the third point.

The answer may be unsimplified, but if simplified it must be done correctly.

Scoring common responses:

The minimal work required to earn all three points is $\frac{1}{3}(240 - 600)$.

The following responses would earn the first two points and be eligible for the answer point if the response continues:

· $\frac{1}{3}S(3t)\big|_0^3$

· $\frac{1}{3}(S(9) - S(0))$

Responses of only the following would earn only the second point:

· $S(9) - S(0)$

· $S(3) - S(0)$

· $\int_0^9 S'(u) du = S(9) - S(0)$

4-1-1

$$\cdot S(3t) \Big|_0^3 \text{ or } S(u) \Big|_0^9$$

The following response would earn no points: -120 in the absence of one of the first two points.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = 3t$. Then $du = 3 \, dt$.

When $t = 0$, $u = 0$.

When $t = 3$, $u = 9$.

$$\begin{aligned} \int_0^3 S'(3t) \, dt &= \frac{1}{3} \int_0^9 S'(u) \, du \\ &= \frac{1}{3}(S(9) - S(0)) \\ &= \frac{1}{3}(240 - 600) = -120 \end{aligned}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point (Approximation)

The response must be a difference and a quotient.

The answer does not need to be simplified, but if it is simplified it must be done correctly.

The minimal work that must be shown to earn this point is either $\frac{20-440}{6}$ or $\frac{-420}{12-6}$.

Second point (Units)

The units must be attached to a correct or incorrect value.

4-1-1

The response will not be penalized for any verbal attempt to discuss the answer after the correct approximation with units is presented.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Approximation

☐ Units

Solution:

$$S'(9) \approx \frac{S(12) - S(6)}{12 - 6} = \frac{20 - 440}{6} = -70 \text{ boxes per day}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The interpretation must include three components:

- indicate the rate of change
- the specific context of the number of boxes remaining
- the specific point in time, $t = 9$.

Responses do not need to include correct units or any units.

The word “decreasing” may be used in place of the word “changing”.

The response is not earned for:

- “the rate of boxes sold on day 9” as this does not indicate a change in the rate of boxes remaining.
- “decreasing at a rate of -70 boxes per day”

“the rate of change of S ” without specifying that S is the number of boxes remaining, does not earn the point.

Acceptable ways to provide the specific point in time:

- On day 9 or after 9 days.
- At $t = 9$ (missing the unit of time)
- On week 9 (incorrect unit of time)

4-1-1

A response that clearly indicates an interval of time (e.g. “during the first 9 days” or “from day 9 on”) does not earn the point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

$S'(9)$ is the rate at which the number of boxes of snacks remaining is changing, in boxes per day, on day 9.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The use of $S(t)$ in place of y earns the point.

The use of x in place of t earns the point.

Tangent lines do not need to be simplified, but any simplification must be done correctly.

Any correct form of the tangent line earns the point. For example:

· $y = 440 - 40(t - 6)$

· $y - 440 = -40(t - 6)$

· $y = -40t + 240 + 440$

· $y = -40t + 680$



0	1
---	---

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$$y = S'(6)(t - 6) + S(6)$$

4-1-1

$$y = -40(t - 6) + 440$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the point the response must include $S''(t) < 0$ and overestimate.

“ $S(t)$ is concave down” can be used instead of $S''(t) < 0$.

Information about $S'(t)$ by itself without reference to $S''(t) < 0$ is not sufficient and does not earn this point.

“Overestimate because it is concave down” is not sufficient by itself because of the use of “it”. The response must clearly, directly or indirectly, indicate to what “it” is referring.

If a response makes the correct conclusion and states, $S''(t) < 0$, and then goes on and makes an incorrect statement or a statement that cannot be verified will not earn this point.

For example: “ $S'(t)$ is concave down, overestimate” does not earn this point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

Because $S''(t) < 0$, the tangent line approximation would produce an overestimate.

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct expression of only $8 \cdot Q'(5)$ earns this point.

Only expression $8 \cdot S'(5)$ also earns this point.

No justifications or explanations are required.

Any incorrect statement such as, $Q'(5) = 8 \cdot S'(5)$, does not earn this point.



0	1
---	---

4-1-1

The student response accurately includes the criteria below.

☐ Expression

Solution:

The rate of change, in ounces per day, of the total weight of the boxes of snacks remaining at $t = 5$ days is $8Q'(5)$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Philosophy: Response clearly references what part of the differential equation is different from the slope field and the response needs only to clearly and explicitly reference one of them.

The response does not need to explicitly reference both the differential equation and the given slope field.

The distinction can be made with explicit reference to one of the two things and an implicit reference to the other

For example:

· “ $\frac{dy}{dt}$ is negative, not positive”. We will infer that negative is referring to the slopes shown in the slope field.

· “This would be correct for a positive derivative, but $\frac{dy}{dt}$ is negative.” We will read “This” as “the slope field” in this sentence.

· “All of the slopes in the slope field are positive, not negative.” We will infer that the values of $\frac{dy}{dt}$ at points in the first quadrant are negative.

A response that lacks even an implicit reference to either the differential equation or the slope field will not earn the point. For example:

· “ $\frac{dy}{dt}$ is negative” lacks any reference to the slope field and so will not earn the point.

· “All the slopes in the graph are positive” lacks any reference to the differential equation, and so will not earn the point.

· Vague answers such as “The slope field has positive points” or “The field is increasing” will not earn the point.



0

1

The student response accurately includes the criteria below.

4-1-1☐

Explanation

Solution:

When $y > 0$, $\frac{dy}{dt} < 0$. However, in the given slope field, the slopes of the line segments for $y > 0$ are positive.

4-1-1

58. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (days)	0	2	5	10
$S(t)$ (number of boxes of snacks remaining)	500	340	200	20

A student organization is selling boxes of snacks to raise money for a school trip. The student organization begins with 500 boxes of snacks and sells them over a ten-day period. For time t days, $0 \leq t \leq 10$, the total number of boxes of snacks remaining to sell is modeled by a decreasing, twice-differentiable function S . Values of $S(t)$ at various times t are shown in the table above.

(a) Using correct units, approximate $S'(5)$ using the average rate of change of S over the interval $2 \leq t \leq 10$.

(b) Explain the meaning of $S'(5)$ in the context of the problem.

(c) It is known that $S'(2) = -55$. Write an equation for the line tangent to the graph of $y = S(t)$ at $t = 2$.

(d) Use a left Riemann sum with the three subintervals indicated by the table to approximate $\int_0^{10} S(t) \, dt$. Show the computations that lead to your answer.

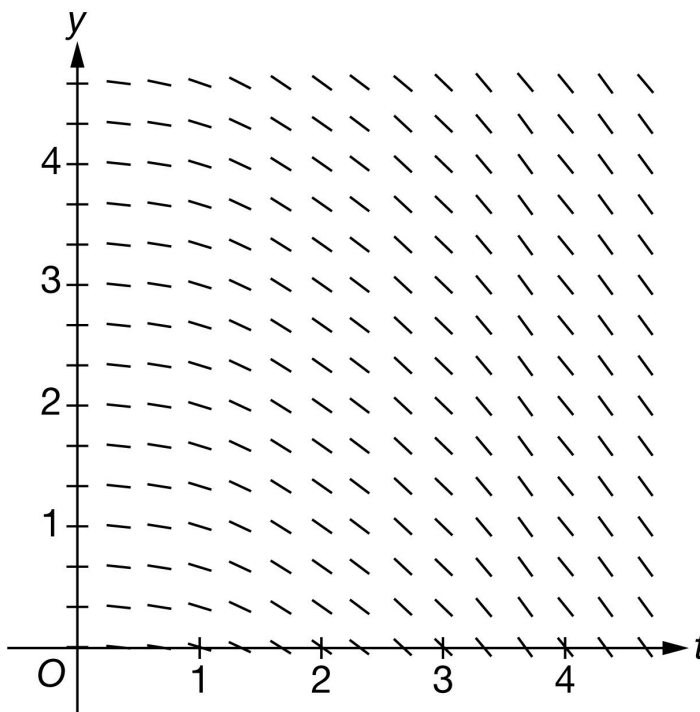
(e) Is the approximation from part (d) an overestimate or an underestimate for $\int_0^{10} S(t) \, dt$? Give a reason for your answer.

(f) Find the value of $\int_0^5 S'(2t) \, dt$. Show the computations that lead to your answer.

(g) For $0 \leq t \leq 10$, the total number of boxes of snacks remaining can also be modeled by the differentiable function M , where M is a function of t . Assume the weight of each box is 7 ounces. Write an expression for the rate of change, in ounces per day, of the total weight of the boxes of snacks remaining at $t = 8$ days.

4-1-1

(h) Explain why the following slope field cannot represent the differential equation $\frac{dy}{dt} = -0.2y$.

**Part A**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for the correct approximation shown with supporting work. The supporting work must be at least a difference and a quotient. Correct values must be taken from the table to earn the point. The difference quotient does not need to be simplified.

Examples that earn the first point include:

$$\begin{aligned} & \cdot \frac{20-340}{10-2} \\ & \cdot \frac{S(10)-S(2)}{8} = \frac{-320}{8} \\ & \cdot \frac{S(10)-S(2)}{8} = -40 \end{aligned}$$

Examples that do not earn the first point include:

$$\begin{aligned} & \cdot \frac{S(10)-S(2)}{8} \\ & \cdot \frac{S(10)-S(2)}{10-2} \end{aligned}$$

Minimal response that earns the point: $\frac{20-340}{10-2}$ or $\frac{20-340}{8}$

4-1-1

Neither $\frac{-320}{8}$ nor -40 , with no supporting work, will earn this point

The second point is for correct units

The correct units must be attached to a numerical value: either correct or incorrect.

The units may be written using words or a combination of words and symbols. For example:

- Boxes per day
- Boxes of snacks per day
- Snacks per day
- Boxes/day

The response will not be penalized for any verbal attempt to discuss the answer after the correct answer has been presented.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$S'(5) \approx \frac{S(10) - S(2)}{10 - 2} = \frac{20 - 340}{8} = -40$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Three components of a correct answer:

- Indicating a rate of change
- Including the context of boxes of snack collection
- Referring to the specific point in time, $t = 5$

Students are not penalized for incorrect units as a modifier in this part. (Whether correct or incorrect, units have no effect on the score in this part.) Example: “The number of coins is decreasing at a rate of 40 coins per day.”

4-1-1

Responses may use “changing” or “decreasing” as long as the value found in part (a) is not brought into part (b) in an incorrect manner. Example: “The number of boxes is decreasing at a rate of 40 boxes per day” is okay, but “The number of boxes is decreasing at a rate of -40 boxes per day” would not earn the point.

Acceptable references to a specific point in time include (but are not limited to):

- On day 5 and after 5 days
- At $t = 5$
- At $t = 5$ weeks or any incorrect unit of time

If the response clearly indicates an interval of time (e.g., “from day 5 on,” “during the first 5 days,” or “for $t = 0$ to $t = 5$ ”), the point is not earned.

Examples that earn the point include:

- $S'(5)$ is the rate of change of the number of boxes of snacks per day at time $t = 5$
- The rate of change of number of boxes of snacks per day is -40 boxes of snacks per day on day 5
- $S'(5)$ is the rate of change in the number of boxes of snacks at time $t = 5$ in boxes of snacks per *month*
- 40 is the rate of decrease in the number of boxes of snacks per day on day 5

Examples that do not earn the point include:

- $S'(5)$ is the average rate of change of the total number of boxes of snacks per day on day 5
- $S'(5)$ is the total number of boxes of snacks per day on day 5
- The rate of change is -40 boxes of snacks per day (Missing “at $t = 5$ ”)
- $S'(5)$ is the slope of the tangent line at $t = 5$
- Rate of change of S at time $t = 5$
- Increase in number of boxes of snacks at $t = 5$
- Boxes per day increases on day 5



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

4-1-1

Solution:

$S'(5)$ is the rate at which the number of boxes of snacks remaining is changing, in boxes per day, on day 5.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned with a correct tangent line equation in any form. The use of x in place of t is acceptable.

Some examples include

· $y - 340 = -55(t - 2), y - 340 = -55(x - 2)$

· $y = -55(t - 2) + 340, y = -55(x - 2) + 340$

· $y = -55t + 450, y = -55x + 450$

The key calculus concept being assessed in this part of the problem is that the value of $S'(2)$ is the slope of the tangent line. In that spirit, we are accepting $S(t) = \dots$ in lieu of $y = \dots$, but not $S(2) = \dots$



0	1
---	---

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$$y = S'(2)(t - 2) + S(2)$$

$$y = -55(t - 2) + 340$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for setting up a left Riemann sum. The setup must be a sum of 3 products.

· A response may still earn this point if there is only one numerical error in the expression for the Riemann sum with values pulled from the table. (See example below).

· Responses with the numerical error described above are not eligible for the second point.

4-1-1

NOTE: either the multiplication or the addition may be indicated implicitly. For example, the response may show appropriate values put into rows and columns. Implicit responses are eligible for both points.

A response presenting a right Riemann sum (instead of left), will not earn any points for part (a).

The second point is for the correct numerical result.

- In order to earn the second point, there must be supporting calculations.
- The numerical values for $S(2)$, $S(5)$, and $S(10)$ must be pulled from the table
- Any simplification shown must be correct.

Use the following as an illustration to score both points. (In most cases, you see scores for which the expression on the left side represents the entirety of relevant work shown for the response.)

- A response of $500 \cdot (2 - 0) + 340 \cdot (5 - 2) + 200 \cdot (10 - 5)$ earns both points.
- A response of $2 \cdot S(0) + 3 \cdot S(2) + 5 \cdot S(5)$ earns the first point. Eligible for second point if more work is on the page.
- A response of $2 \cdot S(0) + 3 \cdot S(2) + 5 \cdot S(5) = 3020$ earns both points.
- A response of $2 \cdot 500 + 3 \cdot 340 + 5 \cdot 200$ earns both points.
- A response of $2 \cdot 500 + 10 \cdot 340 + 5 \cdot 200$ contains one numerical error. Earns the first point. Not eligible for the second point.
- A response of $2 \cdot 500 + 3 \cdot 340 + 5 \cdot 200 = 3200$ earns first point for left side of equation. Does not earn second point due to incorrect simplification.
- A response of $1000 + 1020 + 1000$ or $1000 + 1020 + 1000 = 3020$ does not earn the first point because we cannot see the sum of 3 products. However, this is the correct sum, so we see an answer with supporting work. The second point will be earned.

The response does not need to include “approximately” in any way. We will read $=$ and $\approx$ as the same thing.

Minimalist response: The following expression represents the minimum work that must be seen in a response in order to earn both points.

$$2 \cdot 500 + 10 \cdot 340 + 5 \cdot 200$$



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Left Riemann sum

4-1-1

☐ Answer
Solution:

$$\int_0^{10} S(t) dt \approx S(0)(2-0) + S(2)(5-2) + S(5)(10-5)$$

$$= 500 \cdot 2 + 340 \cdot 3 + 200 \cdot 5 = 3020$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned when the response states that the approximation is an overestimate because S is decreasing.

The response does not need an explicit reference to the approximation in part (d).

The following statements are sufficient for earning this point:

- Overestimate because S is decreasing
- Overestimate because the function is decreasing

NOTE: At this point, there is only one function being discussed in this problem, so we will assume “the function” is S .

When a response includes additional information about concavity, we interpret that as a lack of understanding about the question, so this will not earn the point. When the response references a graph or the given table rather than the function S , the point is not earned.

Examples that earn the point:

- The left Riemann sum overestimates because the function is decreasing.
- The sum is an overestimate because $S(t)$ is decreasing.
- The sum is an overestimate because $S'(t) < 0$.

Examples that do not earn the point:

- The sum is an overestimate because $S'(t)$ is decreasing. ($S'(t)$ decreasing is not known)
- The left Riemann sum overestimates because the graph is decreasing.
- The left Riemann sum overestimates because the function is decreasing and concave up.



0

1

4-1-1

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

Because S is a decreasing function, and this is a left Riemann sum, the answer from part (d) is an overestimate.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with a correct substitution. There are three components of substitution that need to be seen:

- Uses $u = 2t$, which may be done implicitly
- Uses the correct constant of $\frac{1}{2}$
- Uses the correct limits for the substitution (0 and 10). These limits may only be seen when evaluating the antiderivative. These don't need to see them as limits on a definite integral.
- Either of the expressions below will earn the substitution point without any additional supporting work:
 $\frac{1}{2} S(2t) \Big|_0^5$ or $\frac{1}{2} S(u) \Big|_0^{10}$

Special Handling: (Notation Errors)

Errors in notation, such as the examples listed here, will not prevent the response from earning the first or second points. These errors will cause the response to be ineligible for the third point.

- Limits for t attached to an integral with u as the variable of integration (or vice versa),
- Improper equals signs, such as an indefinite integral equal to a definite integral
- Reversed limits of integration, for example: $\int_{10}^0 S'(u) du$

The second point is for the Fundamental Theorem of Calculus. This point can be earned with any of the following, where k is a nonzero constant and $b = 5$ or $b = 10$. No explanation or additional supporting work is necessary.

- $k S(2t) \Big|_0^b$
- $k (S(b) - S(0))$
- This point can be earned without earning the substitution point.

4-1-1

The third point is for the correct numerical value, which is -240 .

Values from the table must be used. $\frac{1}{2}[S(10) - S(0)]$ is not enough to earn the answer point

The expression $\frac{1}{2}(20 - 500)$ close parenthesis is sufficient for this point. No simplification is required. Any simplification that is shown must be correct.

Units are not required, and if shown, they need not be correct in order to earn this point.

The response cannot earn this point without also earning the FTC point.

Minimalist response: The expression $\frac{1}{2}(-480)$ represents the minimum work that must be seen in a response in order to earn all three points.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = 2t$. Then $du = 2 dt$.

When $t = 0$, $u = 0$.

When $t = 5$, $u = 10$.

$$\begin{aligned}\int_0^5 S'(2t) dt &= \frac{1}{2} \int_0^{10} S'(u) du \\ &= \frac{1}{2}(S(10) - S(0)) \\ &= \frac{1}{2}(20 - 500) = -240\end{aligned}$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned for a correct expression at $t = 8$. No supporting work or justification is required.

The correct expression of $7 \cdot S'(8)$ or $7 \cdot M'(8)$ earns this point.

4-1-1

No justifications or explanations are required.

Any incorrect statement such as, $S'(8) = 7 \cdot M'(8)$, does not earn this point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Expression

Solution:

The rate of change, in ounces per day, of the total weight of the boxes of snacks remaining at $t = 8$ days is $7M'(8)$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned for a correct statement identifying the distinction between two things: the differential equation, $\frac{dy}{dt} = -0.2y$, and the given slope field.

The response does not need to explicitly reference both the differential equation and the given slope field, but it must explicitly reference one and implicitly reference the other.

Acceptable responses include

- Identifying an inconsistency between the differential equation and the slope field at one point, e.g., at the point $(4, 0)$, the slope of the segment should be 0, but it is negative.
- Referencing a pattern of slopes, and explaining why that pattern does not fit the given differential equation.

A response that lacks even an implicit reference to the one of two things being compared will not earn the point.

For example, merely stating either $\frac{dy}{dt}$ is negative, or that the slope field is negative does not earn the point.

· “The values for y are less than or equal to zero so $\frac{dy}{dt} \leq 0$ ” lacks any reference to the slope field and will not earn the point.

· “All the slopes in the slope field are negative” lacks any reference to the differential equation and will not earn the point.

Vague answers which have the potential to be incorrect will not earn the point. Some examples are:

- “The field is decreasing”

4-1-1

· Responses that state a specific slope value on the slope field for a general value of x or y that is not correct for all points with that x or y value do not earn the point. (e.g. “the slope field shows a slope of 0 when $y = 4$ ” is an incorrect statement)

A response may correctly solve or attempt to solve the differential equation and then provide a correct argument without reference to their solution. Such a response would earn the point.

A response might actually solve the differential equation and provide a correct argument on the resulting function.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

When $y = 0$, $\frac{dy}{dt} = 0$. However, in the given slope field, for $t > 0$, the slopes of the line segments for $y = 0$ are nonzero.

4-1-1

59. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (days)	0	2	5	10
$S(t)$ (boxes per day)	-60	-40	-30	-5

A student organization is selling boxes of snacks to raise money for a school trip. For time t days, $0 \leq t \leq 10$, the rate of change in the number of boxes of snacks remaining to sell is modeled by an increasing, twice-differentiable function S . Values of $S(t)$ at various times t are shown in the table above.

(a) Use a right Riemann sum with the three subintervals indicated by the table to approximate

$\int_0^{10} S(t) \, dt$. Show the calculations that lead to your answer.

(b) Is your answer from part (a) an overestimate or an underestimate for $\int_0^{10} S(t) \, dt$? Give a reason for your answer.

(c) Using correct units, approximate $S'(2)$ using the average rate of change of S over the interval $0 \leq t \leq 5$.

(d) Explain the meaning of $S'(2)$ in the context of the problem.

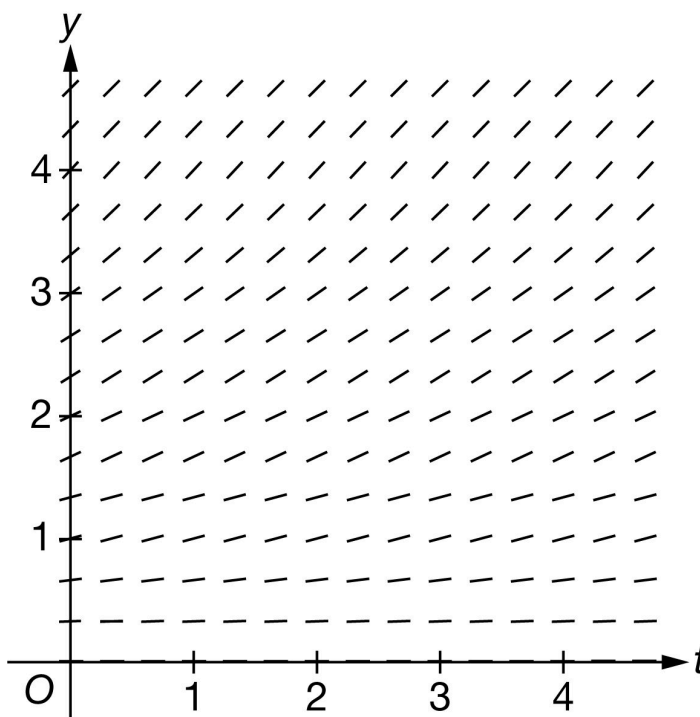
(e) It is known that $S'(5) = 6$. Write an equation for the line tangent to the graph of $y = S(t)$ at $t = 5$.

(f) Find the value of $\int_0^{10} S'\left(\frac{t}{2}\right) \, dt$. Show the calculations that lead to your answer.

(g) Another student organization is also selling boxes of snacks. For $0 \leq t \leq 10$, the total number of boxes of snacks remaining for the second student organization can be modeled by the decreasing and differentiable function M , where M is a function of t . Assume the weight of each box is 7.5 ounces. Write an expression for the rate of change, in ounces per day, of the total weight of the boxes of snacks remaining for the second student organization at $t = 6$ days.

4-1-1

(h) Explain why the following could not be a slope field for the differential equation $\frac{dy}{dt} = M'(t)$ for the function M defined in part (g).



Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for setting up a right Riemann sum. The setup must be a sum of three products.

A response may still earn this point if there is only one numerical error in the expression for the Riemann sum with values pulled from the table. (See example in the table below).

Responses with the numerical error described above are not eligible for the second point.

Note: either the multiplication or the addition may be indicated implicitly. For example, the response may show appropriate values put into rows and columns, like so.

-40	2	-80	or	$(2 - 0)(-40) = -80$
-30	3	-90		$(5 - 2)(-30) = -90$
-5	5	$+(-25)$		$(10 - 5)(-5) = -25$
		-195		-195

Implicit responses such as those above are eligible for both points.

A response presenting a left Riemann sum (instead of right), will not earn any points for part (a).

The second point is for the correct numerical result.

4-1-1

In order to earn the second point, there must be supporting calculations.

The numerical values for $S(2)$, $S(5)$, and $S(10)$ must be pulled from the table

Any simplification shown must be correct.

Use the following as an illustration of how to score both points. (In most cases, you see scores for which the expression represents the entirety of relevant work shown for the response.)

- A response of $(-40) \cdot (2 - 0) + (-30) \cdot (5 - 2) + (-5) \cdot (10 - 5)$ earns both points.
- A response of $2 \cdot S(2) + 3 \cdot S(5) + 5 \cdot S(10)$ earns the first point. Eligible for second point if more work is on the page.
- A response of $2 \cdot S(2) + 3 \cdot S(5) + 5 \cdot S(10) = -195$ earns both points.
- A response of $2 \cdot (-40) + 3 \cdot (-30) + 5 \cdot (-5)$ earns both points.
- A response of $2 \cdot (-40) + 3 \cdot (-30) + 4 \cdot (-5)$ contains one numerical error. Earns the first point. Not eligible for the second point.
- A response of $2 \cdot (-40) + 3 \cdot (-30) + 5 \cdot (-5) = -190$ earns first point for left side of equation. Does not earn second point due to incorrect simplification.
- A response of $(-80) + (-90) + (-25)$ or $(-80) + (-90) + (-25) = -195$ does not earn the first point because we cannot see the sum of three products. However, this is the correct sum, so we are not seeing an unsupported answer. The second point will be earned.

The response does not need to include “approximately” in any way. We will read $=$ and $\approx$ as the same thing.

Minimalist response: The following expression represents the minimum work that must be seen in a response in order to earn both points.

$$2 \cdot (-40) + 3 \cdot (-30) + 5 \cdot (-5)$$



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Right Riemann sum
- ☐ Answer

Solution:

$$\int_0^{10} S(t) \, dt \approx S(2)(2 - 0) + S(5)(5 - 2) + S(10)(10 - 5)$$

4-1-1

$$= (-40) \cdot 2 + (-30) \cdot 3 + (-5) \cdot 5 = -195$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned when the response states that the approximation is an overestimate because S is increasing.

The response does not need an explicit reference to part (a) or to a right Riemann sum.

A response that appeals to concavity as a reason will NOT earn this point.

The following table contains some examples of what WILL or will NOT earn the point.

The following examples earn the point:

- Overestimate because S is increasing
- Overestimate because $S'(t) > 0$
- Overestimate because the graph of S is increasing (we know they are talking about the given function S)
- Overestimate because the function is increasing. (There is only one function in the problem at this point.)

The following examples do not earn the point:

- Overestimate because S is concave down/up
- Overestimate because the graph/curve is increasing. Since no graph/curve is provided, we do not know what graph/curve they are talking about.
- Overestimate because the table [or table values] is/are increasing. The behavior of the values shown in the table is not the same as the behavior of the function S . The table does not tell us what is happening to S between the given points.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

Because S is an increasing function, and this is a right Riemann sum, this is an overestimate.

Part C

4-1-1

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for the correct approximation shown with supporting work.

We need to see both a difference and a quotient. They don't have to be together.

The expression does not need to be simplified, but values for $S(5)$ and $S(0)$ must be pulled from the table.

The response does not need to include “approximately” in any way. We will read $=$ and $\approx$ as the same thing.

The second point is for correct units.

Units must be attached to a numerical value (either correct or incorrect).

Can be written using words or a combination of words and symbols. For example:

- Boxes per day per day or Boxes/day/day
- Boxes per day² or Boxes/day²
- Boxes per day squared

We do not read work shown in part (d) for this point. The units must appear in part (c).

The letter “ B ” is not a standard abbreviation for “boxes.” It will not be interpreted as such.

The response will not be penalized for any attempt to verbally describe their answer after a correct answer has been presented.

Minimalist response: The following expression and units represent the minimum work that must be seen in a response in order to earn both points.

$$\frac{-30+60}{5} \text{ Boxes/day}^2 \text{ or } \frac{-30-(-60)}{5} \text{ Boxes/day}^2$$



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$S'(2) \approx \frac{S(5)-S(0)}{5-0} = \frac{-30-(-60)}{5} = 6 \text{ boxes per day per day}$$

4-1-1

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Three components of a correct answer:

- Indicating a rate of a rate
- Including the context of boxes [remaining to be sold]
- Explicitly referring to the specified point in time $t = 2$

Incorrect or missing units will not be penalized as long as there is a clear connection to the context of the problem.

The words “increasing” and “decreasing” are interpreted as modifiers describing one of the specified rates.

If the numerical value from part (c) is used in this interpretation, then we expect that the choice of “increasing” or “decreasing” is consistent with that value.

The following examples use 6 as the imported value from part (c). If a different [incorrect] value is imported from part (c), then these statements should be consistent with that value. (Remember that we are reading $=$ and $\approx$ as the same thing.)

- “... increasing at a rate of 6 ” is OK [Units not required in part (d)]. (the point is not yet earned with this phrase alone)
- “... decreasing at a rate of 6 ” is OK [Units not required in part (d)]. (the point is not yet earned with this phrase alone)

Note: the context is complicated. $S(t)$ is describing the rate of change of the remaining inventory of boxes. The values in the table indicate that the inventory is decreasing more slowly as time passes. Common sense might tell us that this equates to the idea that the rate at which boxes are sold is decreasing. Explaining the meaning using “boxes sold” rather than “boxes remaining” will not prevent the response from earning this point.

Rate of a rate

To earn this point, the response must explicitly discuss the rate of change of a rate of change. For example:

- The rate of change in the rate at which boxes are changing
- The rate of change in boxes is changing (or decreasing) at a rate of ...
- The rate at which the rate of change of boxes is changing

The words “increasing,” “decreasing,” or “changing” are not interpreted as rates.

Units are not interpreted as rates. (The phrase “boxes per day” is not enough to indicate a rate.)

Slope of tangent line, velocity, and acceleration are not appropriate interpretations for this context.

4-1-1

“Rate of change of S ” is not enough without describing what S represents in this context.

Context

A key component of the context for this explanation is “boxes.”

Any misstatement about boxes sold vs. boxes remaining to be sold will not prevent the response from earning this point.

Explicit reference to $t = 2$

Acceptable references to a specific point in time include (but are not limited to):

- On day 2 (where day could be replaced with any other unit of time)
- At $t = 2$ (with or without units of time)
- After 2 days (where days could be replaced with any other unit of time).

Note: we are choosing to assume that “after” is being used in lieu of “at” unless we have additional evidence to the contrary.

If the response is explicitly indicating an interval of time (e.g., “from day 2 on” or “during the first 2 days”), this point will not be earned.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

$S'(2)$ is the rate at which the rate of change in the number of boxes of snacks remaining is changing, in boxes per day per day, on day 2.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned with a correct tangent line equation in any form. Any of these equations can be written as is without any explanation or additional supporting work. If rewriting of equations is done (transforming to slope-intercept form, perhaps), it must be done correctly.

$$y = 6(t - 5) - 30$$

4-1-1

$$y + 30 = 6(t - 5)$$

$$y = 6t - 60$$

The key calculus concept being assessed in this part of the problem is that the value of $S'(5)$ is the slope of the tangent line. In that spirit...

- The response will not be penalized for using x instead of t .

We are accepting $S(t)$ in lieu of y in all forms of the tangent line equation.



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Tangent line equation

Solution:

$$y = S'(5)(t - 5) + S(5)$$

$$y = 6(t - 5) - 30$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with a correct substitution. There are three components of substitution that we need to see:

- Uses $u = \frac{t}{2}$, which may be done implicitly
- Uses the correct constant of 2
- Uses the correct limits for the substitution (0 and 5). These limits may only be seen when evaluating the antiderivative. We don't need to see them as limits on a definite integral.

Either of the expressions $2S\left(\frac{t}{2}\right)\bigg|_0^{10}$ or $2S(u)\bigg|_0^5$ will earn the substitution point without any additional supporting work.

Special Handling: Notation Errors

Errors in notation, such as the examples listed here, will not prevent the response from earning the first or second points. These errors will cause the response to be ineligible for the third point.

- Limits for t attached to an integral with u as the variable of integration (or vice versa),

4-1-1

· Improper equals signs, such as an indefinite integral equal to a definite integral

· Reversed limits of integration, for example: $\int_5^0 S'(u) \, du$

The second point is for the Fundamental Theorem of Calculus.

This point can be earned with any of the following, where k is a nonzero constant and $b = 5$ or $b = 10$. No explanation or additional supporting work is necessary.

· $kS\left(\frac{t}{2}\right)\Big|_0^b$

· $k(S(b) - S(0))$

This point can be earned without earning the substitution point.

The third point is for the correct numerical value, which is 60.

Values from the table must be used. $2(S(5) - S(0))$ is not enough to earn the answer point.

The expression $2(-30 - (-60))$ is sufficient for this point. No simplification is required. Any simplification that is shown must be correct.

Units are not required, and if shown they do not need to be correct to earn this point.

The response cannot earn this point without also earning the first two points.

Minimalist response: The expression $2(-30 - (-60))$ represents the minimum work that must be seen in a response in order to earn all 3 points.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = \frac{t}{2}$. Then $du = \frac{dt}{2}$.

When $t = 0$, $u = 0$.

When $t = 10$, $u = 5$.

4-1-1

$$\begin{aligned}\int_0^{10} S' \left(\frac{t}{2} \right) dt &= 2 \int_0^5 S' (u) du \\ &= 2 (S (5) - S (0)) \\ &= 2 (-30 - (-60)) = 60\end{aligned}$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct expression at $t = 6$ earns the point. It may be written as $(7.5) \cdot M' (6)$ or $(7.5) \cdot \frac{dM}{dt} \Big|_{t=6}$

The expression $(7.5) \cdot M' (t)$ is not sufficient because it does not use $t = 6$.

The response must use M (not S or any other function name) in order to earn the point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Expression

Solution:

The rate of change, in ounces per day, of the total weight of the boxes of snacks remaining for the second student organization at $t = 6$ days is $7.5M' (6)$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned for a correct statement identifying the distinction between the differential equation

$$\frac{dy}{dt} = M' (t) \text{ and the given slope field.}$$

In order to earn the point, the response must state that the values of the differential equation are negative while the slopes in the given slope field are positive or 0 (nonnegative).

Some ways to say that the values of differential equation are negative:

- M is decreasing
- $M' (t) < 0$
- M has negative slopes

4-1-1

When discussing the slopes in the slope field, the response must explicitly refer to the signs of the slopes.

The following examples could earn the point:

- The slope field has nonnegative slopes.
- The slopes shown are positive or 0.
- The slope field contains positive slopes.
- “The graph” or “The field” could be used in lieu of “The slope field.”

The following examples will not earn the point:

- The slope field is increasing/positive.
- The slope field looks like it represents an increasing function.
- The slope field represents a quadratic [or exponential] function.
- “It” (rather than “the slope field”)

Statements like “ M is a decreasing function, so the signs of the slopes in the slope field should be negative” can earn this point because “should be” is indicating contrast with what is known.

The response will not be penalized for saying that the slopes in the slope field are “always” positive or “all” positive.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

M is a decreasing function, so $\frac{dy}{dt} = M'(t) \leq 0$. However, in the given slope field, the slopes of the line segments for $y > 0$ are positive.

4-1-1

60. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (minutes)	0	8	24	35
$P(t)$ (number of postcards remaining per minute)	-10	-8	-5	-4

A student is delivering postcards to teachers from students on a school trip. For time t minutes, $0 \leq t \leq 35$, the rate at which the total number of postcards remaining to deliver is changing, in number of postcards remaining per minute, is modeled by an increasing, twice-differentiable function P . Values of $P(t)$ at various times t are shown in the table above.

(a) Use a left Riemann sum with the three subintervals indicated by the table to approximate

$$\int_0^{35} P(t) \, dt.$$

Show the calculations that lead to your answer.

(b) Is your answer from part (a) an overestimate or an underestimate for $\int_0^{35} P(t) \, dt$? Give a reason for your answer.

(c) Using correct units, approximate $P'(24)$ using the average rate of change of P over the interval $8 \leq t \leq 35$.

(d) Explain the meaning of $P'(24)$ in the context of the problem.

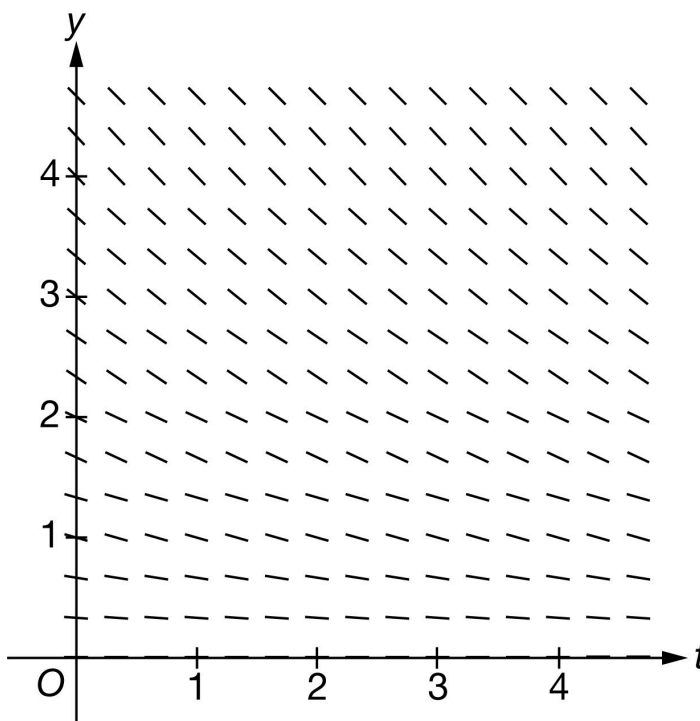
(e) It is known that $P'(8) = 0.2$. Write an equation for the line tangent to the graph of $y = P(t)$ at $t = 8$.

(f) Find the value of $\int_0^{24} P'\left(\frac{t}{3}\right) \, dt$. Show the calculations that lead to your answer.

(g) Another student brings a bag full of postcards to deliver as well. For $0 \leq t \leq 35$, the total number of postcards delivered to teachers by this student can be modeled by the increasing, differentiable function R , where R is a function of t . Assume the weight of each postcard is 4 grams. Write an expression for the rate of change, in grams per minute, of the total weight of postcards delivered by the second student at $t = 11$ minutes.

4-1-1

(h) Explain why the following could not be a slope field for the differential equation $\frac{dy}{dt} = R'(t)$ for the function R defined in part (g).



Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for setting up a left Riemann sum. The setup must be a sum of three products.

A response may still earn this point if there is only 1 numerical error in the expression for the Riemann sum with values pulled from the table. (See example in the table below).

Responses with the numerical error described above are not eligible for the second point.

Note: either the multiplication or the addition may be indicated implicitly. For example, the response may show appropriate values put into rows and columns, like so.

8	(-10)	-80
16	(-8)	-128
11	(-5)	+(-55)
		-263

or

(8 - 0)(-10)	= -80
(24 - 8)(-8)	= -128
(35 - 24)(-5)	= -55
	-263

Implicit responses such as those above are eligible for both points.

A response presenting a right Riemann sum (instead of left), will not earn any points for part (a).

The second point is for the correct numerical result.

4-1-1

In order to earn the second point, there must be supporting calculations.

The numerical values for $P(0)$, $P(8)$, and $P(24)$ must be pulled from the table

Any simplification shown must be correct.

Use the following as an illustration how to score of both points. (In most cases, you see scores for which the expression on the left side represents the entirety of relevant work shown for the response.)

- A response of $(-10) \cdot (8 - 0) + (-8) \cdot (24 - 8) + (-5) \cdot (35 - 24)$ earns both points.
- A response of $8 \cdot P(0) + 16 \cdot P(8) + 11 \cdot P(24)$ earns the first point. Eligible for second point if more work is on the page.
- A response of $8 \cdot P(0) + 16 \cdot P(8) + 11 \cdot P(24) = -263$ earns both points.
- A response of $8 \cdot (-10) + 16 \cdot (-8) + 11 \cdot (-5)$ earns both points.
- A response of $8 \cdot (-10) + 18 \cdot (-8) + 11 \cdot (-5)$ contains one numerical error. Earns the first point. Not eligible for the second point.
- A response of $8 \cdot (-10) + 16 \cdot (-8) + 11 \cdot (-5) = -200$ earns first point for left side of equation. Does not earn second point due to incorrect simplification.
- A response of $(-80) + (-128) + (-55)$ or $(-80) + (-128) + (-55) = -263$ does not earn the first point because we cannot see the sum of three products. However, this is the correct sum, so we are not seeing an unsupported answer. The second point will be earned.

The response does not need to include “approximately” in any way. We will read $=$ and $\approx$ as the same thing.

Minimalist response: The following expression represents the minimum work that must be seen in a response in order to earn both points. $8 \cdot (-10) + 16 \cdot (-8) + 11 \cdot (-5)$



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Left Riemann sum
- ☐ Answer

Solution:

$$\int_0^{35} P(t) dt \approx P(0)(8 - 0) + P(8)(24 - 8) + P(24)(35 - 24)$$

$$= (-10) \cdot 8 + (-8) \cdot 16 + (-5) \cdot 11 = -263$$

4-1-1

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned when the response states that the approximation is an underestimate because P is increasing.

The response does not need an explicit reference to part (a) or to a left Riemann sum.

A response that appeals to concavity as a reason will NOT earn this point.

The following examples earn the point:

- Underestimate because P is increasing
- Underestimate because $P'(t) > 0$
- Underestimate because the graph of P is increasing (we know they are talking about the given function P)
- Underestimate because the function is increasing. (There is only one function in the problem at this point.)

The following examples do not earn the point:

- Underestimate because P is concave down/up
- Underestimate because the graph/curve is increasing. Since no graph/curve is provided, we do not know what graph/curve they are talking about.
- Underestimate because the table [or table values] is/are increasing. The behavior of the values shown in the table is not the same as the behavior of the function P . The table does not tell us what is happening to P between the given points.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

Because P is an increasing function, and this is a left Riemann sum, this is an underestimate.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

4-1-1

The first point is earned for the correct approximation shown with supporting work.

We need to see both a difference and a quotient. They don't have to be together.

The expression does not need to be simplified, but values for $P(35)$ and $P(8)$ must be pulled from the table.

Decimal approximation is not expected, but if presented, then the decimal value must be correct to three decimal places, either rounded or truncated. We do not read anything past the third decimal place.

- If the first three places match 0.148 then the point is earned
- It does not matter whether any subsequent decimal places match the correct value or not. (For example, the value 0.1487 will earn the point because the first three decimal places match the correct value.)
- Responses with fewer than three decimal places (0.14, 0.15, 0.1, etc.) will not earn the point.

The response does not need to include “approximately” in any way. We will read $=$ and $\approx$ as the same thing.

The second point is for correct units.

Units must be attached to a numerical value (either correct or incorrect).

Can be written using words or a combination of words and symbols. For example:

- Postcards per minute per minute or Postcards/min/min
- Postcards per min² or Postcards/min²
- Postcards per minute squared

We do not read work shown in part (d) for this point. The units must appear in part (c).

The letter “ P ” is not a standard abbreviation for “Postcards.” It will not be interpreted as such.

The response will not be penalized for any attempt to verbally describe their answer after a correct answer has been presented.

Minimalist response: The following expression and units represent the minimum work that must be seen in a response in order to earn both points. $\frac{-4+8}{27}$ postcards/min² or $\frac{-4-(-8)}{27}$ postcards/min²



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

4-1-1

Solution:

$$P'(24) \approx \frac{P(35) - P(8)}{35 - 8} = \frac{-4 - (-8)}{27} = \frac{4}{27} \text{ postcard per minute per minute}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Three components of a correct answer:

- Indicating a rate of a rate
- Including the context of postcard delivery (postcards remaining)
- Explicitly referring to the specified point in time $t = 24$

Incorrect or missing units will not be penalized as long as there is a clear connection to the context of the problem.

The words “increasing” and “decreasing” are interpreted as modifiers describing one of the specified rates.

If the numerical value from part (c) is used in this interpretation, then we expect that the choice of “increasing” or “decreasing” is consistent with that value.

The following examples use $\frac{4}{27}$ as the imported value from part (c). If a different [incorrect] value is imported from part (c), then these statements should be consistent with that value. (Remember that we are reading $=$ and $\approx$ as the same thing.)

- “... increasing at a rate of $\frac{4}{27}$ ” is OK [Units not required in part (d)] (the point is not earned with this phrase alone)
- “... decreasing at a rate of $\frac{4}{27}$ ” is OK [Units not required in part (d)] (the point is not earned with this phrase alone)

Note: the context is complicated. $P(t)$ is describing the rate of change of the remaining undelivered postcards. The values in the table indicate that the number of undelivered postcards is decreasing more slowly as time passes. Common sense might tell us that this equates to the idea that the rate at which postcards are delivered is decreasing. Explaining the meaning using “postcards delivered” rather than “postcards remaining” will not prevent the response from earning this point.

Rate of a rate

To earn this point, the response must explicitly discuss the rate of change of a rate of change. For example:

- The rate of change in the rate at which postcards are changing
- The rate of change in postcards is changing/increasing/decreasing at a rate of ...

4-1-1

The rate at which the rate of change of postcards is changing.

The words “increasing,” “decreasing,” or “changing” are not interpreted as rates.

Units are not interpreted as rates. (The phrase “postcards per minute” is not enough to indicate a rate.)

Slope of tangent line, velocity, and acceleration are not appropriate interpretations for this context.

“Rate of change of P ” is not enough without describing what P represents in this context.

Context

A key component of the context for this explanation is “postcards.”

Any misstatement about postcards delivered vs. postcards remaining to be delivered will not prevent the response from earning this point.

Explicit reference to $t = 24$

Acceptable references to a specific point in time include (but are not limited to):

- At minute 24 (where minute could be replaced with any other unit of time)
- At $t = 24$ (with or without units of time)
- After 24 minutes (where minutes could be replaced with any other unit of time).

Note: we are choosing to assume that “after” is being used in lieu of “at” unless we have additional evidence to the contrary.

If the response is explicitly indicating an interval of time (e.g., “from minute 24 on” or “during the first 24 minutes”), this point will not be earned.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

$P'(24)$ is the rate at which the rate of change in the number of postcards remaining is changing, in postcards per minute per minute, at minute 24.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

4-1-1

This point is earned with a correct tangent line equation in any form. Any of these equations can be written as is without any explanation or additional supporting work. If rewriting of equations is done (transforming to slope-intercept form, perhaps), it must be done correctly.

$$y = 0.2(t - 8) - 8$$

$$y + 8 = 0.2(t - 8)$$

$$y = 0.2t - 9.6$$

The key calculus concept being assessed in this part of the problem is that the value of $P'(8)$ is the slope of the tangent line. In that spirit:

- The response will not be penalized for using x instead of t .
- We are accepting $P(t)$ in lieu of y in all forms of the tangent line equation.



0	1
---	---

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$$y = P'(8)(t - 8) + P(8)$$

$$y = 0.2(t - 8) - 8$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with a correct substitution. There are 3 components of substitution that we need to see:

- Uses $u = \frac{t}{3}$, which may be done implicitly
- Uses the correct constant of 3
- Uses the correct limits for the substitution (0 and 8). These limits may only be seen when evaluating the antiderivative. We don't need to see them as limits on a definite integral.

Either of the expressions $3P\left(\frac{t}{3}\right)\bigg|_0^{24}$ or $3P(u)\bigg|_0^8$ will earn the substitution point without any additional supporting work.

Special Handling: Notation Errors

4-1-1

Errors in notation, such as the examples listed here, will not prevent the response from earning the first or second points. These errors will cause the response to be ineligible for the third point.

- Limits for t attached to an integral with u as the variable of integration (or vice versa),
- Improper equals signs, such as an indefinite integral equal to a definite integral
- Reversed limits of integration, for example: $\int_8^0 P'(u) \, du$

The second point is for the Fundamental Theorem of Calculus.

This point can be earned with any of the following, where k is a nonzero constant and $b = 8$ or $b = 24$. No explanation or additional supporting work is necessary.

- $kP\left(\frac{t}{3}\right)\Big|_0^b$
- $k(P(b) - P(0))$

This point can be earned without earning the substitution point.

The third point is for the correct numerical value, which is 6.

Values from the table must be used. $3(P(8) - P(0))$ is not enough to earn the answer point.

The expression $3(-8 - (-10))$ is sufficient for this point. No simplification is required. Any simplification that is shown must be correct.

Units are not required, and if shown, they do not need to be correct to earn this point.

The response cannot earn this point without also earning first two points.

Minimalist response: The expression $3(-8 - (-10))$ represents the minimum work that must be seen in a response in order to earn all three points.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

4-1-1

Let $u = \frac{t}{3}$. Then $du = \frac{dt}{3}$.

When $t = 0$, $u = 0$.

When $t = 24$, $u = 8$.

$$\int_0^{24} P' \left(\frac{t}{3} \right) dt = 3 \int_0^8 P' (u) du$$

$$= 3 (P (8) - P (0))$$

$$= 3 (-8 - (-10)) = 6$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct expression at $t = 11$ earns the point. It may be written as $4 \cdot R' (11)$ or $4 \cdot \frac{dR}{dt} \Big|_{t=11}$

The expression $4 \cdot R' (t)$ is not sufficient because it does not use $t = 11$.

The response must use R (not M or G or any other function name) in order to earn the point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Expression

Solution:

The rate of change, in grams per minute, of the total weight of postcards delivered by the second student at $t = 11$ minutes is $4R' (11)$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned for a correct statement identifying the distinction between the differential equation

$$\frac{dy}{dt} = R' (t) \text{ and the given slope field.}$$

In order to earn the point, the response must state that the values of the differential equation are positive while the slopes in the given slope field are negative or 0 (non-positive).

4-1-1

Some ways to say that the values of differential equation are positive:

- R is increasing
- $R'(t) > 0$
- R has positive slopes

When discussing the slopes in the slope field, the response must explicitly refer to the signs of the slopes.

The following examples could earn the point:

- The slope field has nonpositive slopes.
- The slopes shown are negative or 0.
- The slope field contains negative slopes.
- “The graph” or “The field” could be used in lieu of “The slope field.”

The following examples will not earn the point:

- The slope field is decreasing/negative.
- The slope field looks like it represents a decreasing function.
- The slope field represents a quadratic [or exponential] function.
- “It” (rather than “the slope field”)

Statements like “ R is an increasing function, so the signs of the slopes in the slope field should be positive” can earn this point because “should be” is indicating contrast with what is known.

The response will not be penalized for saying that the slopes in the slope field are “always” negative or “all” negative.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

R is an increasing function, so $\frac{dy}{dt} = R'(t) \geq 0$. However, in the given slope field, the slopes of the line segments for $y > 0$ are negative.

4-1-1

61. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (minutes)	0	9	18	27	36
$P(t)$ (number of postcards remaining)	320	256	100	40	0

A principal is delivering postcards to teachers from students on a school trip. The principal begins with 320 postcards and delivers them over 36 minutes. For time t minutes, $0 \leq t \leq 36$, the total number of postcards remaining to deliver is modeled by a twice-differentiable function P . Values of $P(t)$ at various times t are shown in the table above.

(a) Use a midpoint Riemann sum with the two subintervals indicated by the table to approximate

$$\int_0^{36} P(t) \, dt.$$

Show the computations that lead to your answer.

(b) Find the value of $\int_0^9 P'(4t) \, dt$. Show the computations that lead to your answer.

(c) Using correct units, approximate $P'(18)$ using the average rate of change of P over the interval $9 \leq t \leq 27$.

(d) Explain the meaning of $P'(18)$ in the context of the problem.

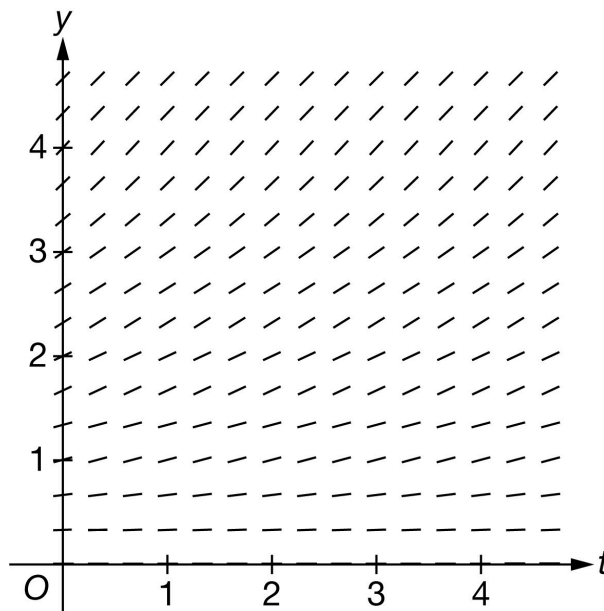
(e) It is known that $P'(9) = -12$. Write an equation for the line tangent to the graph of $y = P(t)$ at $t = 9$.

(f) It is known that $P''(t) < 0$ for $9 \leq t \leq 18$. If $P(15)$ is approximated using the equation from part (e), would the approximation be an overestimate or underestimate for the actual value of $P(15)$? Give a reason for your answer.

(g) For $0 \leq t \leq 36$, the total number of postcards remaining can be modeled by the differentiable function Z , where Z is a function of t . Assume the weight of each postcard is 4.5 grams. Write an expression for the rate of change, in grams per minute, of the total weight of the postcards remaining at $t = 19$ minutes.

4-1-1

(h) Explain why the following slope field cannot represent the differential equation $\frac{dy}{dt} = -0.3y$.

**Part A**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point (Form of midpoint sum): Philosophy: The response should provide clear evidence adding the area of two rectangles whose heights are found using the midpoint of the interval.

Acceptable evidence:

$$\cdot 18 \cdot (P(9) + P(27)) \text{ or } 18 \cdot P(9) + 18 \cdot P(27)$$

$$\cdot (18 - 0) \cdot 256 + (36 - 18) \cdot 40$$

$$\cdot 18 \cdot (256 + 40) \text{ or } 18 \cdot 256 + 18 \cdot 40$$

To earn this point the response can have at most one mistake in the expression or equation. So, expressions of the form $\frac{18}{b} \cdot (P(9) + P(27))$, for any nonzero value of b would earn the point. Note: The value $\frac{18}{b}$ might be distributed across the sum.

Expressions of the form $\frac{a}{b} \cdot (P(9) + P(27))$ for $a \neq 18$ will not earn this point.

Once the first point is earned, it cannot be lost.

Second Point (Answer)

To earn the point, the answer must be correct. It does not need to be simplified, but if it is, the simplification must be correct.

The answer point cannot be earned unless the first (midpoint sum) point was earned.

4-1-1

Minimal complete answers that earn both points:

$$\cdot 18 \cdot (256 + 40)$$

$$\cdot 18 \cdot 256 + 18 \cdot 40$$



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Form of midpoint sum
- ☐ Answer

Solution:

$$\int_0^{36} P(t) dt \approx P(9)(18 - 0) + P(27)(36 - 18)$$

$$= 256 \cdot 18 + 40 \cdot 18 = 5328$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point (Substitution) Philosophy: This point is earned when the response has all three components of the substitution done correctly.

Uses $u = 4t$, perhaps implicitly (any variable could be used for u)

Uses the correct constant of $\frac{1}{4}$

Uses the correct limits for their substitution.

· The limits might not be seen until the final evaluation, and at that point, response may have returned to the variable t .

· Limits for t attached to an integral involving u or vice versa will not earn this substitution point but could still earn the answer point, if the numerical answer is correct.

Second point: (Fundamental Theorem of Calculus) Philosophy: This point is earned when the FTC has been applied correctly to either $\int P'(4t) dt$ or $\int P'(u) du$.

The point is earned for any of the following:

4-1-1

· The equation $\int P'(4t) dt = P(4t)$ or $\int P'(u) du = P(u)$

· $\frac{1}{4}P(4t)\big|_0^9$ or $\frac{1}{4}P(u)\big|_0^{36}$

· $P(36) - P(0)$, $P(36)$, or $P(0)$

This point can be earned if the substitution point is not.

A response that applies the FTC to $P'(t)$ does not earn this point.

Note: $\int_a^b P'(u) du = P(b) - P(a)$ does not earn the point. The FTC point is for the response communicating that the antiderivative of P' is P .

Third Point (Answer)

Must be the correct numerical answer.

Must have earned the second point for the FTC.

Might have not earned the first point for substitution because of an incorrect use of limits, but will still earn the third point for the correct answer.

$P(36)$ and $P(0)$ must be evaluated; the expression $\frac{1}{4}(P(36) - P(0))$ by itself does not earn the third point.

The answer may be unsimplified, but if simplified it must be done correctly.

Scoring common responses:

The minimal work required to earn all three points is $\frac{1}{4}(0 - 320)$ or $-\frac{320}{4}$.

The following responses would earn the first two points and be eligible for the answer point if the response continues:

· $\frac{1}{4}P(4t)\big|_0^9$

· $\frac{1}{4}(P(36) - P(0))$

Responses of only the following would earn only the second point:

· $P(36) - P(0)$

· $P(9) - P(0)$

· $P(36)$

4-1-1

$$\cdot \int_0^9 P'(u) \, du = P(9)$$

$$\cdot P(4t) \Big|_0^9 \text{ or } P(u) \Big|_0^{36}$$

The following responses would earn no points:

· -80 in the absence of one of the first two points.

$$\cdot \int_0^9 P'(4t) \, dt = P(9)$$



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = 4t$. Then $du = 4 \, dt$.

When $t = 0$, $u = 0$.

When $t = 9$, $u = 36$.

$$\begin{aligned} \int_0^9 P'(4t) \, dt &= \frac{1}{4} \int_0^{36} P'(u) \, du \\ &= \frac{1}{4}(P(36) - P(0)) \\ &= \frac{1}{4}(0 - 320) = -80 \end{aligned}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point (Approximation)

The response must be a difference and a quotient.

The answer does not need to be simplified, but if it is simplified it must be done correctly.

4-1-1

The minimal work that must be shown to earn this point is either $\frac{40-256}{18}$ or $\frac{-216}{27-9}$.

Second point (Units)

The units must be attached to a value (either correct or incorrect).

The response will not be penalized for any verbal attempt to discuss the answer after the correct approximation with units is presented.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$P'(18) \approx \frac{P(27)-P(9)}{27-9} = \frac{40-256}{18} = -12 \text{ postcards per minute}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The interpretation must include three components:

- indicate the rate of change
- the specific context of the number of postcards remaining
- the specific point in time, $t = 18$.

Responses do not need to include correct units or any units.

The word “decreasing” may be used in place of the word “changing”.

The response is not earned for:

- “the rate of postcards delivered in minute 18 ” as this does not indicate a change in the rate of delivery.
- “decreasing at a rate of -12 postcards per minute”

“the rate of change of P ” without specifying that P is the number of postcards delivered, does not earn the point.

Acceptable ways to provide the specific point in time:

4-1-1

- In minute 18 or after 18 minutes.
- At $t = 18$ (missing the unit of time)
- After 18 days (incorrect unit of time)

A response that clearly indicates an interval of time (e.g. “during the first 18 minutes” or “from minute 18 on”) does not earn the point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

$P'(18)$ is the rate at which the number of postcards remaining is changing, in postcards per minute, at time $t = 18$ minutes

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The use of $P(t)$ in place of y earns the point.

The use of x in place of t earns the point.

Tangent lines do not need to be simplified, but any simplification must be done correctly.

Any correct form of the tangent line earns the point. For example:

o $y = 256 - 12(t - 9)$

o $y - 256 = 12(t - 9)$

o $y = -12t + 108 + 256$

o $y = -12t + 364$



0	1
---	---

The student response accurately includes the criteria below.

4-1-1

☐ Tangent line equation

Solution:

$$y = P'(9)(t - 9) + P(9)$$

$$y = -12(t - 9) + 256$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point the response must include $P''(t) < 0$ and overestimate.

“ $P(t)$ is concave down” can be used instead of $P''(t) < 0$.

Information about $P'(t)$ by itself without reference to $P''(t) < 0$ is not sufficient and does not earn this point.

“Overestimate because it is concave down” is not sufficient by itself because of the use of “it”. The response must clearly, directly or indirectly, indicate to what “it” is referring.

If a response makes the correct conclusion and states $P''(t) < 0$ and then goes on and makes an incorrect statement or a statement that cannot be verified will not earn this point.

For example: “ $P'(t)$ is concave down, overestimate” does not earn this point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

Because $P''(t) < 0$, the tangent line approximation would produce an overestimate.

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct expression of only $4.5 \cdot Z'(19)$ earns this point.

Only expression $4.5 \cdot P'(19)$ also earns this point.

No justifications or explanations are required.

4-1-1

Any incorrect statement such as, $Z'(19) = 4.5 \cdot P'(19)$, does not earn this point.



0

1

The student response accurately includes the criteria below.

☐ Expression

Solution:

The rate of change, in grams per minute, of the total weight of the postcards remaining at $t = 19$ minutes is $4.5Z'(19)$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Philosophy: Response clearly references what part of the differential equation is different from the slope field and the response needs only to clearly and explicitly reference one of them.

The response does not need to explicitly reference both the differential equation and the given slope field.

The distinction can be made with explicit reference to one of the two things and an implicit reference to the other

For example:

· “ $\frac{dy}{dt}$ is negative, not positive”. We will infer that negative is referring to the slopes shown in the slope field.

· “This would be correct for a positive derivative, but $\frac{dy}{dt}$ is negative.” We will read “This” as “the slope field” in this sentence.

· “All of the slopes in the slope field are positive, not negative.” We will infer that the values of $\frac{dy}{dt}$ at points in the first quadrant are negative.

A response that lacks even an implicit reference to either the differential equation or the slope field will not earn the point. For example:

· “ $\frac{dy}{dt}$ is negative” lacks any reference to the slope field and so will not earn the point.

· “All the slopes in the graph are positive” lacks any reference to the differential equation, and so will not earn the point.

· Vague answers such as “The slope field has positive points” or “The field is increasing” will not earn the point.

4-1-1



0

1

The student response accurately includes the criteria below.

☐

Explanation

Solution:

When $y > 0$, $\frac{dy}{dt} < 0$. However, in the given slope field, the slopes of the line segments for $y > 0$ are positive.

4-1-1

62. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (minutes)	0	8	24
$P(t)$ (postcards per minute)	-6	-10	-12

A student is delivering postcards to teachers from students on a school trip. For time t minutes, $0 \leq t \leq 24$, the rate of change in the number of postcards remaining for the student to deliver is modeled by a twice-differentiable function P , in postcards remaining per minute. Values of $P(t)$ at various times t are shown in the table above.

(a) Using correct units, approximate $P'(8)$ using the average rate of change of P over the interval $0 \leq t \leq 24$.

(b) Explain the meaning of $P'(8)$ in the context of the problem.

(c) The function $N(t)$ models the number of postcards remaining for the student to deliver at time t minutes, where $N'(t) = P(t)$. It is known that $N(8) = 185$. Write an equation for the line tangent to the graph of $y = N(t)$ at $t = 8$.

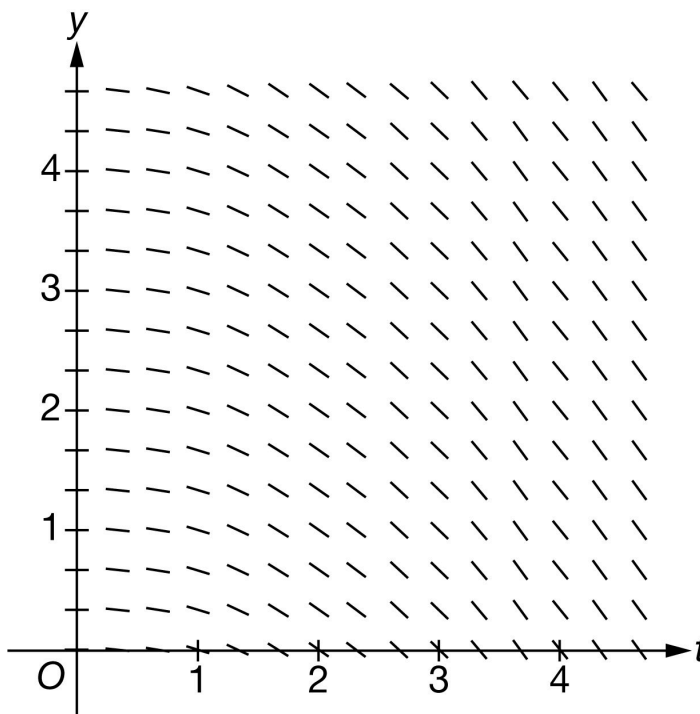
(d) Find the value of $\int_0^8 P'(3t) \, dt$. Show the calculations that lead to your answer.

(e) Use a trapezoidal sum with the two subintervals indicated by the table to approximate $\int_0^{24} P(t) \, dt$. Show the calculations that lead to your answer.

(f) Another student is also delivering postcards. For $0 \leq t \leq 24$, the rate of change in the number of postcards remaining for the second student to deliver, in postcards per minute, can be modeled by the differentiable function R . Assume the weight of each postcard is 4 grams. Write an expression in terms of P and R for the rate of change, in grams per minute, of the combined weight of the postcards remaining for both students at $t = 13$ minutes.

4-1-1

(g) Explain why the following could not be a slope field for the differential equation $\frac{dy}{dt} = -0.25y$.

**Part A**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for the correct approximation shown with supporting work.

Answers without supporting work do not earn this point. Supporting work must be at least a difference and a quotient.

Response does not need to be simplified, but any simplification must be done correctly.

If a decimal is presented, it must be correct to three decimal places. We do not read anything past the third decimal place.

The second point is earned for the correct units.

Units must be attached to a numerical value (either correct or incorrect).

Can be written using words or a combination of words and symbols.

The response will not be penalized for any verbal attempt to discuss the answer after the correct presentation of the approximation.

The minimum response needed to earn both points is: $\frac{-12+6}{24}$ postcards per minute per minute

4-1-1



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$P'(8) \approx \frac{P(24) - P(0)}{24 - 0} = \frac{-12 - (-6)}{24} = \frac{-6}{24}$$

$$= -\frac{1}{4} \text{ postcard per minute per minute}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This part is intended to be independent from part (a), so we do not want or need to see any reference to the value obtained in (a). If that value is referenced, conclusion must be consistent.

Three components of a correct answer:

- Indicating a rate of a rate (look for the word rate to be used correctly twice)
- Including the context of postcards
- Referring to the specific point in time

Units will not be considered for this point. They do not need to be present, and if they are, they do not need to be correct.

Any incorrect statements make the response ineligible for the point. For example:

- $P'(8)$ is the average (or an approximation, or an estimate of) rate of change of
- $P'(8)$ is the acceleration (or rate of change of velocity) of

“Increasing” or “decreasing” may be used in place of “changing” as long as the value found in part (a) is not brought into part (b) in an incorrect manner. Beware of phrases such as “decreasing at a negative rate.”

- Example: $P'(8)$ is the rate of decrease of the rate of change of the number of postcards at 8 minutes. (earns the point)

4-1-1

· Example: $P'(8)$ means the number of postcards per minute is decreasing by $\frac{1}{4}$ postcards/minute/minute. (does not earn the point)

Acceptable references to time include (but are not limited to):

- On (or at) minute 8
- At $t = 8$ (with correct units of time, without correct units of time, or even with incorrect units of time)
- After $t = 8$ (with correct units of time, without correct units of time, or even with incorrect units of time)

If the response clearly indicates an interval of time (e.g. “from minute 0 to minute 8”) the response will not earn the point.

Examples that earn the point include:

- $P'(8)$ represents the rate at which the rate of change of the number of postcards is changing on minute 8.
- $P'(8)$ is the rate of change in the rate of change of the number of postcards at 8 minutes.
- $P'(8)$ represents the rate at which the rate of change of the number of postcards is decreasing at $t = 8$.
- $P'(8)$ equals (answer from part (a)) means the rate of change in the number of postcards per minute is increasing at a rate of (answer from part (a)) at $t = 8$.
- $P'(8)$ represents the rate at which the rate of change of the number of postcards is increasing on the eighth minute.

Examples that do not earn the point include:

- $P'(8)$ represents the rate at which the number of postcards is changing on minute 8.
- $P'(8)$ is the rate of change in the number of postcards per minute at $t = 8$ minutes
- The rate of change in the number of postcards per minute is decreasing at $t = 8$.
- The rate of change in the rate of change of the number of postcards per day is decreasing at $t = 8$.
- $P'(8) = -\frac{1}{4}$ means that the rate of change in the number of postcards is decreasing at a rate of $-\frac{1}{4} t = 8$. (double negative)



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

4-1-1

Solution:

$P'(8)$ is the rate at which the rate of change in the number of postcards remaining is changing, in postcards per minute per minute, at minute 8.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is earned with a correct tangent line equation in any form. Equivalent forms include (but are not limited to):

· $y = -10(t - 8) + 185$

· $y - 185 = -10(t - 8)$

· $y = -10t + 265$

The student may write x in place of t and still earn the point.

No work is required to earn the point.

We will accept $N(t)$ in lieu of y in the equation of the tangent line.



0	1
---	---

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$$y = N'(8)(t - 8) + N(8)$$

$$y = -10(t - 8) + 185$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with a correct substitution.

Two things are required as evidence of correct substitution:

· Coefficient of $\frac{1}{3}$ correctly applied to an integral or an antiderivative

4-1-1

- Evidence that the correct limits of 0 to 24 will be applied

Examples that will earn the first point:

$$\cdot \frac{1}{3} \int_0^{124} P'(u) du$$

$$\cdot \frac{1}{3} P(u) \Big|_0^{24} \text{ or } \frac{1}{3} P(3t) \Big|_0^8$$

Responses that include misstatements such as using t -values for limits with u as the variable of integration in an intermediate step which is later corrected may still earn this point. Intermediate errors in bounds that are subsequently corrected will disqualify the student from earning the third point. If the bounds are never appropriately handled, the response cannot earn the first point. Scratch work that is not connected to the solution will not be read.

Examples:

$$\cdot \int_0^8 P'(3t) dt = \frac{1}{3} \int_0^8 P'(u) du = \frac{1}{3} P(3t) \Big|_0^8 \text{ earns the first point, but will not earn the third point.}$$

$$\cdot \int_0^8 P'(3t) dt = \frac{1}{3} \int_0^8 P'(u) du = \frac{1}{3} [P(8) - P(0)] \text{ does not earn the first point.}$$

The second point is for the Fundamental Theorem of Calculus, demonstrating the knowledge that

$$\int_a^b P'(u) du = P(u) \Big|_a^b = P(b) - P(a)$$

This point can be earned with either $kP(3t) \Big|_a^b$, $kP(t) \Big|_a^b$, or $k(P(b) - P(a))$ where k is a nonzero constant [not necessarily correct], $a = 0$, and $b = 8$ or $b = 24$.

The response can earn this point without earning the first point.

$$\frac{1}{3} [P(24) - P(0)] \text{ earns the first two points.}$$

The third point is for the correct numerical value. Our answer only.

Values from the table must be used (i.e., $\frac{1}{3} [P(24) - P(0)]$ is not sufficient to earn the third point).

If the response includes any linkage error, (i.e., the use of equal signs to equate two expressions that are not equal), the third point will not be earned. If an indefinite integral is linked with a definite integral the response will not be penalized.

$$\cdot \int_0^8 P'(3t) dt = \frac{1}{3} \int P'(u) du = \frac{1}{3} P(3t) \Big|_0^8 = \dots \text{ is eligible for the third point.}$$

Eligibility: Must have earned the first two points.

The minimum response needed to earn all three points is: $\frac{1}{3}(-12 + 6)$.

4-1-1



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = 3t$. Then $du = 3 dt$.

When $t = 0$, $u = 0$.

When $t = 8$, $u = 24$.

$$\begin{aligned}\int_0^8 P'(3t) dt &= \frac{1}{3} \int_0^{24} P'(u) du \\ &= \frac{1}{3}(P(24) - P(0)) \\ &= \frac{1}{3}(-12 - (-6)) = -\frac{6}{3} = -2\end{aligned}$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the form of a trapezoidal sum

Point is earned for the setup of the trapezoidal sum. Numerical values do not need to be pulled from the table for this point, but Δt values must be presented. Responses that contain one incorrect numerical value in the trapezoidal sum setup are still eligible to earn the first point, but will not earn the second.

Response may present a left Riemann sum and a right Riemann sum, then average the two. Again, one incorrect numerical value will not disqualify the response from earning this first point.

Response may compute trapezoids as areas of rectangles and triangles.

Trapezoidal areas may be given in a graph. In order to earn this point, the response must clearly show the computations leading to the values presented in the graph, and must show that these areas are added to obtain the final approximation value.

4-1-1

A response using subtraction instead of addition will not earn the point (e.g.,

$\frac{1}{2}[8(P(0) - P(8)) + 16(P(8) - M(24))]$ earns no points).

Response utilizing “Trapezoidal Rule” will not earn this point, unless this represents only one error in the numerical values.

$\cdot \frac{1}{2}(8)[P(0) + 2P(8) + P(24)]$ is equivalent to $\frac{1}{2}[8(P(0) + P(8)) + 8(P(8) + P(24))]$, thus has only one incorrect numerical value (the second interval width is 8 instead of 16), so can earn the point.

$\cdot \frac{1}{2}(12)[P(0) + 2P(8) + P(24)]$ has two incorrect numerical values (both interval widths are incorrect) so will not earn the point.

The second point is for the correct numerical result.

Our answer only. Response does not need to be simplified, but if it is, it must be simplified correctly.

The answer must be accompanied with supporting work. Supporting work may appear in a graph, with clearly labeled areas. It is possible to earn the second point without earning the first if the supporting work is not sufficient to earn the first point.

Pictures without clear area computations and a clear summation (+) of quantities can earn at most the second point.

Special case: A positive constant multiple of a complete and correctly expressed trapezoidal sum (with the correct values pulled from the table) will only the first point:

$\cdot k \left[\left(\frac{-6-10}{2} \right) \cdot 8 + \left(\frac{-10-12}{2} \right) \cdot 16 \right]$ for $k > 0$ earns just the first point.

The minimum response needed to earn both points: $\left(\frac{-16}{2} \right) \cdot 8 + \left(\frac{-22}{2} \right) \cdot 16$



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Form of trapezoidal sum
- ☐ Answer

Solution:

$$\begin{aligned} \int_0^{24} P(t) dt &\approx \left(\frac{P(0)+P(8)}{2} \right) (8-0) + \left(\frac{P(8)+P(24)}{2} \right) (24-8) \\ &= \left(\frac{-6-10}{2} \right) \cdot 8 + \left(\frac{-10-12}{2} \right) \cdot 16 = -240 \end{aligned}$$

4-1-1

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for a correct constant multiple or correct sum.

Response must contain the constant 4 as a factor of the expression, and/or the correct sum $P(13) + R(13)$.

If the sum is incorrect, then the constant factor of 4 must be applied either to a sum that involves only $P(t)$ and $R(t)$, $P'(t)$ and $R'(t)$, or integrals of $P(t)$ and $R(t)$, or to one of the correct terms of the sum $P(13)$ or $R(13)$, with no additional incorrect terms.

Examples that earn the point include:

$$\cdot 4[P(t) + R(t)]$$

$$\cdot 4 \int [P(t) + R(t)] dt$$

$$\cdot 4[P' + R']$$

$$\cdot 4P(13)$$

Examples that do not earn the point include:

$$\cdot 4[P(t) + R'(t)]$$

$$\cdot 4 \int P(t) dt$$

$$\cdot 4P' + R'$$

$$\cdot 4R(13) + t$$

Special case: $4P(13) + R(13)$ is an error in parentheses, so will earn the first point but is not eligible to earn the second point.

If the constant multiple is missing or incorrect, the sum must be correct. $k[P(13) + R(13)]$ for any nonzero constant k will earn the first point.

The second point is for the correct expression.

$4P(13) + 4R(13)$ (or an equivalent expression) earns both points.

Inclusion of units (correct or incorrect) will not affect the score.

A completely correct expression with one additional constant term added/subtracted would earn only the first point. For example, $4P(13) + 4R(13) + 4R(2)$ earns the first point only.

4-1-1



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Constant multiple or sum
- ☐ Expression

Solution:

The rate of change, in grams per minute, of the combined weight of the postcards remaining for both students at $t = 13$ minutes is $4P(13) + 4R(13)$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is earned for a correct statement identifying specific inconsistencies between the given differential equation and the given slope field. There are many such correct statements, such as:

- The response references a specific point on the y -axis (such as $(0, 4)$), and indicates that the given slope field shows a slope of zero, yet the differential equation would yield a positive slope.
- The response references a pattern of slopes, and explains why that pattern does not fit our differential equation (e.g. “the slope field shows slopes that change as t changes, but our differential equation depends only on y , so slopes should remain constant for a given value of y as t changes.”)

Identifying a mismatch between the differential equation and the slope field at one point is sufficient to earn the point.

Response is not penalized for referencing “ x ” instead of “ t ”.

If the response contains any incorrect statement, the response will not earn the point.

Vague answers which have the potential to be incorrect will not earn the point, nor will responses that present information beyond what we are given. For example:

- Responses that specify that the slope field shows a particular nonzero slope at a particular point will not earn the point, as this cannot actually be determined visually (i.e., the response should not indicate that at $(2, 2)$ the slope shown on the slope field is -1)
- Responses that state a specific slope value on the slope field for a general value of x or y that are not correct for all points with that x or y value do not earn the point. (e.g. “the slope field shows a slope of 0 when $y = 4$ ” is an incorrect statement)

4-1-1

A response may involve a solution to the differential equation, with an argument based on the resulting function. If done correctly, and presented with a correct argument, this will earn the point.



0	1
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The student response accurately includes the criteria below.

☐ Explanation

Solution:

When $y = 0$, $\frac{dy}{dt} = 0$. However, in the given slope field, for $t > 0$, the slopes of the line segments for $y = 0$ are nonzero.

4-1-1

63. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (minutes)	0	8	24	35
$P(t)$ (number of postcards remaining)	316	260	100	0

A principal is delivering postcards to teachers from students on a school trip. The principal begins with 316 postcards and delivers them over 35 minutes. For time t minutes, $0 \leq t \leq 35$, the total number of postcards remaining to deliver is modeled by a decreasing, twice-differentiable function P . Values of $P(t)$ at various times t are shown in the table above.

(a) Using correct units, approximate $P'(8)$ using the average rate of change of P over the interval $0 \leq t \leq 24$.

(b) Explain the meaning of $P'(8)$ in the context of the problem.

(c) It is known that $P'(24) = -11$. Write an equation for the line tangent to the graph of $y = P(t)$ at $t = 24$.

(d) Use a right Riemann sum with the three subintervals indicated by the table to approximate

$\int_0^{35} P(t) \, dt$. Show the computations that lead to your answer.

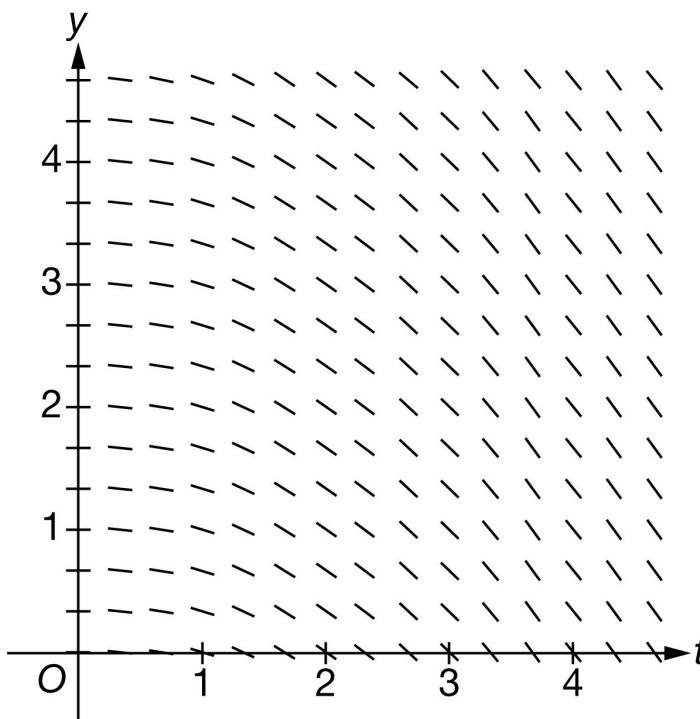
(e) Is the approximation from part (d) an overestimate or an underestimate for $\int_0^{35} P(t) \, dt$? Give a reason for your answer.

(f) Find the value of $\int_0^8 P'(3t) \, dt$. Show the computations that lead to your answer.

(g) For $0 \leq t \leq 35$, the total number of postcards remaining can also be modeled by the differentiable function R , where R is a function of t . Assume the weight of each postcard is 4 grams. Write an expression for the rate of change, in grams per minute, of the total weight of the postcards remaining at $t = 13$ minutes.

4-1-1

(h) Explain why the following slope field cannot represent the differential equation $\frac{dy}{dt} = -0.4y$.



Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for the correct approximation shown with supporting work. The supporting work must be at least a difference and a quotient. Correct values must be taken from the table to earn the point. The difference quotient does not need to be simplified.

Examples that earn the first point include:

$$\cdot \frac{100-316}{24-0}$$

$$\cdot \frac{P(24)-P(0)}{24} = \frac{-216}{24}$$

$$\cdot \frac{P(24)-P(0)}{24} = -9$$

An example that does not earn the first point is $\frac{P(24)-P(0)}{24}$.

Minimal response that earns the first point: $\frac{100-316}{24-0}$ or $\frac{100-316}{24}$

Neither $\frac{-216}{24}$ nor -9 , with no supporting work, will earn this point

The second point is for correct units

4-1-1

The correct units must be attached to a numerical value: either correct or incorrect.

The units may be written using words or a combination of words and symbols. For example:

- Postcards per minute
- Cards per min
- Postcards/min

The response will not be penalized for any verbal attempt to discuss the answer after the correct answer has been presented.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$P'(8) \approx \frac{P(24) - P(0)}{24 - 0} = \frac{100 - 316}{24} = -9 \text{ postcards per minute}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Three components of a correct answer:

- Indicating a rate of change
- Including the context of postcards
- Referring to the specific point in time, $t = 8$

The response is not penalized for incorrect units in this part. (Whether correct or incorrect, units have no effect on the score in this part.)

Responses may use “changing” or “decreasing” as a modifier as long as the value found in part (a) is not brought into part (b) in an incorrect manner. Example: “the number of postcards is decreasing at 9 postcards per minute.” is okay, but “the number of postcards is decreasing at a rate of -9 postcards per day...” would not earn the point.

Acceptable reference to a specific point in time includes (but is not limited to):

4-1-1

- On minute 8 and after 8 minutes
- At $t = 8$
- At $t = 8$ hours or any other incorrect unit of time
- If the response clearly indicates an interval of time (e.g., “from minute 8 on,” “during the first 8 minutes”, or “for $t = 0$ to $t = 8$ ”), the point is not earned.

Examples that earn the point include:

- $P'(8)$ is the rate of change of the number of postcards per minute at time $t = 8$
- The rate of change of number of postcards is -9 postcards per minute at minute 8
- $P'(8)$ is the rate of change in the number of postcards at time $t = 8$ in postcards per hour
- -9 is the rate of decrease in the number of postcards per minute at minute 8

Examples that do not earn the point include:

- $P'(8)$ is the average rate of change of the total number of postcards /minute at minute 8
- $P'(8)$ is the number of postcards per minute at minute 8
- The rate of change is -9 postcards per minute (Missing “at $t = 8$ ”)
- $P'(8)$ is the slope of the tangent line at $t = 8$
- Rate of change of P at time $t = 8$
- Postcards per minute decreases on minute 8



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

$P'(8)$ is the rate at which the number of postcards remaining is changing, in postcards per minute, at time $t = 8$ minutes.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

4-1-1

This point is earned with a correct tangent line equation in any form. The use of x in place of t is acceptable.

Some examples include:

$$\cdot y - 100 = -11(t - 24), y - 100 = -11(x - 24)$$

$$\cdot y = -11(t - 24) + 100, y = -11(x - 24) + 100$$

$$\cdot y = -11t + 364, y = -11x + 364$$

The key calculus concept being assessed in this part of the problem is that the value of $P'(24)$ is the slope of the tangent line. In that spirit, we are accepting $P'(t) = \dots$ in lieu of $y = \dots$, but we are not accepting $P'(24) = \dots$

Note that $P(t) = -11(t - 24) + 100$ earns the point, but $P(24) = -11(t - 24) + 100$ does not earn the point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$$y = P'(24)(t - 24) + P(24)$$

$$y = -11(t - 24) + 100$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for setting up a right Riemann sum. The setup must be a sum of three products.

· A response may still earn this point if there is only one numerical error in the expression for the Riemann sum with values pulled from the table. (See example in the table below).

· Responses with the numerical error described above are not eligible for the second point.

Note: either the multiplication or the addition may be indicated implicitly. For example, the response may show appropriate values put into rows and columns. Implicit responses are eligible for both points.

A response presenting a left Riemann sum (instead of right), will not earn any points for part (d).

The second point is for the correct numerical result.

· In order to earn the second point, there must be supporting calculations.

4-1-1

· The numerical values for $P(8)$, $P(24)$, and $P(35)$ must be pulled from the table

· Any simplification shown must be correct.

Use the following as an illustration how to score both points. (In most cases, you see scores for which the expression on the left side represents the entirety of relevant work shown for the response.)

· A response of $260 \cdot (8 - 0) + 100 \cdot (24 - 8) + 0 \cdot (35 - 24)$ earns both points.

· A response of $8 \cdot P(8) + 16 \cdot P(24) + 11 \cdot P(35)$ earns the first point. Eligible for second point if more work is on the page.

· A response of $8 \cdot P(8) + 16 \cdot P(24) + 11 \cdot P(35) = 3680$ earns both points.

· A response of $8 \cdot 260 + 16 \cdot 100 + 11 \cdot 0$ earns both points.

· A response of $8 \cdot 260 + 12 \cdot 100 + 11 \cdot 0$ contains one numerical error. Earns the first point. Not eligible for the second point.

· A response of $8 \cdot 260 + 16 \cdot 100 + 11 \cdot 0 = 3860$ earns first point for left side of equation. Does not earn second point due to incorrect simplification.

· A response of $2080 + 1600 + 0 = 3680$ or $2080 + 1600 = 3680$ does not earn the first point because we cannot see the sum of 3 products. However, this is the correct sum, so we see an answer with supporting work. The second point will be earned.

The response does not need to include “approximately” in any way. We will read $=$ and $\approx$ as the same thing.

Minimalist response: The following expression represents the minimum work that must be seen in a response in order to earn both points.

$$8 \cdot 260 + 16 \cdot 100 + 11 \cdot 0$$



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Right Riemann sum

☐ Answer

Solution:

$$\int_0^{35} P(t) dt \approx P(8)(8 - 0) + P(24)(24 - 8) + P(35)(35 - 24)$$

$$= 260 \cdot 8 + 100 \cdot 16 + 0 \cdot 11 = 3680$$

4-1-1

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned when the response states that the approximation is an underestimate because P is decreasing.

The response does not need an explicit reference to the approximation in part (d), nor does there need to be an answer given in part (d). The following statements are sufficient for earning this point:

- Underestimate because P is decreasing
- Underestimate because the function is decreasing

Note: At this point, there is only one function being discussed in this problem, so we will assume “the function” is P .

When a response includes additional information about concavity, we interpret that as a lack of understanding about the question, so this will not earn the point. When the response references a graph or the given table rather than the function P , the point is not earned.

Examples that earn the point:

- The right Riemann sum underestimates because the function is decreasing.
- The sum is an underestimate because $P(t)$ is decreasing.
- The sum is an underestimate because $P'(t) < 0$.

Examples that do not earn the point:

- The sum is an underestimate because $P'(t)$ is decreasing. ($P'(t)$ decreasing is not known)
- The right Riemann sum underestimates because the graph is decreasing.
- The right Riemann sum underestimates because the function is decreasing and concave up.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

Because P is a decreasing function, and this is a right Riemann sum, the answer from part (d) is an underestimate.

4-1-1

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with a correct substitution. There are 3 components of substitution that we need to see:

- Uses $u = 3t$, which may be done implicitly
- Uses the correct constant of $\frac{1}{3}$
- Uses the correct limits for the substitution (0 and 24). These limits may only be seen when evaluating the antiderivative. We don't need to see them as limits on a definite integral.
- Either of the expressions below will earn the substitution point without any additional supporting work:

$$\frac{1}{3} P(3t) \Big|_0^8 \text{ or } \frac{1}{3} P(u) \Big|_0^{24}$$

Special Handling: Notation Errors

Errors in notation, such as the examples listed here, will not prevent the response from earning the first or second points. These errors will cause the response to be ineligible for the third point

- Limits for t attached to an integral with u as the variable of integration (or vice versa),
- Improper equals signs, such as an indefinite integral equal to a definite integral
- Reversed limits of integration, for example: $\int_{24}^0 P'(u) du$

The second point is for the Fundamental Theorem of Calculus. This point can be earned with any of the following, where k is a nonzero constant and $b = 8$ or $b = 24$. No explanation or additional supporting work is necessary.

- $kP(3t) \Big|_0^b$
- $k(P(b) - P(0))$

This point can be earned without earning the substitution point.

The third point is for the correct numerical value, which is -72 .

Values from the table must be used. $\frac{1}{3}[P(24) - P(0)]$ is not enough to earn the answer point.

The expression $\frac{1}{3}(100 - 316)$ is sufficient for this point. No simplification is required. Any simplification that is shown must be correct.

Units are not required, and if shown, they need not be correct in order to earn this point.

4-1-1

The response cannot earn this point without also earning the FTC point.

Minimalist response: The expression $\frac{1}{3}(-216)$ represents the minimum work that must be seen in a response in order to earn all three points.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = 3t$. Then $du = 3 dt$.

When $t = 0$, $u = 0$.

When $t = 8$, $u = 24$.

$$\begin{aligned}\int_0^8 P'(3t) dt &= \frac{1}{3} \int_0^{24} P'(u) du \\ &= \frac{1}{3}(P(24) - P(0)) \\ &= \frac{1}{3}(100 - 316) = -\frac{216}{3} = -72\end{aligned}$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned for a correct expression at $t = 13$. No supporting work is required.

The correct expression, $4R'(13)$ or $4P'(13)$, earns this point.

No justification or explanations are required.

Any incorrect statement such as, $P'(13) = 4R'(13)$, does not earn this point.



0	1
---	---

4-1-1

The student response accurately includes the criteria below.

☐ Expression

Solution:

The rate of change, in grams per minute, of the total weight of the postcards remaining at $t = 13$ minutes is $4R'(13)$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned for a correct statement identifying the distinction between two things: the differential equation, $\frac{dy}{dt} = -0.4y$, and the given slope field.

The response does not need to explicitly reference both the differential equation and the given slope field., but it must explicitly reference one and implicitly reference the other.

Acceptable responses include

- Identifying an inconsistency between the differential equation and the slope field at one point, e.g., at the point $(4, 0)$, the slope of the segment should be 0, but it is negative.
- Referencing a pattern of slopes, and explaining why that pattern does not fit the given differential equation.

A response that lacks even an implicit reference to the one of two things being compared will not earn the point. For example, merely stating either $\frac{dy}{dt}$ is negative, or that the slope field is negative discretely does not earn the point.

- “The values for y are less than or equal to zero so $\frac{dy}{dt} \leq 0$ ” lacks any reference to the slope field and will not earn the point.
- “All the slopes in the slope field are negative” lacks any reference to the differential equation and will not earn the point.

Vague answers which have the potential to be incorrect will not earn the point. Some examples are:

- “The field is decreasing”
- Responses that state a specific slope value on the slope field for a general value of x or y that is not correct for all points with that x or y value do not earn the point. (e.g. “the slope field shows a slope of 0 when $y = 4$ ” is an incorrect statement)

A response may correctly solve or attempt to solve the differential equation and then provide a correct argument without reference to their solution. Such a response would earn the point.

4-1-1

A response might actually solve the differential equation and provide a correct argument on the resulting function.



0

1

The student response accurately includes the criteria below.

☐

Explanation

Solution:

When $y = 0$, $\frac{dy}{dt} = 0$. However, in the given slope field, for $t > 0$, the slopes of the line segments for $y = 0$ are nonzero.

4-1-1

64. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (minutes)	0	5	20	30
$D(t)$ (meters)	0	1000	4500	6500

A student on the school track and field team is running laps around a track. For time t minutes, $0 \leq t \leq 30$, the total distance run, in meters, is modeled by an increasing, twice-differentiable function D . Values of $D(t)$ at various times t are shown in the table above.

(a) Using correct units, approximate $D'(20)$ using the average rate of change of D over the interval $5 \leq t \leq 30$.

(b) Explain the meaning of $D'(20)$ in the context of the problem.

(c) It is known that $D'(5) = 200$. Write an equation for the line tangent to the graph of $y = D(t)$ at $t = 5$.

(d) Use a left Riemann sum with the three subintervals indicated by the table to approximate $\int_0^{30} D(t) \, dt$. Show the computations that lead to your answer.

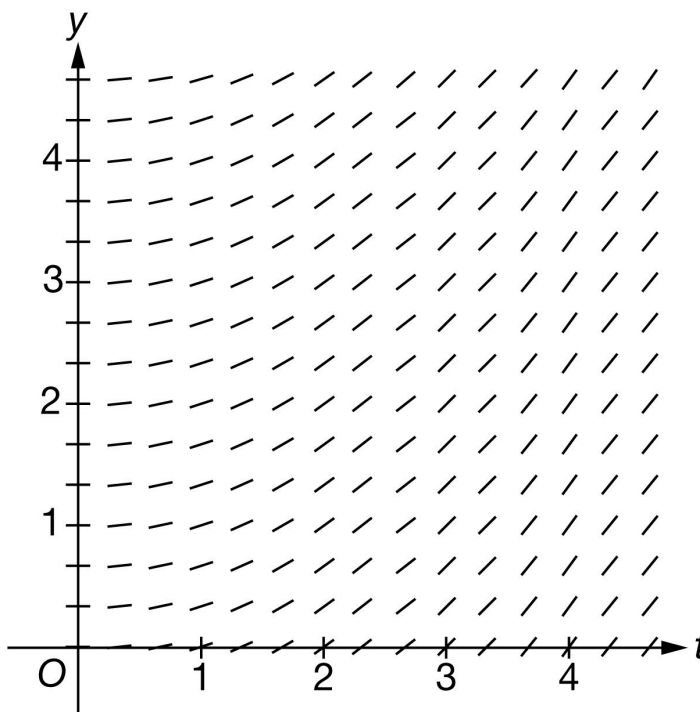
(e) Is the approximation from part (d) an overestimate or an underestimate for $\int_0^{30} D(t) \, dt$? Give a reason for your answer.

(f) Find the value of $\int_0^5 D'(4t) \, dt$. Show the computations that lead to your answer.

(g) The student is using their running practice as a fundraiser. For $0 \leq t \leq 30$, the total number of laps run can be modeled by the differentiable function L , where L is a function of t . For every lap run, the student raises 5 dollars. Write an expression for the rate of change, in dollars per minute, of the total money raised at $t = 22$ minutes.

4-1-1

(h) Explain why the following slope field cannot represent the differential equation $\frac{dy}{dt} = 0.5y$.



Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for the correct approximation shown with supporting work. The supporting work must be at least a difference and a quotient. Correct values must be taken from the table to earn the point. The difference quotient does not need to be simplified.

Examples that earn the first point include:

$$\cdot \frac{6500 - 1000}{30 - 5}$$

$$\cdot \frac{D(30) - D(5)}{25} = \frac{5500}{25}$$

$$\cdot \frac{D(30) - D(5)}{25} = 220$$

An example that does not earn the first point is $\frac{D(30) - D(5)}{30 - 5}$.

Minimal response that earns the point: $\frac{5500 - 1000}{30 - 5}$ or $\frac{6500 - 1000}{25}$

Neither $\frac{5500}{25}$ nor 220, with no supporting work, will earn this point

The second point is for correct units

4-1-1

The correct units must be attached to a numerical value: either correct or incorrect.

The units may be written using words or a combination of words and symbols. For example:

- Meters per minute
- Meters/minute
- m/min

The response will not be penalized for any verbal attempt to discuss the answer after the correct answer has been presented.



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$D'(20) \approx \frac{D(30) - D(5)}{30 - 5} = \frac{6500 - 1000}{25} = 220 \text{ meters per minute}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Three components of a correct answer:

- Indicating a rate of change
- Including the context of student running
- Referring to the specific point in time $t = 20$

The response is not penalized for incorrect units in this part. (Whether correct or incorrect, units have no effect on the score in this part.)

Responses may use “changing” or “increasing” as a modifier as long as the value found in part (a) is not brought into part (b) in an incorrect manner. Example: “The number of meters is increasing at a rate of 220 meters per minute” is okay.

Acceptable reference to a specific point in time includes (but is not limited to):

4-1-1

- On minute 20 and after 20 minutes
- At $t = 20$
- At $t = 20$ hours or any incorrect unit of time

If the response clearly indicates an interval of time (e.g., “from minute 20 on,” “during the first 20 minutes,” or “for $t = 0$ to $t = 20$ ”), the point is not earned.

Examples that earn the point include:

- $D'(20)$ is the rate of change of the distance of meters per minute at time $t = 20$
- The rate of change (or speed/velocity/ how fast) is 220 meters per minute at 20 minutes
- $D'(20)$ is the rate of change in the distance ran at time $t = 20$ in meters per *second*

Examples that do not earn the point include:

- 220 is the rate of increase in the distance of meters per minute at 20 minutes
- $D'(20)$ is the average rate of change of the total meters ran per minute at minute 20
- $D'(20)$ is the number of meters per minute at minute 20
- The rate of change is 220 meters per minute (Missing “at $t = 20$ ”)
- $D'(20)$ is the slope of the tangent line at $t = 20$
- Rate of change of D at time $t = 20$
- Increase in number of meters at $t = 20$



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

$D'(20)$ is the rate at which the total distance run is changing, in meters per minute, at time $t = 20$ minutes.

– OR –

$D'(20)$ is the speed at which the student is running, in meters per minute, at time $t = 20$ minutes.

4-1-1

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned with a correct tangent line equation in any form. The use of x in place of t is acceptable.

Some examples include

· $y = 200(t - 5) + 1000, y = 200(x - 5) + 1000$

· $y - 1000 = 200(t - 5),$

· $y - 1000 = 200(x - 5)$

· $y = 200t, y = 200x$

The key calculus concept being assessed in this part of the problem is that the value of $D'(5)$ is the slope of the tangent line. In that spirit, we are accepting $D'(t) = \dots$ in lieu of $y = \dots$, but we are not accepting $D'(5) = \dots$

Note that $D(t) = 200(t - 5) + 1000$ earns the point, but $D(5) = 200(t - 5) + 1000$ does not earn the point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$$y = D'(5)(t - 5) + D(5)$$

$$y = 200(t - 5) + 1000$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for setting up a left Riemann sum. The setup must be a sum of three products.

· A response may still earn this point if there is only one numerical error in the expression for the Riemann sum with values pulled from the table. (See example in the table below).

· Responses with the numerical error described above are not eligible for the second point.

4-1-1

Note: either the multiplication or the addition may be indicated implicitly. For example, the response may show appropriate values put into rows and columns. Implicit responses are eligible for both points.

A response presenting a right Riemann sum (instead of left), will not earn any points for part (a).

The second point is for the correct numerical result.

- In order to earn the second point, there must be supporting calculations.
- The numerical values for $D(0)$, $D(5)$, and $D(20)$ must be pulled from the table
- Any simplification shown must be correct.

Use the following as an illustration how to score both points. (In most cases, you see scores for which the expression on the left side represents the entirety of relevant work shown for the response.)

- A response of $0 \cdot (5 - 0) + 1000 \cdot (20 - 5) + 4500 \cdot (30 - 20)$ earns both points.
- A response of $5 \cdot D(0) + 15 \cdot D(5) + 10 \cdot D(20)$ earns the first point. Eligible for second point if more work is on the page.
- A response of $5 \cdot D(0) + 15 \cdot D(5) + 10 \cdot D(20) = 6000$ earns both points.
- A response of $5 \cdot 0 + 15 \cdot 1000 + 10 \cdot 4500$ earns both points.
- A response of $5 \cdot 0 + 10 \cdot 1000 + 10 \cdot 4500$ contains one numerical error. Earns the first point. Not eligible for the second point.
- A response of $5 \cdot 0 + 15 \cdot 1000 + 10 \cdot 4500 = 50,000$ earns first point for left side of equation. Does not earn second point due to incorrect simplification.
- A response of $0 + 15,000 + 45,000$ or $0 + 15,000 + 45,000 = 60,000$ does not earn the first point because we cannot see the sum of 3 products. However, this is the correct sum, so we see an answer with supporting work. The second point will be earned.

The response does not need to include “approximately” in any way. We will read $=$ and $\approx$ as the same thing.

Minimalist response: The following expression represents the minimum work that must be seen in a response in order to earn both points.

$$5 \cdot 0 + 15 \cdot 1000 + 10 \cdot 4500$$



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Left Riemann sum

4-1-1

☐ Answer
Solution:

$$\int_0^{30} D(t) dt \approx D(0)(5-0) + D(5)(20-5) + D(20)(30-20)$$

$$= 0 \cdot 5 + 1000 \cdot 15 + 4500 \cdot 10 = 60,000$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned when the response states that the approximation is an underestimate because D is increasing.

The response does not need an explicit reference in part (d). The following statements are sufficient for earning this point:

- Underestimate because D is increasing
- Underestimate because the function is increasing

Note: At this point, there is only one function being discussed in this problem, so we will assume “the function” is D .

When a response includes additional information about concavity, we interpret that as a lack of understanding about the question, so this will not earn the point. When the response references a graph or the given table rather than the function D , the point is not earned.

Examples that earn the point:

- The left Riemann sum underestimates because the function is increasing.
- The sum is an underestimate because $D(t)$ is increasing.
- The sum is an underestimate $D'(t) > 0$.

Examples that do not earn the point:

- The sum is an underestimate because $D'(t)$ is increasing. ($D'(t)$ increasing is not known)
- The left Riemann sum underestimates because the graph is increasing.
- The left Riemann sum underestimates because the function is increasing and concave up.



0	1
---	---

4-1-1

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

Because D is an increasing function, and this is a left Riemann sum, the answer from part (d) is an underestimate.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with a correct substitution. There are 3 components of substitution that we need to see:

- Uses $u = 4t$, which may be done implicitly
- Uses the correct constant of $\frac{1}{4}$
- Uses the correct limits for the substitution (0 and 20). These limits may only be seen when evaluating the antiderivative. We don't need to see them as limits on a definite integral.
- Either of the expressions below will earn the substitution point without any additional supporting work:

$$\frac{1}{4} D(4t) \Big|_0^5 \text{ or } \frac{1}{4} D(u) \Big|_0^{20}$$

Special Handling: Notation Errors

Errors in notation, such as the examples listed here, will not prevent the response from earning the first or second points. These errors will cause the response to be ineligible for the third point

- Limits for t attached to an integral with u as the variable of integration (or vice versa),
- Improper equals signs, such as an indefinite integral equal to a definite integral
- Reversed limits of integration, for example: $\int_{20}^0 D'(u) du$

The second point is for the Fundamental Theorem of Calculus. This point can be earned with any of the following, where k is a nonzero constant and $b = 5$ or $b = 20$. No explanation or additional supporting work is necessary.

- $kD(4t) \Big|_0^b$
- $k(D(b) - D(0))$

The response cannot earn this point without also earning the FTC point.

The third point is for the correct numerical value, which is 1125.

4-1-1

Values from the table must be used. $\frac{1}{4}[D(20) - D(0)]$ is not enough to earn the answer point.

The expression $\frac{1}{4}(4500 - 0)$ is sufficient for this point. No simplification is required. Any simplification that is shown must be correct.

Units are not required, and if shown, they need not be correct in order to earn this point.

The response cannot earn this point without also earning first two points.

Minimalist response: The expression $\frac{1}{4}(4500)$ represents the minimum work that must be seen in a response in order to earn all three points.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = 4t$. Then $du = 4 dt$.

When $t = 0$, $u = 0$.

When $t = 5$, $u = 20$.

$$\begin{aligned}\int_0^5 D'(4t) dt &= \frac{1}{4} \int_0^{20} D'(u) du \\ &= \frac{1}{4}(D(20) - D(0)) \\ &= \frac{1}{4}(4500 - 0) = 1125\end{aligned}$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned for a correct expression at $t = 22$. No supporting work is required.

The correct expression of $5 \cdot L'(22)$ earns this point.

No justifications or explanations are required.

4-1-1

Any incorrect statement such as, $D'(22) = 5 \cdot L'(22)$, does not earn this point.



0

1

The student response accurately includes the criteria below.

☐ Expression

Solution:

The rate of change, in dollars per minute, of the total money raised at $t = 22$ minutes is $5L'(22)$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned for a correct statement identifying the distinction between two things: the differential equation, $\frac{dy}{dt} = 0.5y$, and the given slope field.

The response does not need to explicitly reference both the differential equation and the given slope field, but it must explicitly reference one and implicitly reference the other.

Acceptable responses include

- Identifying an inconsistency between the differential equation and the slope field at one point, e.g., at the point $(4, 0)$, the slope of the segment should be 0, but it is positive.
- Referencing a pattern of slopes, and explaining why that pattern does not fit the given differential equation.

A response that lacks even an implicit reference to the one of two things being compared will not earn the point.

For example, merely stating either $\frac{dy}{dt}$ is positive, or that the slope field is positive does not earn the point.

- “The values for are greater than or equal to zero so $\frac{dy}{dt} \geq 0$ ” lacks any reference to the slope field and will not earn the point.
- “All the slopes in the slope field are positive” lacks any reference to the differential equation and will not earn the point.

Vague answers which have the potential to be incorrect will not earn the point. Some examples are:

- “The field is increasing”
- Responses that state a specific slope value on the slope field for a general value of x or y that is not correct for all points with that x or y value do not earn the point. (e.g. “the slope field shows a slope of 0 when $y = 4$ ” is an

4-1-1

incorrect statement)

A response may correctly solve or attempt to solve the differential equation and then provide a correct argument without reference to their solution. Such a response would earn the point.

A response might actually solve the differential equation and provide a correct argument on the resulting function.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

When $y = 0$, $\frac{dy}{dt} = 0$. However, in the given slope field, for $t > 0$, the slopes of the line segments for $y = 0$ are nonzero.

4-1-1

65. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (minutes)	0	5	20	30
$D(t)$ (meters per minute)	0	100	120	150

A student on the school track and field team is running laps around a track. For time t minutes, $0 \leq t \leq 30$, the rate at which the student runs, in meters per minute, is modeled by an increasing, twice-differentiable function D . Values of $D(t)$ at various times t are shown in the table above.

(a) Use a right Riemann sum with the three subintervals indicated by the table to approximate

$\int_0^{30} D(t) \, dt$. Show the calculations that lead to your answer.

(b) Is your answer from part (a) an overestimate or an underestimate for $\int_0^{30} D(t) \, dt$? Give a reason for your answer.

(c) Using correct units, approximate $D'(5)$ using the average rate of change of D over the interval $0 \leq t \leq 20$.

(d) Explain the meaning of $D'(5)$ in the context of the problem.

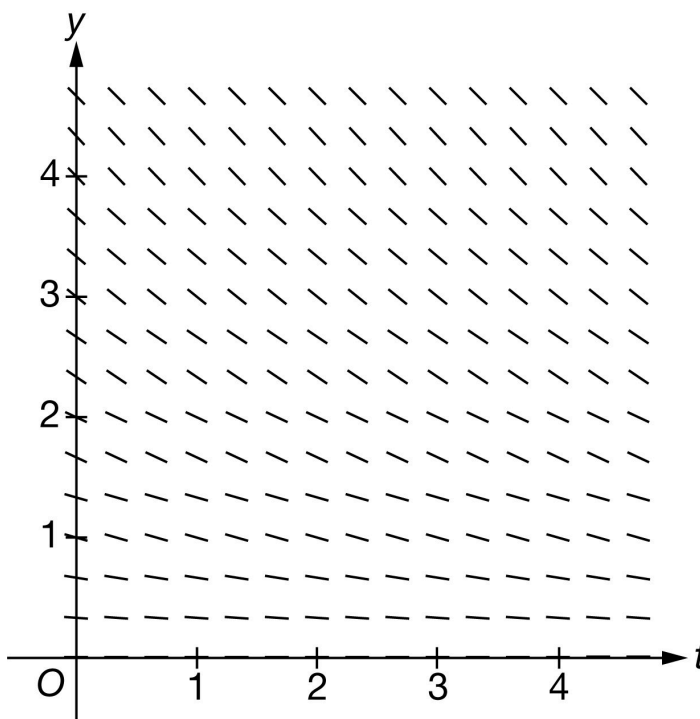
(e) It is known that $D'(20) = 2.5$. Write an equation for the line tangent to the graph of $y = D(t)$ at $t = 20$.

(f) Find the value of $\int_0^{20} D'\left(\frac{t}{4}\right) \, dt$. Show the calculations that lead to your answer.

(g) A friend on the track and field team is using her running practice as a fundraiser. For $0 \leq t \leq 30$, the total number of laps run by the friend can be modeled by the increasing, differentiable function L , where L is a function of t . For every lap run, the friend raises 4 dollars. Write an expression for the rate of change of the money, in dollars per minute, raised by the friend at $t = 19$ minutes.

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(h) Explain why the following could not be a slope field for the differential equation $\frac{dy}{dt} = L'(t)$ for the function L defined in part (g).



Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for setting up a right Riemann sum. The setup must be a sum of three products.

A response may still earn this point if there is only one numerical error in the expression for the Riemann sum with values pulled from the table. (See example in the table below).

Responses with the numerical error described above are not eligible for the second point.

Note: either the multiplication or the addition may be indicated implicitly. For example, the response may show appropriate values put into rows and columns, like so.

100	5	500
120	15	1800
150	10	+1500
		3800

or

$(5 - 0)(100)$	$= 500$
$(20 - 5)(120)$	$= 1800$
$(30 - 20)(150)$	$= 1500$
	3800

Implicit responses such as those above are eligible for both points.

4-1-1

A response presenting a left Riemann sum (instead of right), will not earn any points for part (a).

The second point is for the correct numerical result.

In order to earn the second point, there must be supporting calculations.

The numerical values for $D(5)$, $D(20)$, and $D(30)$ must be pulled from the table.

Any simplification shown must be correct.

Use the following as an illustration of how to score both points. (In most cases, you see scores for which the expression represents the entirety of relevant work shown for the response.)

- A response of $100 \cdot (5 - 0) + 120 \cdot (20 - 5) + 150 \cdot (30 - 20)$ earns both points.
- A response of $5 \cdot D(5) + 15 \cdot D(20) + 10 \cdot D(30)$ earns the first point. Eligible for second point if more work is on the page.
- A response of $5 \cdot D(5) + 15 \cdot D(20) + 10 \cdot D(30) = 3800$ earns both points.
- A response of $5 \cdot 100 + 15 \cdot 120 + 10 \cdot 150$ earns both points.
- A response of $5 \cdot 100 + 25 \cdot 120 + 10 \cdot 150$ contains one numerical error. Earns the first point. Not eligible for the second point.
- A response of $5 \cdot 100 + 15 \cdot 120 + 10 \cdot 150 = 3300$ earns first point for left side of equation. Does not earn second point due to incorrect simplification.
- A response of $500 + 1800 + 1500$ or $500 + 1800 + 1500 = 3800$ does not earn the first point because we cannot see the sum of three products. However, this is the correct sum, so we are not seeing an unsupported answer. The second point will be earned.

The response does not need to include “approximately” in any way. We will read $=$ and $\approx$ as the same thing.

Minimalist response: The following expression represents the minimum work that must be seen in a response in order to earn both points. $5 \cdot 100 + 15 \cdot 120 + 10 \cdot 150$



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Right Riemann sum
- ☐ Answer

Solution:

4-1-1

$$\int_0^{30} D(t) dt \approx D(5)(5-0) + D(20)(20-5) + D(30)(30-20)$$

$$= 100 \cdot 5 + 120 \cdot 15 + 150 \cdot 10 = 3800$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned when the response states that the approximation is an overestimate because D is increasing.

The response does not need an explicit reference to part (a) or to a right Riemann sum.

A response that appeals to concavity as a reason will NOT earn this point.

The following table contains some examples of what WILL or will NOT earn the point.

The following examples earn the point:

- Overestimate because D is increasing
- Overestimate because $D'(t) > 0$
- Overestimate because the graph of D is increasing (we know they are talking about the given function D)
- Overestimate because the function is increasing. (There is only one function in the problem at this point.)

The following examples do not earn the point:

- Overestimate because D is concave down/up
- Overestimate because the graph/curve is increasing. Since no graph/curve is provided, we do not know what graph/curve they are talking about.
- Overestimate because the table [or table values] is/are increasing. The behavior of the values shown in the table is not the same as the behavior of the function D . The table does not tell us what is happening to D between the given points.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

Because D is an increasing function, and this is a right Riemann sum, this is an overestimate.

4-1-1

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for the correct approximation shown with supporting work.

We need to see both a difference and a quotient. They don't have to be together.

The expression does not need to be simplified, but values for $D(20)$ and $D(0)$ must be pulled from the table.

The response does not need to include “approximately” in any way. We will read $=$ and $\approx$ as the same thing.

The second point is for correct units.

Units must be attached to a numerical value (either correct or incorrect).

Can be written using words or a combination of words and symbols. For example:

- Meters per minute per minute or Meters/min/min
- Meters per min^2 or Meters/ min^2
- Meters per minute squared

We do not read work shown in part (d) for this point. The units must appear in part (c).

Use caution when interpreting abbreviations: m should be meters and min should be minutes, but responses may vary. For example,

- if a response includes units “Meters/m/m ” we read “ m ” as minutes; or
- if a response includes units “m/min/min ” we read “ m ” as meters; or
- if a response includes units “m/m/m ” we cannot interpret the units and the response will not earn the second point.

The response will not be penalized for any attempt to verbally describe their answer after a correct answer has been presented.

Minimalist response: The following expression and units represent the minimum work that must be seen in a response in order to earn both points.

$$\frac{120-0}{20} \text{ meters/min}^2 \text{ or } \frac{120}{20-0} \text{ meters/min}^2$$

In the minimalist response, we still need to see subtraction (a difference), so subtracting 0 must be explicit either in the numerator or in the denominator.

4-1-1



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$D'(5) \approx \frac{D(20) - D(0)}{20 - 0} = \frac{120 - 0}{20} = 6 \text{ meters per minute per minute}$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

- Three components of a correct answer:
- Indicating a rate of a rate
- Including the context of running
- Explicitly referring to the specified point in time ($t = 5$)

Incorrect or missing units will not be penalized as long as there is a clear connection to the context of the problem.

The words “increasing” and “decreasing” are interpreted as modifiers describing one of the specified rates.

If the numerical value from part (c) is used in this interpretation, then we expect that the choice of “increasing” or “decreasing” is consistent with that value.

The following examples use 6 as the imported value from part (c). If a different [incorrect] value is imported from part (c), then these statements should be consistent with that value. (Remember that we are reading $=$ and $\approx$ as the same thing.)

- “... increasing at a rate of 6 ” is OK [Units not required in part (d)] (the point is not earned with this phrase alone)
- “... decreasing at a rate of 6 ” is not OK and the point is not earned.

Rate of a rate

To earn this point, the response must explicitly discuss the rate of change of a rate of change in the context of motion [running]. For example:

4-1-1

- The rate of change in the rate at which students are running
- The rate of change in speed is changing (or increasing) at a rate of . . .

Since the runners are unlikely to be running backwards, we interpret speed as being the same as velocity.

- The rate at which the velocity of the runner is changing
- Acceleration of the runner (this is correct for the context)

The words “increasing,” “decreasing,” or “changing” are not interpreted as rates

Units are not interpreted as rates. (The phrase “meters per minute” is not enough to indicate a rate.)

Slope of tangent line is not an appropriate interpretation for this context.

“Rate of change of D ” is not enough without describing what D represents in this context.

Context

The context is running, which could be indicated by words such as (but not limited to): meters, minutes, laps, speed, velocity, runner

Explicit reference to $t = 5$

Acceptable references to a specific point in time include (but are not limited to):

- At minute 5 (where minute could be replaced with any other unit of time)
- At $t = 5$ (with or without units of time)
- After 5 minutes (where minutes could be replaced with any other unit of time).

Note: we are choosing to assume that “after” is being used in lieu of “at” unless we have additional evidence to the contrary.

If the response is explicitly indicating an interval of time (e.g., “from minute 5 on” or “during the first 5 minutes”), this point will not be earned.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

4-1-1

$D'(5)$ is the rate at which the rate at which the student runs is changing, in meters per minute per minute, at minute 5.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This point is earned with a correct tangent line equation in any form. Any of these equations can be written as is without any explanation or additional supporting work. If rewriting of equations is done (transforming to slope-intercept form, perhaps), it must be done correctly.

· $y = 2.5(t - 20) + 120$

· $y - 120 = 2.5(t - 20)$

· $y = 2.5t + 70$

The key calculus concept being assessed in this part of the problem is that the value of $D'(20)$ is the slope of the tangent line. In that spirit:

- The response will not be penalized for using x instead of t .
- We are accepting $D(t)$ in lieu of y in all forms of the tangent line equation.



0	1
---	---

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$$y = D'(20)(t - 20) + D(20)$$

$$y = 2.5(t - 20) + 120$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with a correct substitution. There are three components of substitution that we need to see:

- Uses $u = \frac{t}{4}$, which may be done implicitly
- Uses the correct constant of 4

4-1-1

· Uses the correct limits for the substitution (0 and 5). These limits may only be seen when evaluating the antiderivative. We don't need to see them as limits on a definite integral.

Either of the expressions $4D\left(\frac{t}{4}\right)\bigg|_0^{20}$ or $4D(u)\bigg|_0^5$ will earn the substitution point without any additional supporting work.

Special Handling: Notation Errors

Errors in notation, such as the examples listed here, will not prevent the response from earning the first or second points. These errors will cause the response to be ineligible for the third point.

- Limits for t attached to an integral with u as the variable of integration (or vice versa),
- Improper equals signs, such as an indefinite integral equal to a definite integral
- Reversed limits of integration, for example: $\int_5^0 D'(u) du$

The second point is for the Fundamental Theorem of Calculus.

This point can be earned with any of the following, where k is a nonzero constant and $b = 5$ or $b = 20$. No explanation or additional supporting work is necessary.

- $kD\left(\frac{t}{4}\right)\bigg|_0^b$
- $k(D(b) - D(0))$ or $kD(b)$ (because $D(0) = 0$)

This point can be earned without earning the substitution point.

The third point is for the correct numerical value, which is 400.

Values from the table must be used. $4(D(5) - D(0))$ is not enough to earn the answer point.

The expressions $4(100 - 0)$ or $4(100)$ are sufficient for this point. No simplification is required. Any simplification that is shown must be correct.

Units are not required, and if shown, they do not need to be correct to earn this point.

The response cannot earn this point without also earning the first two points.

Minimalist response: The expression $4(100)$ represents the minimum work that must be seen in a response in order to earn all three points.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

4-1-1

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = \frac{t}{4}$. Then $du = \frac{dt}{4}$.

When $t = 0$, $u = 0$.

When $t = 20$, $u = 5$.

$$\begin{aligned}\int_0^{20} D' \left(\frac{t}{4} \right) dt &= 4 \int_0^5 D' (u) du \\ &= 4 (D (5) - D (0)) \\ &= 4 (100 - 0) = 400\end{aligned}$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct expression at $t = 19$ earns the point. It may be written as $4 \cdot L' (19)$ or $4 \cdot \frac{dL}{dt} \Big|_{t=19}$

The expression $4 \cdot L' (t)$ is not sufficient because it does not use $t = 19$.

The response must use L (not M or D or any other function name) in order to earn the point.



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Expression

Solution:

The rate of change of the money, in dollars per minute, raised by the friend at $t = 19$ minutes is $4L' (19)$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

4-1-1

This point is earned for a correct statement identifying the distinction between the differential equation

$$\frac{dy}{dt} = L'(t) \text{ and the given slope field.}$$

In order to earn the point, the response must:

State that the values of the differential equation are positive while the slopes in the given slope field are negative or 0.

Some ways to say that the values of differential equation are positive:

- L is increasing
- $L'(t) > 0$
- L has positive slopes

When discussing the slopes in the slope field, the response must explicitly refer to the signs of the slopes.

The following examples could earn the point:

- The slope field has negative (non-positive) slopes.
- The slopes shown are negative or 0.
- “The graph” or “The field” could be used in lieu of “The slope field.”

The following examples will not earn the point:

- The slope field is decreasing/negative.
- The slope field looks like it represents a decreasing function.
- The slope field represents a quadratic [or exponential] function.
- “It” (rather than “the slope field”)

Statements like “ L is an increasing function, so the signs of the slopes in the slope field should be positive” can earn this point because “should be” is indicating contrast with what is known.

The response will not be penalized for saying that the slopes in the slope field are “always” negative or “all” negative.



0	1
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The student response accurately includes the criteria below.

☐ Explanation

4-1-1**Solution:**

L is an increasing function, so $\frac{dy}{dt} = L'(t) \geq 0$. However, in the given slope field, the slopes of the line segments for $y > 0$ are negative.

4-1-1

66. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (minutes)	0	5	20
$R(t)$ (meters per minute)	40	200	160

A student on the school track and field team is running on the track. For time t minutes, $0 \leq t \leq 20$, the rate at which the student runs, in meters per minute, is modeled by a twice-differentiable function, $R(t)$. Values of $R(t)$ at various times t are shown in the table above.

(a) Using correct units, approximate $R'(5)$ using the average rate of change of R over the interval $0 \leq t \leq 20$.

(b) Explain the meaning of $R'(5)$ in the context of the problem.

(c) The function $D(t)$ models the number of meters run by the student at time t minutes, where $D'(t) = R(t)$. It is known that $D(5) = 800$. Write an equation for the line tangent to the graph of $y = D(t)$ at $t = 5$.

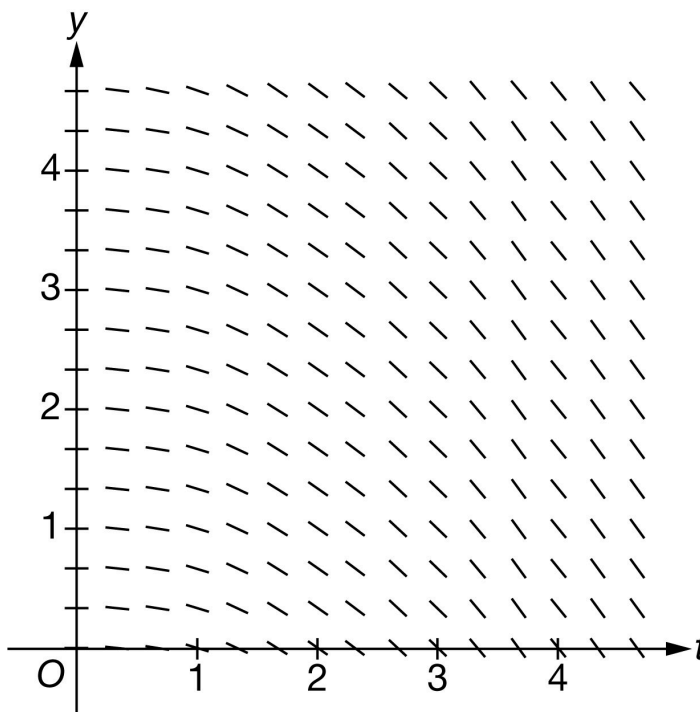
(d) Find the value of $\int_0^5 R'(4t) \, dt$. Show the calculations that lead to your answer.

(e) Use a trapezoidal sum with the two subintervals indicated by the table to approximate $\int_0^{20} R(t) \, dt$. Show the calculations that lead to your answer.

(f) The student and a friend team up to run on the school track for a fundraiser for 20 minutes. For $0 \leq t \leq 20$, the total number of laps run by the student is modeled by $M(t)$ and the total number of laps run by the friend is modeled by $L(t)$. Both M and L are differentiable functions of t . Each person raises 5 dollars for each lap run. Write an expression for the rate of change, in dollars per minute, of the total money raised at time $t = 18$ minutes.

4-1-1

(g) Explain why the following could not be a slope field for the differential equation $\frac{dy}{dt} = -0.35y$.

**Part A**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for the correct approximation shown with supporting work.

Answers without supporting work do not earn this point. Supporting work must be at least a difference and a quotient.

Response does not need to be simplified, but any simplification must be done correctly.

If a decimal is presented, it must be correct to three decimal places. We do not read anything past the third decimal place.

The second point is earned for the correct units.

Units must be attached to a numerical value (either correct or incorrect).

Can be written using words or a combination of words and symbols.

The response will not be penalized for any verbal attempt to discuss the answer after the correct presentation of the approximation.

The minimum response needed to earn both points is: $\frac{160-40}{20}$ meters per minute per minute.

4-1-1



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$R'(5) \approx \frac{R(20) - R(0)}{20 - 0} = \frac{160 - 40}{20} = 6 \text{ meters per minute per minute}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

This part is intended to be independent from part (a), so we do not want or need to see any reference to the value obtained in (a). If that value is referenced, conclusion must be consistent.

Three components of a correct answer:

- Indicating a rate of a rate (look for the word rate to be used correctly twice)
- Including the context of the student running
- Referring to the specific point in time

Because the context is one of motion, a description of $R'(5)$ as the acceleration of the runner at time $t = 5$ earns the point.

Units will not be considered for this point. They do not need to be present, and if they are, they do not need to be correct.

Any incorrect statement makes the response ineligible for the point. For example:

$R'(5)$ is the average (or an approximation of, or an estimate of) rate of change of

“Increasing” or “decreasing” may be used in place of “changing” as long as the value found in part (a) is not brought into part (b) in an incorrect manner. Beware of phrases such as “decreasing at a negative rate.”

- Example: $R'(5)$ is the rate of increase of the rate at which the student runs at 5 minutes. (earns the point)
- Example: $R'(5)$ means the rate at which the student runs is increasing by 6 meters / min / min. at time $t = 5$ (does not earn the point)

4-1-1

Acceptable references to time include (but are not limited to):

- At the fifth minute
- At $t = 5$ (with correct units of time, without correct units of time, or even with incorrect units of time)
- After $t = 5$ (with correct units of time, without correct units of time, or even with incorrect units of time)

If the response clearly indicates an interval of time (e.g. “during the first 5 minutes”) the response will not earn the point.

Examples that earn the point include:

- $R'(5)$ represents the rate at which the rate at which the student runs is changing on day 5.
- $R'(5)$ is the rate of change in the velocity of the runner at 5 days.
- $R'(5)$ represents the rate at which the velocity of the runner is increasing at $t = 5$.
- $R'(5)$ equals (answer from part (a)) means the velocity of the runner is increasing at a rate of (answer from part (a)) at $t = 5$.

Examples that do not earn the point include:

- $R'(5)$ represents the rate at which the meters per minute is changing on day 5.
- $R'(5)$ is the rate of change in the meters run at $t = 5$ days.
- The rate at which the velocity changes is increasing at $t = 5$.
- The rate of change in the rate of change of the runner is increasing at $t = 5$.



0	1
---	---

The student response accurately includes the criteria below.

☐ Explanation

Solution:

$R'(5)$ is the rate of change in the rate at which the student runs, in meters per minute per minute, at time $t = 5$ minutes.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

4-1-1

The point is earned with a correct tangent line equation in any form. Equivalent forms include (but are not limited to):

o $y = 200(t - 5) + 800$

o $y - 800 = 200(t - 5)$

o $y = 200t - 200$

The student may write x in place of t and still earn the point.

No work is required to earn the point.

We will accept $D(t)$ in lieu of y in the equation of the tangent line.



0	1
---	---

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$$y = D'(5)(t - 5) + D(5)$$

$$y = 200(t - 5) + 800$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with a correct substitution.

Two things are required to award this point:

- Coefficient of $\frac{1}{4}$ correctly applied to an integral or an antiderivative
- Evidence that the correct limits of 0 to 20 will be or have been applied

Examples that will earn the first point (these are just a few):

$$\cdot \frac{1}{4} \int_0^{20} R'(u) \, du$$

$$\cdot \frac{1}{4} R(u) \Big|_0^{20} \text{ or } \frac{1}{4} R(4t) \Big|_0^5$$

4-1-1

Responses that include misstatements such as using t -values for limits with u as the variable of integration in an intermediate step which is later corrected may still earn this point. Intermediate errors in bounds that are subsequently corrected will disqualify the student from earning the third point. If the bounds are never appropriately handled, the response cannot earn the first point. Scratch work that is not connected to the solution will not be read.

Examples:

$$\int_0^5 R'(4t) dt = \frac{1}{4} \int_0^5 R'(u) du = \frac{1}{4} R(4t) \Big|_0^5 \text{ earns the first point, but will not earn the third point.}$$

$$\int_0^5 R'(4t) dt = \frac{1}{4} \int_0^5 R'(u) du = \frac{1}{4} [R(5) - R(0)] \text{ does not earn the first point.}$$

The second point is for the Fundamental Theorem of Calculus, demonstrating the knowledge that

$$\int_a^b R'(u) du = R(u) \Big|_a^b = R(b) - R(a)$$

This point can be earned with either $kR(4t) \Big|_a^b$, $kR(t) \Big|_a^b$, or $k(R(b) - R(a))$ where k is a nonzero constant [not necessarily correct], $a = 0$, and $b = 5$ or $b = 20$.

A response can earn this point without earning the first point.

$$\frac{1}{4} [R(20) - M(0)] \text{ earns the first two points.}$$

The third point is for the correct numerical value. Our answer only.

Values from the table must be used (i.e., $\frac{1}{4} [R(20) - M(0)]$ is not sufficient to earn the third point).

If the response includes any linkage error, (i.e., the use of equal signs to equate two expressions that are not equal), the third point will not be earned. If an indefinite integral is linked with a definite integral the response will not be penalized.

$$\int_0^5 R'(4t) dt = \frac{1}{4} \int R'(u) du = \frac{1}{4} R(4t) \Big|_0^5 = \dots \text{ is eligible for the third point.}$$

Eligibility: Must have earned the first two points.

The minimum response needed to earn all three points is: $\frac{1}{4}(160 - 40)$.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

Substitution

4-1-1

☐☐

Fundamental Theorem of Calculus

☐

Answer

Solution:

Let $u = 4t$. Then $du = 4 dt$.

When $t = 0$, $u = 0$.

When $t = 5$, $u = 20$.

$$\begin{aligned}\int_0^5 R'(4t) dt &= \frac{1}{4} \int_0^{20} R'(u) du \\ &= \frac{1}{4}(R(20) - R(0)) \\ &= \frac{1}{4}(160 - 40) = 30\end{aligned}$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The **first point** is for the form of a trapezoidal sum

Point is earned for the setup of the trapezoidal sum. Numerical values do not need to be pulled from the table for this point, but Δt values must be presented. Responses that contain one incorrect numerical value in the trapezoidal sum setup are still eligible to earn the first point, but will not earn the second.

Response may present a left Riemann sum and a right Riemann sum, then average the two. Again, one incorrect numerical value will not disqualify the response from earning this first point.

Response may compute trapezoids as areas of rectangles and triangles.

Trapezoidal areas may be given in a graph. In order to earn this point, the response must clearly show the computations leading to the values presented in the graph, and must show that these areas are added to obtain the final approximation value.

A response using subtraction instead of addition will not earn the point

(e.g. $\frac{1}{2}[5(R(0) - R(5)) + 15(R(5) - R(20))]$ earns no points).

Response utilizing “Trapezoidal Rule” will not earn this point, unless this represents only one incorrect numerical value.

4-1-1

$\cdot \frac{1}{2}(5) [R(0) + 2R(5) + R(20)]$ is equivalent to $\frac{1}{2}[5(R(0) + R(5)) + 5(M(5) + R(20))]$, thus has only one incorrect numerical value (the second interval width is 5 instead of 15), so can earn the point.

$\cdot \frac{1}{2}(10) [R(0) + 2R(5) + R(20)]$ has two incorrect numerical values (both interval widths are incorrect) so will not earn the point.

The second point is for the correct numerical result.

Our answer only. Response does not need to be simplified, but if it is, it must be simplified correctly.

The answer must be accompanied with supporting work. Supporting work may appear in a graph, with clearly labeled areas. It is possible to earn the second point without earning the first point if the supporting work is not sufficient to earn the first point.

Pictures without clear area computations and a clear summation (+) of quantities can earn at most the second point.

Special case: A positive constant multiple of a complete and correctly expressed trapezoidal sum (with the correct values pulled from the table) will earn the first but not the second point:

$\cdot k \left[\left(\frac{40+200}{2} \right) \cdot 5 + \left(\frac{200+160}{2} \cdot 15 \right) \right]$ for $k > 0$ earns only the first point.

The minimum response needed to earn both points: $\left(\frac{240}{2} \right) \cdot 5 + \left(\frac{360}{2} \right) \cdot 15$



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Form of trapezoidal sum

☐ Answer

Solution:

$$\begin{aligned} \int_0^{20} R(t) \, dt &\approx \left(\frac{R(0)+R(5)}{2} \right) (5-0) + \left(\frac{R(5)+R(20)}{2} \right) (20-5) \\ &= \left(\frac{40+200}{2} \right) \cdot 5 + \left(\frac{200+160}{2} \right) \cdot 15 = 3300 \end{aligned}$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for a correct constant multiple or correct sum.

4-1-1

Response must contain the constant 5 as a factor of the expression, and/or the correct sum $L'(18) + M'(18)$.

If the sum is incorrect, then the constant factor of 5 must be applied either to a sum that involves only $L(t)$ and $M(t)$, $L'(t)$ and $M'(t)$, or integrals of $L(t)$ and $M(t)$, or to one of the correct terms of the sum $L'(18)$ or $M'(18)$, with no additional incorrect terms.

Examples that earn the point include:

- $5[M'(t) + L'(t)]$
- $5 \int [M(t) + L(t)] dt$
- $5[M + L]$
- $5M'(18)$

Examples that do not earn the point include:

- $5[M(t) + L'(t)]$
- $5 \int M(t) dt$
- $5M' + L$
- $5L'(18) + t$

Special case: $5L'(18) + M'(18)$ is an error in parentheses, so will earn the first point but is not eligible to earn the second point.

If the constant multiple is missing or incorrect, the sum must be correct. $k[L'(18) + M'(18)]$ for any nonzero constant k will earn the first point.

The second point is for the correct expression.

$5L'(18) + 5M'(18)$ (or an equivalent expression) earns both points.

Inclusion of units (correct or incorrect) will not affect the score.

A completely correct expression with one additional constant term added/subtracted would earn a just the first point. For example, $5L'(18) + 5M'(18) + 5(2)$ earns only the first point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

Constant multiple or sum

4-1-1

☐☐

Expression

Solution:

The rate of change, in dollars per minute, of the total money raised at time $t = 18$ minutes is $5M'(18) + 5L'(18)$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is earned for a correct statement identifying specific inconsistencies between the given differential equation and the given slope field. There are many such correct statements, such as:

- The response references a specific point on the y -axis (such as $(0, 4)$), and indicates that the given slope field shows a slope of zero, yet the differential equation would yield a positive slope.
- The response references a pattern of slopes, and explains why that pattern does not fit our differential equation (e.g. “the slope field shows slopes that change as t changes, but our differential equation depends only on y , so slopes should remain constant for a given value of y as t changes.”)

Identifying a mismatch between the differential equation and the slope field at one point is sufficient to earn the point.

Response is not penalized for referencing “ x ” instead of “ t ”.

If the response contains any incorrect statement, the response will not earn the point.

Vague answers which have the potential to be incorrect will not earn the point, nor will responses that present information beyond what we are given. For example:

- Responses that specify that the slope field shows a particular nonzero slope at a particular point will not earn the point, as this cannot actually be determined visually (i.e., the response should not indicate that at $(2, 2)$ the slope shown on the slope field is -1)
- Responses that state a specific slope value on the slope field for a general value of x or y that are not correct for all points with that x or y value do not earn the point. (e.g. “the slope field shows a slope of 0 when $y = 4$ ” is an incorrect statement)

A response may involve a solution to the differential equation, with an argument based on the resulting function. If done correctly, and presented with a correct argument, this will earn the point.



0

1

4-1-1

The student response accurately includes the criteria below.

☐ Explanation

Solution:

When $y = 0$, $\frac{dy}{dt} = 0$. However, in the given slope field, for $t > 0$, the slopes of the line segments for $y = 0$ are nonzero.

4-1-1

67. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (minutes)	0	8	16	24	32
$D(t)$ (meters)	0	1500	2800	3900	5000

A student on the school track and field team is running laps around a track. For time t minutes, $0 \leq t \leq 32$, the total distance run, in meters, is modeled by a twice-differentiable function D . Values of $D(t)$ at various times t are shown in the table above.

(a) Use a midpoint Riemann sum with the two subintervals indicated by the table to approximate

$\int_0^{32} D(t) \, dt$. Show the computations that lead to your answer.

(b) Find the value of $\int_0^{16} D'(2t) \, dt$. Show the computations that lead to your answer.

(c) Using correct units, approximate $D'(16)$ using the average rate of change of D over the interval $8 \leq t \leq 24$.

(d) Explain the meaning of $D'(16)$ in the context of the problem.

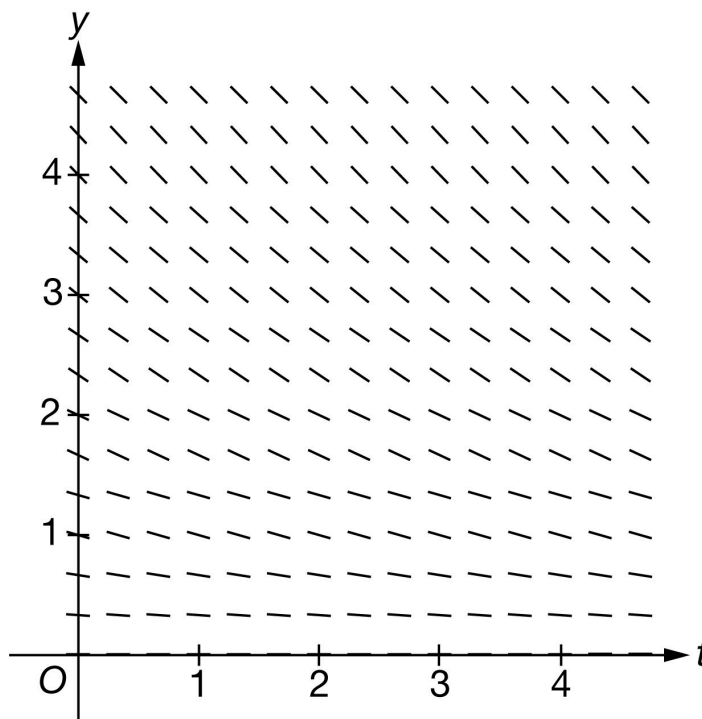
(e) It is known that $D'(24) = 140$. Write an equation for the line tangent to the graph of $y = D(t)$ at $t = 24$.

(f) It is known that $D''(t) < 0$ for $24 \leq t \leq 31$. If $D(30)$ is approximated using the equation from part (e), would the approximation be an overestimate or underestimate for the actual value of $D(30)$? Give a reason for your answer.

(g) A second student is raising funds by running laps for 60 minutes. For $0 \leq t \leq 60$, the total number of laps run by this student is modeled by $L(t)$, a differentiable function of t . For each lap run, the student raises 4 dollars. Write an expression for the rate of change, in dollars per minute, of the total money raised at time $t = 45$ minutes.

4-1-1

(h) Explain why the following slope field cannot represent the differential equation $\frac{dy}{dt} = 0.25y$.



Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point (Form of midpoint sum): Philosophy: The response should provide clear evidence adding the area of two rectangles whose heights are found using the midpoint of the interval.

Acceptable evidence:

$$\cdot 16 \cdot (D(8) + D(24)) \text{ or } 16 \cdot D(8) + 16 \cdot D(24)$$

$$\cdot (16 - 0) \cdot 1500 + (32 - 16) \cdot 3900$$

$$\cdot 16 \cdot (1500 + 3900) \text{ or } 16 \cdot 1500 + 16 \cdot 3900$$

To earn the point the response can have at most one mistake in the expression or equation. So, expressions of the form $\frac{16}{b} \cdot (D(8) + D(24))$, for any nonzero value of b would earn the point. Note: The value $\frac{16}{b}$ might be distributed across the sum.

Expressions of the form $\frac{a}{b} \cdot (D(8) + D(24))$, for $a \neq 16$ will not earn this point.

Once the first point is earned, it cannot be lost.

Second Point (Answer)

4-1-1

To earn the point, the answer must be correct. It does not need to be simplified, but if it is, the simplification must be correct.

The answer point cannot be earned unless the first (midpoint sum) point was earned.

Minimal complete answers that earn both points:

$$\cdot 16 \cdot (1500 + 3900)$$

$$\cdot 16 \cdot 1500 + 16 \cdot 3900$$



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Form of midpoint sum
- ☐ Answer

Solution:

$$\begin{aligned} \int_0^{32} D(t) dt &\approx D(8)(16 - 0) + D(24)(32 - 16) \\ &= 1500 \cdot 16 + 3900 \cdot 16 = 86,400 \end{aligned}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point (Substitution) Philosophy: This point is earned when the response has all 3 components of the substitution done correctly.

Uses $u = 2t$, perhaps implicitly (any variable could be used for u)

Uses the correct constant of $\frac{1}{2}$

Uses the correct limits for their substitution.

· The limits might not be seen until the final evaluation, and at that point, response may have returned to the variable t .

· Limits for t attached to an integral involving u or vice versa will not earn this substitution point but could still earn the answer point, if the numerical answer is correct.

4-1-1

Second point: (Fundamental Theorem of Calculus) Philosophy: This point is earned when the FTC has been applied correctly to either $\int D'(2t) dt$ or $\int D'(u) du$.

The point is earned for any of the following:

- The equation $\int D'(2t) dt = D(2t)$ or $\int D'(u) du = D(u)$
- $\frac{1}{2}D(2t)\big|_0^{16}$ or $\frac{1}{2}D(u)\big|_0^{32}$
- $D(32) - D(0)$, $D(32)$, or $D(0)$

This point can be earned if the substitution point is not.

Note: $\int_a^b D'(u) du = D(b) - D(a)$ does not earn the point. The FTC point is for the response communicating that the antiderivative of D' is D .

Third Point (Answer)

Must be the correct numerical answer.

Must have earned the second point for the FTC.

Might have not earned the first point for substitution because of an incorrect use of limits, but will still earn the third point for the correct answer.

$D(32)$ and $D(0)$ must be evaluated; the expression $\frac{1}{2}(D(32) - D(0))$ by itself does not earn the third point.

The answer may be unsimplified, but if simplified it must be done correctly.

Scoring common responses:

The minimal work required to earn all three points is $\frac{1}{2}(5000 - 0)$ or $\frac{1}{2}(5000)$.

The following responses would earn the first two points and be eligible for the answer point if the response continues:

- $\frac{1}{2}D(2t)\big|_0^{16}$
- $\frac{1}{2}(D(32) - D(0))$

Responses of only the following would earn only the second point:

- $D(32) - D(0)$
- $D(16) - D(0)$

4-1-1

$$\cdot D(32)$$

$$\cdot \int_0^{16} D'(u) \, du = D(16)$$

$$\cdot D(2t)|_0^{16} \text{ or } D(u)|_0^{32}$$

The following responses would earn no points:

· 2500 in the absence of one of the first two points.

$$\cdot \int_0^{16} D'(2t) \, dt = D(16)$$



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = 2t$. Then $du = 2 \, dt$.

When $t = 0$, $u = 0$.

When $t = 16$, $u = 32$.

$$\int_0^{16} D'(2t) \, dt = \frac{1}{2} \int_0^{32} D'(u) \, du$$

$$= \frac{1}{2}(D(32) - D(0))$$

$$= \frac{1}{2}(5000 - 0) = 2500$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point (Approximation)

The response must be a difference and a quotient.

4-1-1

The answer does not need to be simplified, but if it is simplified it must be done correctly.

The minimal work that must be shown to earn this point is either $\frac{3900-1500}{16}$ or $\frac{2400}{24-8}$.

Second point (Units)

The units must be attached to a value (either correct or incorrect).



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$D'(16) \approx \frac{D(24)-D(8)}{24-8} = \frac{3900-1500}{16}$$

= 150 meters per minute

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The interpretation must include three components

- indicate the rate of change, the velocity, or the speed
- the specific context of the total distance run
- the specific point in time, $t = 16$.

Responses do not need to include correct units or any units.

The phrase “the student is running” may be used in place of “the total distance run is changing”.

The phrase “the student is running” is not enough. The response must attach “running” to the word “rate”, as in “the student is running at a rate of”.

The response is not earned for “the rate of total distance run” as this does not indicate a change in the rate of distance.

“the rate of change of D ” without specifying that D is the total distance run, does not earn the point.

4-1-1

Acceptable ways to provide the specific point in time:

- At minute 16 or after 16 minutes.
- At $t = 16$ (missing the unit of time)
- At 16 seconds (incorrect unit of time)

A response that clearly indicates an interval of time (e.g. “during the first 16 minutes” or “from minute 16 on”) does not earn the point.



0	1
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The student response accurately includes the criteria below.

☐ Explanation

Solution:

$D'(16)$ is the rate at which the total distance run is changing, in meters per minute, at time $t = 16$ minutes.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The use of $D(t)$ in place of y earns the point.

The use of x in place of t earns the point.

Tangent lines do not need to be simplified, but any simplification must be done correctly.

Any correct form of the tangent line earns the point. For example:

- $y = 3900 + 140(t - 24)$
- $y - 3900 = 140(t - 24)$
- $y = 140t - 3360 + 3900$
- $y = 140t + 540$



0	1
---	---

The student response accurately includes the criteria below.

4-1-1

☐ Tangent line equation
Solution:

$$y = D'(24)(t - 24) + D(24)$$

$$y = 140(t - 24) + 3900$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point the response must include $D''(t) < 0$ and overestimate.

“ $D(t)$ is concave down” can be used instead of $D''(t) < 0$.

Information about $D'(t)$ by itself without reference to $D''(t) < 0$ is not sufficient and does not earn this point.

“Overestimate because it is concave down” is not sufficient by itself because of the use of “it”. The response must clearly, directly or indirectly, indicate to what “it” is referring.

If a response makes the correct conclusion and states $D''(t) < 0$ and then goes on and makes an incorrect statement or a statement that cannot be verified will not earn this point.

For example: “ $D'(t)$ is concave down, overestimate” does not earn this point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason
Solution:

Because $D''(t) < 0$, the tangent line approximation would produce an overestimate.

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The correct expression of only $4 \cdot L'(45)$ earns this point.

No justifications or explanations are required.

The expression $4 \cdot D'(45)$ does not earn this point.

4-1-1

Any incorrect statement such as, $L'(45) = 4 \cdot D'(45)$, does not earn this point.



0

1

The student response accurately includes the criteria below.

☐ Expression

Solution:

The rate of change, in dollars per minute, of the total money raised at time $t = 45$ minutes is $4L'(45)$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Philosophy: Response clearly references what part of the differential equation is different from the slope field and the response needs only to clearly and explicitly reference one of them.

The response does not need to explicitly reference both the differential equation and the given slope field.

The distinction can be made with explicit reference to one of the two things and an implicit reference to the other

For example:

· “ $\frac{dy}{dt}$ is positive, not negative”. We will infer that negative is referring to the slopes shown in the slope field.

· “This would be correct for a negative derivative, but $\frac{dy}{dt}$ is positive.” We will read “This” as “the slope field” in this sentence.

· “All of the slopes in the slope field are negative, not positive.” We will infer that the values of $\frac{dy}{dt}$ at points in the first quadrant are positive.

A response that lacks even an implicit reference to either the differential equation or the slope field will not earn the point. For example:

· “ $\frac{dy}{dt}$ is positive” lacks any reference to the slope field and so will not earn the point.

· “All the slopes in the graph are negative” lacks any reference to the differential equation, and so will not earn the point.

· Vague answers such as “The slope field has negative points” or “The field is decreasing” will not earn the point.

4-1-1



0

1

The student response accurately includes the criteria below.

Solution:

When $y > 0$, $\frac{dy}{dt} > 0$. However, in the given slope field, the slopes of the line segments for $y > 0$ are negative.

4-1-1

68. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (seconds)	0	3	6	9	12
$p(t)$ (feet)	-5	-8.5	-12.5	-15	-20

The position of a particle, A , moving along the x -axis is modeled by a twice-differentiable function p , where time t is measured in seconds and $p(t)$ is measured in feet. Selected values of $p(t)$ are shown in the table above.

- (a) Use a midpoint Riemann sum with two subintervals of equal length and values from the table to approximate the value of $\int_0^{12} p(t) \, dt$. Show the computations that lead to your answer.
- (b) Use the data in the table to approximate $p'(5)$ using the average rate of change of $p(t)$ over the interval $3 \leq t \leq 6$. Show the computations that lead to your answer. Indicate units of measure.
- (c) Interpret the meaning of $p'(5)$ in the context of the problem.
- (d) Must there exist a value of k , for $0 < k < 12$, such that $p'(k)$ is equal to the average rate of change of p over the interval $0 \leq x \leq 12$? Justify your answer.
- (e) Find $\int_0^3 p'(2t + 6) \, dt$. Show the computations that lead to your answer.
- (f) Let $h(x) = \int_1^{2x} p(3t) \, dt$. Find $h'(2)$. Show the computations that lead to your answer.
- (g) The velocity of a second particle, Q , can be modeled by a twice-differentiable function g . It is known that $g(3) = -1$, $g'(3) = -5$, and $g''(3) = 7$. Is the speed of particle Q increasing or decreasing at time $t = 3$? Give a reason for your answer.
- (h) Let $y = f(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = 12x^2 - y^3$ with initial condition $f\left(\frac{1}{2}\right) = -2$. Write an equation for the line tangent to the graph of f at the point $\left(\frac{1}{2}, -2\right)$.

Part A

4-1-1

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for correct midpoint Riemann sum, with appropriate values taken from the table.

Need to see sum of two terms, each as a product of two factors, i.e. $\text{base}_1 \cdot \text{height}_1 + \text{base}_2 \cdot \text{height}_2$

$$p(3) \cdot (6 - 0) + p(9) \cdot (12 - 6) \text{ or } p(3) \cdot (6) + p(9) \cdot (6)$$

If at least three of the four base/height values are correct in the sum of products, the expression earns the first point. However, if one of these values is incorrect, the response does not earn the second point.

Second Point

Earned for an answer of -141 or equivalent.

Can be earned without the first point for the Riemann sum setup demonstration. With credible evidence that a midpoint Riemann sum attempt was made, but not enough work shown, this point can be earned in the presence of the correct answer. For example, a sum of two terms with no evidence of products $(-51 - 90)$, or a graph with areas labeled and summed to -141 , can earn this point.

Examples:

- $-141 = -51 - 90$ by itself earns second point, not first point
- -141 declared, with $-8.5 \cdot 6$ and $-15 \cdot 6$ identified earns second point, not first point (perhaps graphically)
- $k(-8.5 \cdot 6 - 15 \cdot 6)$, nonzero constant $k \neq 1$ by itself earns first point, not second point
- $p(3) \cdot (6 - 0) + p(9) \cdot (12 - 6)$ by itself earns first point, not second point



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Form of midpoint Riemann sum
- ☐ Answer

Solution:

$$\int_0^{12} p(t) dt \approx p(3)(6 - 0) + p(9)(12 - 6)$$

$$= -8.5 \cdot 6 + (-15) \cdot 6 = -141$$

4-1-1

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for a correct difference quotient, with appropriate values from the table.

The response must show both a difference and a quotient.

$\frac{p(6)-p(3)}{6-3}$ by itself doesn't earn first point; however $\frac{-12.5-(-8.5)}{6-3}$ or $A \frac{-12.5-(-8.5)}{3}$ by itself earns the first point

Second Point

Earned for correct units of ft per sec or ft/sec or f/s.

Correct units must be tied to a number to earn second point.

Correct units must appear in part (b); correct units in part (c) only do not earn the second point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Approximation

☐ Units

Solution:

$$p'(5) \approx \frac{-12.5 - (-8.5)}{6-3} = \frac{-4}{3} \text{ feet per second}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Philosophy: Earned for correct $p'(5)$ interpretation.

Need two components in the response:

· “rate of change of position” or “velocity” of the particle

· at $t = 5$

4-1-1

Correct ways to describe the rate of change include:

- rate at which position (of the particle) is changing/decreasing
- rate at which particle is moving left
- instantaneous velocity or velocity of the particle

Ambiguous/incorrect ways to describe rate include:

- rate at which particle is changing/decreasing
- rate at which particle is moving

The response does not need to include the value of the approximation found in part (b), but will not be penalized for doing so (e.g. "the rate at which the particle's position is decreasing is $\frac{4}{3}$ at time $t = 5$ " earns the point).

If the response includes the approximation value from part (b) and makes an incorrect statement such as "the rate at which the position of the particle is decreasing is $-\frac{4}{3}$ at time $t = 5$ ", the point is not earned.

Units are not required in part (c). If units are presented for t , they must be in seconds.



0	1
---	---

The student response accurately includes the criteria below.

☐ Interpretation

Solution:

$p'(5)$ is the rate at which the position of the particle is changing, in feet per second, at time $t = 5$ seconds.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for both Mean Value Theorem (MVT) conditions with the appropriate implication that p is continuous because we know that p is differentiable.

Ways to earn first point include:

- p differentiable $\Rightarrow p$ continuous
- p differentiable, hence p continuous

4-1-1

· p differentiable, so p continuous

p differentiable and p continuous does not earn first point.

Second Point

Philosophy: Earned for concluding “yes” and invoking Mean Value Theorem or its equivalent formulation, with consideration of the continuity of p .

MVT title does not have to be stated to earn second point.

This point is not earned if other theorems (IVT, etc.) are invoked.

If presented, the value for $p'(c) = -\frac{5}{4}$ must be correct to earn second point.

Examples:

- “By MVT, yes since p differentiable” earns neither point
- “By MVT, yes since p continuous” earns second point only
- “By MVT, yes since p differentiable and continuous” earns second point only
- “By MVT, yes since p differentiable hence continuous” earns both points
- “By MVT, yes since p twice differentiable hence continuous” earns both points
- “By MVT, yes since function differentiable hence continuous” earns both points



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Conditions
- ☐ Conclusion using Mean Value Theorem

Solution:

p is differentiable. $\Rightarrow p$ is continuous.

Therefore, the Mean Value Theorem can be applied to p on the interval $0 \leq x \leq 12$ to guarantee that there exists a value of k , for $0 < k < 12$, such that $p'(k)$ equals the average rate of change of p over the interval $0 \leq x \leq 12$.

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

4-1-1

First Point

Philosophy: Earned for correct substitution.

A correct substitution requires three elements (with consistent use of u or another substitution variable):

- $u = 2t + 6$ (can be explicit or implicit)
- $du = 2 dt$ (yielding integral coefficient of $\frac{1}{2}$)
- Correct integral limits throughout, with t from 0 to 3 and u from 6 to 12

The substitution could be implicit: $\int_0^3 p'(2t + 6) dt = \frac{1}{2}p(2t + 6) \Big|_0^3$

Omitting the coefficient of $\frac{1}{2}$ or using some other constant will not earn the substitution or answer points; the response could still earn the second point.

In the third element above, if response shows an otherwise correct substitution with unchanged limits of integration on a definite integral from 0 to 3, then point is not earned.

Alternately, the integral can be worked with as an indefinite integral with u returned to $2t + 6$ in the antiderivative before evaluating as a means to eliminate the need to change limits. Such a response can earn the first point, if handled correctly.

Second Point

Philosophy: Earned for correct Fundamental Theorem of Calculus application, with evidence of connection between $\int p'(u)$ and $p(u)$.

Ways to earn second point include:

- $k(p(12) - p(6))$ or $k(p(2 \cdot 3 + 6) - p(2 \cdot 0 + 6))$ for k a nonzero constant
- $k p(2t + 6) \Big|_0^3$ or $k p(u) \Big|_6^{12}$ for k a nonzero constant
- $\int p'(2t + 6) dt = k p(2t + 6)$ or $\int p'(u) du = k p(u)$ for any nonzero constant k , with or without appropriate limits of integration on either integral

The first two points can be earned simultaneously by responses such as $\int_0^3 p'(2t + 6) dt = \frac{1}{2}p(2t + 6) \Big|_0^3$

Third Point

Earned only for an answer of -3.75 or equivalent value.

The value does not need to be simplified, but if it is, the simplification must be correct to earn this point.

4-1-1

Scoring Examples:

$$\cdot \frac{1}{2} \int p'(u) \, du = \frac{1}{2} p(u) \text{ earns } 0 - 1 - 0$$

$$\cdot \frac{1}{2} \int p'(u) \, du = \frac{1}{2} p(u) \Big|_0^3 = \frac{1}{2} (p(3) - p(0)) \text{ earns } 0 - 1 - 0$$

$$\cdot \int p'(u) \, du = p(u) \Big|_6^{12} = (p(12) - p(6)) = -7.5 \text{ earns } 0 - 1 - 0$$

$$\cdot 2 \int p'(u) \, du = 2p(u) \Big|_0^3 = 2[-8.5 + 5] \text{ earns } 0 - 1 - 0$$

$$\cdot \int_0^3 p'(2t + 6) \, dt = p(t^2 + 6t) \Big|_0^3 = p(27) - p(0) \text{ earns } 0 - 0 - 0$$

$$\cdot \frac{1}{2} \int_0^3 p'(u) \, du = \frac{1}{2} (p(12) - p(6)) = -3.75 \text{ earns } 0 - 1 - 1$$

$$\cdot \int_0^3 p'(2t + 6) \, dt = \frac{1}{2} (p(12) - p(6)) = -3.75 \text{ earns } 1 - 1 - 1$$

$$\cdot \int_0^3 p'(2t + 6) \, dt = \frac{1}{2} p(u) \Big|_6^{12} = \frac{1}{2} (p(12) - p(6)) = -3.75 \text{ earns } 1 - 1 - 1$$

$$\cdot \int_0^3 p'(2t + 6) \, dt = \frac{1}{2} p(2t + 6) \Big|_0^3 = \frac{1}{2} (p(12) - p(6)) \text{ earns } 1 - 1 - 0$$

$$\cdot \frac{1}{2} \int p'(u) = \frac{1}{2} p(u); \text{ and } \frac{1}{2} p(2t + 6) \Big|_0^3 = \frac{1}{2} (-20 - (12.5)) \text{ earns } 1 - 1 - 1$$



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:Let $u = 2t + 6$. Then $du = 2 \, dt$.When $t = 0$, $u = 6$.

4-1-1

When $t = 3$, $u = 12$.

$$\begin{aligned}\int_0^3 p'(2t + 6) dt &= \frac{1}{2} \int_6^{12} p'(u) du \\ &= \frac{1}{2}(p(12) - p(6)) \\ &= \frac{1}{2}(-20 - (-12.5)) = -3.75\end{aligned}$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for correct Fundamental Theorem of Calculus application, resulting in $a \cdot p(6x)$ for some nonzero constant a .

Ways to earn first point include responses of $p(3 \cdot 2x)$ or $p(6x)$ (Note $p(6t)$ would also earn this point.)

The expression $p(3 \cdot 2x) - p(3 \cdot 1)$ or $(p(3 \cdot 2x) - p(3 \cdot 1)) \cdot 3$ earns this point. The $p(3 \cdot 1)$ is viewed as a chain rule error.

Any response correctly presenting and evaluating the derivative of the integral in terms of the substitution $u = 3t$ earns first point. Part of this response may include $\int_1^{2x} p(3t) dt = \frac{1}{3} \int_3^{6x} p(u) du$.

Second Point

Philosophy: This point is earned for correctly applying the chain rule to find $h'(x)$ or $h'(2)$. That is accomplished by multiplying the term involving $p(x)$ by 2.

Can only be earned if response earned first (FTC) point.

The first two points can be, and often are, earned simultaneously.

The point is not earned if the presented derivative is $a \cdot p(6x)$ for any $a \neq 2$.

The point is not earned for responses of $2p(3t) \big|_1^{2x}$ or $2(p(6x) - p(b))$, for any value of b .

Third Point

Can only be earned if response earned first (FTC) and second (chain rule) points.

Earned only for answer of -40 or equivalent value.

The value does not need to be simplified, but if it is, the simplification must be correct to earn this point.

Scoring examples:

4-1-1

$$\cdot p(3 \cdot 2x) - p(3); p(12) - p(3) = -20 + 8.5 \text{ earns } 1 - 0 - 0$$

$$\cdot p(3 \cdot 2x) \cdot 2 - p(3) \cdot 0; -40 \text{ earns } 1 - 1 - 1$$

$$\cdot 2p(3t) \Big|_1^{2x} = 2(p(12) - p(3)) \text{ earns } 1 - 0 - 0$$

$$\cdot p(3x) \cdot 2; p(6) \cdot 2 = -12.5 \cdot 2 \text{ earns } 0 - 0 - 0$$

$$\cdot p(6x) \cdot 2; p(12) \cdot 2 \text{ earns } 1 - 1 - 0$$



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Fundamental Theorem of Calculus
- ☐ Chain rule
- ☐ Answer

Solution:

$$h'(x) = \frac{d}{dx} \int_1^{2x} p(3t) dt = p(3 \cdot 2x) \cdot 2$$

$$h'(2) = p(12) \cdot 2 = -20 \cdot 2 = -40$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for narrowing consideration of given information to g and g' .

g and g' need not be evaluated to earn the first point.

Considering velocity and acceleration not sufficient to earn first point, unless acceleration is linked to g' .

If response considers g and g' and g'' , then eventually g'' must be eliminated to earn first point.

Second Point

Philosophy: Earned for “increasing” with reason, using the opposite signs of g and g' .

Ways to earn second point include:

$$\cdot \text{“increasing since } g < 0 \text{ and } g' < 0 \text{”}$$

4-1-1

- “particle speeding up since g and g' have the same signs”

Can only be earned if response earned first point.

A response that calls out all three given values ($g(3)$, $g'(3)$, and $g''(3)$) initially might still earn one or both points. If the response goes on to focus upon/reason from the values of g and g' only, both points can be earned.

Examples:

- “increasing since velocity and acceleration have the same signs” earns neither point
- “increasing since velocity < 0 and acceleration < 0 ” earns neither point
- “increasing since $(-1) \cdot (-5) = 5 > 0$ ” earns both points



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Narrows consideration to g and g'
- ☐ Answer with reason

Solution:

Because $g(3) = -1 < 0$, the velocity of the particle is negative. Because $g'(3) = -5 < 0$, the acceleration of the particle is negative. Because the velocity and acceleration have the same signs, the speed of the particle is increasing at time $t = 3$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Philosophy: Earned for correct tangent line equation, written in any form.

Correct responses include:

- $y = 11\left(x - \frac{1}{2}\right) - 2$ or $y + 2 = 11\left(x - \frac{1}{2}\right)$
- $y = 11x - \frac{15}{2}$
- $y = \left(12 \cdot \left(\frac{1}{2}\right)^2 - (-2)^3\right)\left(x - \frac{1}{2}\right) - 2$

Incorrect responses include:

- $y = (12x^2 - y^3)\left(x - \frac{1}{2}\right) - 2$

4-1-1

$$y = 11x - 2$$

$f(x) = 11x - \frac{15}{2}$. ($f(x)$ was defined in this problem as the particular solution to the differential equation, so it cannot also be a tangent line.



0	1
---	---

The student response accurately includes the criteria below.

☐ Tangent line equation

Solution:

$$\left. \frac{dy}{dx} \right|_{\left(\frac{1}{2}, -2\right)} = 12 \cdot \left(\frac{1}{2}\right)^2 - (-2)^3 = 11$$

$$y = 11\left(x - \frac{1}{2}\right) - 2$$

4-1-1

69. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (seconds)	0	4	8	12	16
$p(t)$ (feet)	3	9	16.5	26	36

The position of a particle, A , moving along the x -axis is modeled by a twice-differentiable function p , where time t is measured in seconds and $p(t)$ is measured in feet. Selected values of $p(t)$ are shown in the table above.

- (a) Use a midpoint Riemann sum with two subintervals of equal length and values from the table to approximate the value of $\int_0^{16} p(t) \, dt$. Show the computations that lead to your answer.
- (b) Use the data in the table to approximate $p'(11)$ using the average rate of change of $p(t)$ over the interval $8 \leq t \leq 12$. Show the computations that lead to your answer. Indicate units of measure.
- (c) Interpret the meaning of $p'(11)$ in the context of the problem.
- (d) Must there exist a value of k , for $0 < k < 16$, such that $p'(k)$ is equal to the average rate of change of p over the interval $0 \leq x \leq 16$? Justify your answer.
- (e) Find $\int_0^4 p'(2t + 4) \, dt$. Show the computations that lead to your answer.
- (f) Let $h(x) = \int_1^{4x} p(2t) \, dt$. Find $h'(1)$. Show the computations that lead to your answer.
- (g) The velocity of a second particle, Q , can be modeled by a twice-differentiable function g . It is known that $g(3) = 2$, $g'(3) = -5$, and $g''(3) = 11$. Is the speed of particle Q increasing or decreasing at time $t = 3$? Give a reason for your answer.
- (h) Let $y = f(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = -xy^2 + 3$ with initial condition $f(5) = 2$. Write an equation for the line tangent to the graph of f at the point $(5, 2)$.

Part A

4-1-1

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for correct Midpoint Riemann sum, with appropriate values taken from the table.

Need to see sum of two terms, each as a product of two factors, i.e. $\text{base}_1 \cdot \text{height}_1 + \text{base}_2 \cdot \text{height}_2$

$p(4) \cdot (8 - 0) + p(12) \cdot (16 - 8)$ or $p(3) \cdot (6) + p(9) \cdot (6)$ earns first point.

If at least three of the four base/height values are correct in the sum of products, the expression earns the first point.

Second Point

280 or equivalent

Can be earned without first point with Riemann sum setup demonstration. With credible evidence that a midpoint Riemann sum attempt was made without not enough work shown, this point can be earned in the presence of the correct answer. For example, a sum of two terms with no evidence of products ($72 + 208$) or a graph with areas labeled and summed to 280 or equivalent, can earn this point.

Examples:

- $280 = 72 + 208$ by itself earns second point, not first point
- 280 declared, with $9 \cdot 8$ and $26 \cdot 8$ identified (perhaps graphically) earns second point, not first point
- $k(9 \cdot 8 + 26 \cdot 8)$, nonzero constant $k \neq 1$ by itself earns first point, not second point
- $p(4) \cdot (8) + p(12) \cdot (16 - 8)$ by itself earns first point, not second point



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Form of midpoint Riemann sum
- ☐ Answer

Solution:

$$\int_0^{16} p(t) dt \approx p(4)(8 - 0) + p(12)(16 - 8)$$

$$= 9 \cdot 8 + 26 \cdot 8 = 280$$

4-1-1

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for correct difference quotient, with appropriate values from the table.

Need to see both a difference and a quotient

$\frac{p(12)-p(8)}{12-8}$ by itself doesn't earn first point; however $\frac{26-16.5}{12-8}$ or $\frac{26-16.5}{4}$ by itself earns first point

Second Point

Earned for correct units of ft per sec or ft/sec or f/s.

Correct units must be tied to a number to earn second point.

Correct units must appear in part (b); correct units in part (c) only doesn't earn second point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$p'(11) \approx \frac{26-16.5}{12-8} = \frac{9.5}{4} = 2.375 \text{ feet per second}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Philosophy: Earned for correct $p'(11)$ interpretation.

Need two components in response:

- “rate of change of position” or “velocity” of the particle
- at $t = 11$

Correct ways to describe rate at time at $t = 11$ include:

4-1-1

- Rate at which position is changing/increasing
- Rate at which particle is moving right
- Instantaneous velocity or velocity of particle

Ambiguous/incorrect ways to describe rate include:

- Rate at which particle is changing/increasing
- Rate at which particle is moving

Units are not required in part (c). If units are presented for t , they must be in seconds.



0	1
---	---

The student response accurately includes the criteria below.

☐ Interpretation

Solution:

$p'(11)$ is the rate at which the position of the particle is changing, in feet per second, at time $t = 11$ seconds.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for both Mean Value Theorem (MVT) conditions with the appropriate implication that p is continuous because we know that p is differentiable.

Ways to earn first point include:

- p differentiable $\Rightarrow p$ continuous
- p differentiable, hence p continuous
- p differentiable, so p continuous

p differentiable and p continuous does not earn first point.

Second Point

Philosophy: Earned for concluding “yes” and invoking Mean Value Theorem or its equivalent formulation, with consideration of the continuity of p .

4-1-1

MVT title does not have to be stated to earn second point.

This point is not earned if other theorems (IVT, etc.) are invoked.

If presented, the value for $p'(k) = \frac{33.5}{16}$ must be correct to earn second point.

Examples:

- “By MVT, yes since p differentiable” earns neither point
- “By MVT, yes since p continuous” earns second point only
- “By MVT, yes since p differentiable and continuous” earns second point only
- “By MVT, yes since p differentiable hence continuous” earns both points
- “By MVT, yes since p twice differentiable hence continuous” earns both points
- “By MVT, yes since function differentiable hence continuous” earns both points



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Conditions
- ☐ Conclusion using Mean Value Theorem

Solution:

p is differentiable. $\Rightarrow p$ is continuous.

Therefore, the Mean Value Theorem can be applied to p on the interval $0 \leq x \leq 16$ to guarantee that there exists a value of k , for $0 < k < 16$, such that $p'(k)$ equals the average rate of change of p over the interval $0 \leq x \leq 16$.

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for correct substitution application.

Need three elements in response (with consistent use of u or another substitution variable):

- $u = 2t + 4$ (can be explicit or implicit)

4-1-1

- $du = 2 dt$ (yielding integral coefficient of $\frac{1}{2}$)
- Correct integral limits throughout, with t from 0 to 4 and u from 4 to 12

Omitting the coefficient of $\frac{1}{2}$ or using some other value will not earn first point; response could earn second point, but not third point.

In the third element above, if response shows an otherwise correct substitution with unchanged limits of integration on a definite integral from 0 to 4, then point is not earned.

Alternately, the integral can be worked with as an indefinite integral with u returned to $2t + 4$ in the antiderivative before evaluating as a means to eliminate the need to change limits. Such a response can earn the first point if handled correctly.

For example: $\int_0^4 p'(2t + 4) dt = \frac{1}{2}p(2t + 4) \Big|_0^4$ earns the substitution point.

Second Point

Philosophy: Earned for correct Fundamental Theorem of Calculus application, with evidence of connection between $\int p'(u)$ and $p(u)$.

Ways to earn second point include:

- $k(p(12) - p(4))$ or $k(p(2 \cdot 4 + 4) - p(2 \cdot 0 + 4))$ for k a nonzero constant
- $k p(2t + 4) \Big|_0^3$ or $k p(u) \Big|_4^{12}$ for k a nonzero constant
- $\int p'(2t + 4) dt = k p(2t + 4)$ or $\int p'(u) du = k p(u)$ for any nonzero constant k , with or without appropriate limits of integration on either integral

First and second points can be earned simultaneously by responses such as $\int_0^4 p'(2t + 4) dt = \frac{1}{2}p(2t + 4) \Big|_0^4$

Third Point

Earned only for an answer of 8.5 or equivalent value.

Value does not need to be simplified, but if it is, the simplification must be correct to earn this point.

Scoring Examples:

- $\frac{1}{2} \int p'(u) du = \frac{1}{2}p(u)$ earns 0 – 1 – 0
- $\frac{1}{2} \int p'(u) du = \frac{1}{2}p(u) \Big|_0^4 = \frac{1}{2}(p(4) - p(0))$ earns 0 – 1 – 0

4-1-1

$$\cdot \int p'(u) \, du = p(u) \Big|_4^{12} = (p(12) - p(4)) = 17 \text{ earns } 0 - 1 - 0$$

$$\cdot 2 \int p'(u) \, du = 2p(u) \Big|_0^4 = 2[26 - 9] \text{ earns } 0 - 1 - 0$$

$$\cdot \int_0^4 p'(2t + 4) \, dt = p(t^2 + 4t) \Big|_0^4 = p(32) - p(0) \text{ earns } 0 - 0 - 0$$

$$\cdot \frac{1}{2} \int_0^4 p'(u) \, du = \frac{1}{2}(p(12) - p(4)) = 8.5 \text{ earns } 0 - 1 - 1$$

$$\cdot \int_0^4 p'(2t + 4) \, dt = \frac{1}{2}(p(12) - p(4)) = 8.5 \text{ earns } 1 - 1 - 1$$

$$\cdot \int_0^4 p'(2t + 4) \, dt = \frac{1}{2}p(u) \Big|_4^{12} = \frac{1}{2}(p(12) - p(4)) = 8.5 \text{ earns } 1 - 1 - 1$$

$$\cdot \int_0^4 p'(2t + 4) \, dt = \frac{1}{2}p(2t + 4) \Big|_0^4 = \frac{1}{2}(p(12) - p(4)) \text{ earns } 1 - 1 - 0$$

$$\cdot \frac{1}{2} \int p'(u) = \frac{1}{2}p(u); \text{ and } \frac{1}{2}p(2t + 4) \Big|_0^4 = \frac{1}{2}(26 - 9) \text{ earns } 1 - 1 - 1$$



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = 2t + 4$. Then $du = 2 \, dt$.

When $t = 0$, $u = 4$.

When $t = 4$, $u = 12$.

$$\begin{aligned} \int_0^4 p'(2t + 4) \, dt &= \frac{1}{2} \int_4^{12} p'(u) \, du \\ &= \frac{1}{2}(p(12) - p(4)) \end{aligned}$$

4-1-1

$$= \frac{1}{2}(26 - 9) = 8.5$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for correct Fundamental Theorem of Calculus application, resulting in $a \cdot p(8x)$ for some nonzero constant a .

Ways to earn first point include responses of $p(2 \cdot 4x)$ or $p(8x)$ (Note $p(8t)$ would also earn this point.)

The expression $p(2 \cdot 4x) - p(2 \cdot 1)$ or $(p(2 \cdot 4x) - p(2 \cdot 1)) \cdot 4$ earns this point. The $p(2 \cdot 1)$ is viewed as a chain rule error.

Any response correctly presenting and evaluating the derivative of the integral in terms of the substitution $u = 2t$ earns first point. Part of this response may include $\int_1^{4x} p(2t) dt = \frac{1}{3} \int_2^{8x} p(u) du$.

Second Point

Philosophy: This point is earned for correctly applying the chain rule to find $h'(x)$ or $h'(1)$. That is accomplished by multiplying the term involving $p(x)$ by 4.

Can only be earned if response earned first (FTC) point.

The first two points can be, and often are, earned simultaneously.

The point is not earned if the presented derivative is $a \cdot p(8x)$ for any $a \neq 4$.

The point is not earned for responses of $4p(2t) \big|_1^{4x}$ or $4(p(8x) - p(b))$, for any value of b .

Third Point

Can only be earned if response earned first (FTC) and second (chain rule) points.

Earned only for answer of 66 or equivalent value.

The value does not need to be simplified, but if it is, the simplification must be correct to earn this point.

Scoring examples:

$$\cdot p(2 \cdot 4x) - p(2); p(8) - p(2) \text{ earns } 1 - 0 - 0$$

$$\cdot p(2 \cdot 4x) \cdot 4 - p(2) \cdot 0; 66 \text{ earns } 1 - 1 - 1$$

$$\cdot 4p(2t) \big|_1^{4x} = 4(p(8) - p(2)) \text{ earns } 1 - 0 - 0$$

$$\cdot p(2x) \cdot 2; p(2) \cdot 2 \text{ earns } 0 - 0 - 0$$

4-1-1

· $p(2 \cdot 4x) \cdot 4$; $p(8) \cdot 4$ earns $1 - 1 - 0$



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Fundamental Theorem of Calculus
- ☐ Chain rule
- ☐ Answer

Solution:

$$h'(x) = \frac{d}{dx} \int_1^{4x} p(2t) dt = p(2 \cdot 4x) \cdot 4$$

$$h'(1) = p(8) \cdot 4 = 16.5 \cdot 4 = 66$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for narrowing consideration of given information to g and g' .

g and g' need not be evaluated to earn the first point.

Considering velocity and acceleration not sufficient to earn first point, unless acceleration is linked to g' .

If response considers g and g' and g'' , then eventually g'' must be eliminated to earn first point.

Second Point

Philosophy: Earned for “decreasing” with reason, using the opposite signs of g and g' .

Ways to earn second point include:

- “decreasing since $g > 0$ and ” $g' < 0$ ”
- “particle slowing down since g and g' have opposite signs”

Can only be earned if response earned first point.

Examples:

- “decreasing since velocity and acceleration have opposite signs” earns neither point

4-1-1

· “decreasing since velocity > 0 and acceleration < 0 ” earns neither point

· “decreasing since $(2) \cdot (-5) = -10 < 0$ ” earns both points



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Narrows consideration to g and g'
- ☐ Answer with reason

Solution:

Because $g(3) = 2 > 0$, the velocity of the particle is positive. Because $g'(3) = -5 < 0$, the acceleration of the particle is negative. Because the velocity and acceleration have opposite signs, the speed of the particle is decreasing at time $t = 3$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Philosophy: Earned for correct tangent line equation.

Correct responses include:

· $y = -17(x - 5) + 2$ or $y - 2 = -17(x - 5)$

· $y = -17x + 87$

· $y = (-5 \cdot 2^2 + 3)(x - 5) + 2$

Incorrect responses include:

· $y = (-xy^2 + 3)(x - 5) + 2$

· $y = 17x + 87$

· $f(x) = -17x + 87$



0	1
---	---

The student response accurately includes the criteria below.

4-1-1☐ Tangent line equation**Solution:**

$$\left. \frac{dy}{dx} \right|_{(5,2)} = -5 \cdot 2^2 + 3 = -17$$

$$y = -17(x - 5) + 2$$

4-1-1

70. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (seconds)	0	4	8	12	16
$p(t)$ (feet)	25	21	16	12	7.5

The position of a particle, A , moving along the x -axis is modeled by a twice-differentiable function p , where time t is measured in seconds and $p(t)$ is measured in feet. Selected values of $p(t)$ are shown in the table above.

- (a) Use a midpoint Riemann sum with two subintervals of equal length and values from the table to approximate the value of $\int_0^{16} p(t) \, dt$. Show the computations that lead to your answer.
- (b) Use the data in the table to approximate $p'(2)$ using the average rate of change of $p(t)$ over the interval $0 \leq t \leq 4$. Show the computations that lead to your answer. Indicate units of measure.
- (c) Interpret the meaning of $p'(2)$ in the context of the problem.
- (d) Must there exist a value of d , for $0 < d < 16$, such that $p'(d)$ is equal to the average rate of change of p over the interval $0 \leq x \leq 16$? Justify your answer.
- (e) Find $\int_0^4 p'(3t + 4) \, dt$. Show the computations that lead to your answer.
- (f) Let $h(x) = \int_1^{2x} p(3t) \, dt$. Find $h'(2)$. Show the computations that lead to your answer.
- (g) The velocity of a second particle, Q , can be modeled by a twice-differentiable function g . It is known that $g(3) = 2$, $g'(3) = -4$, and $g''(3) = -8$. Is the speed of particle Q increasing or decreasing at time $t = 3$? Give a reason for your answer.
- (h) Let $y = f(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = -2x^2 + y^2$ with initial condition $f(3) = 5$. Write an equation for the line tangent to the graph of f at the point $(3, 5)$.

Part A

4-1-1

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for correct Midpoint Riemann sum, with appropriate values taken from the table.

Need to see sum of two terms, each as a product of two factors, i.e. $\text{base}_1 \cdot \text{height}_1 + \text{base}_2 \cdot \text{height}_2$

$p(4) \cdot (8 - 0) + p(12) \cdot (16 - 8)$ or $p(4) \cdot (8) + p(12) \cdot (8)$ earns first point.

If at least three of the four base/height values are correct in the sum of products, the expression earns the first point.

Second Point

264 or equivalent

Can be earned without first point with Riemann sum setup demonstration. With credible evidence that a midpoint Riemann sum attempt was made without not enough work shown, this point can be earned in the presence of the correct answer. For example, a sum of two terms with no evidence of products ($168 + 96$) or a graph with areas labeled and summed to 264 or equivalent, can earn this point.

Examples:

- $264 = 168 + 96$ by itself earns second point, not first point
- 264 declared, with $21 \cdot 8$ and $12 \cdot 8$ identified (perhaps graphically) earns second point, not first point
- $k(21 \cdot 8 + 21 \cdot 8)$, nonzero constant $k \neq 1$ by itself earns first point, not second point
- $p(4) \cdot (8) + p(12) \cdot (16 - 8)$ by itself earns first point, not second point



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Form of midpoint Riemann sum
- ☐ Answer

Solution:

$$\int_0^{16} p(t) dt \approx p(4)(8 - 0) + p(12)(16 - 8)$$

$$= 21 \cdot 8 + 12 \cdot 8 = 264$$

4-1-1

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for correct difference quotient, with appropriate values from the table.

Need to see both a difference and a quotient

$\frac{p(4)-p(0)}{4-0}$ by itself doesn't earn first point; however $\frac{21-25}{4-0}$ or $\frac{21-25}{4}$ by itself earns first point

Second Point

Earned for correct units of ft per sec or ft/sec or f/s.

Correct units must be tied to a number to earn second point.

Correct units must appear in part (b); correct units in part (c) only doesn't earn second point.



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$p'(2) \approx \frac{21-25}{4-0} = \frac{-4}{4} = -1 \text{ foot per second}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Philosophy: Earned for correct $p'(2)$ interpretation.

Need two components in response:

- “rate of change of position” or “velocity” of the particle
- at $t = 2$

Correct ways to describe rate at time at $t = 2$ include:

4-1-1

- Rate at which position is changing/decreasing
- Rate at which particle is moving left
- Instantaneous velocity or velocity of particle

Ambiguous/incorrect ways to describe rate include:

- Rate at which particle is changing/decreasing
- Rate at which particle is moving

Units are not required in part (c). If units are presented for t , they must be in seconds.



0	1
---	---

The student response accurately includes the criteria below.

☐ Interpretation

Solution:

$p'(2)$ is the rate at which the position of the particle is changing, in feet per second, at time $t = 2$ seconds.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for both Mean Value Theorem (MVT) conditions with the appropriate implication that p is continuous because we know that p is differentiable.

Ways to earn first point include:

- p differentiable $\Rightarrow p$ continuous
- p differentiable, hence p continuous
- p differentiable, so p continuous

p differentiable and p continuous does not earn first point.

Second Point

Philosophy: Earned for concluding “yes” and invoking Mean Value Theorem or its equivalent formulation, with consideration of the continuity of p .

4-1-1

MVT title does not have to be stated to earn second point.

This point is not earned if other theorems (IVT, etc.) are invoked.

If presented, the value for $p'(d) = -\frac{17.5}{16}$ must be correct to earn second point.

Examples:

- “By MVT, yes since p differentiable” earns neither point
- “By MVT, yes since p continuous” earns second point only
- “By MVT, yes since p differentiable and continuous” earns second point only
- “By MVT, yes since p differentiable hence continuous” earns both points
- “By MVT, yes since p twice differentiable hence continuous” earns both points
- “By MVT, yes since function differentiable hence continuous” earns both points



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Conditions
- ☐ Conclusion using Mean Value Theorem

Solution:

p is differentiable. $\Rightarrow p$ is continuous.

Therefore, the Mean Value Theorem can be applied to p on the interval $0 \leq x \leq 16$ to guarantee that there exists a value of d , for $0 < d < 16$, such that $p'(d)$ equals the average rate of change of p over the interval $0 \leq x \leq 16$.

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for correct substitution application.

Need three elements in response (with consistent use of u or another substitution variable):

- $u = 3t + 4$ (can be explicit or implicit)

4-1-1

- $du = 3 dt$ (yielding integral coefficient of $\frac{1}{3}$)
- Correct integral limits throughout, with t from 0 to 4 and u from 4 to 16

Omitting the coefficient of $\frac{1}{3}$ or using some other value will not earn first point; response could earn second point, but not third point.

In the third element above, if response shows an otherwise correct substitution with unchanged limits of integration on a definite integral from 0 to 4, then point is not earned.

Alternately, the integral can be worked with as an indefinite integral with u returned to $3t + 4$ in the antiderivative before evaluating as a means to eliminate the need to change limits. Such a response can earn the first point if handled correctly.

For example: $\int_0^4 p'(3t + 4) dt = \frac{1}{3}p(3t + 4) \Big|_0^4$ earns the substitution point.

Second Point

Philosophy: Earned for correct Fundamental Theorem of Calculus application, with evidence of connection between $\int p'(u)$ and $p(u)$.

Ways to earn second point include:

- $k(p(16) - p(4))$ or $k(p(3 \cdot 4 + 4) - p(3 \cdot 0 + 4))$ for k a nonzero constant
- $k p(3t + 4) \Big|_0^4$ or $k p(u) \Big|_4^{16}$ for k a nonzero constant
- $\int p'(3t + 4) dt = k p(3t + 4)$ or $\int p'(u) du = k p(u)$ for any nonzero constant k , with or without appropriate limits of integration on either integral

First and second points can be earned simultaneously by responses such as $\int_0^4 p'(3t + 4) dt = \frac{1}{2}p(3t + 4) \Big|_0^4$

Third Point

Earned only for an answer of -4.5 or equivalent value.

Value does not need to be simplified, but if it is, the simplification must be correct to earn this point.

Scoring Examples:

- $\frac{1}{3} \int p'(u) du = \frac{1}{3}p(u)$ earns $0 - 1 - 0$
- $\frac{1}{3} \int p'(u) du = \frac{1}{3}p(u) \Big|_0^4 = \frac{1}{3}(p(4) - p(0))$ earns $0 - 1 - 0$

4-1-1

$$\cdot \int p'(u) \, du = p(u) \Big|_4^{16} = (p(16) - p(4)) = -13.5 \text{ earns } 0 - 1 - 0$$

$$\cdot 3 \int p'(u) \, du = 3p(u) \Big|_4^{16} = 3[7.5 - 21] \text{ earns } 0 - 1 - 0$$

$$\cdot \int_0^4 p'(3t + 4) \, dt = p(1.5t^2 + 4t) \Big|_0^4 = p(40) - p(0) \text{ earns } 0 - 0 - 0$$

$$\cdot \frac{1}{3} \int_0^4 p'(u) \, du = \frac{1}{3}(p(16) - p(4)) = -4.5 \text{ earns } 0 - 1 - 1$$

$$\cdot \int_0^4 p'(3t + 4) \, dt = \frac{1}{3}(p(16) - p(4)) = -4.5 \text{ earns } 1 - 1 - 1$$

$$\cdot \int_0^4 p'(3t + 4) \, dt = \frac{1}{3}p(u) \Big|_4^{16} = \frac{1}{3}(p(16) - p(4)) = -4.5 \text{ earns } 1 - 1 - 1$$

$$\cdot \int_0^4 p'(3t + 4) \, dt = \frac{1}{3}p(3t + 4) \Big|_0^4 = \frac{1}{3}(p(16) - p(4)) \text{ earns } 1 - 1 - 0$$

$$\cdot \frac{1}{3} \int p'(u) = \frac{1}{3}p(u); \text{ and } \frac{1}{3}p(3t + 4) \Big|_0^4 = \frac{1}{3}(7.5 - 21) \text{ earns } 1 - 1 - 1$$



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = 3t + 4$. Then $du = 3 \, dt$.

When $t = 0$, $u = 4$.

When $t = 4$, $u = 16$.

$$\begin{aligned} \int_0^4 p'(3t + 4) \, dt &= \frac{1}{3} \int_4^{16} p'(u) \, du \\ &= \frac{1}{3}(p(16) - p(4)) \end{aligned}$$

4-1-1

$$= \frac{1}{3}(7.5 - 21) = -4.5$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for correct Fundamental Theorem of Calculus application, resulting in $a \cdot p(6x)$ for some nonzero constant a .

Ways to earn first point include responses of $p(3 \cdot 2x)$ or $p(6x)$ (Note $p(6t)$ would also earn this point.)

The expression $p(3 \cdot 2x) - p(3 \cdot 1)$ or $(p(3 \cdot 2x) - p(3 \cdot 1)) \cdot 2$ earns this point. The $p(3 \cdot 1)$ is viewed as a chain rule error.

Any response correctly presenting and evaluating the derivative of the integral in terms of the substitution $u = 3t$ earns first point. Part of this response may include $\int_1^{2x} p(3t) dt = \frac{1}{3} \int_3^{6x} p(u) du$.

Second Point

Philosophy: This point is earned for correctly applying the chain rule to find $h'(x)$ or $h'(2)$. That is accomplished by multiplying the term involving $p(x)$ by 2.

Can only be earned if response earned first (FTC) point.

The first two points can be, and often are, earned simultaneously.

The point is not earned if the presented derivative is $a \cdot p(6x)$ for any $a \neq 2$.

The point is not earned for responses of $2p(3t) \big|_1^{2x}$ or $2(p(6x) - p(b))$, for any value of b .

Third Point

Can only be earned if response earned first (FTC) and second (chain rule) points.

Earned only for answer of 24 or equivalent value.

The value does not need to be simplified, but if it is, the simplification must be correct to earn this point.

Scoring examples:

$$\cdot p(3 \cdot 2x) - p(3); p(12) - p(3) \text{ earns } 1 - 0 - 0$$

$$\cdot p(3 \cdot 2x) \cdot 2 - p(3) \cdot 0; 24 \text{ earns } 1 - 1 - 1$$

$$\cdot 2p(3t) \big|_1^{2x} = 2(p(12) - p(3)) \text{ earns } 1 - 0 - 0$$

$$\cdot p(3x) \cdot 2; p(6) \cdot 2 \text{ earns } 0 - 0 - 0$$

4-1-1

$\cdot p(6x) \cdot 2$; $p(12) \cdot 2$ earns $1 - 1 - 0$



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Fundamental Theorem of Calculus
- ☐ Chain rule
- ☐ Answer

Solution:

$$h'(x) = \frac{d}{dx} \int_1^{2x} p(3t) dt = p(3 \cdot 2x) \cdot 2$$

$$h'(2) = p(12) \cdot 2 = 12 \cdot 2 = 24$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for narrowing consideration of given information to g and g' .

g and g' need not be evaluated to earn the first point.

Considering velocity and acceleration not sufficient to earn first point, unless acceleration is linked to g' .

If response considers g and g' and g'' , then eventually g'' must be eliminated to earn first point.

Second Point

Philosophy: Earned for “decreasing” with reason, using the opposite signs of g and g' .

Ways to earn second point include:

- $\cdot$ “decreasing since $g < 0$ and $g' > 0$ ”
- $\cdot$ “particle slowing down since g and g' have opposite signs”

Can only be earned if response earned first point.

Examples:

- $\cdot$ “decreasing since velocity and acceleration have opposite signs” earns neither point

4-1-1

· “decreasing since velocity < 0 and acceleration > 0 ” earns neither point

“decreasing since $(2) \cdot (-4) = -8 < 0$ ” earns both points



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Narrows consideration to g and g'
- ☐ Answer with reason

Solution:

Because $g(3) = 2 > 0$, the velocity of the particle is positive. Because $g'(3) = -4 < 0$, the acceleration of the particle is negative. Because the velocity and acceleration have opposite signs, the speed of the particle is decreasing at time $t = 3$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Philosophy: Earned for correct tangent line equation.

Correct responses include:

$$y = 7(x - 3) + 5 \text{ or } y - 5 = 7(x - 3)$$

$$y = 7x - 16$$

$$y = (-2 \cdot 3^2 + 5^2)(x - 3) + 5$$

Incorrect responses include:

$$y = (-2x^2 + y^2)(x - 3) + 5$$

$$y = -7x - 16$$

$$f(x) = 7x - 16$$



0	1
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The student response accurately includes the criteria below.

4-1-1☐ Tangent line equation**Solution:**

$$\left. \frac{dy}{dx} \right|_{(3,5)} = -2 \cdot 3^2 + 5^2 = 7$$

$$y = 7(x - 3) + 5$$

4-1-1

71. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (seconds)	0	3	6	9	12
$p(t)$ (feet)	3	5.5	7	13	17

The position of a particle, A , moving along the x -axis is modeled by a twice-differentiable function p , where time t is measured in seconds and $p(t)$ is measured in feet. Selected values of $p(t)$ are shown in the table above.

- Use a midpoint Riemann sum with two subintervals of equal length and values from the table to approximate the value of $\int_0^{12} p(t) \, dt$. Show the computations that lead to your answer.
- Use the data in the table to approximate $p'(10)$ using the average rate of change of $p(t)$ over the interval $9 \leq t \leq 12$. Show the computations that lead to your answer. Indicate units of measure.
- Interpret the meaning of $p'(10)$ in the context of the problem.
- Must there exist a value of c , for $0 < c < 12$, such that $p'(c)$ is equal to the average rate of change of p over the interval $0 \leq x \leq 12$? Justify your answer.
- Find $\int_0^3 p'(2t+3) \, dt$. Show the computations that lead to your answer.
- Let $h(x) = \int_2^{3x} p(3t) \, dt$. Find $h'(1)$. Show the computations that lead to your answer.
- The velocity of a second particle, Q , can be modeled by a twice-differentiable function g . It is known that $g(2) = -6$, $g'(2) = 4$, and $g''(2) = -3$. Is the speed of particle Q increasing or decreasing at time $t = 2$? Give a reason for your answer.
- Let $y = f(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = 4x^2y^3$ with initial condition $f\left(\frac{1}{2}\right) = -2$. Write an equation for the line tangent to the graph of f at the point $\left(\frac{1}{2}, -2\right)$.

Part A

4-1-1

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for correct midpoint Riemann sum, with appropriate values taken from the table.

Need to see sum of two terms, each as a product of two factors, i.e. $\text{base}_1 \cdot \text{height}_1 + \text{base}_2 \cdot \text{height}_2$

$p(3) \cdot (6 - 0) + p(9) \cdot (12 - 6)$ or $p(3) \cdot (6) + p(9) \cdot (6)$ earns first point.

If at least three of the four base/height values are correct in the sum of products, the expression earns the first point.

Second Point

111 or equivalent

Can be earned without first point with Riemann sum setup demonstration. With credible evidence that a midpoint Riemann sum attempt was made without not enough work shown, this point can be earned in the presence of the correct answer. For example, a sum of two terms with no evidence of products ($33 + 78$) or a graph with areas labeled and summed to 111 or equivalent, can earn this point.

Examples:

- $111 = 33 + 78$ by itself earns second point, not first point
- 111 declared, with $5.5 \cdot 6$ and $13 \cdot 6$ identified (perhaps graphically) earns second point, not first point
- $k(5.5 \cdot 6 + 13 \cdot 6)$, nonzero constant $k \neq 1$ by itself earns first point, not second point
- $p(3) \cdot (6) + p(9) \cdot (12 - 6)$ by itself earns first point, not second point



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Form of midpoint Riemann sum
- ☐ Answer

Solution:

$$\int_0^{12} p(t) dt \approx p(3)(6 - 0) + p(9)(12 - 6)$$

$$= 5.5 \cdot 6 + 13 \cdot 6 = 111$$

4-1-1

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for correct difference quotient, with appropriate values from the table.

Need to see both a difference and a quotient

$\frac{p(12)-p(9)}{12-9}$ by itself doesn't earn first point; however $\frac{17-13}{12-9}$ or $\frac{17-13}{3}$ by itself earns first point

Second Point

Earned for correct units of ft per sec or ft/sec or f/s.

Correct units must be tied to a number to earn second point.

Correct units must appear in part (b); correct units in part (c) only doesn't earn second point.



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$p'(10) \approx \frac{17-13}{12-9} = \frac{4}{3} \text{ feet per second}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Philosophy: Earned for correct $p'(10)$ interpretation.

Need two components in response:

- “rate of change of position” or “velocity” of the particle
- at $t = 10$

Correct ways to describe rate at time at $t = 10$ include:

4-1-1

- Rate at which position is changing/increasing
- Rate at which particle is moving right
- Instantaneous velocity or velocity of particle

Ambiguous/incorrect ways to describe rate include:

- Rate at which particle is changing/increasing
- Rate at which particle is moving

Units are not required in part (c). If units are presented for t , they must be in seconds.



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Interpretation

Solution:

$p'(10)$ is the rate at which the position of the particle is changing, in feet per second, at time $t = 10$ seconds.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for both Mean Value Theorem (MVT) conditions with the appropriate implication that p is continuous because we know that p is differentiable.

Ways to earn first point include:

- p differentiable $\Rightarrow p$ continuous
- p differentiable, hence p continuous
- p differentiable, so p continuous

p differentiable and p continuous does not earn first point.

Second Point

Philosophy: Earned for concluding “yes” and invoking Mean Value Theorem or its equivalent formulation, with consideration of the continuity of p .

4-1-1

MVT title does not have to be stated to earn second point.

This point is not earned if other theorems (IVT, etc.) are invoked.

If presented, the value for $p'(c) = \frac{7}{6}$ must be correct to earn second point.

Examples:

- “By MVT, yes since p differentiable” earns neither point
- “By MVT, yes since p continuous” earns second point only
- “By MVT, yes since p differentiable and continuous” earns second point only
- “By MVT, yes since p differentiable hence continuous” earns both points
- “By MVT, yes since p twice differentiable hence continuous” earns both points
- “By MVT, yes since function differentiable hence continuous” earns both points



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Conditions
- ☐ Conclusion using Mean Value Theorem

Solution:

p is differentiable. $\Rightarrow p$ is continuous.

Therefore, the Mean Value Theorem can be applied to p on the interval $0 \leq x \leq 12$ to guarantee that there exists a value of c , for $0 < c < 12$, such that $p'(c)$ equals the average rate of change of p over the interval $0 \leq x \leq 12$.

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for correct substitution application.

Need three elements in response (with consistent use of u or another substitution variable):

- $u = 2t + 3$ (can be explicit or implicit)

4-1-1

- $du = 2 dt$ (yielding integral coefficient of $\frac{1}{2}$)
- Correct integral limits throughout, with t from 0 to 3 and u from 3 to 9

Omitting the coefficient of $\frac{1}{2}$ or using some other value will not earn first point; response could earn second point, but not third point.

In the third element above, if response shows an otherwise correct substitution with unchanged limits of integration on a definite integral from 0 to 3, then point is not earned.

Alternately, the integral can be worked with as an indefinite integral with u returned to $2t + 3$ in the antiderivative before evaluating as a means to eliminate the need to change limits. Such a response can earn the first point if handled correctly.

For example: $\int_0^3 p'(2t + 3) dt = \frac{1}{2}p(2t + 3) \Big|_0^3$ earns the substitution point.

Second Point

Philosophy: Earned for correct Fundamental Theorem of Calculus application, with evidence of connection between $\int p'(u)$ and $p(u)$.

Ways to earn second point include:

- $k(p(9) - p(3))$ or $k(p(2 \cdot 3 + 3) - p(2 \cdot 0 + 3))$ for k a nonzero constant
- $k p(2t + 3) \Big|_0^3$ or $k p(u) \Big|_3^9$ for k a nonzero constant
- $\int p'(2t + 3) dt = k p(2t + 3)$ or $\int p'(u) du = k p(u)$ for any nonzero constant k , with or without appropriate limits of integration on either integral

First and second points can be earned simultaneously by responses such as $\int_0^3 p'(2t + 3) dt = \frac{1}{2}p(2t + 3) \Big|_0^3$

Third Point

Earned only for an answer of 3.75 or equivalent value.

Value does not need to be simplified, but if it is, the simplification must be correct to earn this point.

Scoring Examples:

- $\frac{1}{2} \int p'(u) du = \frac{1}{2}p(u)$ earns 0 – 1 – 0
- $\frac{1}{2} \int p'(u) du = \frac{1}{2}p(u) \Big|_0^3 = \frac{1}{2}(p(3) - p(0))$ earns 0 – 1 – 0

4-1-1

$$\cdot \int p'(u) \, du = p(u) \Big|_3^9 = (p(9) - p(3)) = 7.5 \text{ earns } 0 - 1 - 0$$

$$\cdot 2 \int p'(u) \, du = 2p(u) \Big|_0^3 = 2[13 - 5.5] \text{ earns } 0 - 1 - 0$$

$$\cdot \int_0^3 p'(2t + 3) \, dt = p(t^2 + 3t) \Big|_0^3 = p(18) - p(0) \text{ earns } 0 - 0 - 0$$

$$\cdot \frac{1}{2} \int_0^3 p'(u) \, du = \frac{1}{2}(p(9) - p(3)) = 3.75 \text{ earns } 0 - 1 - 1$$

$$\cdot \int_0^3 p'(2t + 3) \, dt = \frac{1}{2}(p(9) - p(3)) = 3.75 \text{ earns } 1 - 1 - 1$$

$$\cdot \int_0^3 p'(2t + 3) \, dt = \frac{1}{2}p(u) \Big|_3^9 = \frac{1}{2}(p(9) - p(3)) = 3.75 \text{ earns } 1 - 1 - 1$$

$$\cdot \int_0^3 p'(2t + 3) \, dt = \frac{1}{2}p(2t + 3) \Big|_0^3 = \frac{1}{2}(p(9) - p(3)) \text{ earns } 1 - 1 - 0$$

$$\cdot \frac{1}{2} \int p'(u) = \frac{1}{2}p(u); \frac{1}{2}p(2t + 3) \Big|_0^3 = \frac{1}{2}(13 - 5.5) \text{ earns } 1 - 1 - 1$$



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = 2t + 3$. Then $du = 2 \, dt$.

When $t = 0$, $u = 3$.

When $t = 3$, $u = 9$.

$$\int_0^3 p'(2t + 3) \, dt = \frac{1}{2} \int_3^9 p'(u) \, du$$

$$= \frac{1}{2}(p(9) - p(3))$$

$$= \frac{1}{2}(13 - 5.5) = 3.75$$

4-1-1

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for correct Fundamental Theorem of Calculus application, resulting in $a \cdot p(9x)$ for some nonzero constant a .

Ways to earn first point include responses of $p(3 \cdot 3x)$ or $p(9x)$ (Note $p(9t)$ would also earn this point.)

The expression $p(3 \cdot 3x) - p(3 \cdot 2)$ or $(p(3 \cdot 3x) - p(3 \cdot 2)) \cdot 3$ earns this point. The $p(3 \cdot 2)$ is viewed as a chain rule error.

Any response correctly presenting and evaluating the derivative of the integral in terms of the substitution $u = 3t$ earns first point. Part of this response may include $\int_2^{3x} p(3t) dt = \frac{1}{3} \int_6^{9x} p(u) du$.

Second Point

Philosophy: This point is earned for correctly applying the chain rule to find $h'(x)$ or $h'(1)$. That is accomplished by multiplying the term involving $p(x)$ by 3.

Can only be earned if response earned first (FTC) point.

The first two points can be, and often are, earned simultaneously.

The point is not earned if the presented derivative is $a \cdot p(9x)$ for any $a \neq 3$.

The point is not earned for responses of $3p(3t) \big|_2^{3x}$ or $3(p(9x) - p(b))$, for any value of b .

Third Point

Can only be earned if response earned first (FTC) and second (chain rule) points.

Earned only for answer of 39 or equivalent value.

The value does not need to be simplified, but if it is, the simplification must be correct to earn this point.

Scoring examples:

$\cdot p(3 \cdot 3x) - p(6); p(9) - p(6) = 13 - 7$ earns $1 - 0 - 0$

$\cdot p(3 \cdot 3x) \cdot 3 - p(6) \cdot 0; 39$ earns $1 - 1 - 1$

$\cdot 3p(3t) \big|_2^{3x} = 3(p(9) - p(6))$ earns $1 - 0 - 0$

$\cdot p(3x) \cdot 3; p(3) \cdot 3 = 5.5 \cdot 3$ earns $0 - 0 - 0$

4-1-1

· $p(9x) \cdot 3$; $p(9) \cdot 3$ earns $1 - 1 - 0$



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Fundamental Theorem of Calculus
- ☐ Chain rule
- ☐ Answer

Solution:

$$h'(x) = \frac{d}{dx} \int_2^{3x} p(3t) dt = p(3 \cdot 3x) \cdot 3$$

$$h'(1) = p(9) \cdot 3 = 13 \cdot 3 = 39$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for narrowing consideration of given information to g and g' .

g and g' need not be evaluated to earn the first point.

Considering velocity and acceleration not sufficient to earn first point, unless acceleration is linked to g' .

If response considers g and g' and g'' , then eventually g'' must be eliminated to earn first point.

Second Point

Philosophy: Earned for “decreasing” with reason, using the opposite signs of g and g' .

Ways to earn second point include:

- “decreasing since $g < 0$ and $g' > 0$ ”
- “particle slowing down since g and g' have opposite signs”

Can only be earned if response earned first point.

Examples:

- “decreasing since velocity and acceleration have opposite signs” earns neither point

4-1-1

· “decreasing since velocity < 0 and acceleration > 0 ” earns neither point

· “decreasing since $(-6) \cdot 4 = -24 < 0$ ” earns both points



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Narrows consideration to g and g'
- ☐ Answer with reason

Solution:

Because $g(2) = -6 < 0$, the velocity of the particle is negative. Because $g'(2) = 4 > 0$, the acceleration of the particle is positive. Because the velocity and acceleration have opposite signs, the speed of the particle is decreasing at time $t = 2$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Philosophy: Earned for correct tangent line equation.

Correct responses include:

· $y = -8\left(x - \frac{1}{2}\right) - 2$ or $y + 2 = -8\left(x - \frac{1}{2}\right)$

· $y = -8x + 2$

· $y = \left(4 \cdot \left(\frac{1}{2}\right)^2 \cdot (-2)^3\right)\left(x - \frac{1}{2}\right) - 2$

Incorrect responses include:

· $y = (4x^2y^3)\left(x - \frac{1}{2}\right) - 2$

· $y = 8x + 2$

· $f(x) = -8x + 2$



0	1
---	---

The student response accurately includes the criteria below.

4-1-1☐ Tangent line equation**Solution:**

$$\frac{dy}{dx} \Big|_{\left(\frac{1}{2}, -2\right)} = 4 \cdot \left(\frac{1}{2}\right)^2 \cdot (-2)^3 = -8$$

$$y = -8 \left(x - \frac{1}{2}\right) - 2$$

4-1-1

72. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (seconds)	0	5	10	15	20
$p(t)$ (feet)	-25	-20	-14	-8	-2.5

The position of a particle, A , moving along the x -axis is modeled by a twice-differentiable function p , where time t is measured in seconds and $p(t)$ is measured in feet. Selected values of $p(t)$ are shown in the table above.

- (a) Use a midpoint Riemann sum with two subintervals of equal length and values from the table to approximate the value of $\int_0^{20} p(t) \, dt$. Show the computations that lead to your answer.
- (b) Use the data in the table to approximate $p'(7)$ using the average rate of change of $p(t)$ over the interval $5 \leq t \leq 10$. Show the computations that lead to your answer. Indicate units of measure.
- (c) Interpret the meaning of $p'(7)$ in the context of the problem.
- (d) Must there exist a value of b , for $0 < b < 20$, such that $p'(b)$ is equal to the average rate of change of p over the interval $0 \leq x \leq 20$? Justify your answer.
- (e) Find $\int_0^5 p'(2t + 5) \, dt$. Show the computations that lead to your answer.
- (f) Let $h(x) = \int_3^{5x} p(4t) \, dt$. Find $h'(1)$. Show the computations that lead to your answer.
- (g) The velocity of a second particle, Q , can be modeled by a twice-differentiable function g . It is known that $g(1) = -2$, $g'(1) = 4$, and $g''(1) = 7$. Is the speed of particle Q increasing or decreasing at time $t = 1$? Give a reason for your answer.
- (h) Let $y = f(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = -2x^3y^2 - 5$ with initial condition $f(-1) = 2$. Write an equation for the line tangent to the graph of f at the point $(-1, 2)$.

Part A

4-1-1

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for correct Midpoint Riemann sum, with appropriate values taken from the table.

Need to see sum of two terms, each as a product of two factors, i.e. $\text{base}_1 \cdot \text{height}_1 + \text{base}_2 \cdot \text{height}_2$

$p(5) \cdot (10 - 0) + p(15) \cdot (20 - 10)$ or $p(5) \cdot (10) + p(15) \cdot (10)$ earns first point.

If at least three of the four base/height values are correct in the sum of products, the expression earns the first point.

Second Point

−280 or equivalent

Can be earned without first point with Riemann sum setup demonstration. With credible evidence that a midpoint Riemann sum attempt was made without not enough work shown, this point can be earned in the presence of the correct answer. For example, a sum of two terms with no evidence of products ($-200 - 80$) or a graph with areas labeled and summed to -280 or equivalent, can earn this point.

Examples:

- $-280 = -200 - 80$ by itself earns second point, not first point
- -280 declared, with $-20 \cdot 10$ and $-8 \cdot 10$ identified (perhaps graphically) earns second point, not first point
- $k(-20 \cdot 10 - 8 \cdot 10)$, nonzero constant $k \neq 1$ by itself earns first point, not second point
- $p(3) \cdot (10) + p(15) \cdot (20 - 10)$ by itself earns first point, not second point



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Form of midpoint Riemann sum
- ☐ Answer

Solution:

$$\int_0^{20} p(t) dt \approx p(5)(10 - 0) + p(15)(20 - 10)$$

$$= -20 \cdot 10 + (-8) \cdot 10 = -280$$

4-1-1

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for correct difference quotient, with appropriate values from the table.

Need to see both a difference and a quotient

$\frac{p(10)-p(5)}{10-5}$ by itself doesn't earn first point; however $\frac{-14-(-20)}{10-5}$ or $\frac{-14-(-20)}{5}$ by itself earns first point

Second Point

Earned for correct units of ft per sec or ft/sec or f/s.

Correct units must be tied to a number to earn second point.

Correct units must appear in part (b); correct units in part (c) only doesn't earn second point.



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Approximation
- ☐ Units

Solution:

$$p'(7) \approx \frac{-14-(-20)}{10-5} = \frac{6}{5} \text{ feet per second}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Philosophy: Earned for correct $p'(7)$ interpretation.

Need two components in response:

- “rate of change of position” or “velocity” of the particle
- at $t = 7$

Correct ways to describe rate at time at $t = 7$ include:

4-1-1

- Rate at which position is changing/increasing
- Rate at which particle is moving right
- Instantaneous velocity or velocity of particle

Ambiguous/incorrect ways to describe rate include:

- Rate at which particle is changing/increasing
- Rate at which particle is moving

Units are not required in part (c). If units are presented for t , they must be in seconds.



0	1
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The student response accurately includes the criteria below.

☐ Interpretation

Solution:

$p'(7)$ is the rate at which the position of the particle is changing, in feet per second, at time $t = 7$ seconds.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for both Mean Value Theorem (MVT) conditions with the appropriate implication that p is continuous because we know that p is differentiable.

Ways to earn first point include:

- p differentiable $\Rightarrow p$ continuous
- p differentiable, hence p continuous
- p differentiable, so p continuous

p differentiable and p continuous does not earn first point.

Second Point

Philosophy: Earned for concluding “yes” and invoking Mean Value Theorem or its equivalent formulation, with consideration of the continuity of p .

4-1-1

MVT title does not have to be stated to earn second point.

This point is not earned if other theorems (IVT , etc.) are invoked.

If presented, the value for $p'(b) = \frac{22.5}{20}$ must be correct to earn second point.

Examples:

- “By MVT , yes since p differentiable” earns neither point
- “By MVT , yes since p continuous” earns second point only
- “By MVT , yes since p differentiable and continuous” earns second point only
- “By MVT , yes since p differentiable hence continuous” earns both points
- “By MVT , yes since p twice differentiable hence continuous” earns both points
- “By MVT , yes since function differentiable hence continuous” earns both points



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Conditions
- ☐ Conclusion using Mean Value Theorem

Solution:

p is differentiable. $\Rightarrow p$ is continuous.

Therefore, the Mean Value Theorem can be applied to p on the interval $0 \leq x \leq 20$ to guarantee that there exists a value of b , for $0 < b < 20$, such that $p'(b)$ equals the average rate of change of p over the interval $0 \leq x \leq 20$.

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for correct substitution application.

Need three elements in response (with consistent use of u or another substitution variable):

- $u = 2t + 5$ (can be explicit or implicit)

4-1-1

- $du = 2dt$ (yielding integral coefficient of $\frac{1}{2}$)
- Correct integral limits throughout, with t from 0 to 5 and u from 5 to 15

Omitting the coefficient of $\frac{1}{2}$ or using some other value will not earn first point; response could earn second point, but not third point.

In the third element above, if response shows an otherwise correct substitution with unchanged limits of integration on a definite integral from 0 to 5, then point is not earned.

Alternately, the integral can be worked with as an indefinite integral with u returned to $2t + 5$ in the antiderivative before evaluating as a means to eliminate the need to change limits. Such a response can earn the first point if handled correctly.

For example: $\int_0^5 p'(2t + 5)dt = \frac{1}{2}p(2t + 5)\big|_0^5$ earns the substitution point.

Second Point

Philosophy: Earned for correct Fundamental Theorem of Calculus application, with evidence of connection between $\int p'(u)$ and $p(u)$.

Ways to earn second point include:

- $k(p(15) - p(5))$ or $k(p(2 \cdot 5 + 5) - p(2 \cdot 0 + 5))$ for k a nonzero constant
- $kp(2t + 5)\big|_0^5$ or $kp(u)\big|_5^{15}$ for k a nonzero constant
- $\int p'(2t + 5)dt = kp(2t + 5)$ or $\int p'(u)du = kp(u)$ for any nonzero constant k , with or without appropriate limits of integration on either integral

First and second points can be earned simultaneously by responses such as $\int_0^5 p'(2t + 5)dt = \frac{1}{2}p(2t + 5)\big|_0^5$

Third Point

Earned only for an answer of 6 or equivalent value.

Value does not need to be simplified, but if it is, the simplification must be correct to earn this point.

Scoring Examples:

- $\frac{1}{2} \int p'(u)du = \frac{1}{2}p(u)$ earns 0 – 1 – 0
- $\frac{1}{2} \int p'(u)du = \frac{1}{2}p(u)\big|_0^5 = \frac{1}{2}(p(5) - p(0))$ earns 0 – 1 – 0

4-1-1

$$\cdot \int p'(u) du = p(u) \Big|_5^{15} = (p(15) - p(5)) = 12 \text{ earns } 0 - 1 - 0$$

$$\cdot 2 \int p'(u) du = 2p(u) \Big|_0^5 = 2[-8 - (-20)] \text{ earns } 0 - 1 - 0$$

$$\cdot \int_0^5 p'(2t+5) dt = p(t^2+5t) \Big|_0^5 = p(50) - p(0) \text{ earns } 0 - 0 - 0$$

$$\cdot \frac{1}{2} \int_0^5 p'(u) du = \frac{1}{2}(p(15) - p(5)) = 6 \text{ earns } 0 - 1 - 1$$

$$\cdot \int_0^5 p'(2t+5) dt = \frac{1}{2}(p(15) - p(5)) = 6 \text{ earns } 1 - 1 - 1$$

$$\cdot \int_0^5 p'(2t+5) dt = \frac{1}{2}p(u) \Big|_5^{15} = \frac{1}{2}(p(15) - p(5)) = 6 \text{ earns } 1 - 1 - 1$$

$$\cdot \int_0^5 p'(2t+5) dt = \frac{1}{2}p(2t+5) \Big|_0^5 = \frac{1}{2}(p(15) - p(5)) \text{ earns } 1 - 1 - 0$$

$$\cdot \frac{1}{2} \int p'(u) = \frac{1}{2}p(u); \text{ and } \frac{1}{2}p(2t+5) \Big|_0^5 = \frac{1}{2}(-8 - (-20)) \text{ earns } 1 - 1 - 1$$



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Substitution
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Let $u = 2t + 5$. Then $du = 2 dt$.

When $t = 0$, $u = 5$.

When $t = 5$, $u = 15$.

$$\int_0^5 p'(2t+5) dt = \frac{1}{2} \int_5^{15} p'(u) du$$

$$= \frac{1}{2}(p(15) - p(5))$$

$$= \frac{1}{2}(-8 - (-20)) = 6$$

4-1-1

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for correct Fundamental Theorem of Calculus application, resulting in $a \cdot p(20x)$ for some nonzero constant a .

Ways to earn first point include responses of $p(4 \cdot 5x)$ or $p(20x)$ (Note $p(20t)$ would also earn this point.)

The expression $p(4 \cdot 5x) - p(4 \cdot 3)$ or $(p(4 \cdot 5x) - p(4 \cdot 3)) \cdot 5$ earns this point. The $p(4 \cdot 3)$ is viewed as a chain rule error.

Any response correctly presenting and evaluating the derivative of the integral in terms of the substitution $u = 4t$ earns first point. Part of this response may include $\int_3^{5x} p(4t) dt = \frac{1}{4} \int_{12}^{20x} p(u) du$.

Second Point

Philosophy: This point is earned for correctly applying the chain rule to find $h'(x)$ or $h'(1)$. That is accomplished by multiplying the term involving $p(x)$ by 5.

Can only be earned if response earned first (FTC) point.

The first two points can be, and often are, earned simultaneously.

The point is not earned if the presented derivative is $a \cdot p(20x)$ for any $a \neq 5$.

The point is not earned for responses of $5p(4t) \big|_3^{5x}$ or $5(p(20x) - p(b))$, for any value of b .

Third Point

Can only be earned if response earned first (FTC) and second (chain rule) points.

Earned only for answer of -12.5 or equivalent value.

The value does not need to be simplified, but if it is, the simplification must be correct to earn this point.

Scoring examples:

$\cdot p(4 \cdot 5x) - p(12); p(20) - p(12)$ earns $1 - 0 - 0$

$\cdot p(4 \cdot 5x) \cdot 5 - p(12) \cdot 0; -12.5$ earns $1 - 1 - 1$

$\cdot 5p(4t) \big|_3^{5x} = 5(p(20) - p(12))$ earns $1 - 0 - 0$

$\cdot p(4x) \cdot 5; p(4) \cdot 5$ earns $0 - 0 - 0$

4-1-1

· $p(4 \cdot 5x) \cdot 5$; $p(20) \cdot 5$ earns $1 - 1 - 0$



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Fundamental Theorem of Calculus
- ☐ Chain rule
- ☐ Answer

Solution:

$$h'(x) = \frac{d}{dx} \int_3^{5x} p(4t) dt = p(4 \cdot 5x) \cdot 5$$

$$h'(1) = p(20) \cdot 5 = -2.5 \cdot 5 = -12.5$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point

Philosophy: Earned for narrowing consideration of given information to g and g' .

g and g' need not be evaluated to earn the first point.

Considering velocity and acceleration not sufficient to earn first point, unless acceleration is linked to g' .

If response considers g and g' and g'' , then eventually g'' must be eliminated to earn first point.

Second Point

Philosophy: Earned for “decreasing” with reason, using the opposite signs of g and g' .

Ways to earn second point include:

- “decreasing since $g < 0$ and $g' > 0$ ”
- “particle slowing down since g and g' have opposite signs”

Can only be earned if response earned first point.

Examples:

- “decreasing since velocity and acceleration have opposite signs” earns neither point

4-1-1

· “decreasing since velocity < 0 and acceleration > 0 ” earns neither point

“decreasing since $(-2) \cdot 4 = -8 < 0$ ” earns both points



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Narrows consideration to g and g'
- ☐ Answer with reason

Solution:

Because $g(1) = -2 < 0$, the velocity of the particle is negative. Because $g'(1) = 4 > 0$, the acceleration of the particle is positive. Because the velocity and acceleration have opposite signs, the speed of the particle is decreasing at time $t = 1$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Philosophy: Earned for correct tangent line equation.

Correct responses include:

- $y = 3(x + 1) + 2$ or $y - 2 = 3(x + 1)$
- $y = 3x + 5$
- $y = (-2 \cdot (-1)^3 \cdot 2^2 - 5)(x + 1) + 2$

Incorrect responses include:

- $y = (-2x^3y^2 - 5)(x + 1) + 2$
- $y = -3x + 5$
- $f(x) = 3x + 5$



0	1
---	---

The student response accurately includes the criteria below.

4-1-1

☐ Tangent line equation**Solution:**

$$\left. \frac{dy}{dx} \right|_{(-1, 2)} = -2(-1)^3 \cdot 2^2 - 5 = 3$$

$$y = 3(x + 1) + 2$$

73.  Let f be the function defined by $f(x) = \frac{1}{4}x^4 - \frac{2}{3}x^3 + \frac{1}{2}x^2 - \frac{1}{2}x$. For how many values of x in the open interval $(0, 1.565)$ is the instantaneous rate of change of f equal to the average rate of change of f on the closed interval $[0, 1.565]$?

(A) Zero

(B) One

(C) Three

(D) Four

**Answer C**

Correct. The average rate of change of f on the closed interval $[0, 1.565]$ is $\frac{f(1.565) - f(0)}{1.565 - 0} = -0.39206$. The instantaneous rate of change of f is the derivative, $f'(x) = x^3 - 2x^2 + x - \frac{1}{2}$. The graph of f' , produced using the calculator, intersects the horizontal line $y = -0.39206$ three times in the open interval $(0, 1.565)$.

74.  Let g be the function with first derivative $g'(x) = \sqrt{x^3 + x}$ for $x > 0$. If $g(2) = -7$, what is the value of $g(5)$?

(A) 4.402

(B) 11.402

(C) 13.899

(D) 20.899

**Answer C**

Correct. By the Fundamental Theorem of Calculus, $g(5) - g(2) = \int_2^5 g'(x) dx$. Therefore,

$$g(5) = g(2) + \int_2^5 \sqrt{x^3 + x} dx = -7 + \int_2^5 \sqrt{x^3 + x} dx = 13.899, \text{ where the evaluation of the definite integral is done with the calculator.}$$

4-1-1

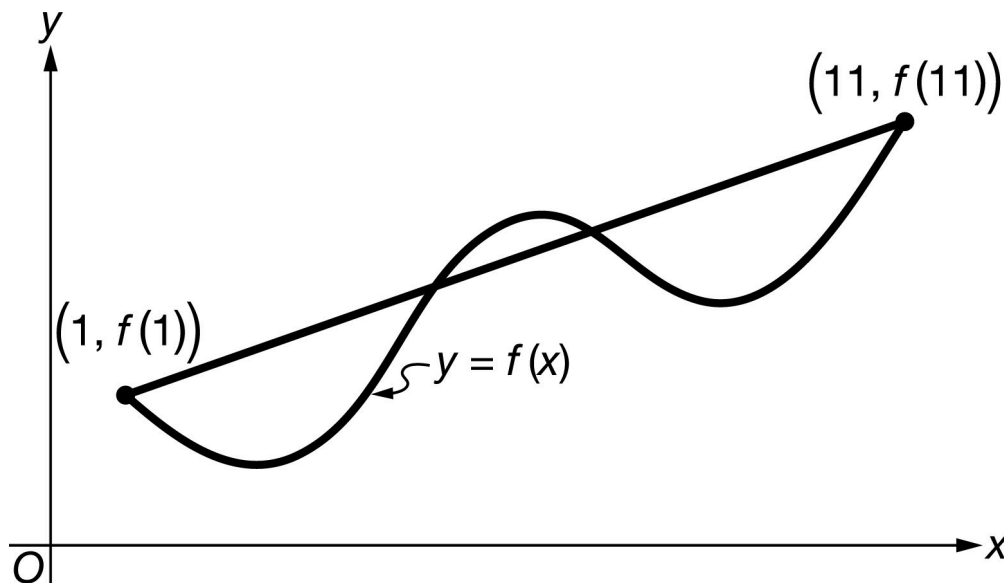
75. The function C gives the cost, in dollars, to produce a particular product, where $C(x)$ is the cost, in dollars, to produce x units of the product. The function M defined by $M(x) = C(x+1) - C(x)$ gives the marginal cost, in dollars, to produce unit number $x+1$. Which of the following gives the best estimate for the marginal cost, in dollars, to produce the 57th unit of the product?
- (A) $\frac{C(56)}{56}$
- (B) $\frac{C(57)}{57} - \frac{C(56)}{56}$
- (C) $C'(56)$ ✓
- (D) $C'(57) - C'(56)$

Answer C

Correct. The marginal cost, in dollars, to produce the 57th unit is given by $M(56) = C(56+1) - C(56)$. This is equal to $\frac{C(57)-C(56)}{57-56}$, which is the average rate of change of the cost function C . This is an approximation for the derivative of C at $x = 56$. Therefore, the marginal cost to produce the 57th unit is best estimated by $C'(56)$.

4-1-1

76.



The figure above shows the graph of the differentiable function f for $1 \leq x \leq 11$ and the secant line through the points $(1, f(1))$ and $(11, f(11))$. For how many values of x in the closed interval $[1, 11]$ does the instantaneous rate of change of f at x equal the average rate of change of f over that interval?

- (A) Zero
- (B) Two
- (C) Three
- (D) Four



Answer C

Correct. The graph of f intersects the secant line at $x = 1$, $x = a$, $x = b$, and $x = 11$, where $1 < a < b < 11$. In each of the intervals $(1, a)$, (a, b) , and $(b, 11)$, there will be exactly one value of x at which the instantaneous rate of change of f is equal to the average rate of change of f over the interval $[1, 11]$. In other words, there will be three values of x at which the slope of the line tangent to the graph of f will equal the slope of the secant line.

4-1-1

77. Oil is spilled onto a kitchen floor. The area covered by the oil at time t is given by the function A , where $A(t)$ is measured in square centimeters and t is measured in seconds. Which of the following gives the rate at which the area covered by the oil is changing at time $t = 10$?
- (A) $A(10)$
- (B) $A(11) - A(9)$
- (C) $\frac{A(10)}{10}$
- (D) $A'(10)$ ✓

Answer D

Correct. The derivative of the function A evaluated at $t = 10$ gives the rate of change of the area covered by the oil at time $t = 10$.

78.  The mass of a colony of bacteria, in grams, is modeled by the function P given by $P(t) = 2 + 5\tan^{-1}\left(\frac{t}{2}\right)$, where t is measured in days. What is the instantaneous rate of change of the mass of the colony, in grams per day, at the moment the colony reaches a mass of 6 grams?
- (A) -0.606
- (B) 0.250
- (C) 1.214 ✓
- (D) 1.942

Answer C

Correct. Find the time when the colony reaches a mass of 6 grams by using the calculator to solve the equation $P(t) = 2 + 5\tan^{-1}\left(\frac{t}{2}\right) = 6$. The solution is $t = 2.059277$. Since the derivative P' can be interpreted as the instantaneous rate of change of P with respect to time, the calculator is used to numerically calculate the derivative of P at this value of t .

4-1-1

79.

 $Q(t) = 3e^{-0.7t} \sin\left(\frac{t}{3}\right) + 0.01 \sin(4t) - 0.02 \cos(4t)$

The function Q defined above models the electric charge, measured in coulombs, inside a lightbulb t seconds after it is turned on. Which of the following presents the method for finding the instantaneous rate of change of the lightbulb's electric charge, in coulombs per second, at time $t = 4$?

(A) $Q''(4) = -0.213$

(B) $Q'(4) = -0.171$ ✓

(C) $\frac{Q(4) - Q(0)}{4 - 0} = 0.053$

(D) $Q(4) = 0.194$

Answer B

Correct. The derivative of Q can be interpreted as the instantaneous rate of change of Q with respect to t . The calculator is used to numerically calculate the derivative of Q at $t = 4$.

80. The function $t = f(A)$ models the time, in minutes, for a chemical reaction to occur as a function of the amount A of catalyst used, measured in milliliters. What are the units for $f''(A)$?

(A) minutes per milliliter

(B) milliliters per minute

(C) minutes per milliliter per milliliter ✓

(D) milliliters per minute per minute

Answer C

Correct. The first derivative $f'(A)$ is the limit of the difference quotient $\frac{\Delta t}{\Delta A}$. The units of the dependent variable t in the numerator are minutes, and the units of the independent variable A in the denominator are milliliters. The first derivative, therefore, has units in minutes per milliliters. If $y = f'(A)$, then the second derivative $f''(A)$ is the limit of the difference quotient $\frac{\Delta y}{\Delta A}$. The units of the dependent variable y in the numerator are the units for the first derivative, minutes per milliliter, and the units of the independent variable A in the denominator are milliliters. The second derivative, therefore, has units in minutes per milliliter per milliliter.

4-1-1

81.  A weight suspended by a spring vibrates vertically according to the function D given by $D(t) = 2 \sin\left(4\pi\left(t + \frac{1}{8}\right)\right)$, where $D(t)$, in centimeters, is the directed distance of the weight from its central position t seconds after the start of the motion. Assume the positive direction is upward. What is the instantaneous rate of change of the weight's position, in centimeters per second, at the moment the weight is first 1 centimeter above its central position?
- (A) -157.914
- (B) -21.765 ✓
- (C) -12
- (D) 0

Answer B

Correct. Find the time when the weight is first 1 centimeter above its central position by using the calculator to solve $D(t) = 2 \sin\left(4\pi\left(t + \frac{1}{8}\right)\right) = 1$. The solution is $t = 0.083333$ (or $t = \frac{1}{12}$). Store this value in the calculator. Because the derivative D' can be interpreted as the instantaneous rate of change of D with respect to time, the calculator is used to numerically calculate the derivative of D at this value of t . This is the value of $D'(0.083333)$.

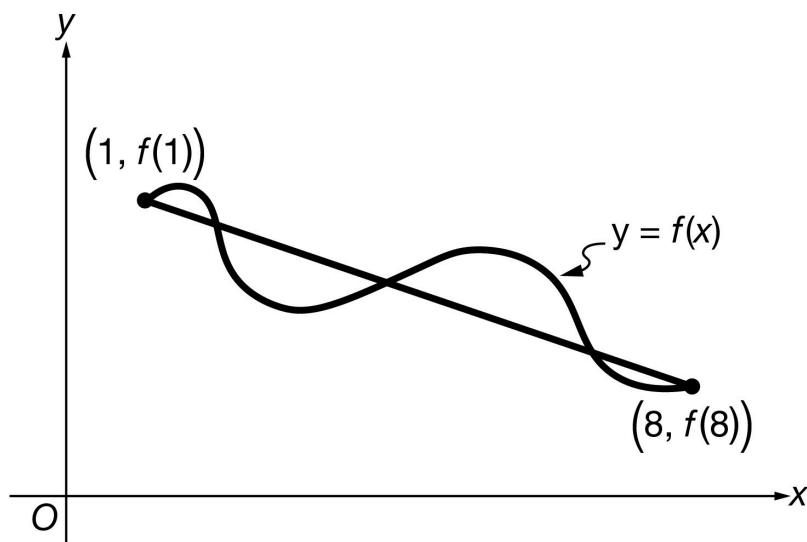
82. A city planner developed the function P to model the population of a city over the next 25 years, where $P(t)$ is the projected population of the city t years from now. According to the model, which of the following gives the rate at which the population of the city will be changing at time $t = 5$?
- (A) $P(5)$
- (B) $P(6) - P(4)$
- (C) $\frac{P(5)}{5}$
- (D) $P'(5)$ ✓

Answer D

Correct. The derivative of the function P evaluated at $t = 5$ gives the rate of change of the population at time $t = 5$.

4-1-1

83.



The figure above shows the graph of the differentiable function f for $1 \leq x \leq 8$ and the secant line through the points $(1, f(1))$ and $(8, f(8))$. For how many values of x in the closed interval $[1, 8]$ does the instantaneous rate of change of f at x equal the average rate of change of f over that interval?

- (A) Zero
- (B) Three
- (C) Four
- (D) Five



Answer C

Correct. The graph of f intersects the secant line at $x = 1$, $x = a$, $x = b$, $x = c$, and $x = 8$, where $1 < a < b < c < 8$. In each of the intervals $(1, a)$, (a, b) , (b, c) , and $(c, 8)$, there is exactly one value of x at which the instantaneous rate of change of f is equal to the average rate of change of f over the interval $[1, 8]$. In other words, there are four values of x at which the slope of the line tangent to the graph of f is equal to the slope of the secant line.

4-1-1

84.

 $D(t) = 10 + 4.9 \cos\left(\frac{\pi}{6}t\right)$

The function D defined above models the depth, in feet, of the water t hours after 12 A.M. in a certain harbor. Which of the following presents the method for finding the instantaneous rate of change of the depth of the water, in feet per hour, at 6 A.M. ?

(A) $\frac{D(6)-D(0)}{6-0} = -1.633$

(B) $D'(6) = 0$

(C) $D''(6) = 1.343$

(D) $D(6) = 5.100$

**Answer B**

Correct. The derivative of D can be interpreted as the instantaneous rate of change of D with respect to t . The calculator is used to numerically calculate the derivative of D at $t = 6$.

85. The function C gives the production cost for a bakery to produce cakes of a certain type, where $C(x)$ is the cost, in dollars, to produce x of the cakes. The function M defined by $M(x) = C(x+1) - C(x)$ gives the marginal cost, in dollars, to produce cake number $x+1$. Which of the following gives the best estimate for the marginal cost, in dollars, to produce the 40th cake?

(A) $\frac{C(39)}{39}$

(B) $\frac{C(40)}{40} - \frac{C(39)}{39}$

(C) $C'(39)$

(D) $C'(40) - C'(39)$

**Answer C**

Correct. The marginal cost, in dollars, to produce the 40th cake is given by $M(39) = C(39+1) - C(39)$. This is equal to $\frac{C(40)-C(39)}{40-39}$, which is the average rate of change of the cost function C . This is an approximation for the derivative of C at $x = 39$. Therefore, the marginal cost to produce the 40th cake is best estimated by $C'(39)$.

4-1-1

86. The function $t = f(S)$ models the time, in hours, for a sample of water to evaporate as a function of the size S of the sample, measured in milliliters. What are the units for $f''(S)$?
- (A) hours per milliliter
 - (B) milliliters per hour
 - (C) hours per milliliter per milliliter ✓
 - (D) milliliters per hour per hour

Answer C

Correct. The first derivative $f'(S)$ is the limit of the difference quotient $\frac{\Delta t}{\Delta S}$. The units of the dependent variable t in the numerator are hours, and the units of the independent variable S in the denominator are milliliters. The first derivative, therefore, has units in hours per milliliter. If $y = f'(S)$, then the second derivative $f''(S)$ is the limit of the difference quotient $\frac{\Delta y}{\Delta S}$. The units of the dependent variable y in the numerator are the units for the first derivative, hours per milliliter, and the units of the independent variable S in the denominator are milliliters. The second derivative, therefore, has units in hours per milliliter per milliliter.

4-1-1

87.  A 12,000-liter tank of water is filled to capacity. At time $t = 0$, water begins to drain out of the tank at a rate modeled by $r(t)$, measured in liters per hour, where r is given by the piecewise-defined function

$$r(t) = \begin{cases} \frac{600t}{t+3} & \text{for } 0 \leq t \leq 5 \\ 1000e^{-0.2t} & \text{for } t > 5 \end{cases}$$

- (a) Is r continuous at $t = 5$? Show the work that leads to your answer.
 (b) Find the average rate at which water is draining from the tank between time $t = 0$ and time $t = 8$ hours.
 (c) Find $r'(3)$. Using correct units, explain the meaning of that value in the context of this problem.
 (d) Write, but do not solve, an equation involving an integral to find the time A when the amount of water in the tank is 9000 liters.

Part A

2 points are earned for conclusion with analysis

$$\lim_{t \rightarrow 5^-} r(t) = \lim_{t \rightarrow 5^-} \left(\frac{600t}{t+3} \right) = 375 = r(5)$$

$$\lim_{t \rightarrow 5^+} r(t) = \lim_{t \rightarrow 5^+} (1000e^{-0.2t}) = 367.879$$

Because the left-hand and right-hand limits are not equal, r is not continuous at $t = 5$.



0	1	2
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The student response earns all of the following points:

2 points are earned for conclusion with analysis

$$\lim_{t \rightarrow 5^-} r(t) = \lim_{t \rightarrow 5^-} \left(\frac{600t}{t+3} \right) = 375 = r(5)$$

$$\lim_{t \rightarrow 5^+} r(t) = \lim_{t \rightarrow 5^+} (1000e^{-0.2t}) = 367.879$$

Because the left-hand and right-hand limits are not equal, r is not continuous at $t = 5$.

Part B

1 point is earned for the integrand

1 point is earned for the limits and constant

1 point is earned for the answer

4-1-1

$$\frac{1}{8} \int_0^8 r(t) dt = \frac{1}{8} \left(\int_0^5 \frac{600t}{t+3} dt + \int_5^8 1000e^{-0.2t} dt \right)$$

$$= 258.052 \text{ or } 258.053$$



0	1	2	3
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The student response earns all of the following points:

1 point is earned for the integrand

1 point is earned for the limits and constant

1 point is earned for the answer

$$\frac{1}{8} \int_0^8 r(t) dt = \frac{1}{8} \left(\int_0^5 \frac{600t}{t+3} dt + \int_5^8 1000e^{-0.2t} dt \right)$$

$$= 258.052 \text{ or } 258.053$$

Part C

1 point is earned for $r'(3)$

1 point is earned for the meaning of $r'(3)$

$$r'(3) = 50$$

The rate at which water is draining out of the tank at time $t = 3$ hours is increasing at 50 liters/hour².



0	1	2
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The student response earns all of the following points:

1 point is earned for $r'(3)$

1 point is earned for the meaning of $r'(3)$

$$r'(3) = 50$$

4-1-1

The rate at which water is draining out of the tank at time $t = 3$ hours is increasing at 50 liters/hour².

Part D

1 point is earned for the integral

1 point is earned for the equation

$$12,000 - \int_0^A r(t)dt = 9000$$



0	1	2
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The student response earns all of the following points:

1 point is earned for the integral

1 point is earned for the equation

$$12,000 - \int_0^A r(t)dt = 9000$$

4-1-1

88.  A storm washed away sand from a beach, causing the edge of the water to get closer to a nearby road. The rate at which the distance between the road and the edge of the water was changing during the storm is modeled by $f(t) = \sqrt{t} + \cos t - 3$ meters per hour, t hours after the storm began. The edge of the water was 35 meters from the road when the storm began, and the storm lasted 5 hours. The derivative of $f(t)$ is $f'(t) = \frac{1}{2\sqrt{t}} - \sin t$.
- What was the distance between the road and the edge of the water at the end of the storm?
 - Using correct units, interpret the value $f'(4) = 1.007$ in terms of the distance between the road and the edge of the water.
 - At what time during the 5 hours of the storm was the distance between the road and the edge of the water decreasing most rapidly? Justify your answer.
 - After the storm, a machine pumped sand back onto the beach so that the distance between the road and the edge of the water was growing at a rate of $g(p)$ meters per day, where p is the number of days since pumping began. Write an equation involving an integral expression whose solution would give the number of days that sand must be pumped to restore the original distance between the road and the edge of the water.

Part A

1 point is earned for the integral

1 point is earned for the answer

$$35 + \int_0^5 f(t) dt = 26.494 \text{ or } 26.495 \text{ meters}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the integral

1 point is earned for the answer

$$35 + \int_0^5 f(t) dt = 26.494 \text{ or } 26.495 \text{ meters}$$

Part B

1 point is earned for interpretation of $f'(4)$

4-1-1

1 point is earned for units

Four hours after the storm began, the rate of change of the distance between the road and the edge of the water is increasing at a rate of 1.007 meters/hours².



0	1	2
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The student response earns all of the following points:

1 point is earned for interpretation of $f'(4)$

1 point is earned for units

Four hours after the storm began, the rate of change of the distance between the road and the edge of the water is increasing at a rate of 1.007 meters/hours².

Part C

1 point is earned if considers $f'(t) = 0$

1 point is earned for the answer

1 point is earned for justification

$f'(t) = 0$ when $t = 0.66187$ and $t = 2.84038$

The minimum of f for $0 \leq t \leq 5$ may occur at 0, 0.66187, 2.84038, or 5.

$f(0) = -2$

$f(0.66187) = -1.39760$

$f(2.84038) = -2.26963$

$f(5) = -0.48027$

The distance between the road and the edge of the water was decreasing most rapidly at time $t = 2.840$ hours after the storm began.



0	1	2	3
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4-1-1

The student response earns all of the following points:

1 point is earned if considers $f'(t) = 0$

1 point is earned for the answer

1 point is earned for justification

$f'(t) = 0$ when $t = 0.66187$ and $t = 2.84038$

The minimum of f for $0 \leq t \leq 5$ may occur at 0, 0.66187, 2.84038, or 5.

$f(0) = -2$

$f(0.66187) = -1.39760$

$f(2.84038) = -2.26963$

$f(5) = -0.48027$

The distance between the road and the edge of the water was decreasing most rapidly at time $t = 2.840$ hours after the storm began.

Part D

1 point is earned for the integral of g

1 point is earned for the answer

$$-\int_0^5 f(t)dt = \int_0^x g(p)dp$$



0	1	2
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The student response earns all of the following points:

1 point is earned for the integral of g

1 point is earned for the answer

$$-\int_0^5 f(t)dt = \int_0^x g(p)dp$$

4-1-1

89.  There is no snow on Janet's driveway when snow begins to fall at midnight. From midnight to 9 A.M., snow accumulates on the driveway at a rate modeled by $f(t) = 7te^{\cos t}$ cubic feet per hour, where t is measured in hours since midnight. Janet starts removing snow at 6 A.M. ($t = 6$). The rate $g(t)$, in cubic feet per hour, at which Janet removes snow from the driveway at time t hours after midnight is modeled by

$$g(t) = \begin{cases} 0 & \text{for } 0 \leq t < 6 \\ 125 & \text{for } 6 \leq t < 7 \\ 108 & \text{for } 7 \leq t \leq 9. \end{cases}$$

- (a) How many cubic feet of snow have accumulated on the driveway by 6 A.M.?
 (b) Find the rate of change of the volume of snow on the driveway at 8 A.M.
 (c) Let $h(t)$ represent the total amount of snow, in cubic feet, that Janet has removed from the driveway at time t hours after midnight. Express h as a piecewise-defined function with domain $0 \leq t \leq 9$.
 (d) How many cubic feet of snow are on the driveway at 9 A.M.?

Part A

1 point is earned for integral

1 point is earned for the answer

$$\text{Area} = \int_0^6 f(t) dt = 142.274 \text{ or } 142.275 \text{ cubic feet}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for integral

1 point is earned for the answer

$$\text{Area} = \int_0^6 f(t) dt = 142.274 \text{ or } 142.275 \text{ cubic feet}$$

Part B

1 point is earned for the answer

Rate of change is $f(8) - g(8) = -59.582$ or -59.583 cubic feet per hour.

4-1-1



0	1
---	---

The student response earns all of the following points:

1 point is earned for the answer

Rate of change is $f(8) - g(8) = -59.582$ or -59.583 cubic feet per hour.

Part C

1 point is earned for $0 \leq t \leq 6$

1 point is earned for $6 < t \leq 7$

1 point is earned for $7 < t \leq 9$

$$h(0) = 0$$

$$\text{For } 0 < t \leq 6, h(t) = h(0) + \int_0^t g(s) ds = 0 + \int_0^t 0 ds = 0.$$

$$\text{For } 6 < t \leq 7, h(t) = h(6) + \int_6^t g(s) ds = 0 + \int_6^t 125 ds = 125(t - 6).$$

$$\text{For } 7 < t \leq 9, h(t) = h(7) + \int_7^t g(s) ds = 125 + \int_7^t 108 ds = 125 + 108(t - 7).$$

$$\text{Thus, } h(t) = \begin{cases} 0 & \text{for } 0 \leq t \leq 6 \\ 125(t - 6) & \text{for } 6 < t \leq 7 \\ 125 + 108(t - 7) & \text{for } 7 < t \leq 9 \end{cases}$$



0	1	2	3
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The student response earns all of the following points:

1 point is earned for $0 \leq t \leq 6$

4-1-1

1 point is earned for $6 < t \leq 7$

1 point is earned for $7 < t \leq 9$

$$h(0) = 0$$

$$\text{For } 0 < t \leq 6, h(t) = h(0) + \int_0^t g(s) ds = 0 + \int_0^t 0 ds = 0.$$

$$\text{For } 6 < t \leq 7, h(t) = h(6) + \int_6^t g(s) ds = 0 + \int_6^t 125 ds = 125(t - 6).$$

$$\text{For } 7 < t \leq 9, h(t) = h(7) + \int_7^t g(s) ds = 125 + \int_7^t 108 ds = 125 + 108(t - 7).$$

$$\text{Thus, } h(t) = \begin{cases} 0 & \text{for } 0 \leq t \leq 6 \\ 125(t - 6) & \text{for } 6 < t \leq 7 \\ 125 + 108(t - 7) & \text{for } 7 < t \leq 9 \end{cases}$$

Part D

1 point is earned for integral

1 point is earned for $h(9)$

1 point is earned for the answer

$$\text{Amount of snow is } \int_0^9 f(t) dt - h(9) = 26.334 \text{ or } 26.335 \text{ cubic feet.}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for integral

1 point is earned for $h(9)$

1 point is earned for the answer

4-1-1

Amount of snow is $\int_0^9 f(t) dt - h(9) = 26.334$ or 26.335 cubic feet.

90. The temperature of a cup of coffee t minutes after being removed from a microwave oven to cool is modeled by the differentiable function L , where $L(t)$ is measured in degrees Fahrenheit ($^{\circ}\text{F}$). Which of the following is the best interpretation of the statement $L'(4) = -12$?

- (A) At time $t = 4$ minutes, the temperature of the coffee is decreasing at a rate of 12°F per minute. ✓
- (B) At time $t = 4$ minutes, the rate of change of the temperature of the coffee is decreasing at a rate of 12°F per minute per minute.
- (C) During the first four minutes, the temperature of the coffee decreases by 12°F .
- (D) During the first four minutes, the temperature of the coffee decreases at an average rate of 12°F per minute.

Answer A

Correct. In the expression $L'(4)$, the 4 represents the value of the independent variable and is therefore the number of minutes after the coffee is removed from the microwave oven. $L'(4)$, being the value of a derivative, is the rate of change of the temperature. The units for the derivative would be the units of L per unit of t , hence $^{\circ}\text{F}$ per minute. Because $L'(4)$ is negative, the temperature was decreasing at the rate of 12°F per minute.

4-1-1

91.



t (days)	$W(t)$ (°C)
0	20
3	31
6	28
9	24
12	22
15	21

The temperature, in degrees Celsius (°C), of the water in a pond is a differentiable function W of time t . The table above shows the water temperature as recorded every 3 days over a 15-day period.

- (a) Use data from the table to find an approximation for $W'(12)$. Show the computations that lead to your answer. Indicate units of measure.
- (b) Approximate the average temperature, in degrees Celsius, of the water over the time interval $0 \leq t \leq 15$ days by using a trapezoidal approximation with subintervals of length $\Delta t = 3$ days.
- (c) A student proposes the function P , given by $P(t) = 20 + 10te^{(-t/3)}$, as a model for the temperature of the water in the pond at time t , where t is measured in days and $P(t)$ is measured in degrees Celsius. Find $P'(12)$. Using appropriate units, explain the meaning of your answer in terms of water temperature.
- (d) Use the function P defined in part (c) to find the average value, in degrees Celsius, of $P(t)$ over the time interval $0 \leq t \leq 15$ days.

Part A

1 point is earned for the difference quotient

1 point is earned for the answer (with units)

Difference quotient; e.g.

$$W'(12) \approx \frac{W(15) - W(12)}{15 - 12} = -\frac{1}{3}^{\circ}\text{C/day or}$$

$$W'(12) \approx \frac{W(12) - W(9)}{12 - 9} = -\frac{2}{3}^{\circ}\text{C/day or}$$

$$W'(12) \approx \frac{W(15) - W(9)}{15 - 9} = -\frac{1}{2}^{\circ}\text{C/day}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the difference quotient

1 point is earned for the answer (with units)

4-1-1

Difference quotient; e.g.

$$W'(12) \approx \frac{W(15) - W(12)}{15 - 12} = -\frac{1}{3}^{\circ}\text{C/day or}$$

$$W'(12) \approx \frac{W(12) - W(9)}{12 - 9} = -\frac{2}{3}^{\circ}\text{C/day or}$$

$$W'(12) \approx \frac{W(15) - W(9)}{15 - 9} = -\frac{1}{2}^{\circ}\text{C/day}$$

Part B

1 point is earned for the trapezoidal method

1 point is earned for the answer

$$\frac{3}{2}(20 + 2(31) + 2(28) + 2(24) + 2(22) + 21) = 376.5$$

$$\text{Average temperature} \approx \frac{1}{15}(376.5) = 25.1^{\circ}\text{C}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the trapezoidal method

1 point is earned for the answer

$$\frac{3}{2}(20 + 2(31) + 2(28) + 2(24) + 2(22) + 21) = 376.5$$

$$\text{Average temperature} \approx \frac{1}{15}(376.5) = 25.1^{\circ}\text{C}$$

Part C

1 point is earned for $P'(12)$ (with or without units)

1 point is earned for the interpretation

$$\begin{aligned} P'(12) &= 10e^{-t/3} - \frac{10}{3}te^{-t/3} \Big|_{t=12} \\ &= -30e^{-4} = -0.549^{\circ}\text{C/day} \end{aligned}$$

4-1-1

This means the the temperature is decreasing at the rate of $0.549^{\circ}\text{C}/\text{day}$ when $t = 12$ days.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $P'(12)$ (with or without units)

1 point is earned for the interpretation

$$\begin{aligned} P'(12) &= 10e^{-t/3} - \frac{10}{3}te^{-t/3} \Big|_{t=12} \\ &= -30e^{-4} = -0.549^{\circ}\text{C}/\text{day} \end{aligned}$$

This means the the temperature is decreasing at the rate of $0.549^{\circ}\text{C}/\text{day}$ when $t = 12$ days.

Part D

1 point is earned for the integrand

1 point is earned for the limits and average value constant

1 point is earned for the answer

$$\frac{1}{15} \int_0^{15} (20 + 10te^{-t/3}) dt = 25.757^{\circ}\text{C}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the integrand

1 point is earned for the limits and average value constant

1 point is earned for the answer

4-1-1

$$\frac{1}{15} \int_0^{15} (20 + 10te^{-t/3}) dt = 25.757^\circ\text{C}$$

4-1-1

92.



t (minutes)	0	4	9	15	20
$W(t)$ (degrees Fahrenheit)	55.0	57.1	61.8	67.9	71.0

The temperature of water in a tub at time t is modeled by a strictly increasing, twice-differentiable function W , where $W(t)$ is measured in degrees Fahrenheit and t is measured in minutes. At time $t = 0$, the temperature of the water is 55°F . The water is heated for 30 minutes, beginning at time $t = 0$. Values of $W(t)$ at selected times t for the first 20 minutes are given in the table above.

(a) Use the data in the table to estimate $W'(12)$. Show the computations that lead to your answer. Using correct units, interpret the meaning of your answer in the context of this problem.

(b) Use the data in the table to evaluate $\int_0^{20} W'(t) dt$. Using correct units, interpret the meaning of $\int_0^{20} W'(t) dt$ in the context of this problem.

(c) For $0 \leq t \leq 20$, the average temperature of the water in the tub is $\frac{1}{20} \int_0^{20} W(t) dt$. Use a left Riemann sum with the four subintervals indicated by the data in the table to approximate $\frac{1}{20} \int_0^{20} W(t) dt$. Does this approximation overestimate or underestimate the average temperature of the water over these 20 minutes? Explain your reasoning.

(d) For $20 \leq t \leq 25$, the function W that models the water temperature has first derivative given by $W'(t) = 0.4\sqrt{t} \cos(0.06t)$. Based on the model, what is the temperature of the water at time $t = 25$?

Part A

1 point is earned for the estimation

1 point is earned for the interpretation with units

$$W'(12) \approx \frac{W(15) - W(9)}{15 - 9} = \frac{67.9 - 61.8}{6}$$

$$= 1.017 \text{ (or } 1.016)$$

The water temperature is increasing at a rate of approximately 1.017°F per minute at time $t = 12$ minutes.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the estimation

1 point is earned for the interpretation with units

4-1-1

$$W'(12) \approx \frac{W(15) - W(9)}{15 - 9} = \frac{67.9 - 61.8}{6}$$

$$= 1.017 \text{ (or } 1.016)$$

The water temperature is increasing at a rate of approximately 1.017°F per minute at time $t = 12$ minutes.

Part B

1 point is earned for the value

1 point is earned for the interpretation with units

$$\int_0^{20} W'(t) dt = W(20) - W(0) = 71.0 - 55.0 = 16$$

The water has warmed by 16°F over the interval from $t = 0$ to $t = 20$ minutes.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the value

1 point is earned for the interpretation with units

$$\int_0^{20} W'(t) dt = W(20) - W(0) = 71.0 - 55.0 = 16$$

The water has warmed by 16°F over the interval from $t = 0$ to $t = 20$ minutes.

Part C

1 point is earned for the left Riemann sum

1 point is earned for the approximation

1 point is earned for underestimate with reason

4-1-1

$$\begin{aligned}
 \frac{1}{20} \int_0^{20} W(t) dt &\approx \frac{1}{20} (4 \cdot W(0) + 5 \cdot W(4) + 6 \cdot W(9) + 5 \cdot W(15)) \\
 &= \frac{1}{20} (4 \cdot 55.0 + 5 \cdot 57.1 + 6 \cdot 61.8 + 5 \cdot 67.9) \\
 &= \frac{1}{20} \cdot 1215.8 = 60.79
 \end{aligned}$$

This approximation is an underestimate, because a left Riemann sum is used and the function W is strictly increasing.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the left Riemann sum

1 point is earned for the approximation

1 point is earned for underestimate with reason

$$\begin{aligned}
 \frac{1}{20} \int_0^{20} W(t) dt &\approx \frac{1}{20} (4 \cdot W(0) + 5 \cdot W(4) + 6 \cdot W(9) + 5 \cdot W(15)) \\
 &= \frac{1}{20} (4 \cdot 55.0 + 5 \cdot 57.1 + 6 \cdot 61.8 + 5 \cdot 67.9) \\
 &= \frac{1}{20} \cdot 1215.8 = 60.79
 \end{aligned}$$

This approximation is an underestimate, because a left Riemann sum is used and the function W is strictly increasing.

Part D

1 point is earned for the integrand

1 point is earned for the answer

$$W(25) = 71.0 + \int_{20}^{25} W'(t) dt$$

$$= 71.0 + 2.043155 = 73.043$$

4-1-1



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the integrand

1 point is earned for the answer

$$W(25) = 71.0 + \int_{20}^{25} W'(t) dt$$

$$= 71.0 + 2.043155 = 73.043$$

4-1-1

93.  The wind chill is the temperature, in degrees Fahrenheit ($^{\circ}\text{F}$), a human feels based on the air temperature, in degrees Fahrenheit, and the wind velocity v , in miles per hour (mph). If the air temperature is 32°F , then the wind chill is given by $W(v) = 55.6 - 22.1v^{0.16}$ and is valid for $5 \leq v \leq 60$.
- (a) Find $W'(20)$. Using correct units, explain the meaning of $W'(20)$ in terms of the wind chill.
- (b) Find the average rate of change of W over the interval $5 \leq v \leq 60$. Find the value of v at which the instantaneous rate of change of W is equal to the average rate of change of W over the interval $5 \leq v \leq 60$.
- (c) Over the time interval $0 \leq t \leq 4$ hours, the air temperature is a constant 32°F . At time $t = 0$, the wind velocity is $v = 20$ mph. If the wind velocity increases at a constant rate of 5 mph per hour, what is the rate of change of the wind chill with respect to time at $t = 3$ hours? Indicate units of measure.

Part A

1 point is earned for the value

1 point is earned for the explanation

$$W'(20) = -22.1 \cdot 0.16 \cdot 20^{-0.84} = -0.285 \text{ or } -0.286$$

When $v = 20$ mph, the wind chill is decreasing at $0.286^{\circ}\text{F}/\text{mph}$.

1 point is earned for units in (a) and (c)

Units of $^{\circ}\text{F}/\text{mph}$ in (a) and $^{\circ}\text{F}/\text{hr}$ in (c)



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the value

1 point is earned for the explanation

$$W'(20) = -22.1 \cdot 0.16 \cdot 20^{-0.84} = -0.285 \text{ or } -0.286$$

When $v = 20$ mph, the wind chill is decreasing at $0.286^{\circ}\text{F}/\text{mph}$.

1 point is earned for units in (a) and (c)

Units of $^{\circ}\text{F}/\text{mph}$ in (a) and $^{\circ}\text{F}/\text{hr}$ in (c)

4-1-1

Part B

1 point is earned for the average rate of change.

1 point is earned for $W'(v)$ = average rate of change

1 point is earned for the value of v .

The average rate of change of W over the interval $5 \leq v \leq 60$ is $\frac{W(60) - W(5)}{60 - 5} = -0.253$ or -0.254 .

$$W'(v) = \frac{W(60) - W(5)}{60 - 5} \text{ when } v = 23.011.$$



0	1	2	3
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The student response earns all of the following points:

1 point is earned for the average rate of change.

1 point is earned for $W'(v)$ = average rate of change

1 point is earned for the value of v .

The average rate of change of W over the interval $5 \leq v \leq 60$ is $\frac{W(60) - W(5)}{60 - 5} = -0.253$ or -0.254 .

$$W'(v) = \frac{W(60) - W(5)}{60 - 5} \text{ when } v = 23.011.$$

Part C

1 point is earned for $\frac{dv}{dt} = 5$

1 point is earned if uses $v(3) = 35$ or uses $v(5) = 20 + 5t$

1 point is earned for answer

4-1-1

$$\left. \frac{dW}{dt} \right|_{t=3} = \left(\frac{dW}{dv} \cdot \frac{dv}{dt} \right) \bigg|_{t=3} = W'(35) \cdot 5 = -0.892^\circ\text{F/hr}$$

OR

$$W = 55.6 - 22.1(20 + 5t)^{0.16}$$

$$\left. \frac{dW}{dt} \right|_{t=3} = -0.892^\circ\text{F/hr}$$

1 point is earned for units in (a) and (c)

Units of $^\circ\text{F}/\text{mph}$ in (a) and $^\circ\text{F}/\text{hr}$ in (c)

0	1	2	3	4
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The student response earns all of the following points:

1 point is earned for $\frac{dv}{dt} = 5$

1 point is earned if uses $v(3) = 35$ or uses $v(5) = 20 + 5t$

1 point is earned for answer

$$\left. \frac{dW}{dt} \right|_{t=3} = \left(\frac{dW}{dv} \cdot \frac{dv}{dt} \right) \bigg|_{t=3} = W'(35) \cdot 5 = -0.892^\circ\text{F/hr}$$

OR

$$W = 55.6 - 22.1(20 + 5t)^{0.16}$$

$$\left. \frac{dW}{dt} \right|_{t=3} = -0.892^\circ\text{F/hr}$$

1 point is earned for units in (a) and (c)

4-1-1

Units of $^{\circ}\text{F}/\text{mph}$ in (a) and $^{\circ}\text{F}/\text{hr}$ in (c)